

MERGED PPTS OF BT605

VU Medical Zone

EFFORT BY: Amaan Khan

BIOSAFETY & BIOETHICS

LESSONS (1 – 87 Midterm) FOR



Bioethics

Lesson # 1



Bioethics

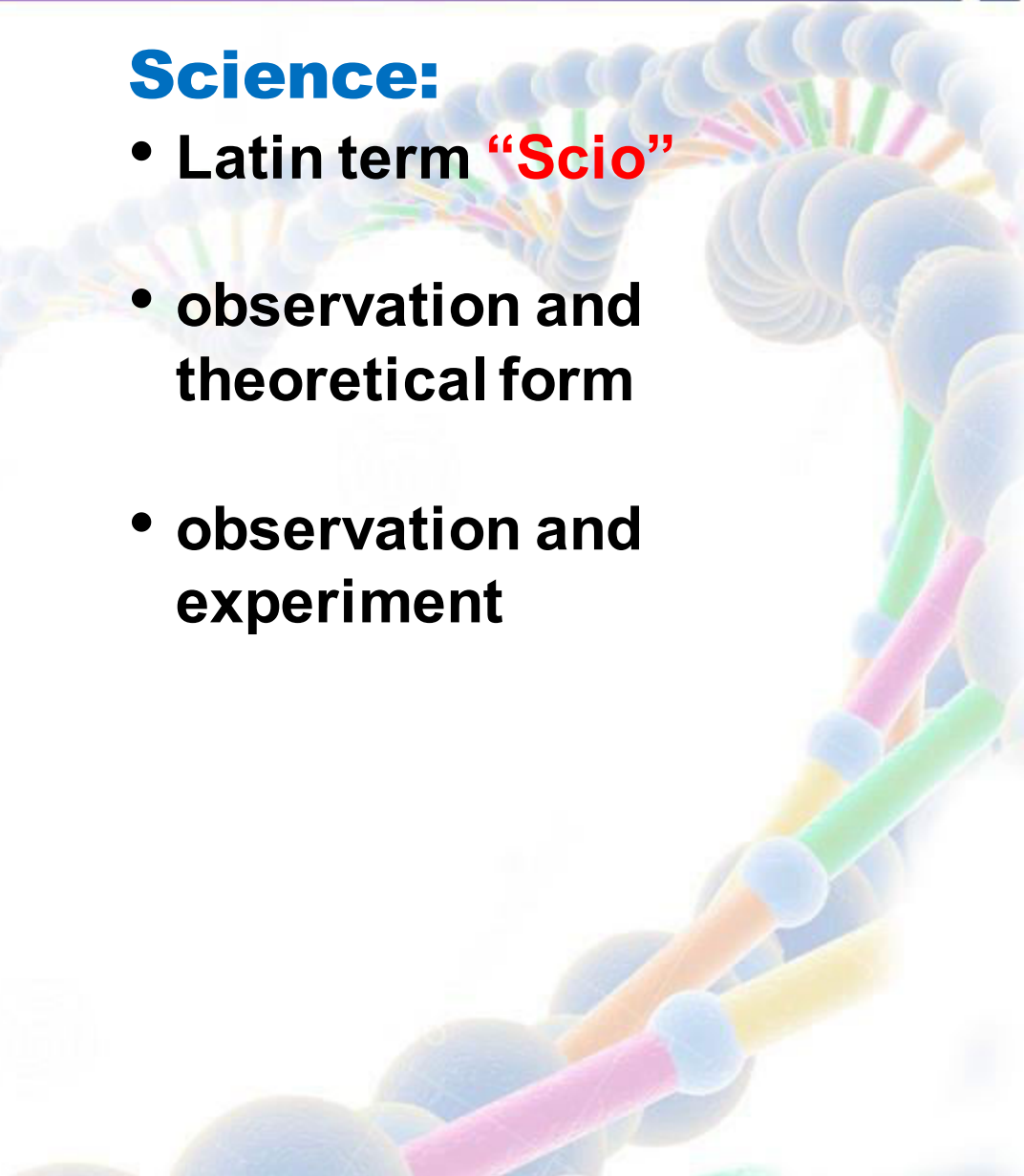


**Science, Ethics
and Values**

Science, Ethics and Values

Science:

- Latin term “Scio”
- observation and theoretical form
- observation and experiment



Science, Ethics and Values

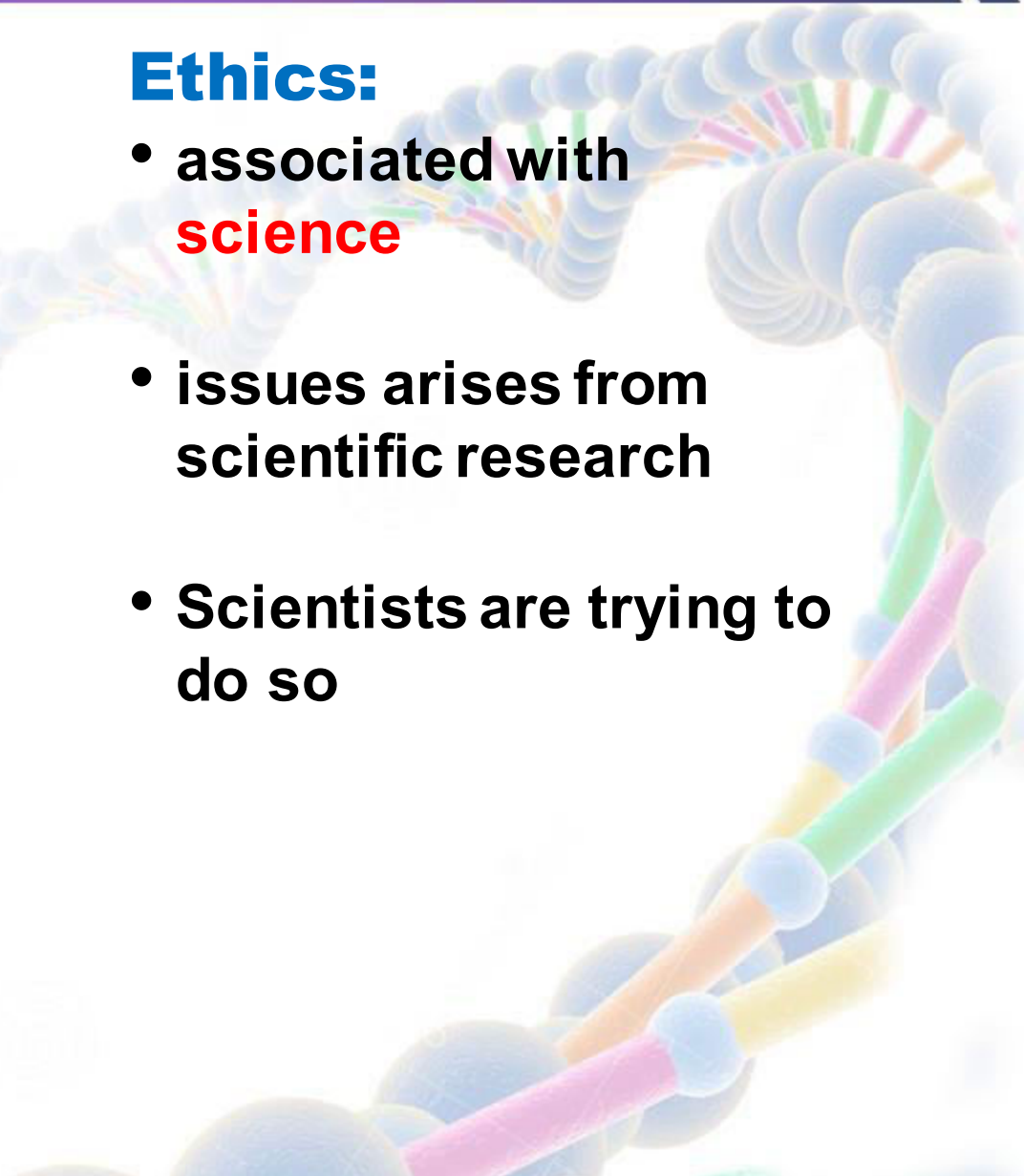
Science:

- investigation of the universe by a **set of methodologies**
- progress made by scientific methods
- Step-wise, not a single activity, not a value free

Science, Ethics and Values

Ethics:

- associated with **science**
- issues arises from scientific research
- Scientists are trying to do so



Science, Ethics and Values

Values:

- Science has entered in to our **daily lives**
- Proper resource allocation
- Reflects what society at the time deems to be valuable



Bioethics

Lesson # 2



Bioethics



Attitudes to Science

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Attitudes to Science

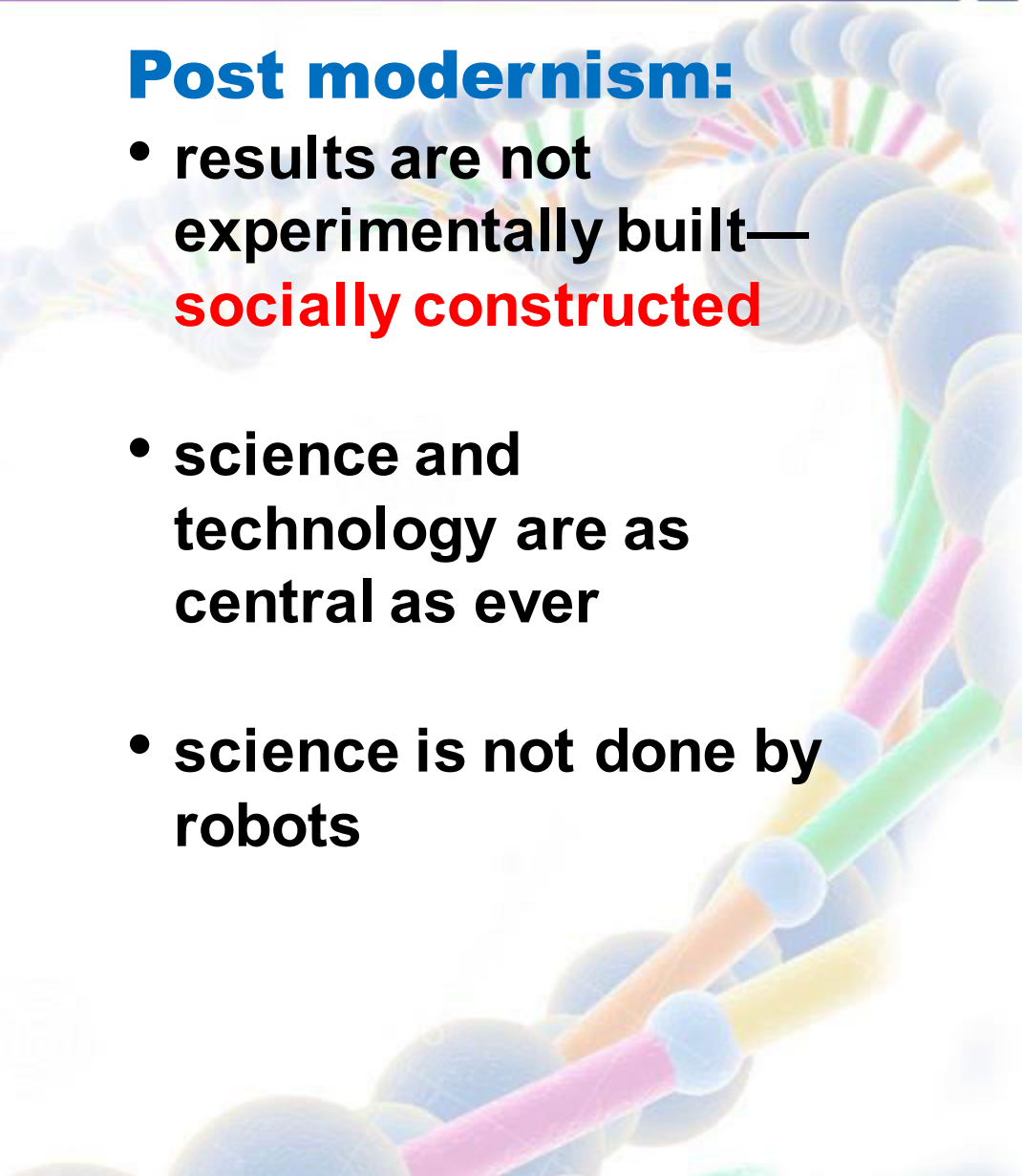
Attitudes:

- Nietzsche “**God is dead**”
- Wittgenstein “scientific terms---interpreted in social context”
- contribution to the economy growth

Attitudes to Science

Post modernism:

- results are not experimentally built—**socially constructed**
- science and technology are as central as ever
- science is not done by robots



Attitudes to Science

Understand the public:

- define science, think about **research and policies**
- Impact on the public/
explore as a subject or career.
- scientific institutions---
public confidence

Bioethics

Lesson # 3



Bioethics

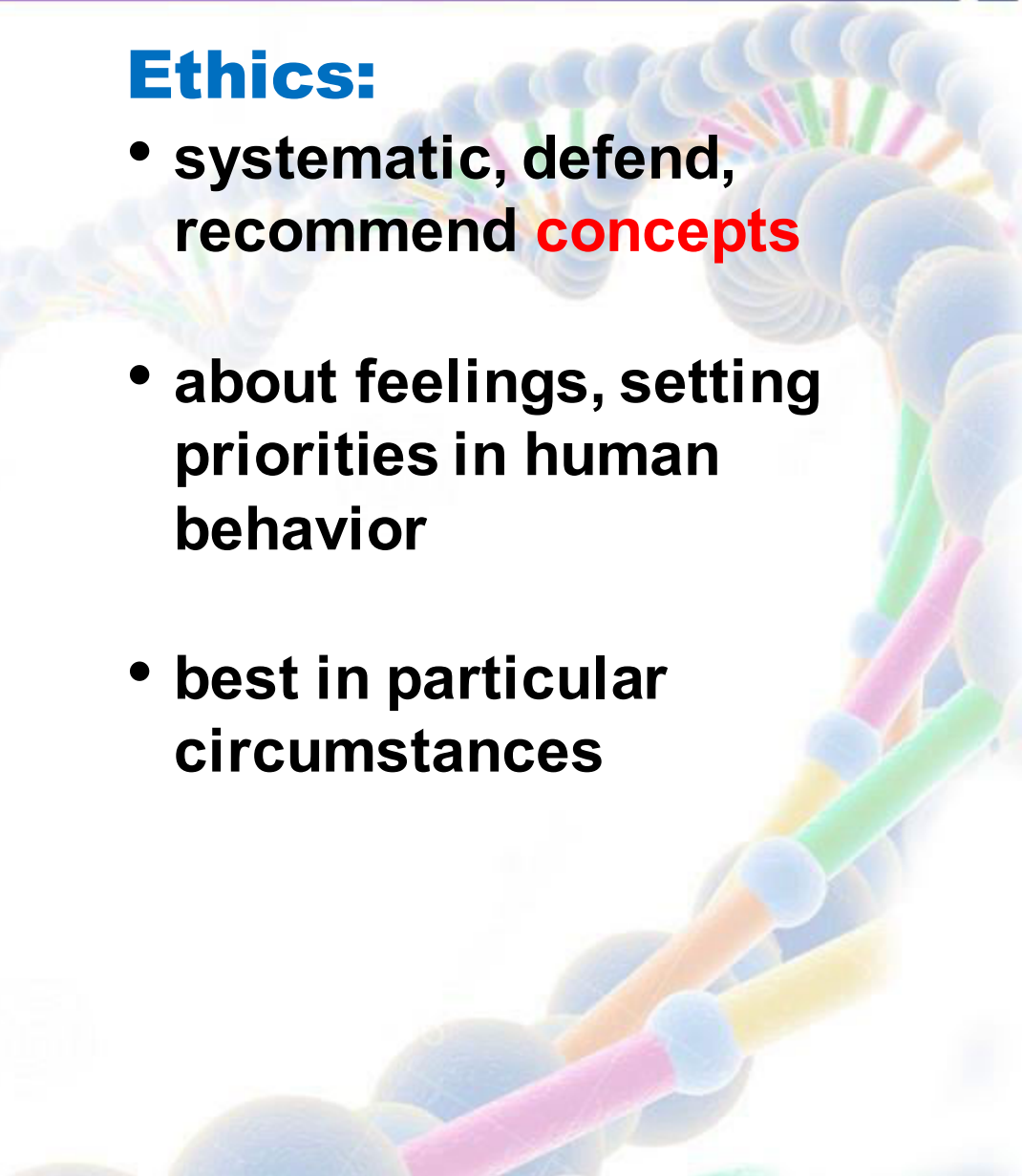


What is Ethics?

What is Ethics?

Ethics:

- systematic, defend, recommend **concepts**
- about feelings, setting priorities in human behavior
- best in particular circumstances



What is Ethics?

Religion:

- set high ethical **standards**
- intense motivation for ethical behavior
- has to do with religious beliefs but not confined to religion

What is Ethics?

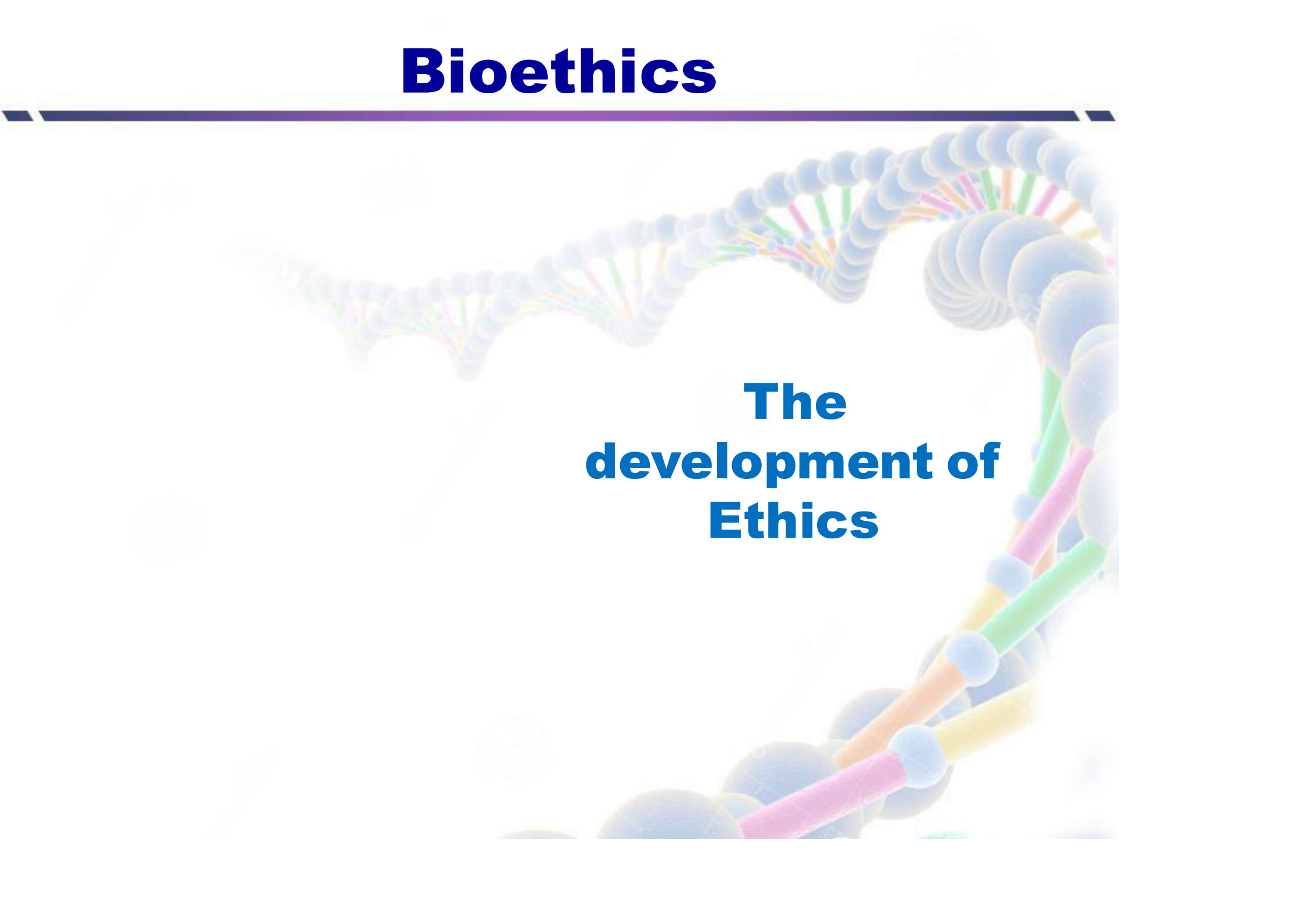
Types of Ethics:

- Meta ethics-----
theoretical meaning
- normative ethics-----
practical meaning
- applied ethics--domain
of action

Bioethics

Lesson # 4

Bioethics



The development of Ethics

The development of Ethics

Ancient Greece:

- Plato “**everything has its own form**”
- Aristotle---function
- happiness is about expressing a virtue

The development of Ethics

Jewish/Christian thinking:

- God had spoken through his ten **commandments**
- codes of conduct
- follow the life of Jesus Christ

The development of Ethics

Natural Law:

- Thomas broke -----
Aristotle idea
- function of every part
of the human body
- basis of catholic
ethical teaching

The development of Ethics

Consequentialism:

- modern science—
Newton/ Galileo
- Consequences of the action
- Example: Saddam Hussein

Bioethics

Lesson # 5



Bioethics



**The growth
of
bioethics**

The growth of bioethics

Origin of notion of bioethics:

- Potter **“Bioethics, the survival of science”**
- Callahan’s “Bioethics as a discipline”
- Kennedy institute of Ethics

The growth of bioethics

Origin of academic discipline and institutionalization:

- goes **hand in hand**
- Informed consent
- follow the criteria all over the world



The growth of bioethics

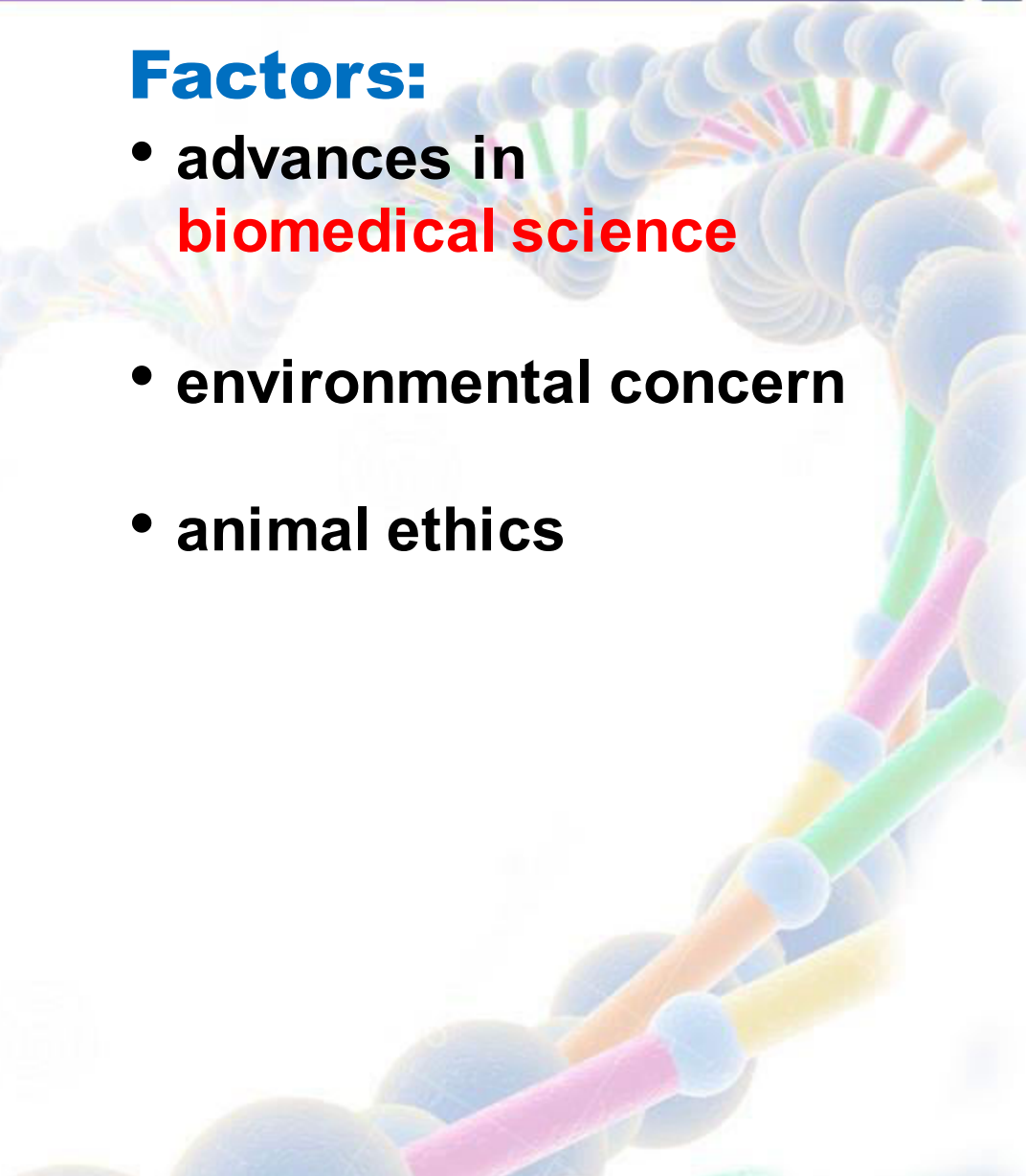
Origin of bioethics a Phenomenon:

- idea of protecting **environment**
- virtue of public concern
- development of new technologies

The growth of bioethics

Factors:

- advances in **biomedical science**
- environmental concern
- animal ethics



Bioethics

Lesson # 6



Bioethics

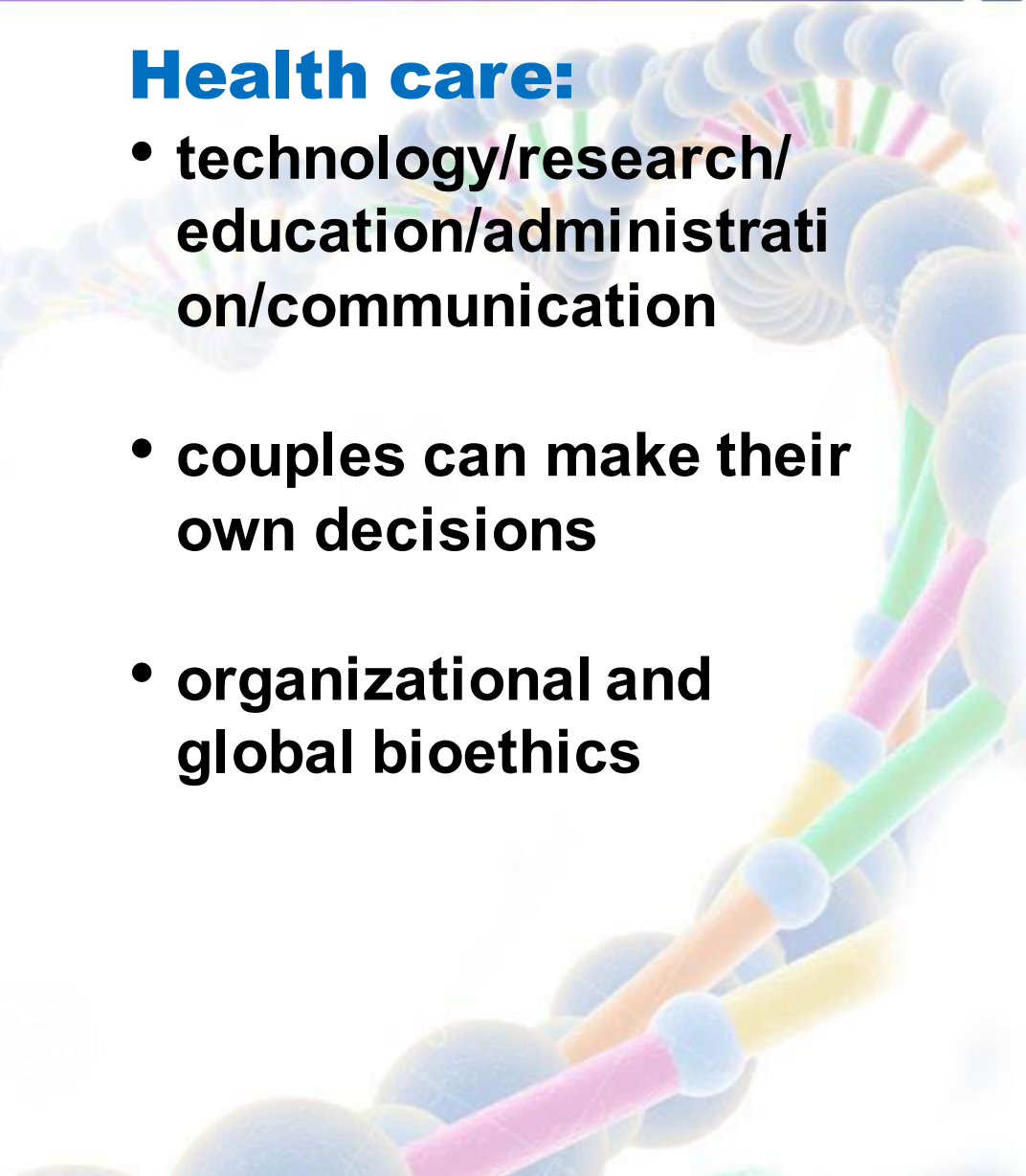


Bioethics in 21st Century

Bioethics in 21 century

Health care:

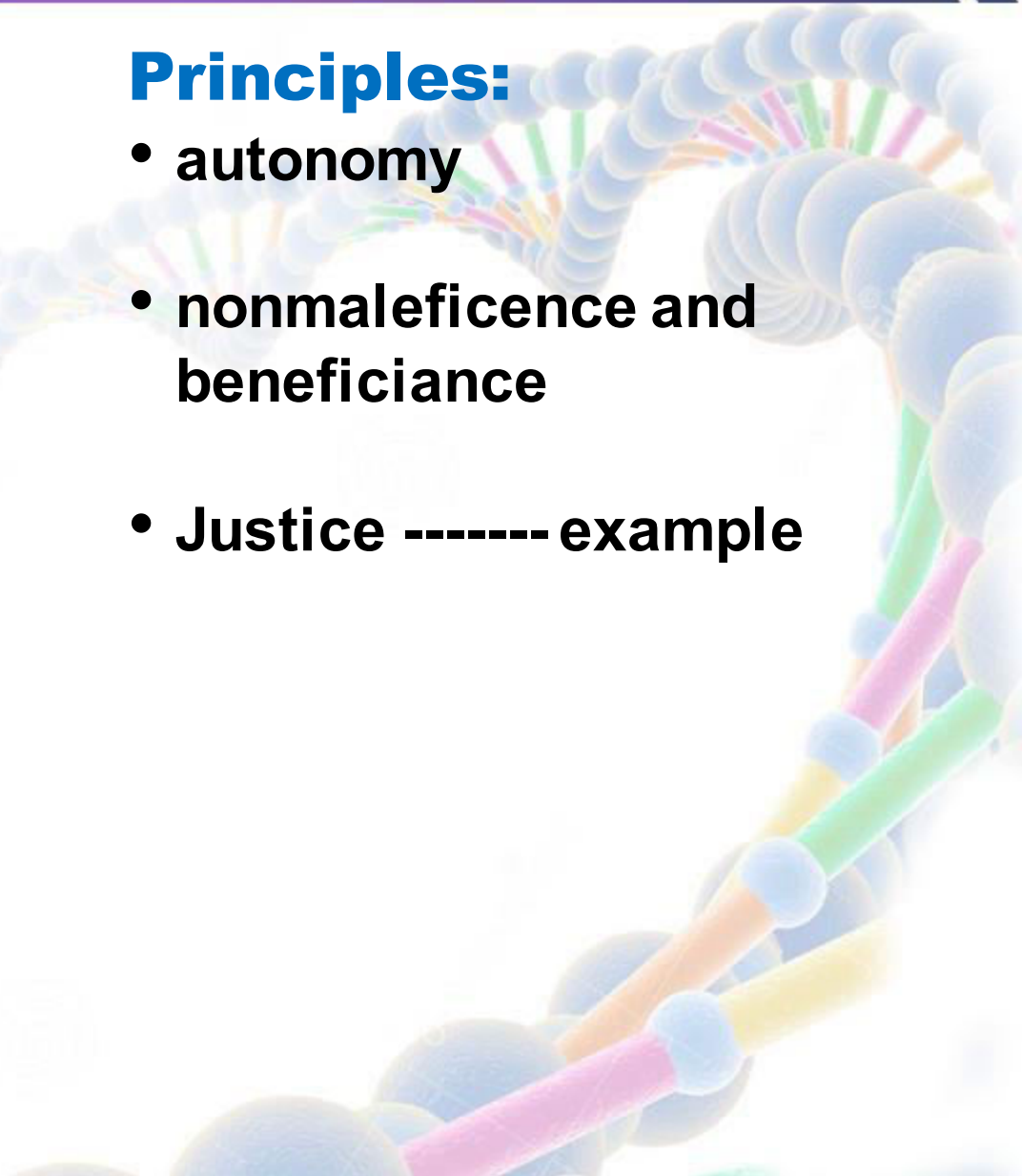
- technology/research/
education/administrati
on/communication
- couples can make their
own decisions
- organizational and
global bioethics



Bioethics in 21 century

Principles:

- autonomy
- nonmaleficence and beneficance
- Justice ----- example



Bioethics

Lesson # 7



Bioethics



Making ethical decisions

3/1/2018

Making ethical decisions

decisions:

- long and complex **history**
- ethics is about decisions and making choices
- our daily conversation has an ethical component

Making ethical decisions

Virtue ethics:

- What is most **virtuous**?
- expression of the individuals than keeping the rules
- we become virtuous by practising virtue

Making ethical decisions

Virtue ethics:

- What is most **virtuous**?
- expression of the individuals than keeping the rules
- we become virtuous by practising virtue

Bioethics

Lesson # 8

3/1/2018

Bioethics



Place of humans in nature

3/1/2018

Place of humans in nature

Human beings:

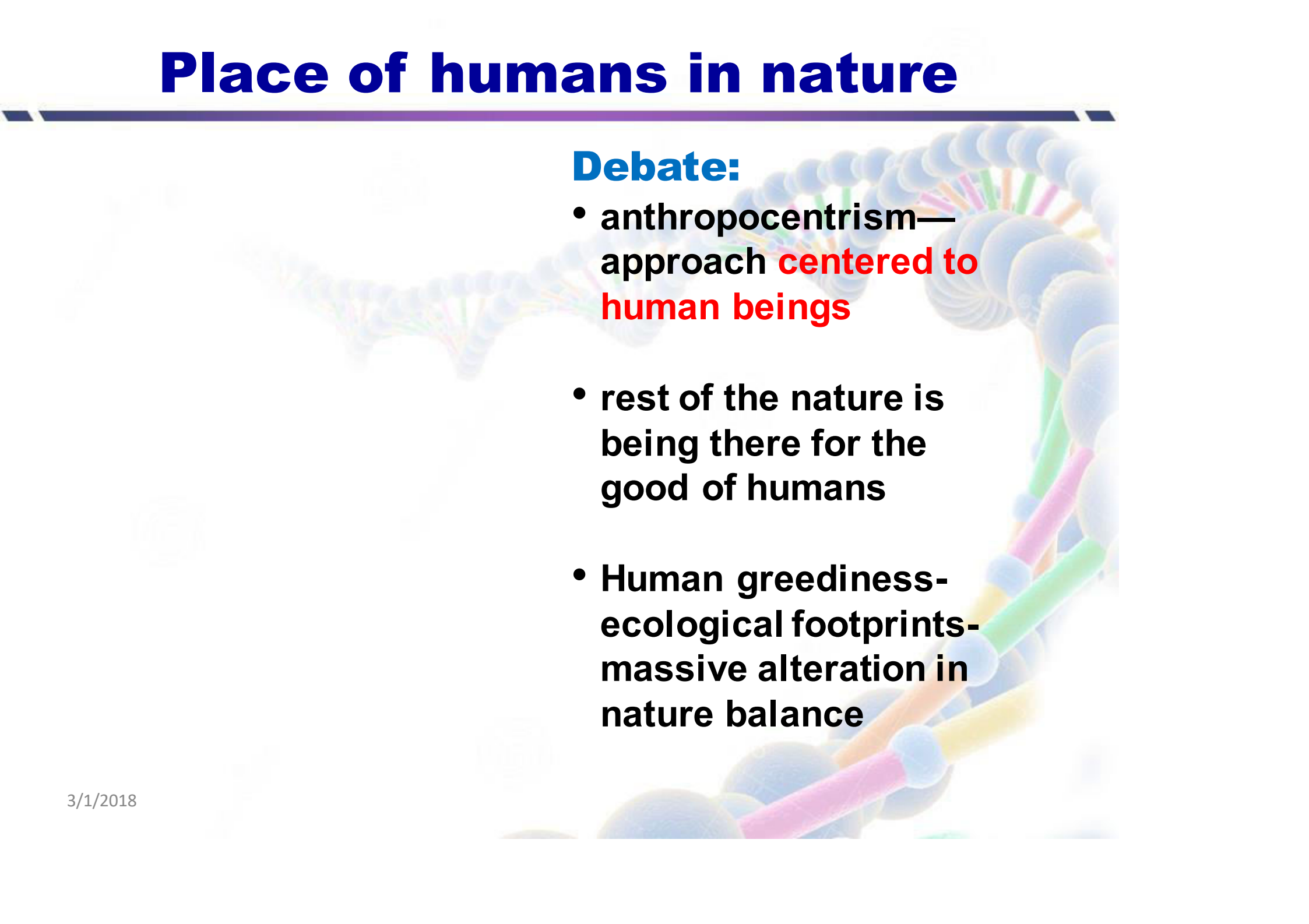
- humans occupy a unique **position**
- brain power--- aspects of nature
- make extensive use of natural resources
- every element of nature is not in human control

Place of humans in nature

Developed societies:

- humans are the part of **natural order**
- moral relationship between humans and natural environment
- misuse of nature for our needs

Place of humans in nature



Debate:

- anthropocentrism—
approach **centered to
human beings**
- rest of the nature is
being there for the
good of humans
- Human greediness-
ecological footprints-
massive alteration in
nature balance

Place of humans in nature

Ecocentrism:

- Centered--**ecosystem**
- Includes soil, water, air, forest, mountains
- biotic components are dependent on non-living
- Human ill treatment---altering nature—threatening for humans

The place of humans in nature

Ecocentrism:

- humans are responsible for all **biological life**
- ability of thinking and perceiving world as a whole
- Rolston---ecosystem is much more than the sum of its parts

The place of humans in nature

biocentrism:

- centered--**biosphere**
- humans are one of many millions of species
- doesn't prevent humans--using natural resources if other living organisms doesn't matter--consequentialism

The place of humans in nature

Theocentrism:

- God centered approach to the **world**
- Comes from the religious faith---God is a creator
- Environment—belongs to God

Bioethics

Lesson # 9

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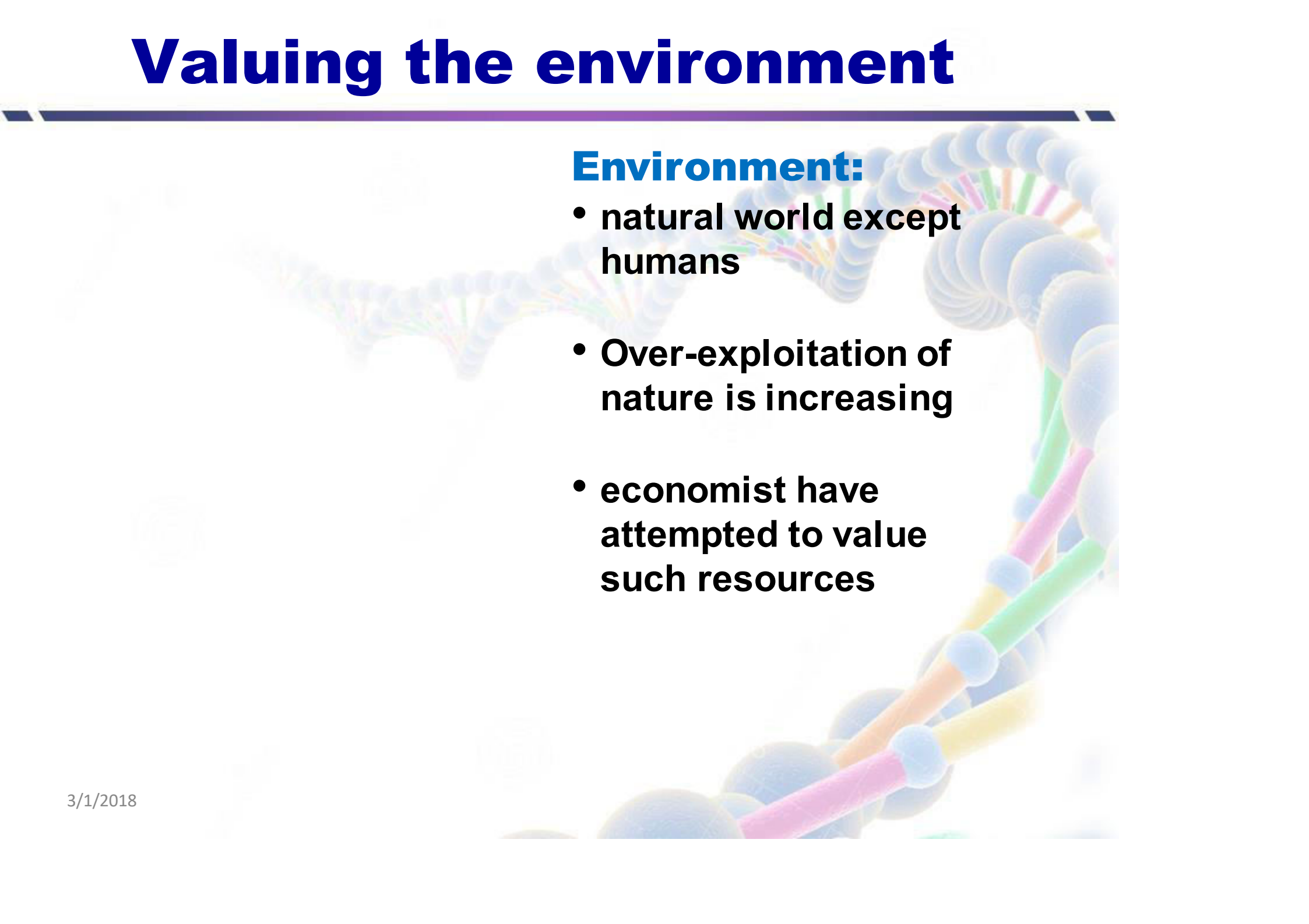
Bioethics



Valuing the environment

3/1/2018

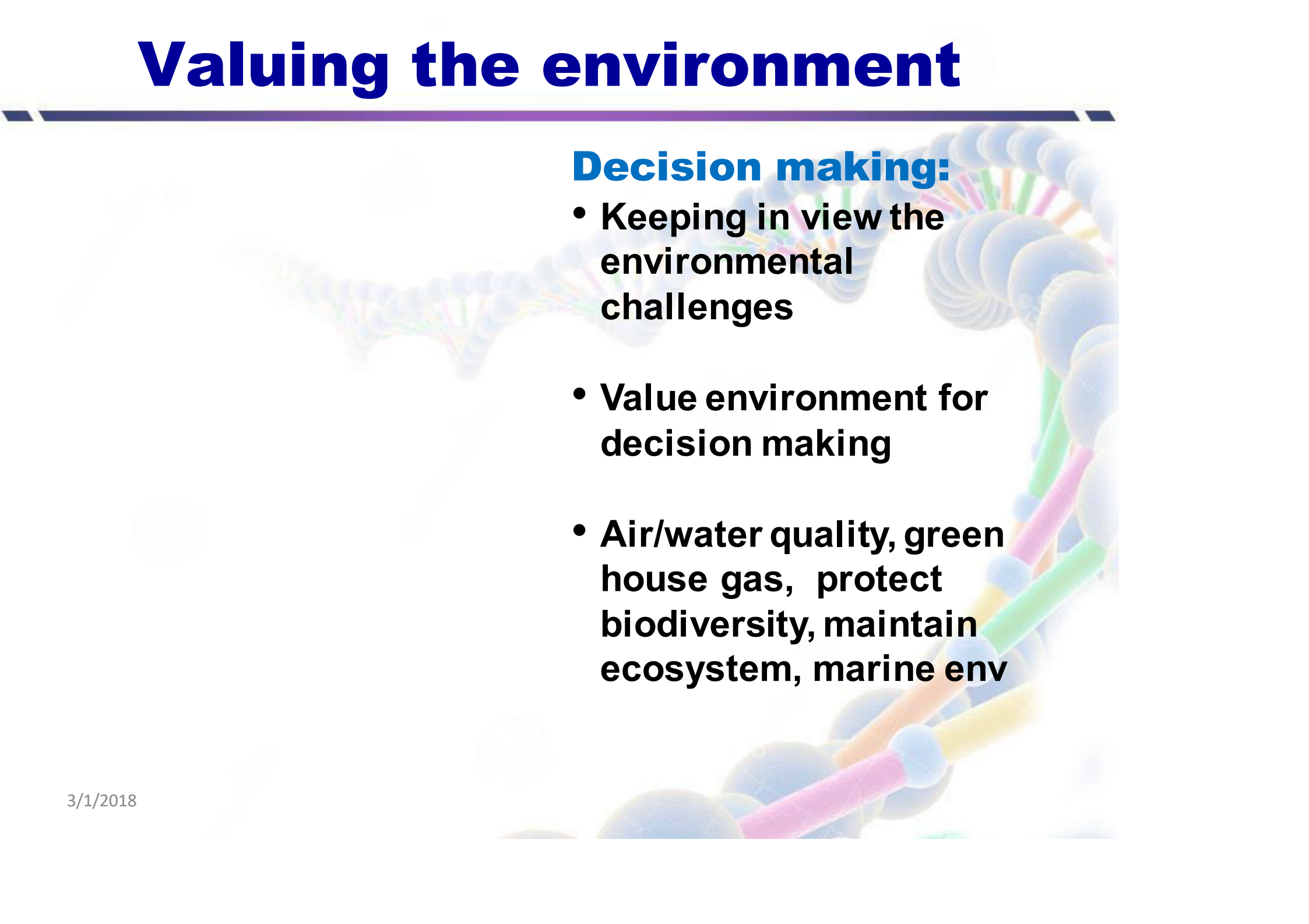
Valuing the environment



Environment:

- natural world except humans
- Over-exploitation of nature is increasing
- economist have attempted to value such resources

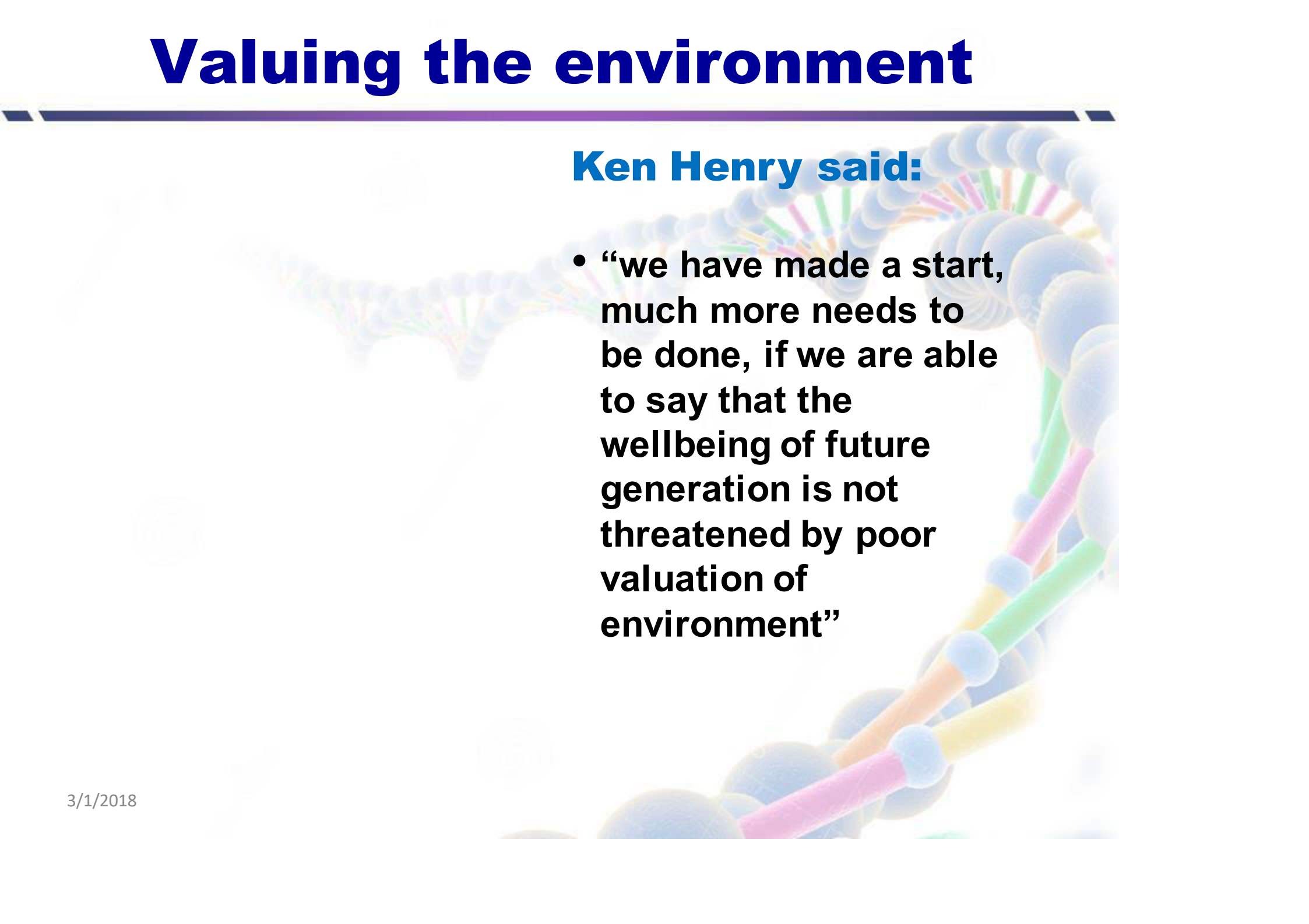
Valuing the environment



Decision making:

- Keeping in view the environmental challenges
- Value environment for decision making
- Air/water quality, green house gas, protect biodiversity, maintain ecosystem, marine env

Valuing the environment



Ken Henry said:

- “we have made a start, much more needs to be done, if we are able to say that the wellbeing of future generation is not threatened by poor valuation of environment”

Valuing the environment

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- “we have made a start, much more needs to be done, if we are able to say that the wellbeing of future generation is not threatened by poor valuation of environment”

Valuing the environment

Intrinsic value:

- value that environment and living forms have their **own rights**
- intrinsic value of birds/green and pleasant places have their own values
- mainly involves religion

Valuing the environment

Instrumental value:

- Supply of human's material **needs**
- Actual and potential use in supplying resources for human living
- debate

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Lesson # 10

3/1/2018

Bioethics



Themes in environmental ethics

3/1/2018

Themes in environmental ethics

Population load:

- human population has put and putting lots of load----- environment
- pressure on the natural resources---humans to live
- activity of humans can damage the environment

Themes in environmental ethics

Rio Declaration:

- Two current themes in environmental ethics (1992)
- precautionary principle
- sustainability

Themes in environmental ethics

Precautionary principle:

- Old concept---applied to different areas
- Deontological
- Consequentialist ethical thinking

Themes in environmental ethics

Precautionary principle:

- Old concept---applied to different areas
- Deontological
- Consequentialist ethical thinking

Themes in environmental ethics

Sustainability:

- Activity should be conducted repeatedly without accumulating environmental damage
- Agriculture has no lasting affect on the environment
- Local/large level

Bioethics

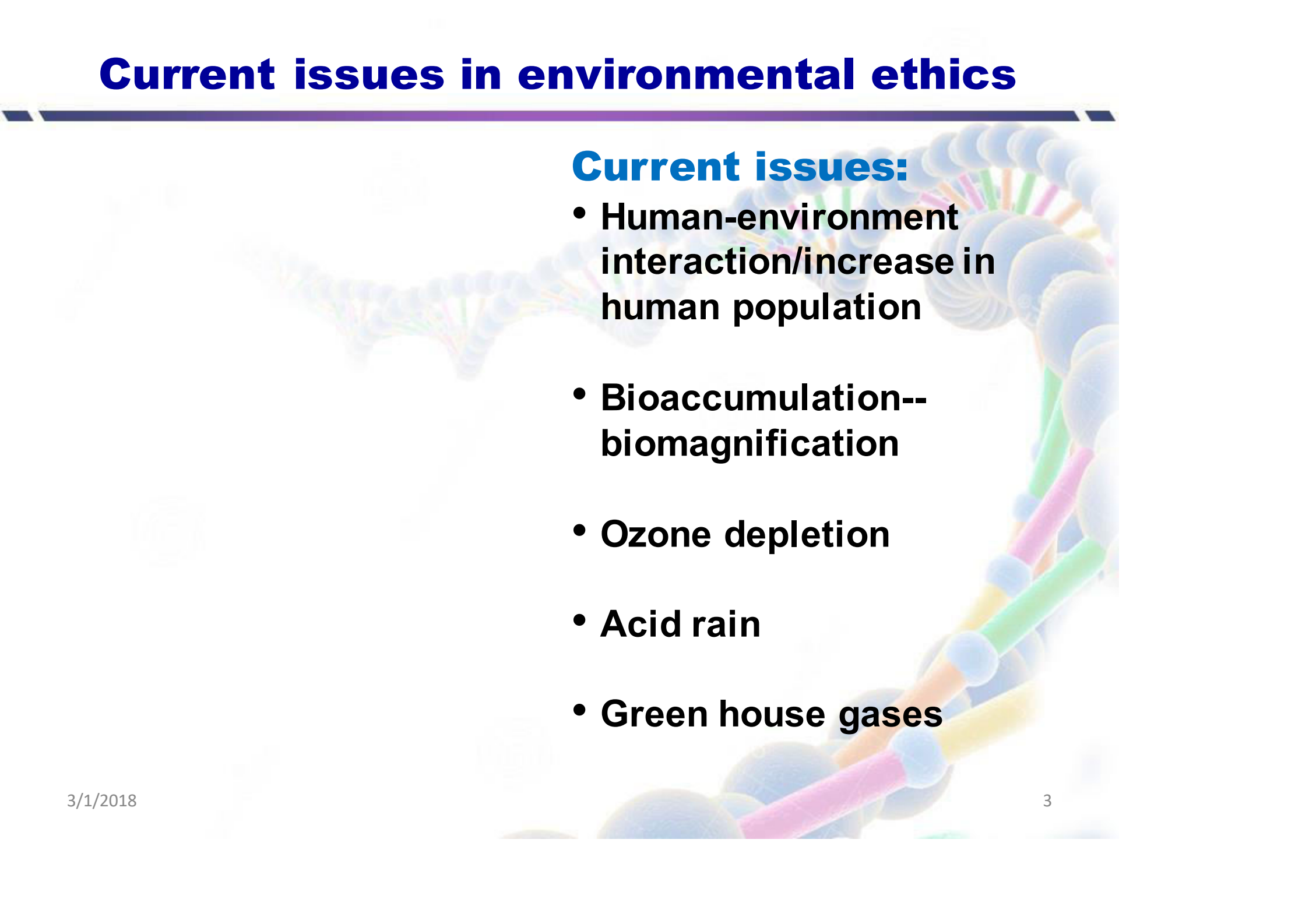
Lesson # 11

Bioethics



Current issues in environmental ethics

Current issues in environmental ethics



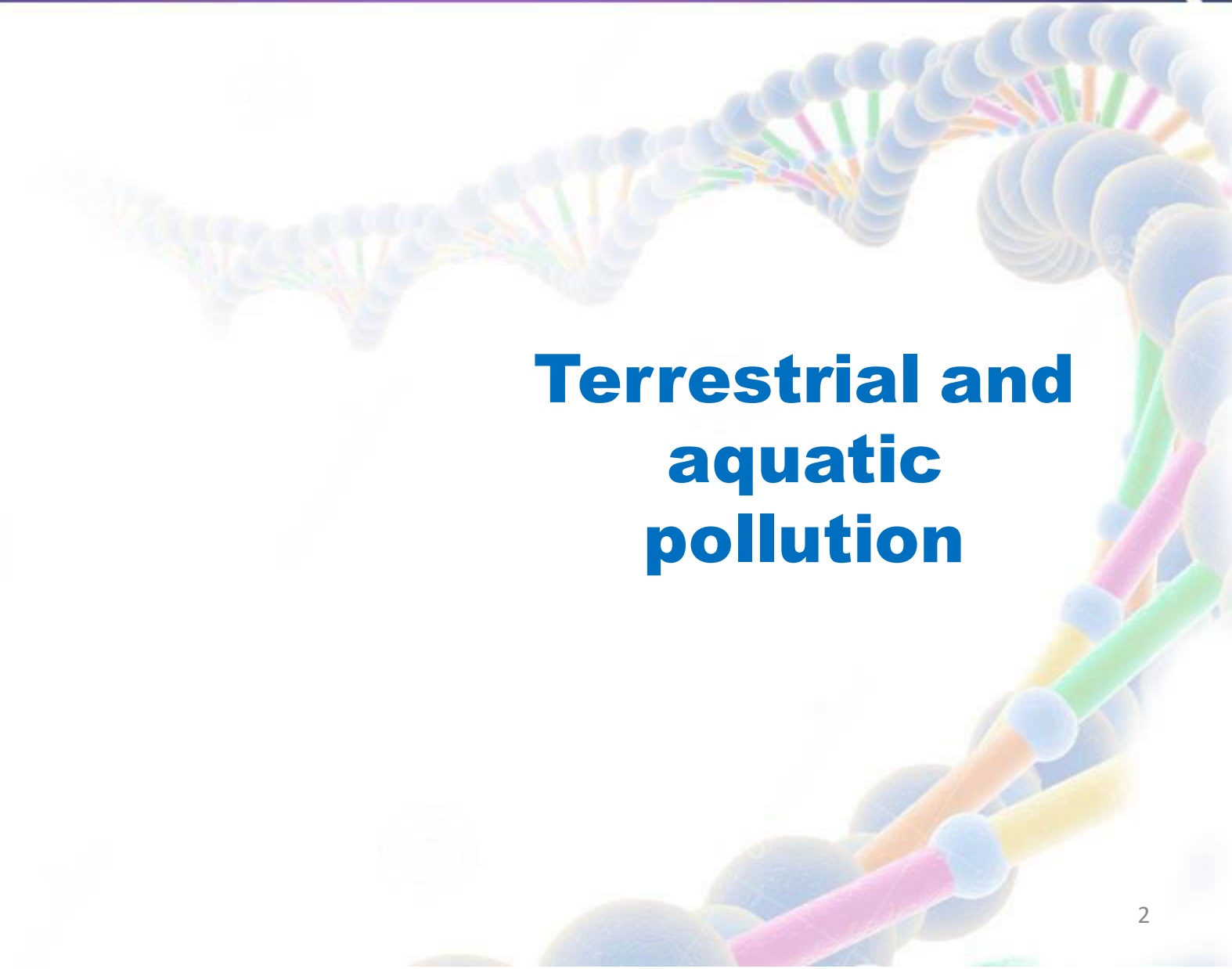
Current issues:

- Human-environment interaction/increase in human population
- Bioaccumulation--biomagnification
- Ozone depletion
- Acid rain
- Green house gases

Bioethics

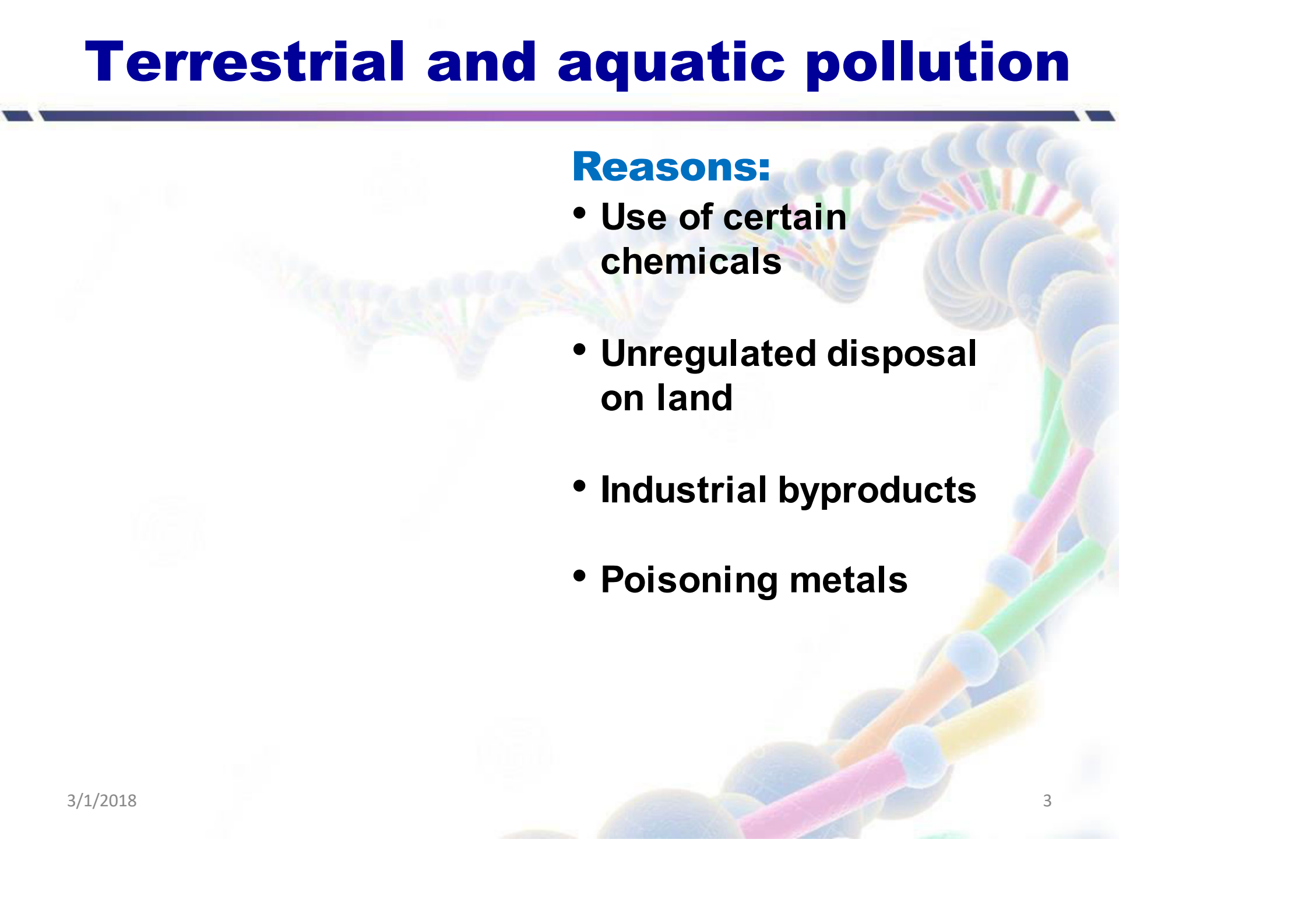
Lesson # 12

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Terrestrial and aquatic pollution

Terrestrial and aquatic pollution



Reasons:

- Use of certain chemicals
- Unregulated disposal on land
- Industrial byproducts
- Poisoning metals

Terrestrial and aquatic pollution

Silent spring:

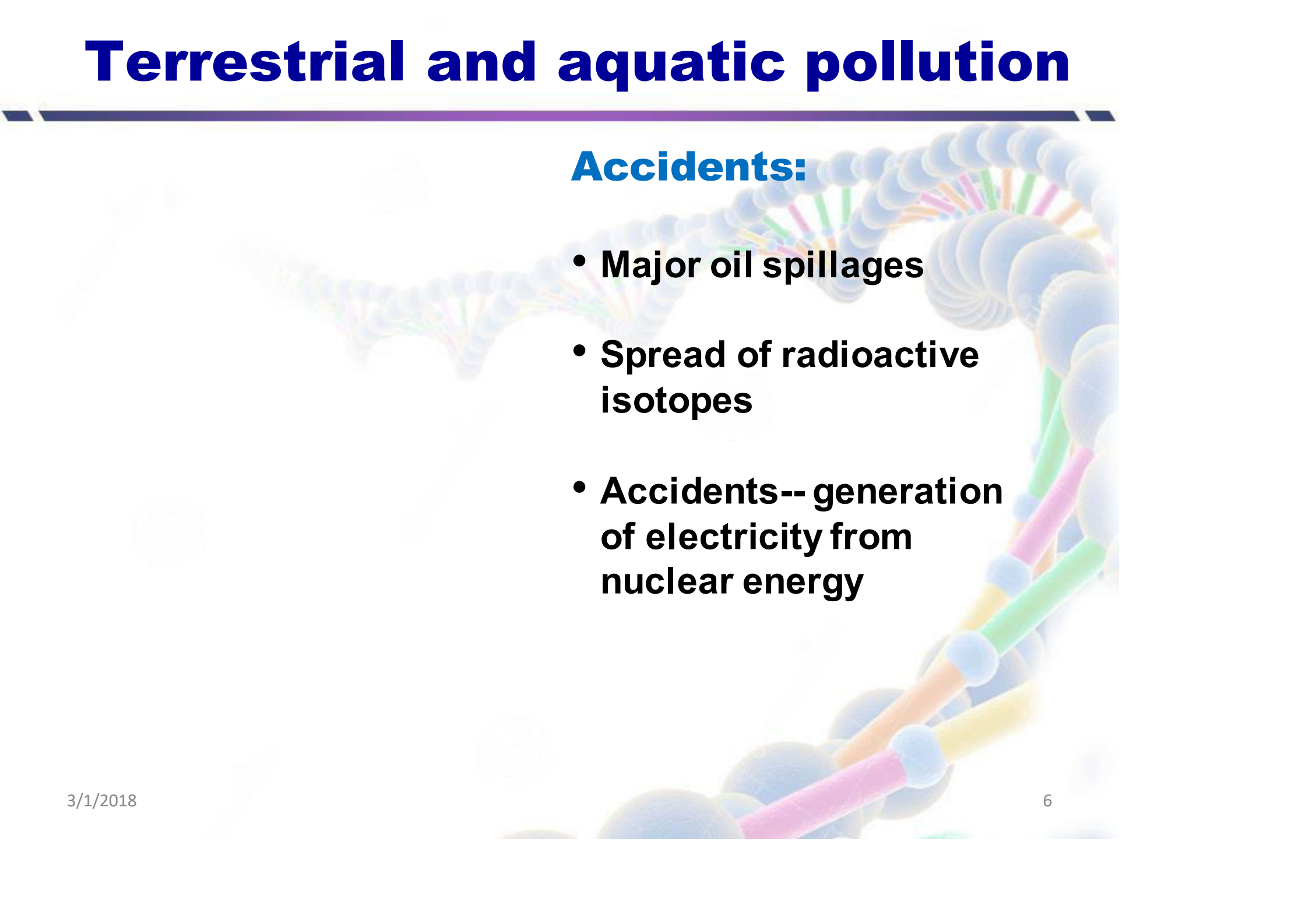
- Rachel Carson—silent spring
- First one to introduce the chemical-pollution
- Agri-chemicals have accumulated in our food chain
- Environment protection laws are very weak

Terrestrial and aquatic pollution

Ozone depletion:

- Chemical reaction in the atmosphere----- aerosols
- Aerosol sprays are used in refrigerator as a coolant
- Destroy ozone layer

Terrestrial and aquatic pollution



Accidents:

- Major oil spillages
- Spread of radioactive isotopes
- Accidents-- generation of electricity from nuclear energy

Bioethics

Lesson # 13

Bioethics



Global climate change

Global climate change

Factors increasing atmospheric CO₂:

- industrial **revolution**
- burning of fossil fuels
- burning of wood
- CO₂ is a greenhouse gas
- trapped infrared rays from the sun

Global climate change

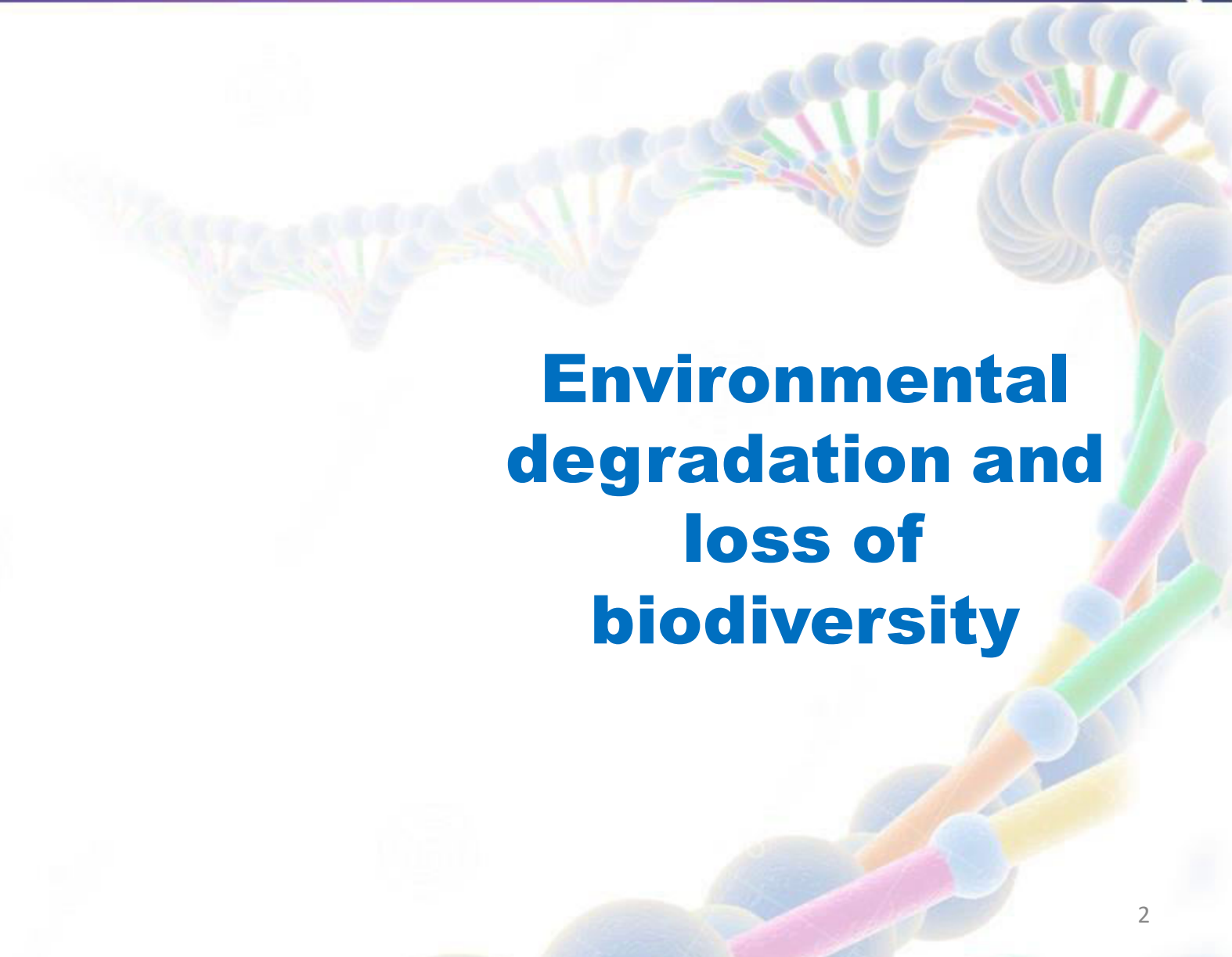
Global warming:

- earth climate zone is **shifting**
- Polar ice start melting
- Sea level increases
- metabolic rate of methane producing bacteria increases
- Species may extinct

Bioethics

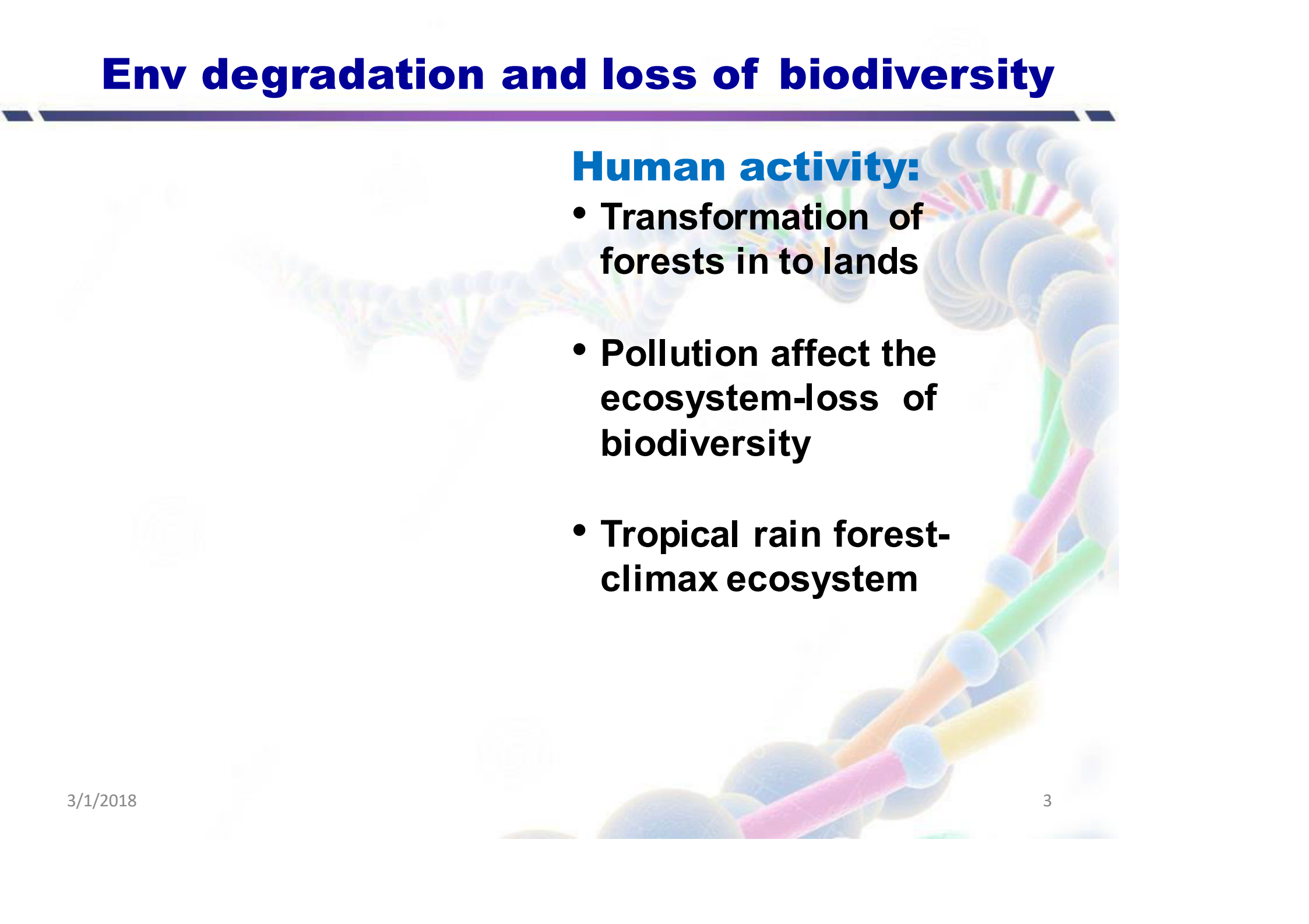
Lesson # 14

Bioethics



Environmental degradation and loss of biodiversity

Env degradation and loss of biodiversity



Human activity:

- Transformation of forests into lands
- Pollution affects the ecosystem-loss of biodiversity
- Tropical rain forest-climax ecosystem

Env degradation and loss of biodiversity

Clearance of tropical forest:

- Use of wood
- Need of a land
- We are losing 7 million hectares per year
- Soil is degrading without trees

Bioethics

Lesson # 15



Bioethics



Ethics of animal research

Ethics of animal research

Animals for research:

- 26 million animals-----
research
- vital role in scientific and medical advances
- animals--- used in ethical framework

Ethics of animal research

UK-cost benefit analysis:

- analysis of procedure and experiments
- number and type of animals used
- must be weighed against the potential benefits of the project.

Ethics of animal research

Animal welfare:

- application for project licenses
- standards of animal care and welfare
- accept the use of animals in medical research



Ethics of animal research

Benefits of animal research:

- **benefits of animal research is enormous**
- **good experiments reduce the number of animals**
- **reduce the pain experienced by animals**

Bioethics

Lesson # 16



Bioethics



**Animals as
recreation**

Animals as recreation

Use of animals:

- animals in sports, companionship, leisure and **fashion**
- race horses-peak fitness
- injuries in sports—save horses for breeding

Animals as recreation

Companion-animals:

- **pets—status of friends and children**
- **too much pampered-form of cruelty**
- **breeds-people aesthetics satisfaction**
- **difficulty in breathing/giving birth naturally**

Animals as recreation

Use of animal fur:

- luxury item for clothing
- issues- animal welfare and conservation
- leopard and jaguar are protected
- Mink breed for fur

Bioethics

Lesson # 17



Bioethics



**Animals for food
and draughting**

Animals for food and draughting

Draft animals:

- **beast of burden**
- **trained to perform task**
- **perform light harness work**
- **become a part of rural development-agriculture**

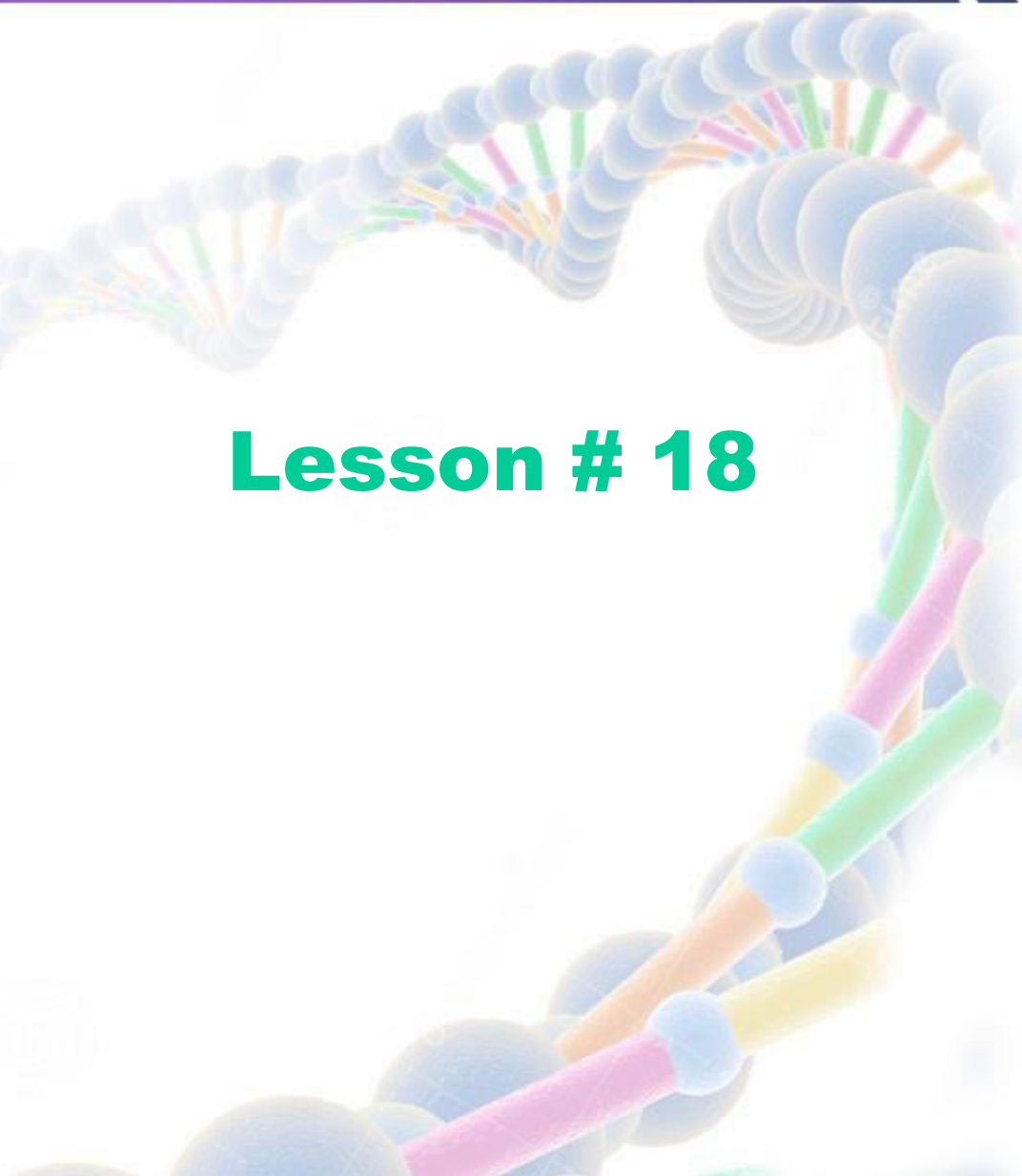
Animals for food and draughting

Slaughter:

- **Muslims and Jewish—
cut the neck without
stunning**
- **electricity is a cruel
method**
- **industrial method**

Bioethics

Lesson # 18



Bioethics



**Code of ethics
for biologist**

Code of ethics for biologist

codes:

- scientists--perform experiment --described in their **experiments**
- best interpretation
- summarize honestly
- acknowledge the contributors for publications

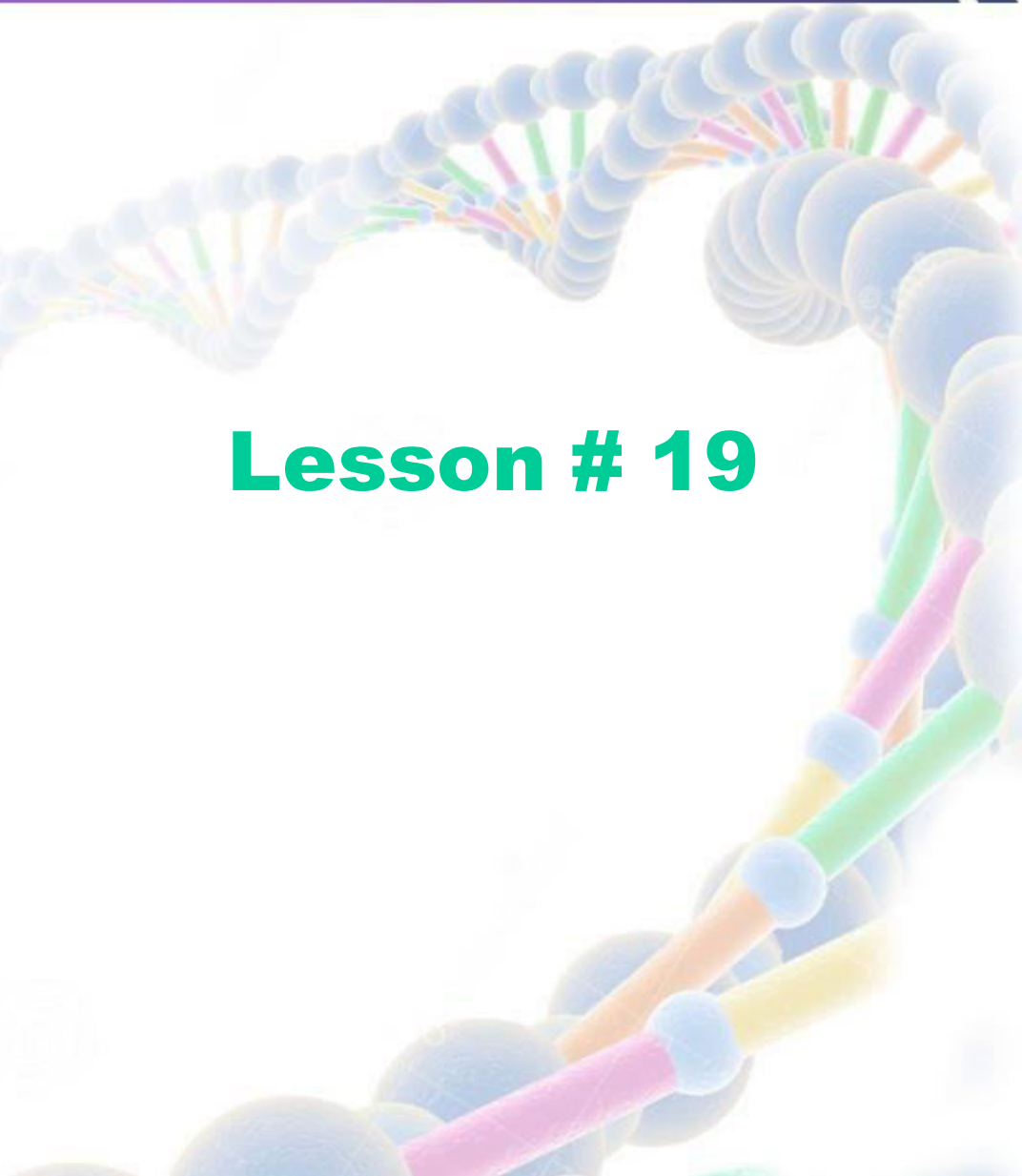
Code of ethics for biologist

codes:

- **treat manuscript confidentially**
- **no inaccurate or misleading information**
- **disclose financial resources**
- **help colleagues/
support professional
organization**

Bioethics

Lesson # 19



Bioethics



**Patient-
physician
relationship**

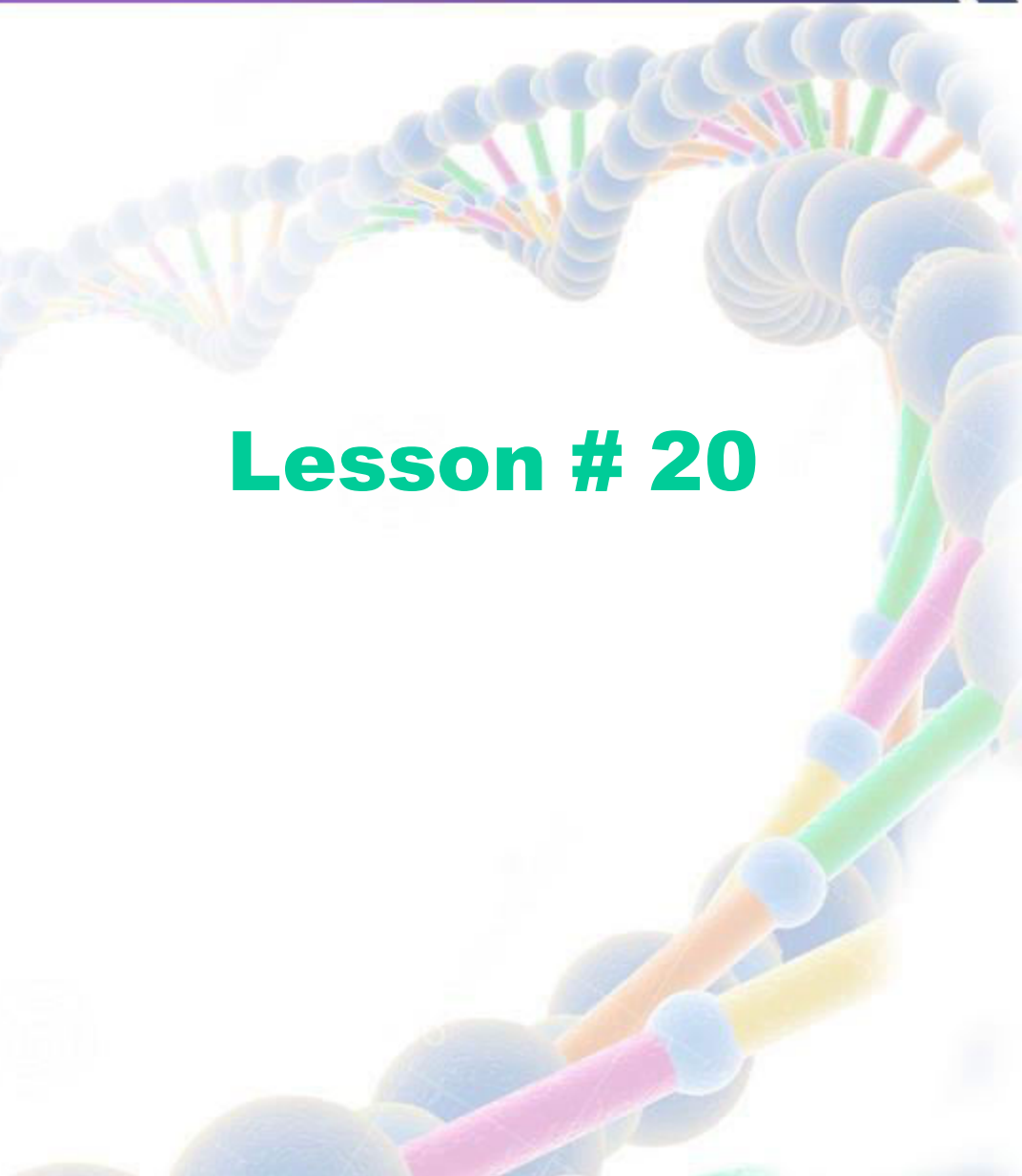
Patient-physician relationship

Fundamental elements:

- patient has a right to receive **information**
- patient has a right to make decisions
- patient has a right to confidentiality
- Continuity-- availability of health care

Bioethics

Lesson # 20



Bioethics



**Codes for
nurses**

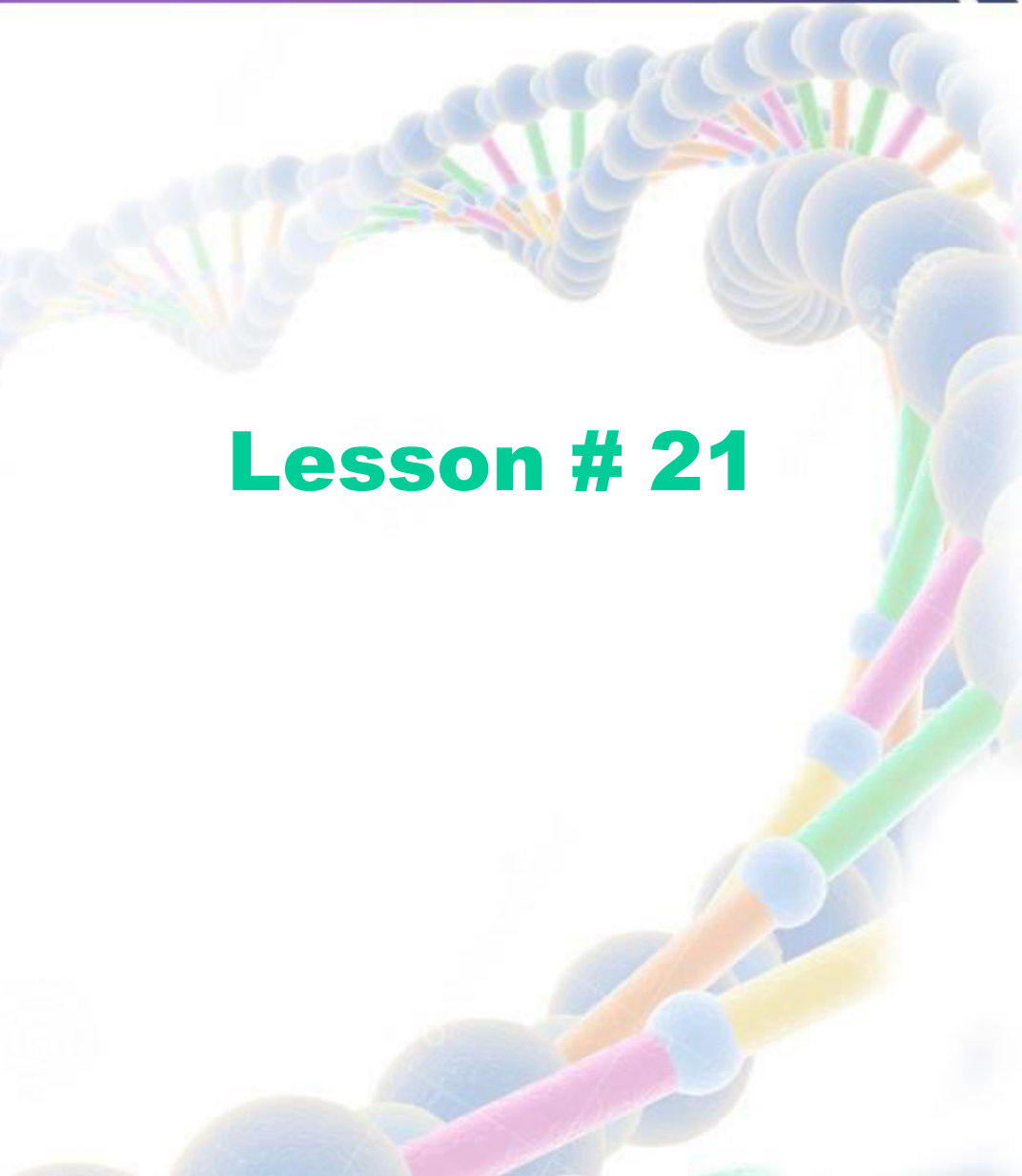
Codes for nurses

Codes:

- nurses and **people**
- nurses and practice
- nurses and profession
- nurses and co-workers

Bioethics

Lesson # 21



Bioethics



**Patient
rights/responsibilities**

Patient rights/responsibilities

Responsibilities:

- give correct/complete **information**
- ask questions
- cooperate with your caregivers
- accept health consequences

Patient rights/responsibilities

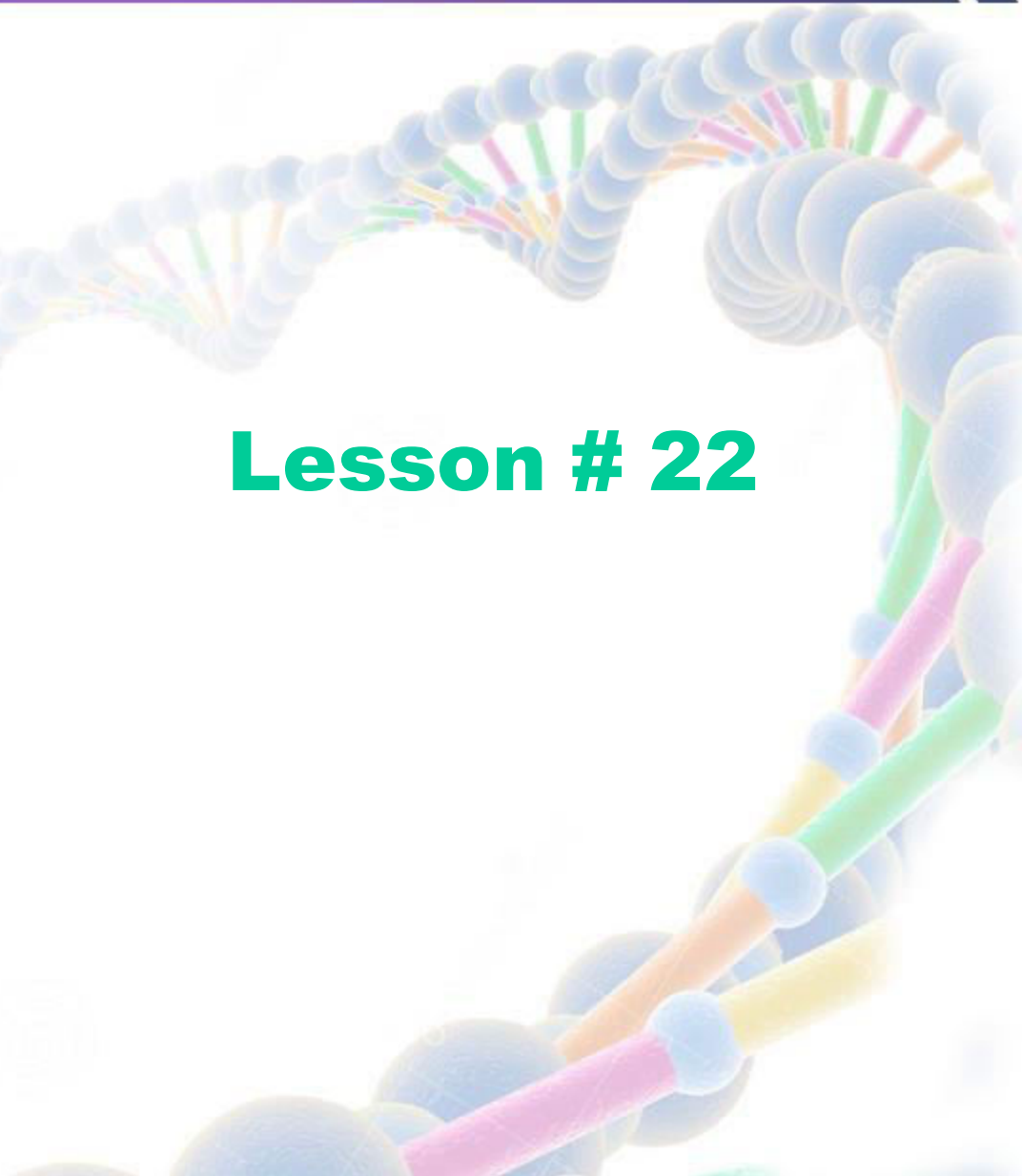
Rights:

- respect and privacy
- quality care
- information and communication
- make decisions



Bioethics

Lesson # 22



Bioethics

Truth telling



Truth telling

Bad news:

- common cold
- unpleasant information
- objective bad news
- subjective bad news



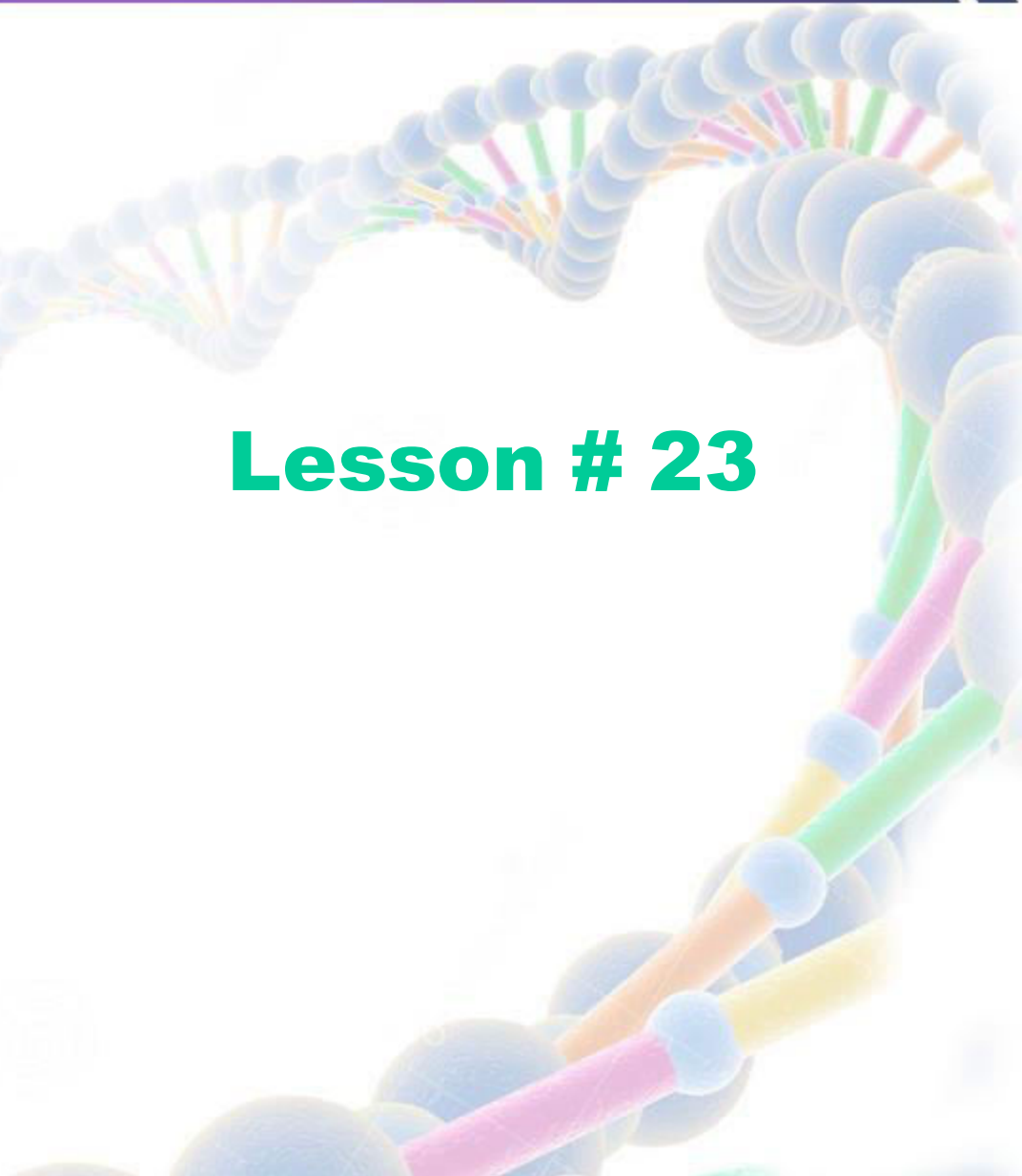
Truth telling

Breaking bad news:

- amount of bad news to deliver
- attending to cultural and ethical issues
- managing psychological distress
- producing competent messengers of bad news

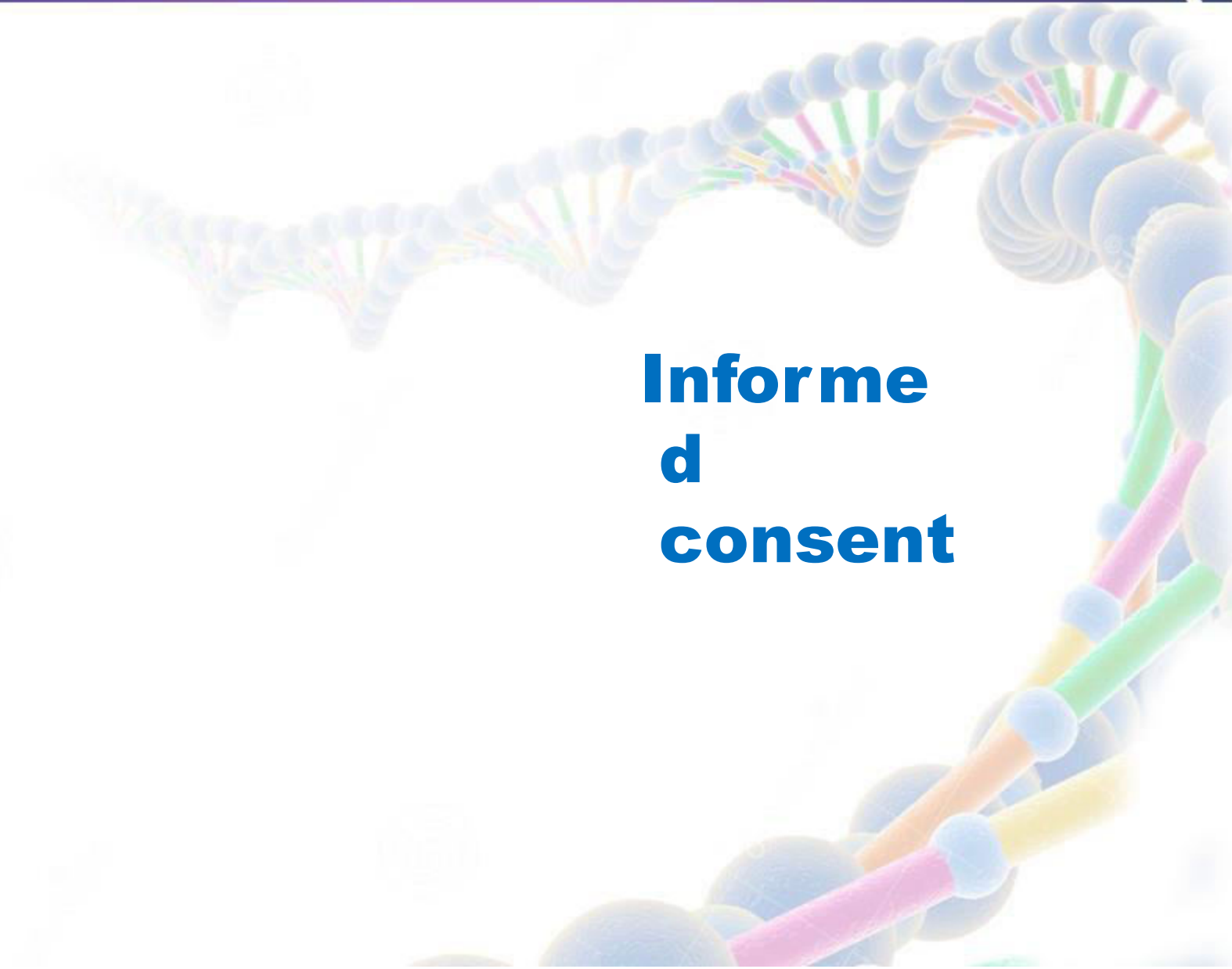
Bioethics

Lesson # 23



Bioethics

**Inform
ed
consent**



Informed consent

Consent??? :

- legal and ethical right of the **patient**
- permission before getting the healthcare



Informed consent

Elements:

- Nature of decision-
patient is participating
in decision making
- Relative risk/benefits
- Assessment of patient
understanding
- Acceptance of
intervention by the
patient

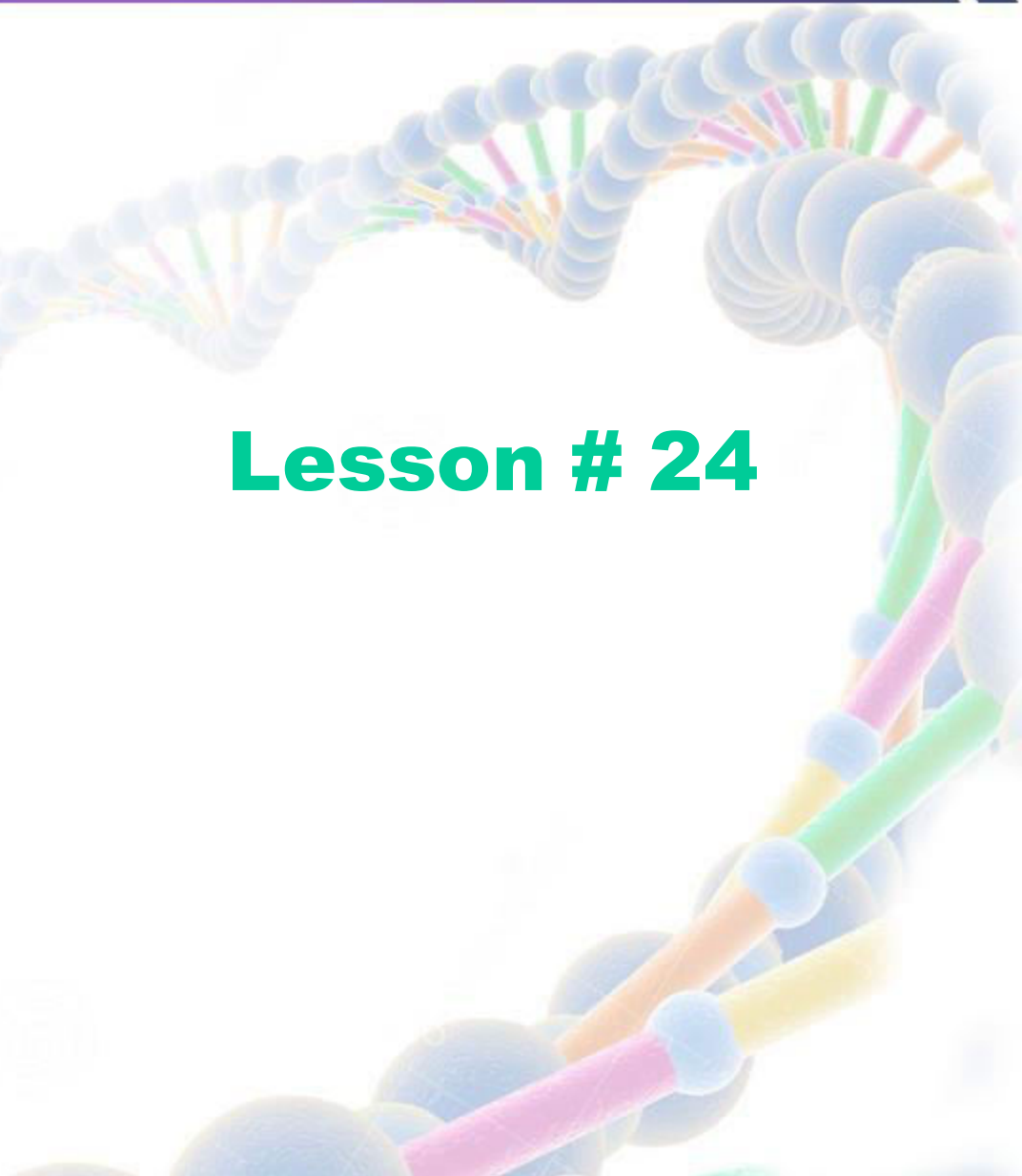
Informed consent

Adequate information:

- Reasonable physician standard- decide that which information is adequate
- Reasonable patient standard—complete information—decision
- Subjective standard

Bioethics

Lesson # 24



Bioethics



**Patients
advance
directives**

Patients advance directives

Advance directives:

- appoint someone to make **decisions**
- legal document-tell physician about your wishes
- general (donation) or detailed (treatment plan)

Patients advance directives

Types:

- living will-applies to treatment such as dialysis –limited
- oral
- terminal illness—if patient die shortly

Patients advance directives

Types:

- health care power of attorney
- durable power of attorney
- agent/proxy
- agent make decisions

Patients advance directives

Patient self determination act:

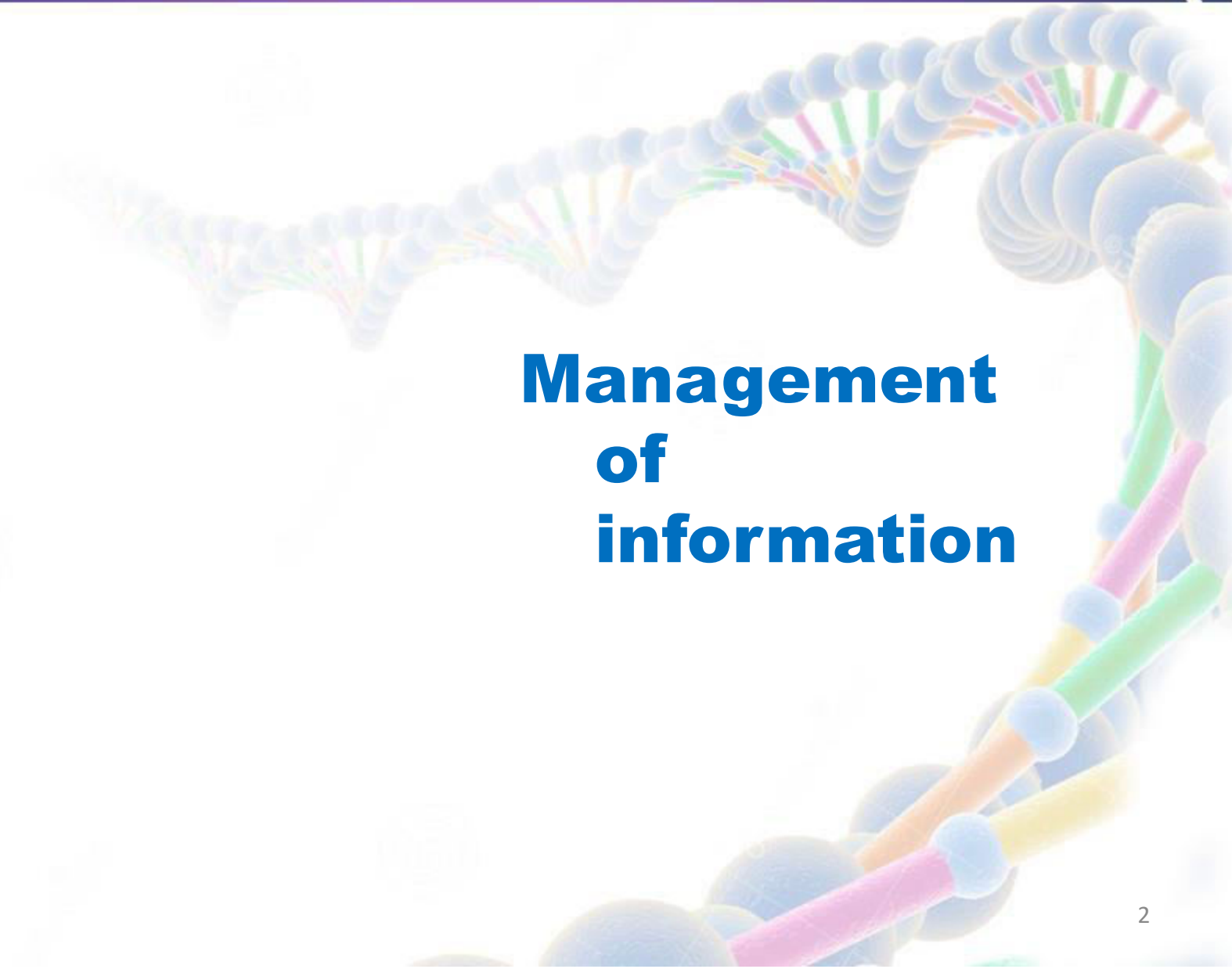
- encourages everyone to decide
- hospital medical care
- extended medical care



Bioethics

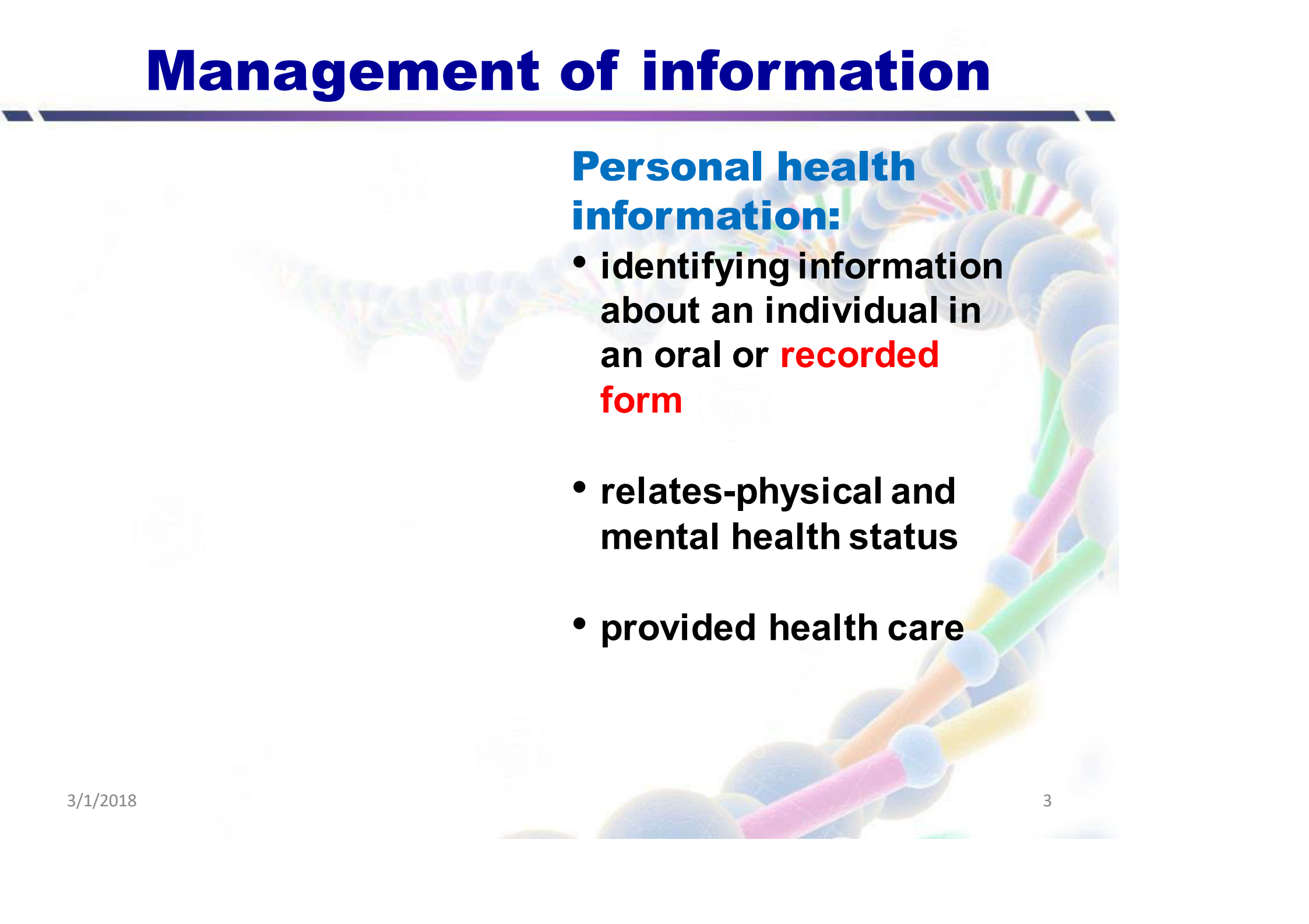
Lesson # 25

Bioethics



Management of information

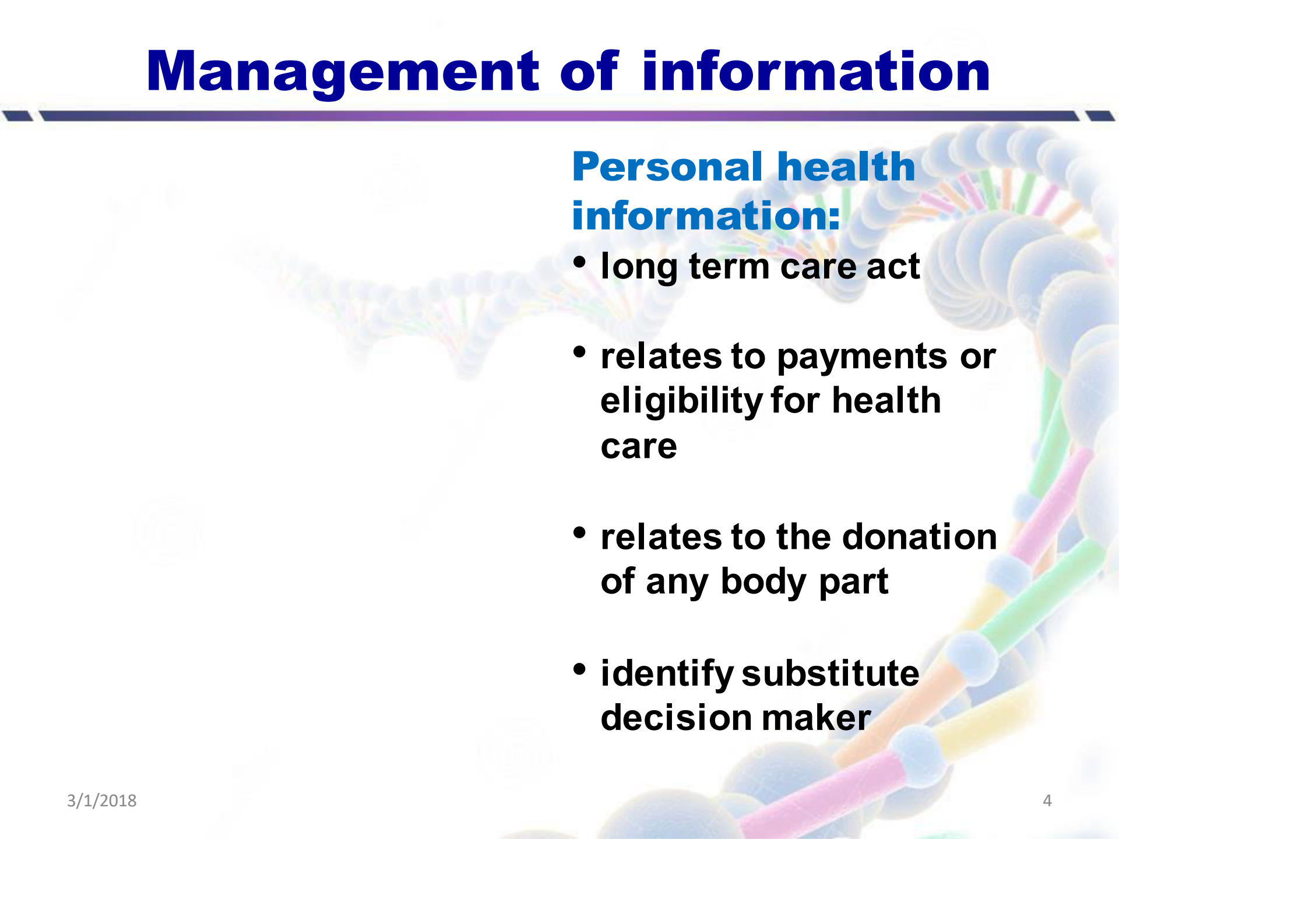
Management of information



Personal health information:

- identifying information about an individual in an oral or **recorded form**
- relates-physical and mental health status
- provided health care

Management of information



Personal health information:

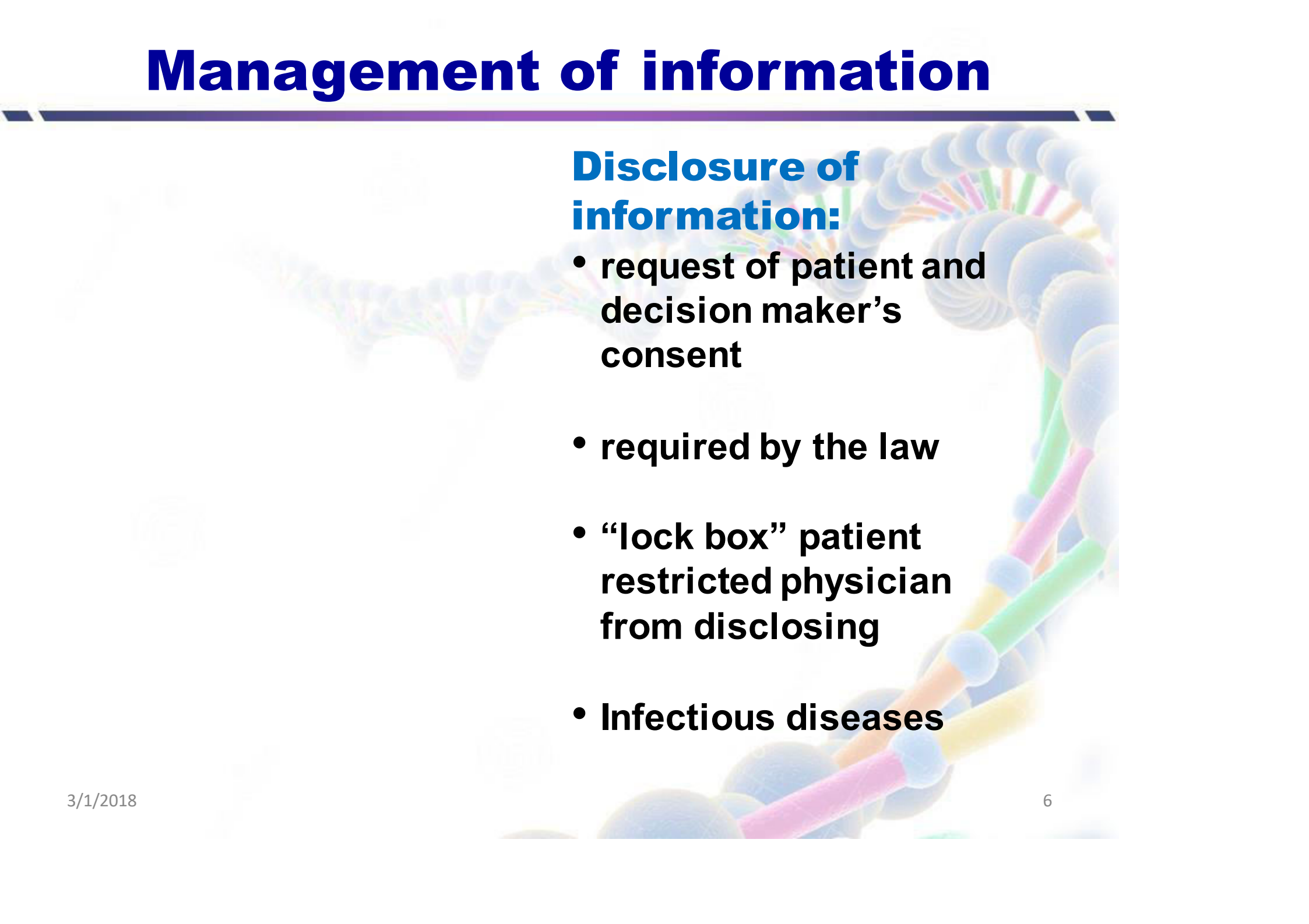
- long term care act
- relates to payments or eligibility for health care
- relates to the donation of any body part
- identify substitute decision maker

Management of information

Principles:

- physicians act in accordance legally and professionally
- establish and preserve physician patient relationship
- High standard of patient care----patient give complete and accurate information

Management of information

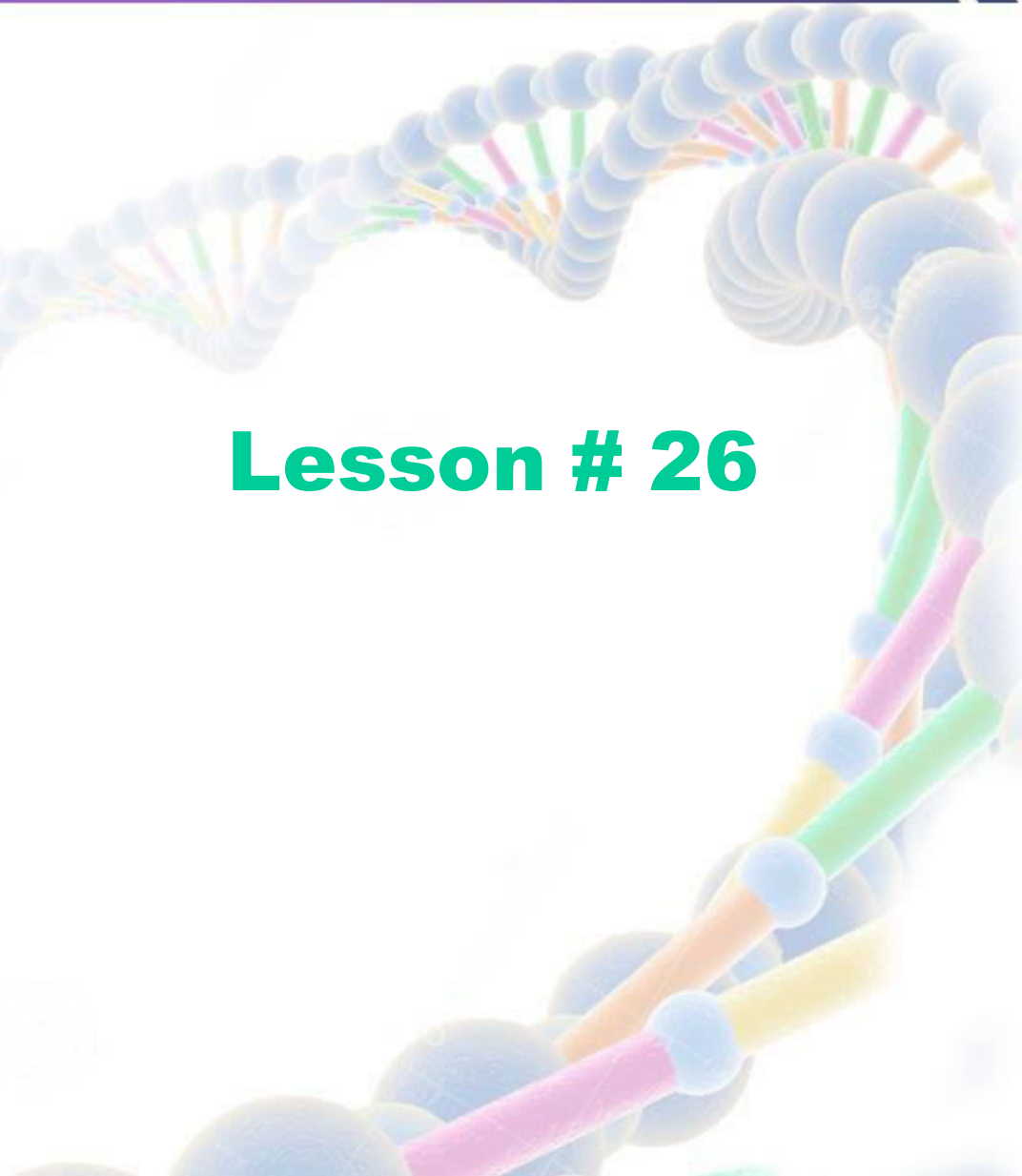


Disclosure of information:

- request of patient and decision maker's consent
- required by the law
- “lock box” patient restricted physician from disclosing
- Infectious diseases

Bioethics

Lesson # 26



Bioethics



**Problems of
moral
justification**

Problems of moral justification

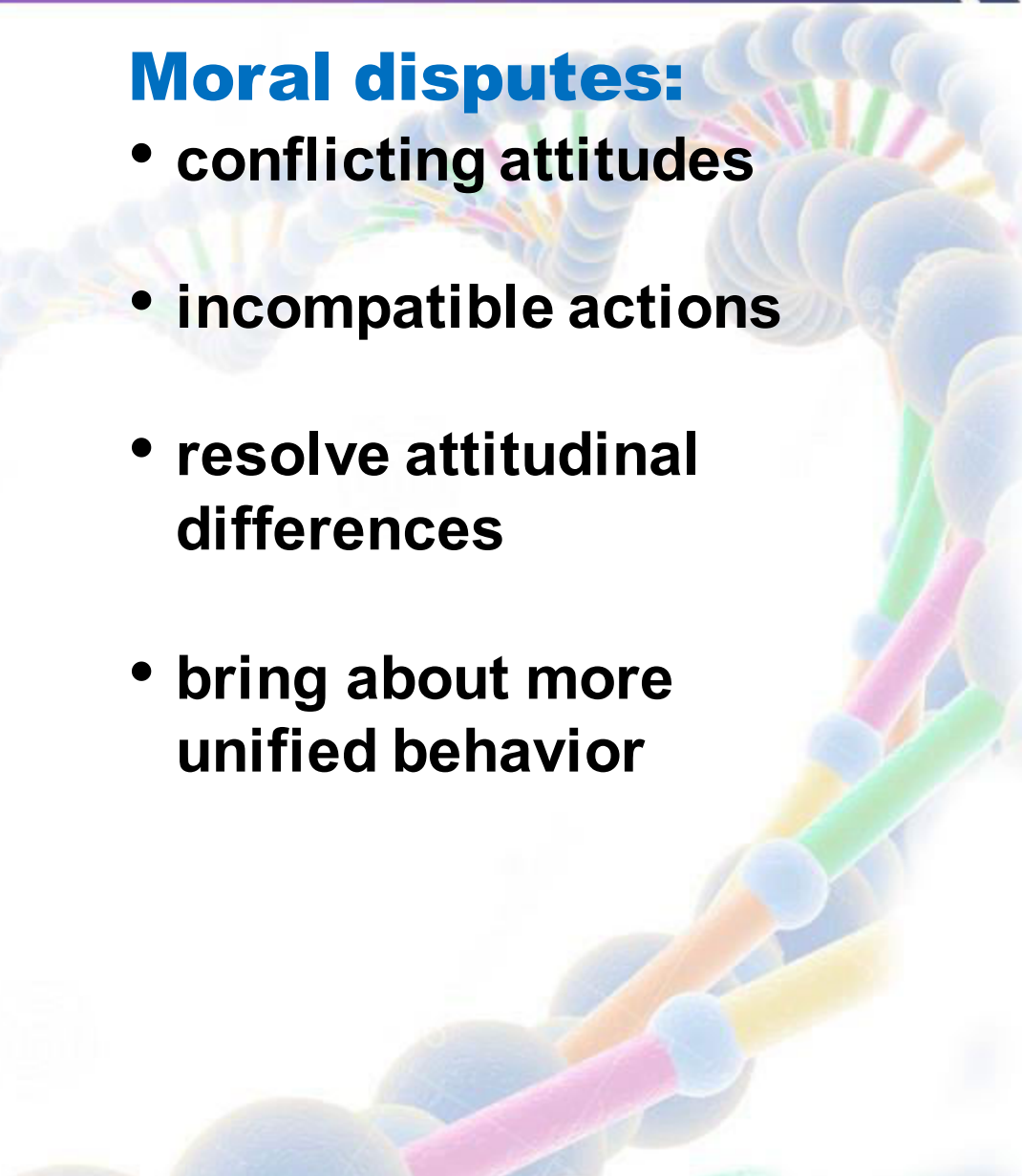
Moral theory:

- provide an account of truth and falsity of **moral judgments**
- provide an account of justification for moral views
- solution to a problem is to be convincing

Problems of moral justification

Moral disputes:

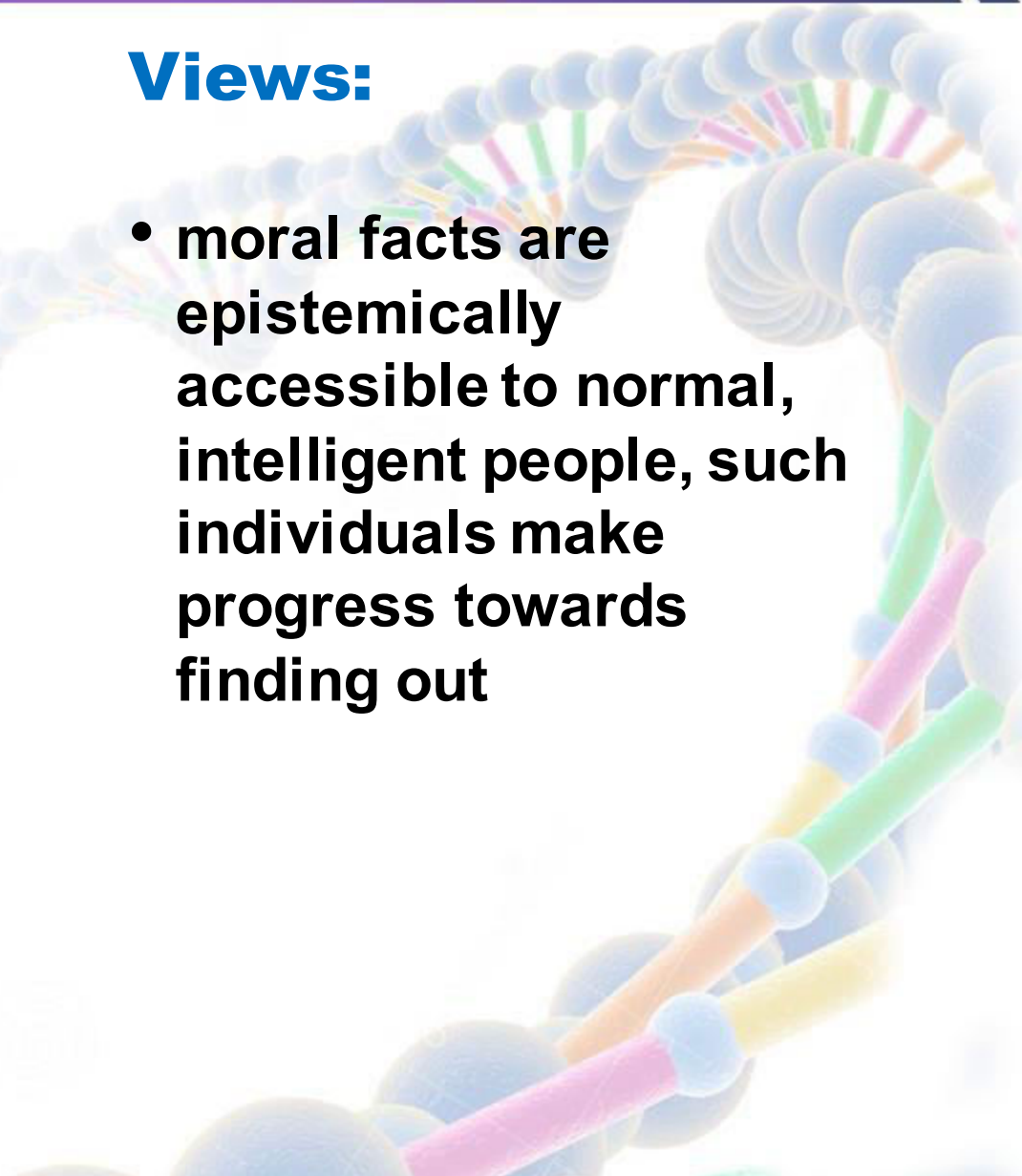
- **conflicting attitudes**
- **incompatible actions**
- **resolve attitudinal differences**
- **bring about more unified behavior**



Problems of moral justification

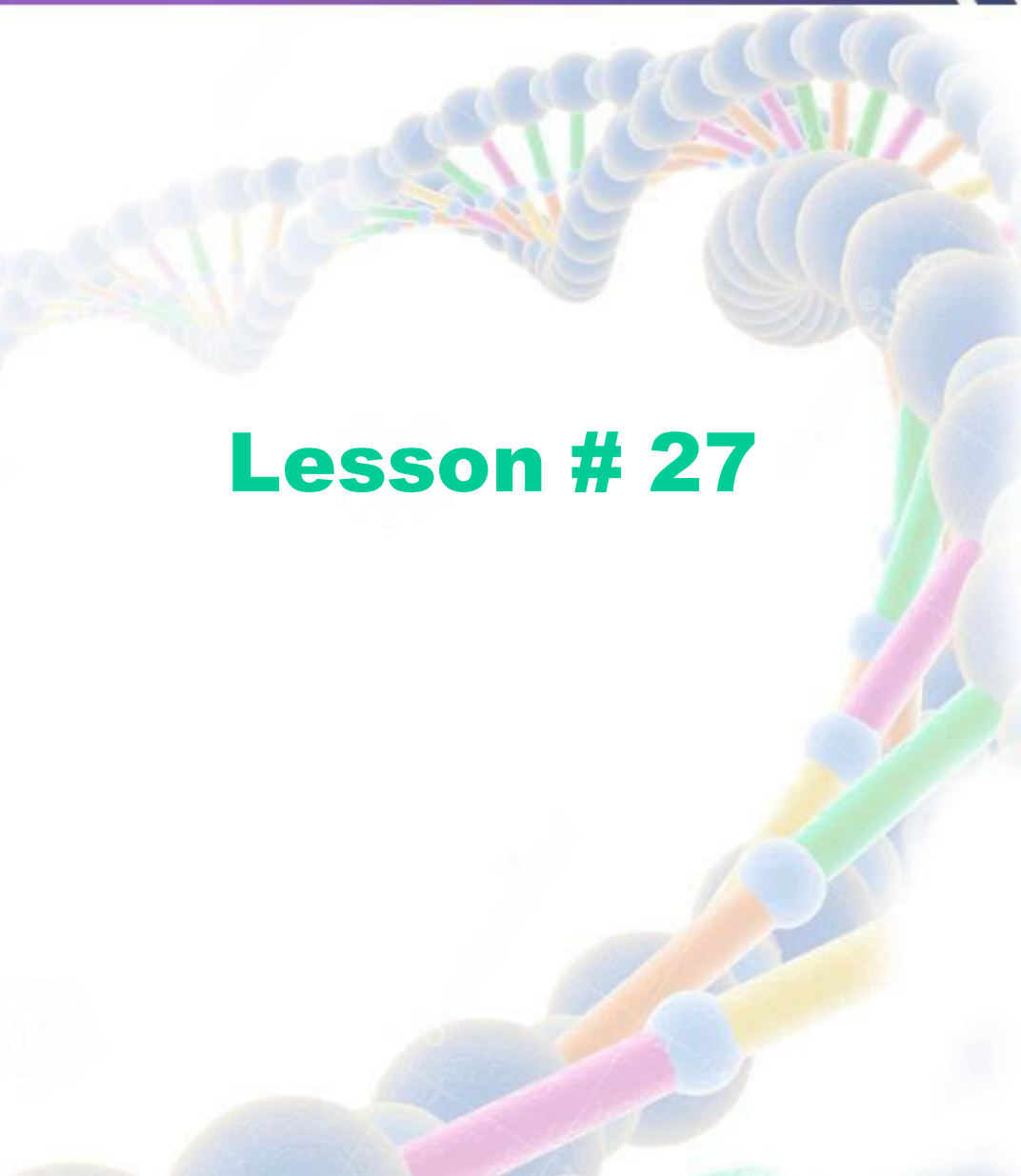
Views:

- moral facts are epistemically accessible to normal, intelligent people, such individuals make progress towards finding out



Bioethics

Lesson # 27



Bioethics



Maternal-fetal relationship

Maternal-fetal relationship

Biologically linked people:

- physicians take pregnant women as two individuals who are **biologically linked**
- most mothers accept the risk to their own health
- refuses-----medical therapy for saving fetal life----ethical issues

Maternal-fetal relationship

Maternal-fetal conflict:

- advances in medical technology—direct procedures towards the fetus
- physicians—medically best for each individual
- unethical—harming one individual to benefit other

Maternal-fetal relationship

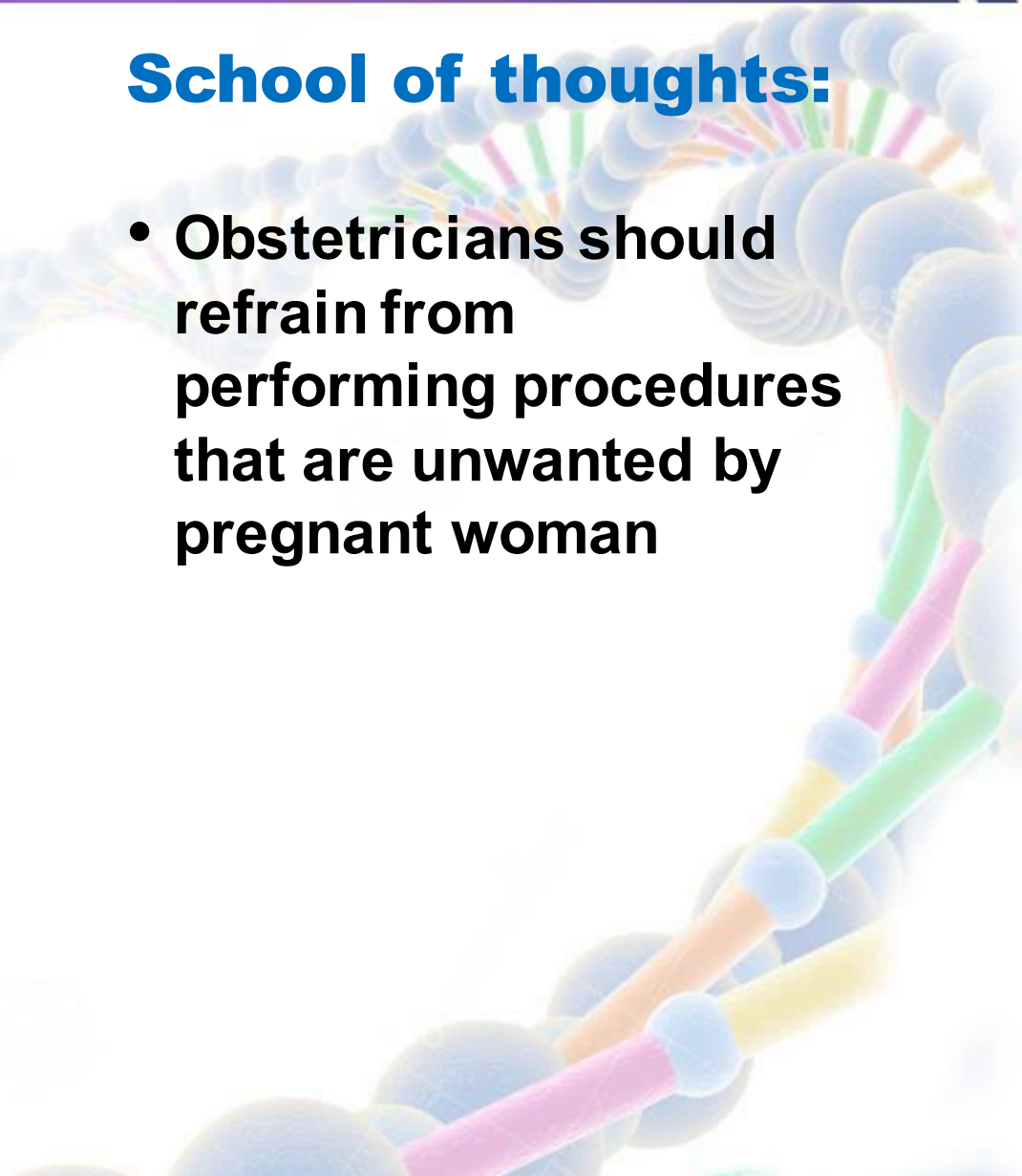
US Law:

- fetus has the right—
begin his life with
sound body and mind
- charges of fetal abuse
- refusal of
hospitalization,
intrauterine
transfusion or surgical
delivery

Maternal-fetal relationship

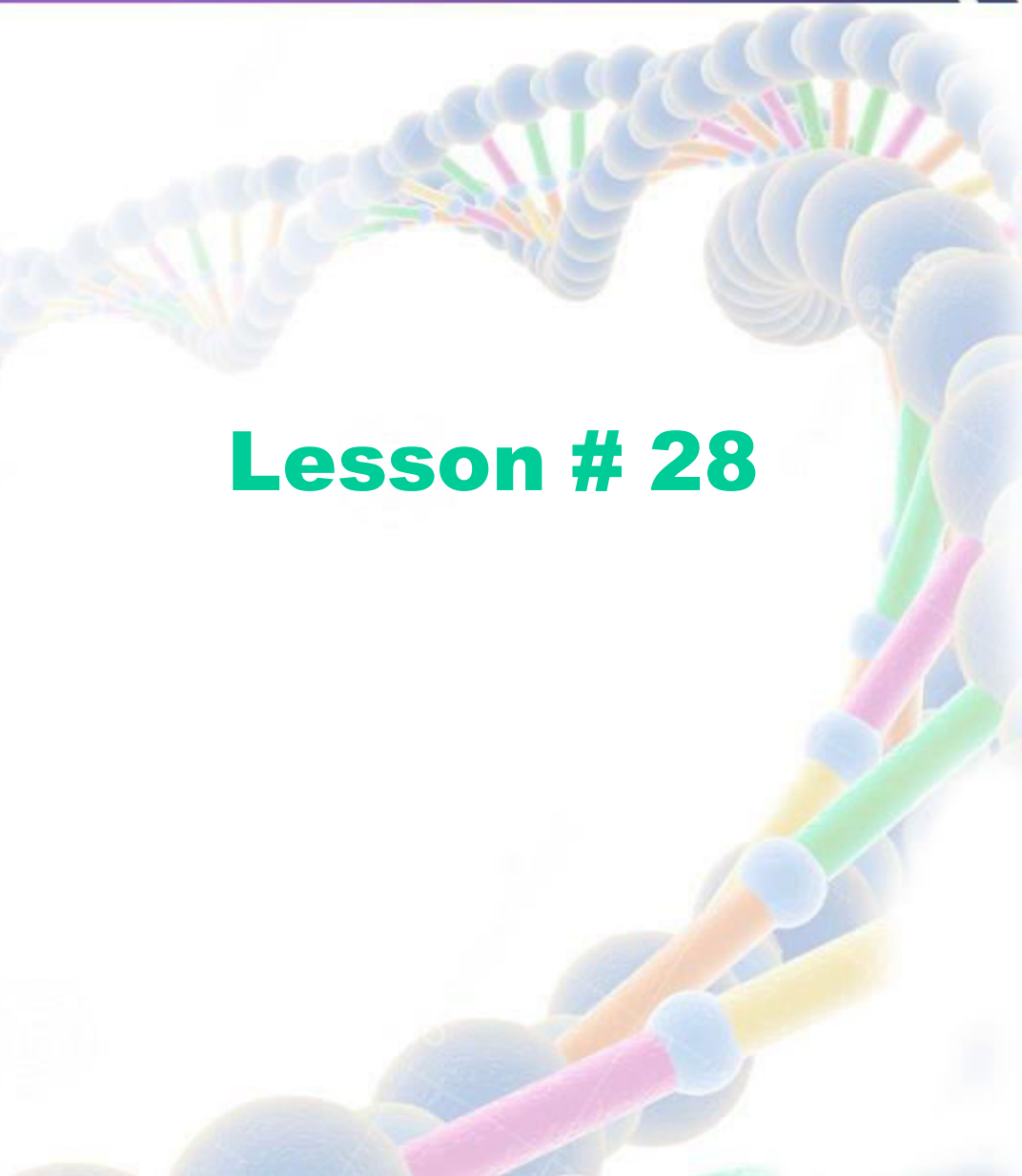
School of thoughts:

- **Obstetricians should refrain from performing procedures that are unwanted by pregnant woman**



Bioethics

Lesson # 28



Bioethics

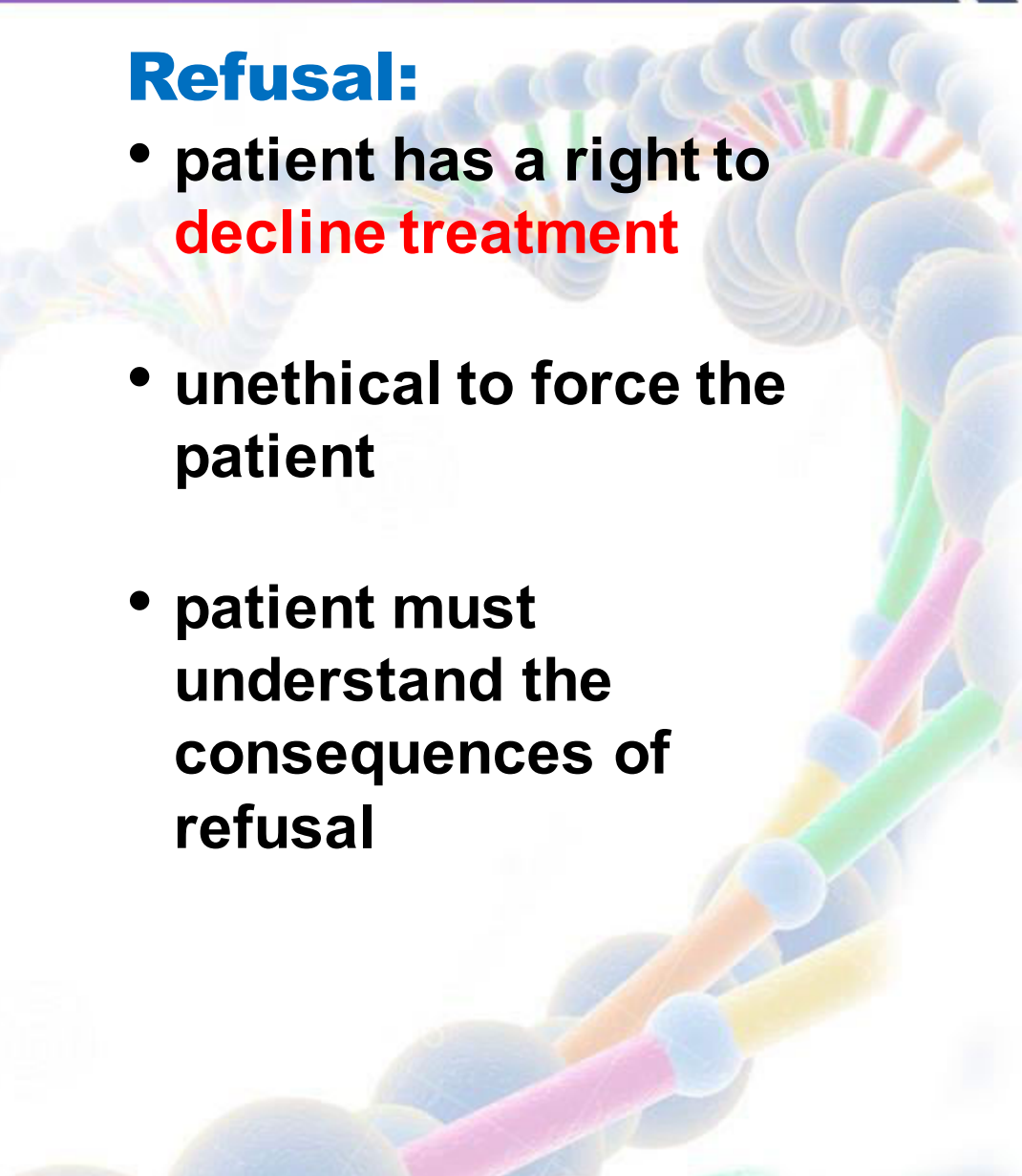


**Refusal of
treatment**

Refusal of treatment

Refusal:

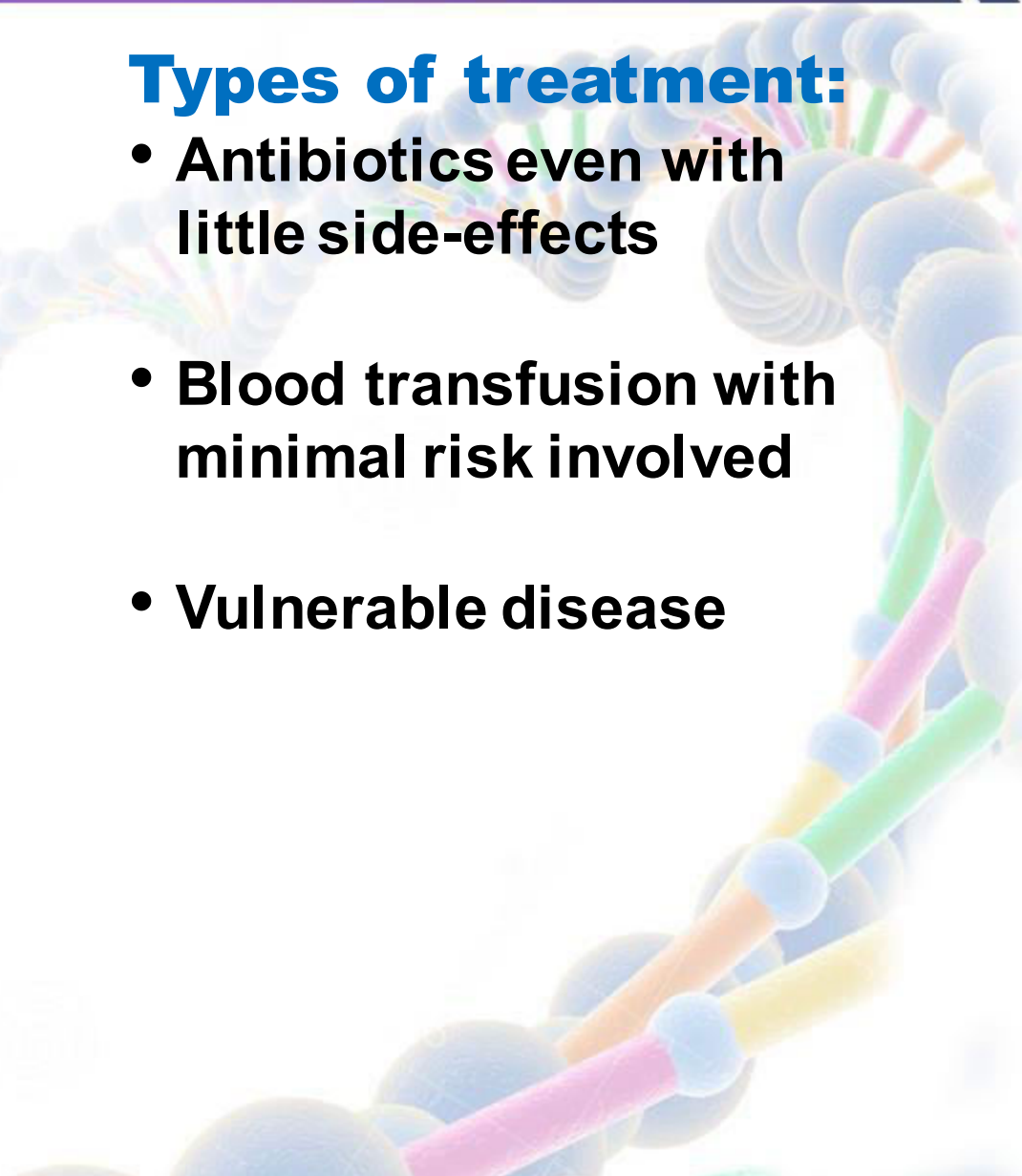
- patient has a right to **decline treatment**
- unethical to force the patient
- patient must understand the consequences of refusal



Refusal of treatment

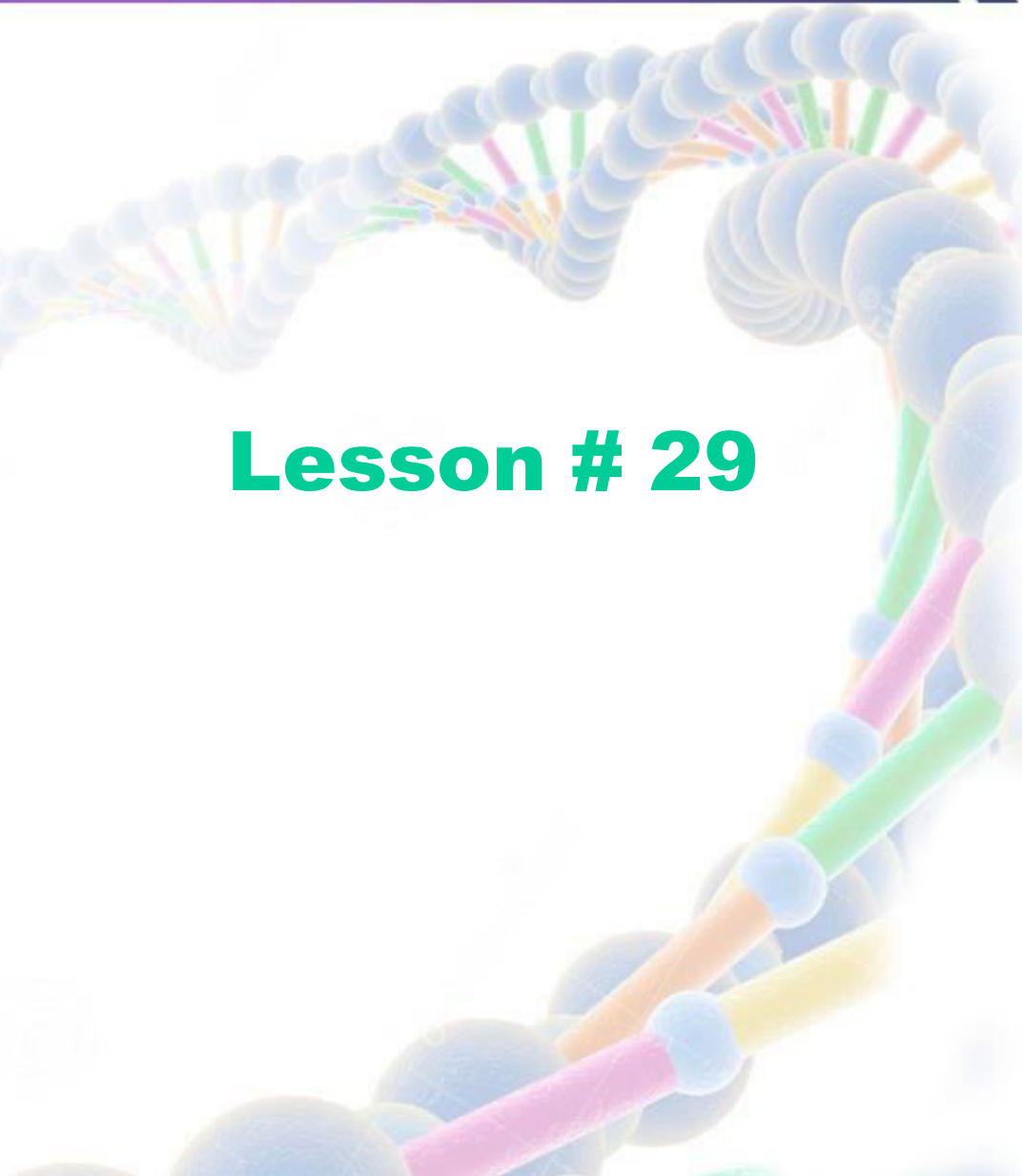
Types of treatment:

- **Antibiotics even with little side-effects**
- **Blood transfusion with minimal risk involved**
- **Vulnerable disease**



Bioethics

Lesson # 29



Bioethics

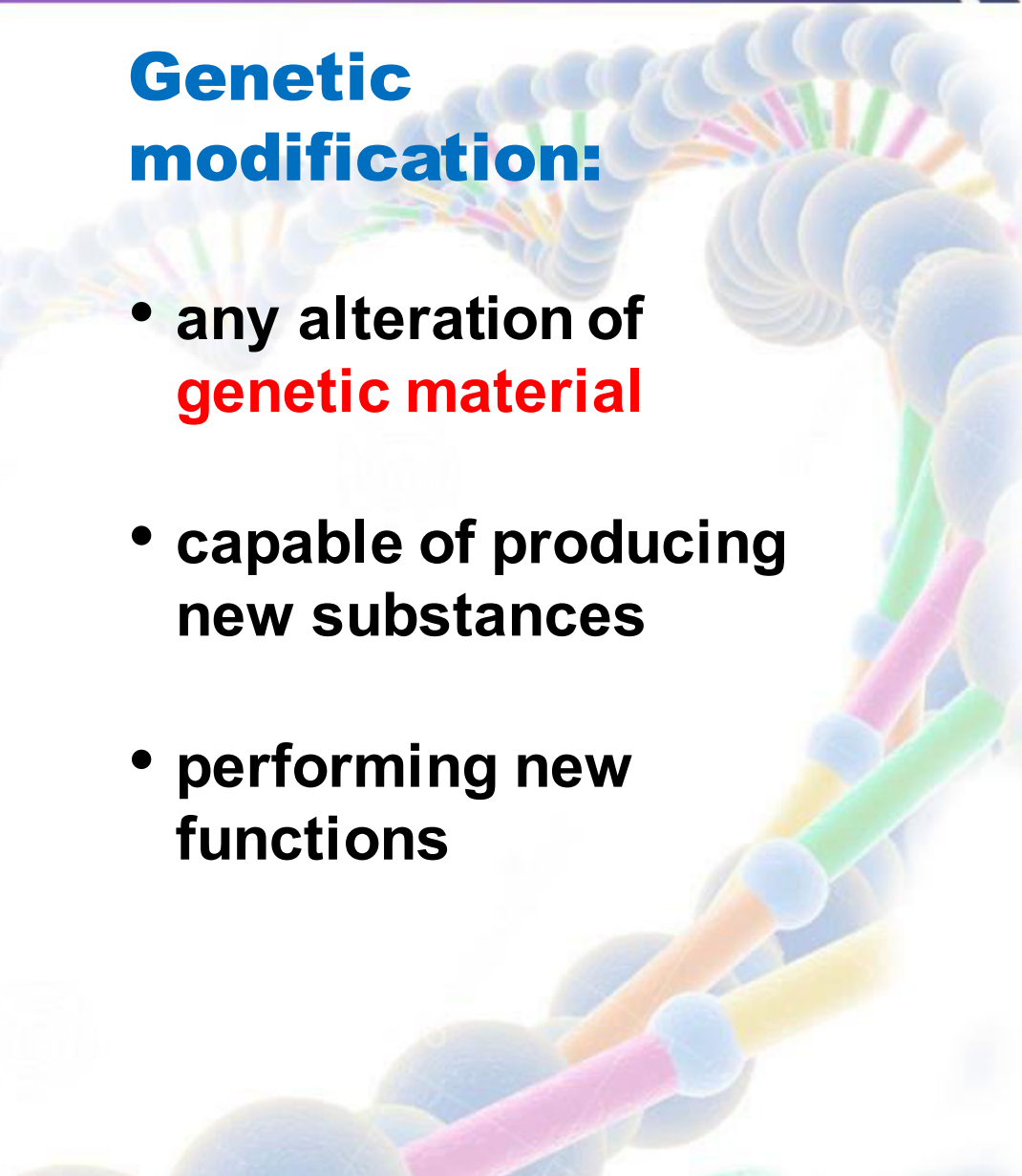


Ethics and genetic modification

Ethics and genetic modification

Genetic modification:

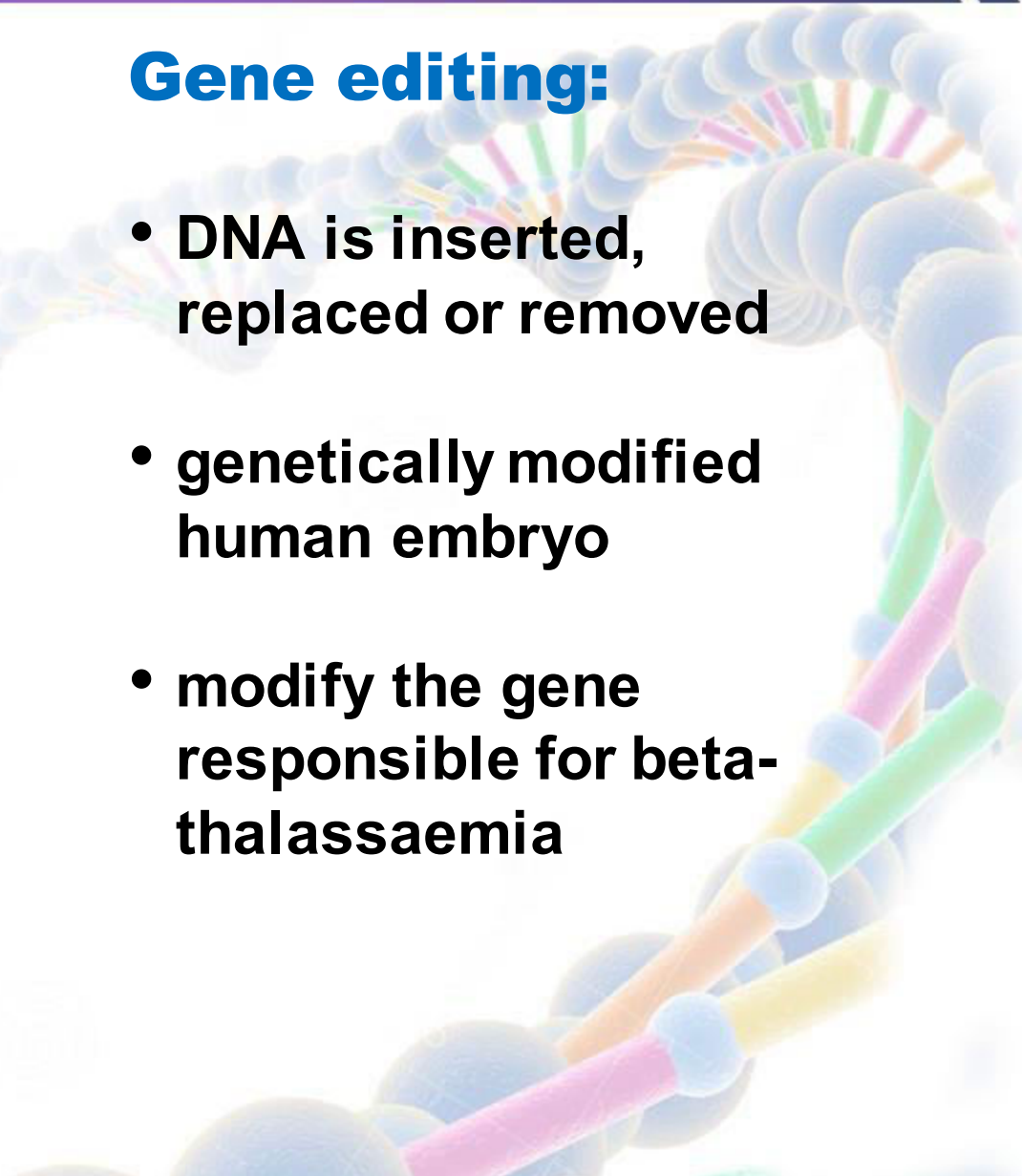
- any alteration of **genetic material**
- capable of producing new substances
- performing new functions



Ethics and genetic modification

Gene editing:

- DNA is inserted, replaced or removed
- genetically modified human embryo
- modify the gene responsible for beta-thalassaemia



Ethics and genetic modification

Transgenic:

- potatoes with high protein/Rice with high vitamin A level
- Mule
- DNA of human tumor fragment is inserted into the tobacco plant
- Flu vaccine

Ethics and genetic modification

Favr Savr tomato (1994):

- **genetically modified tomato**
- **no alien gene**
- **block the gene involved in ripening**
- **longer shelf life**

Ethics and genetic modification

Ethical issues:

- potential risk to the environment
- potential risk to human health
- socio-economic effects



Ethics and genetic modification

Ethical issues:

- entities have the rights and protections
- personal, social and cultural consequences
- fundamental issues in creating new individuals



Bioethics

Lesson # 30

Bioethics



Biotechnology and risk factors

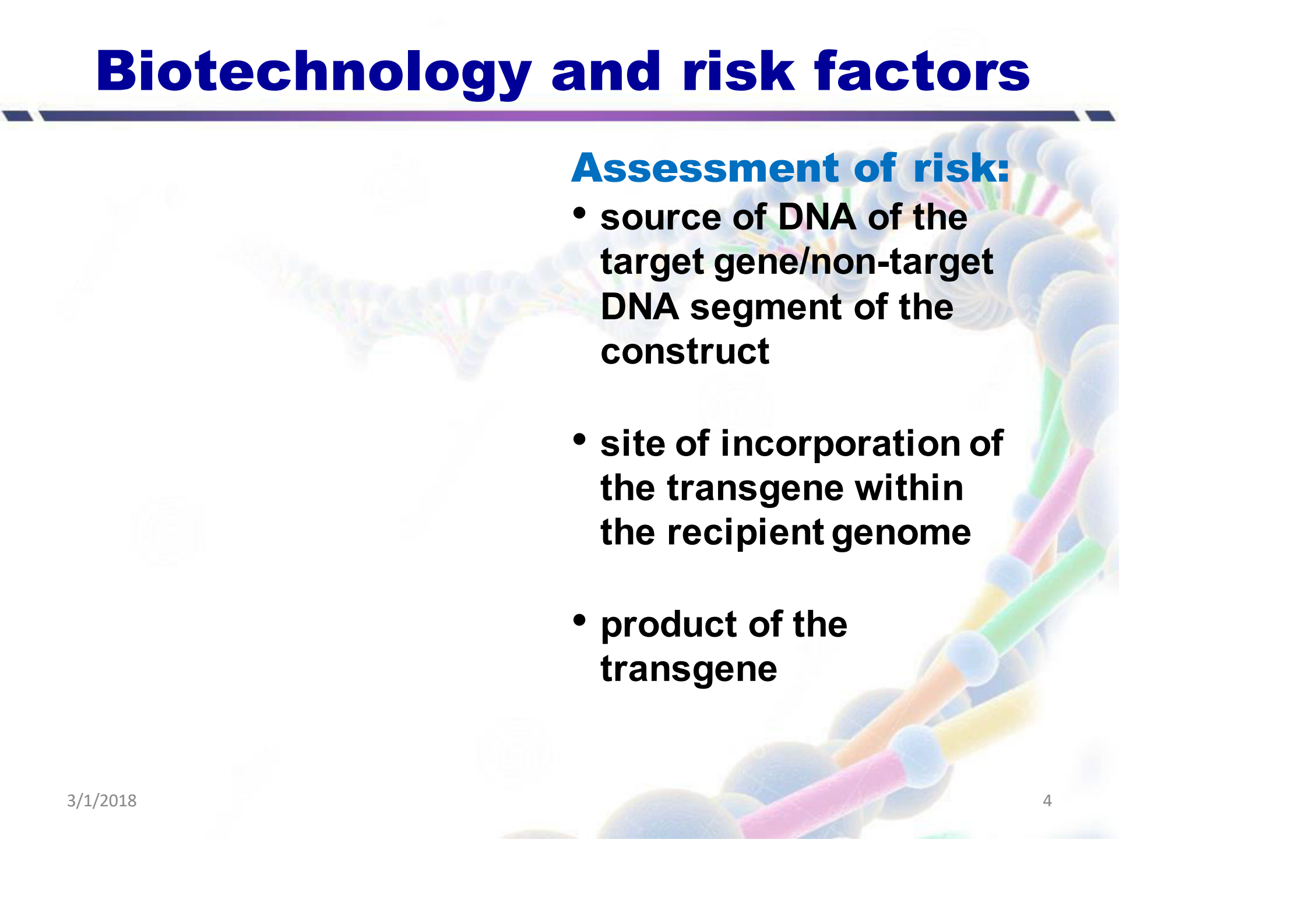
Biotechnology and risk factors



Areas of risk:

- human health
- biodiversity
- animal welfare
- poor communities

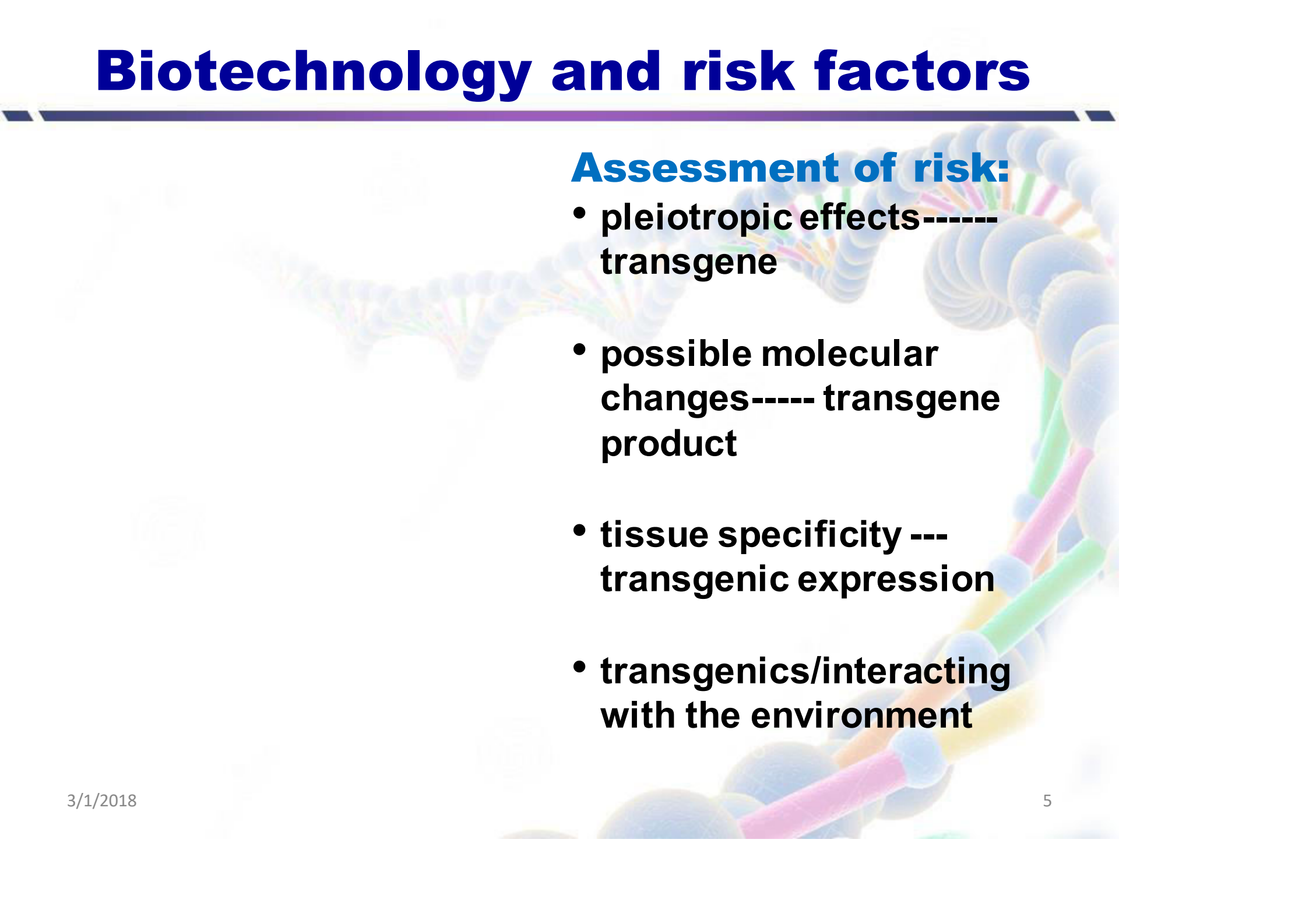
Biotechnology and risk factors



Assessment of risk:

- source of DNA of the target gene/non-target DNA segment of the construct
- site of incorporation of the transgene within the recipient genome
- product of the transgene

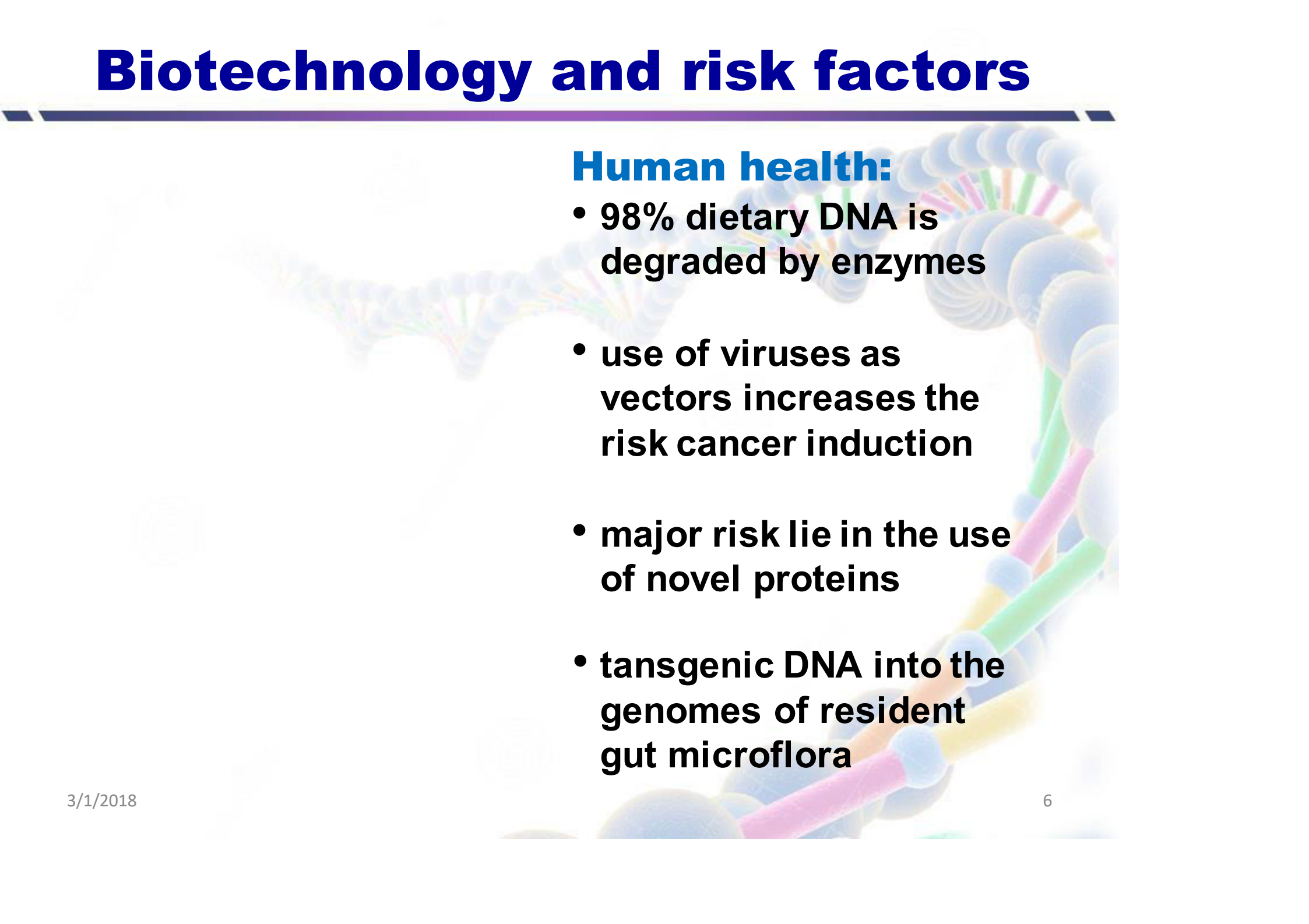
Biotechnology and risk factors



Assessment of risk:

- pleiotropic effects----- transgene
- possible molecular changes----- transgene product
- tissue specificity --- transgenic expression
- transgenics/interacting with the environment

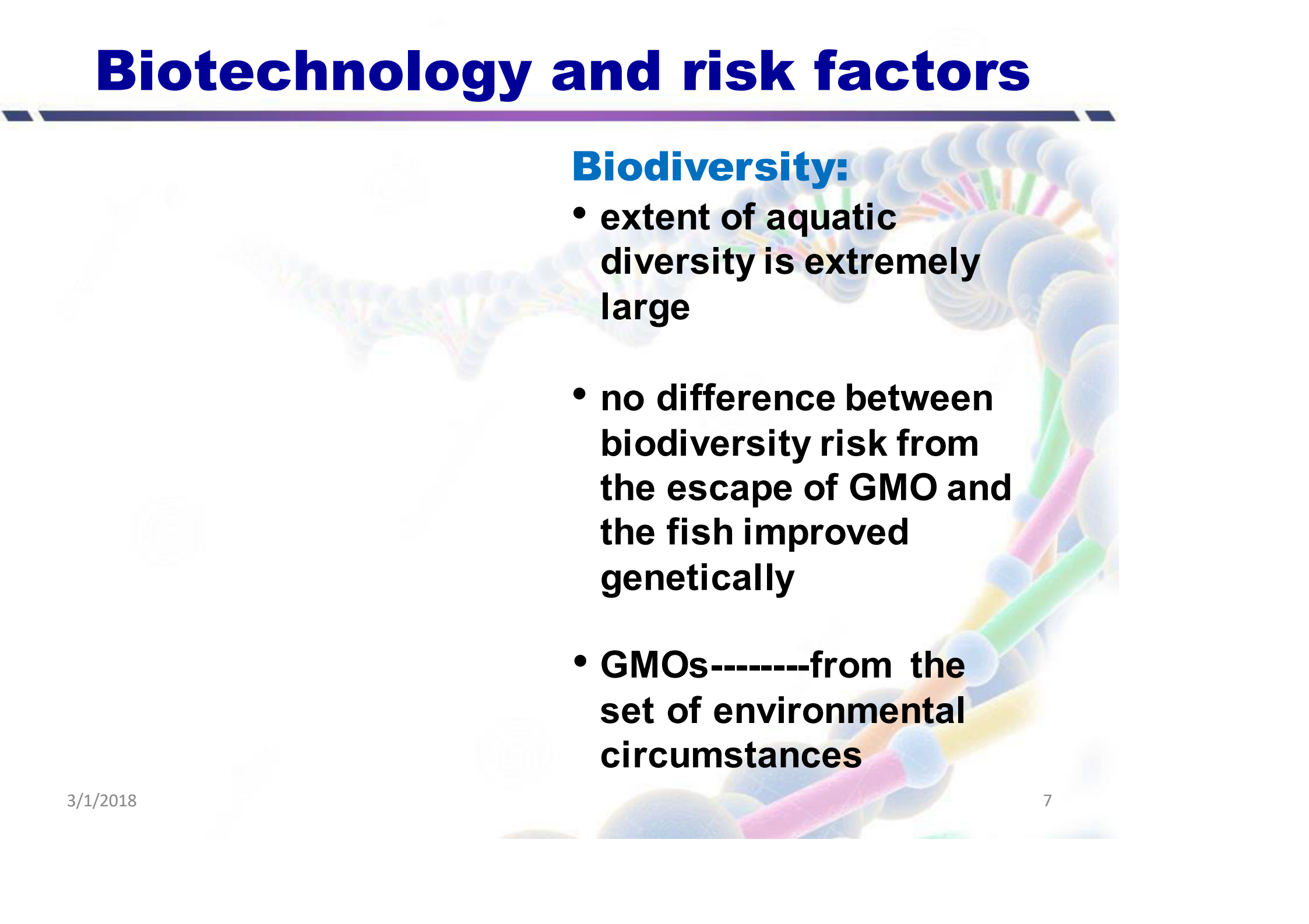
Biotechnology and risk factors



Human health:

- 98% dietary DNA is degraded by enzymes
- use of viruses as vectors increases the risk cancer induction
- major risk lie in the use of novel proteins
- transgenic DNA into the genomes of resident gut microflora

Biotechnology and risk factors



Biodiversity:

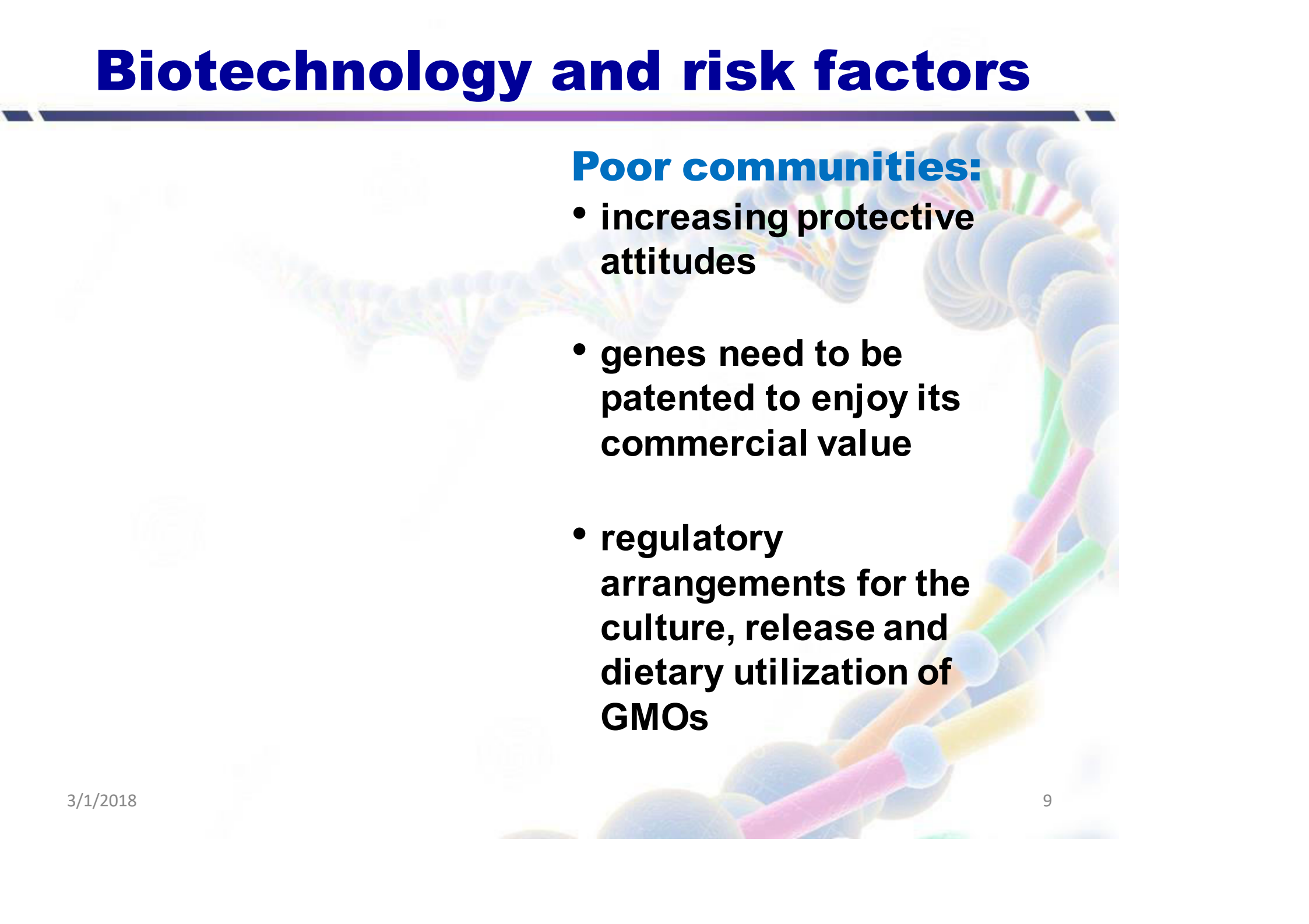
- extent of aquatic diversity is extremely large
- no difference between biodiversity risk from the escape of GMO and the fish improved genetically
- GMOs-----from the set of environmental circumstances

Biotechnology and risk factors

Animal welfare:

- changes in coloration, cranial deformities, acromegaly, infertility
- reduced viability
- nutritional levels can be improved

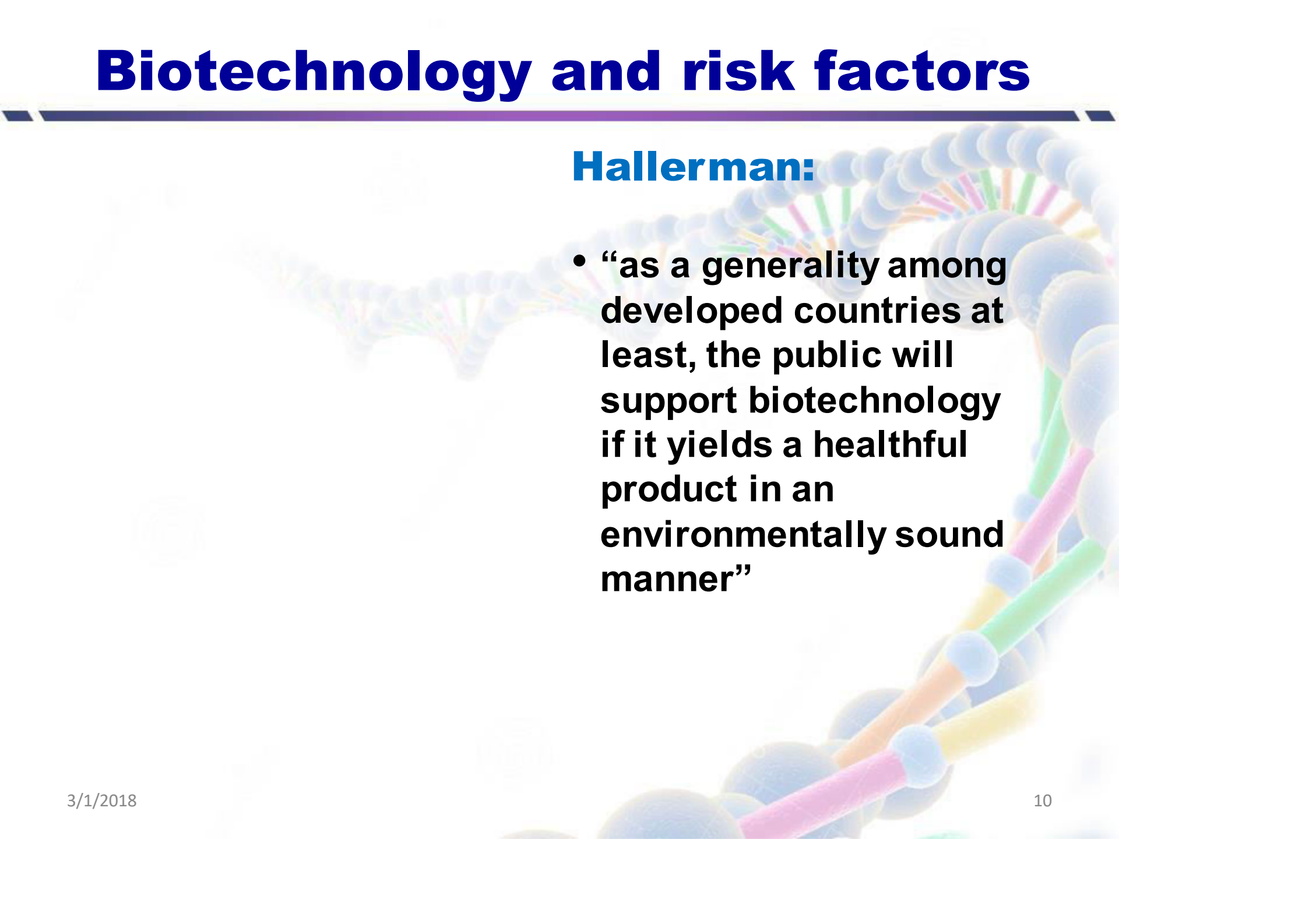
Biotechnology and risk factors



Poor communities:

- increasing protective attitudes
- genes need to be patented to enjoy its commercial value
- regulatory arrangements for the culture, release and dietary utilization of GMOs

Biotechnology and risk factors



Hallerman:

- “as a generality among developed countries at least, the public will support biotechnology if it yields a healthful product in an environmentally sound manner”

Bioethics

Lesson # 31



Bioethics

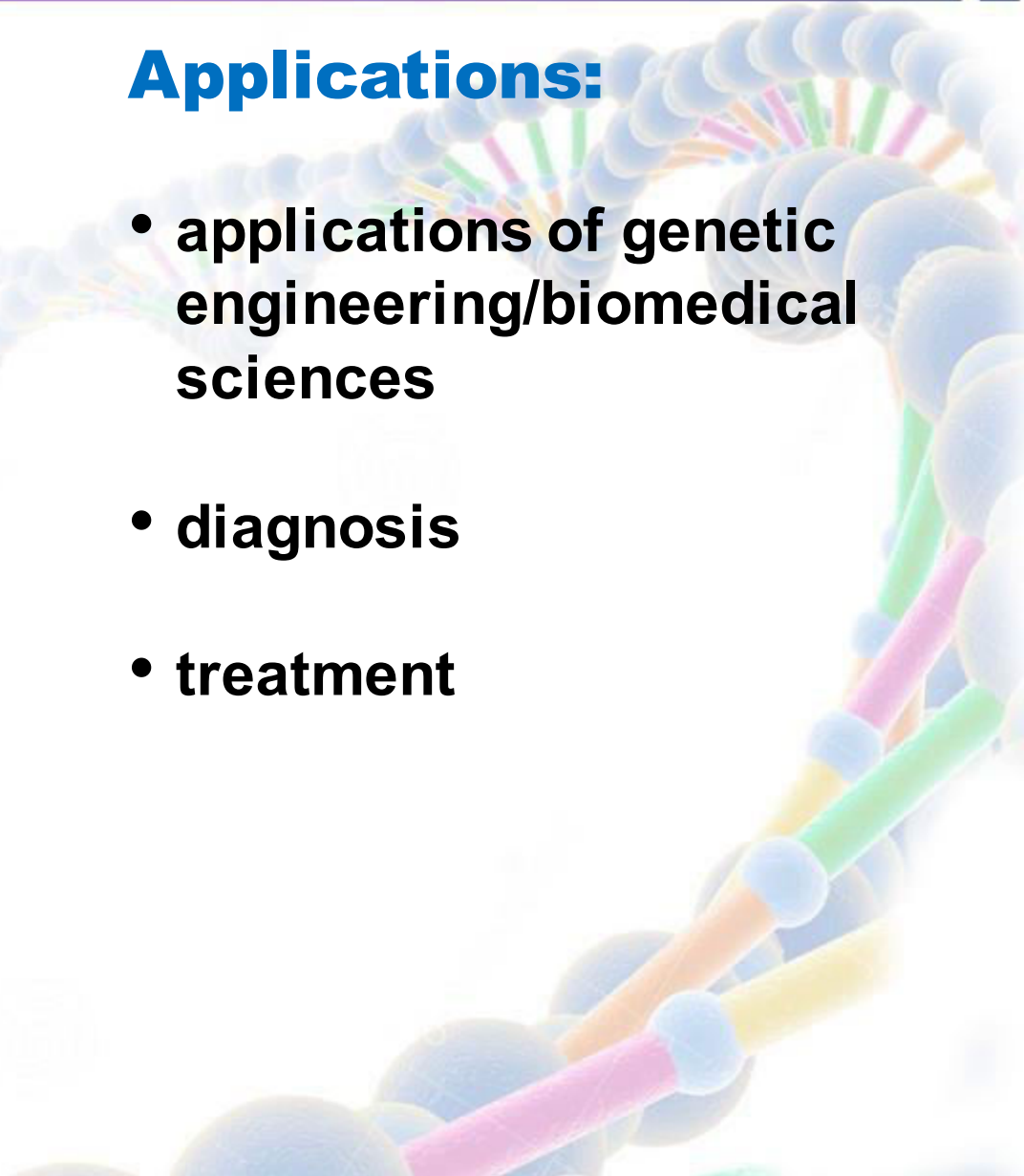


**Misuse of
biotechnology**

Misuse of biotechnology

Applications:

- applications of genetic engineering/biomedical sciences
- diagnosis
- treatment

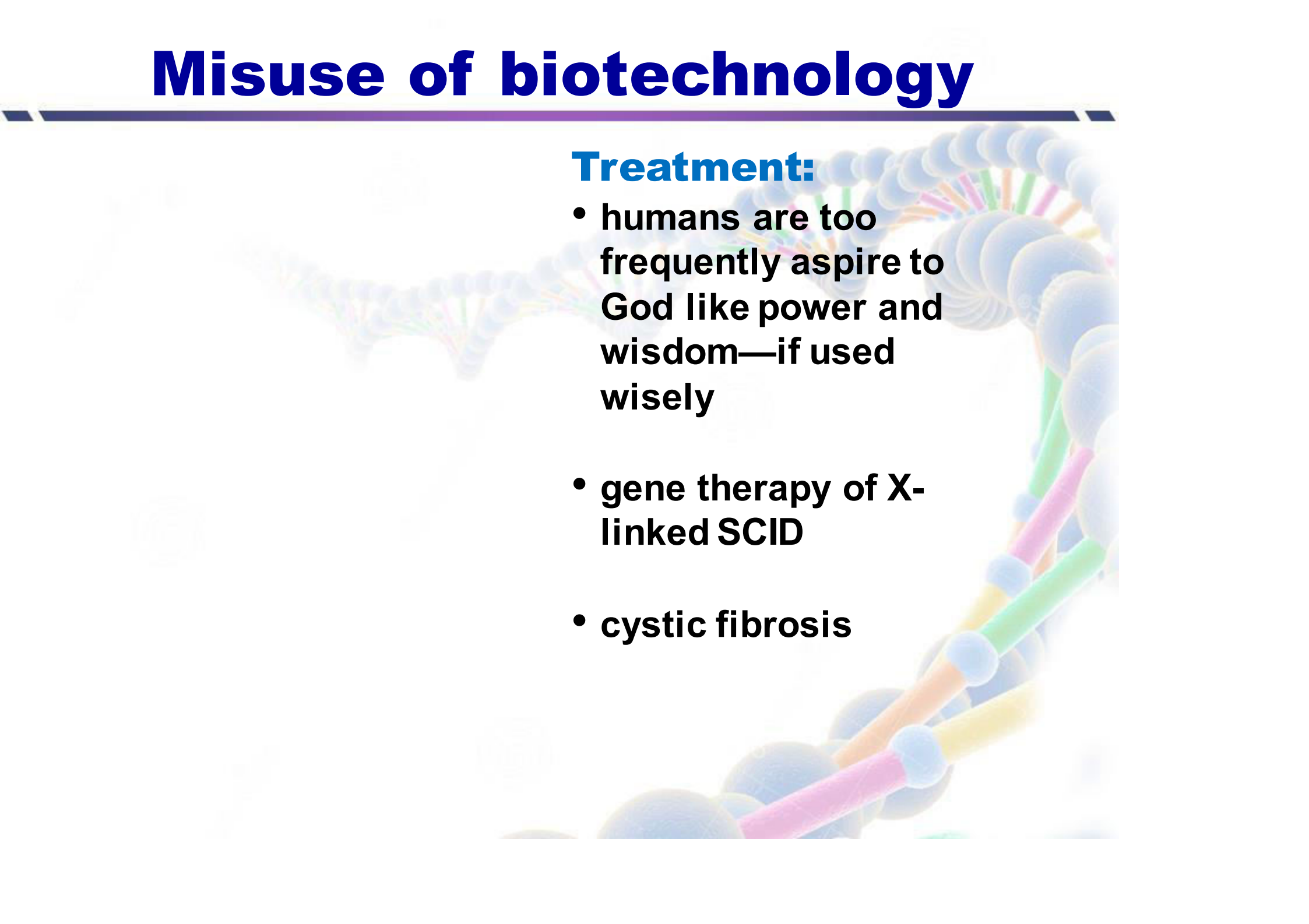


Misuse of biotechnology

Diagnosis:

- 99% homology between human genes and the mouse genome
- gene function is not necessarily identical
- prenatal/postnatal diagnosis/cancer studies

Misuse of biotechnology



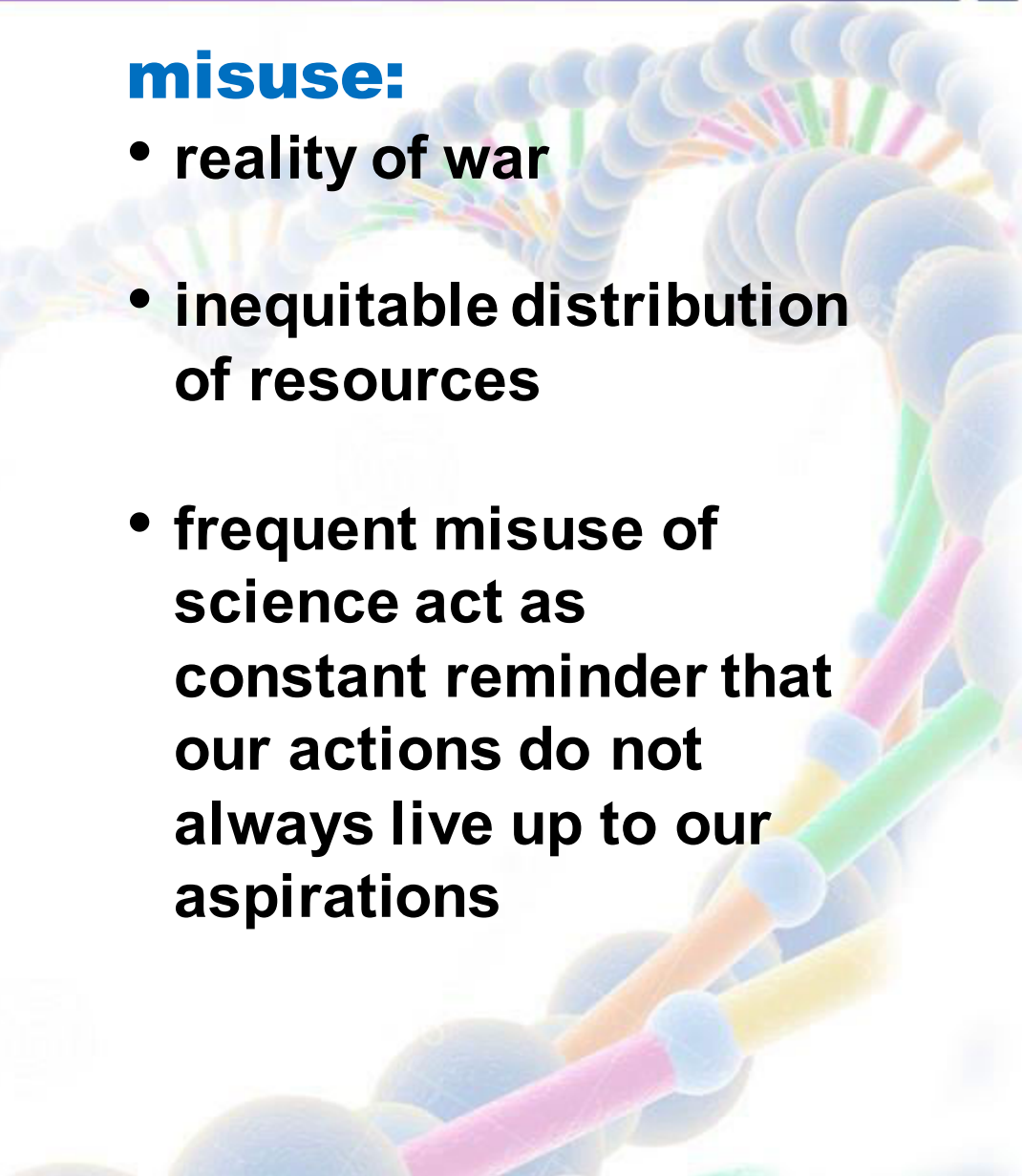
Treatment:

- humans are too frequently aspire to God like power and wisdom—if used wisely
- gene therapy of X-linked SCID
- cystic fibrosis

Misuse of biotechnology

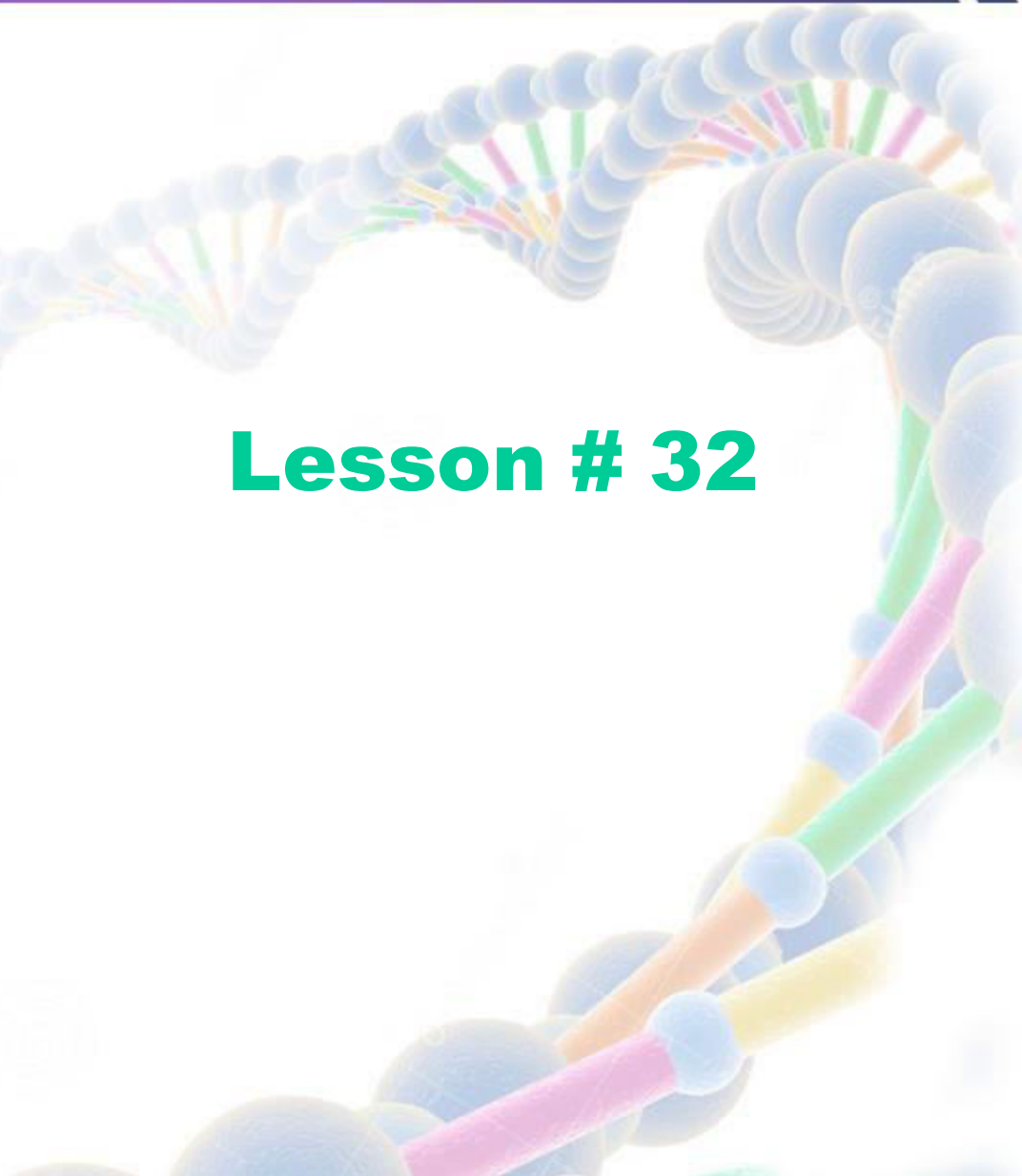
misuse:

- reality of war
- inequitable distribution of resources
- frequent misuse of science act as constant reminder that our actions do not always live up to our aspirations



Bioethics

Lesson # 32



Bioethics

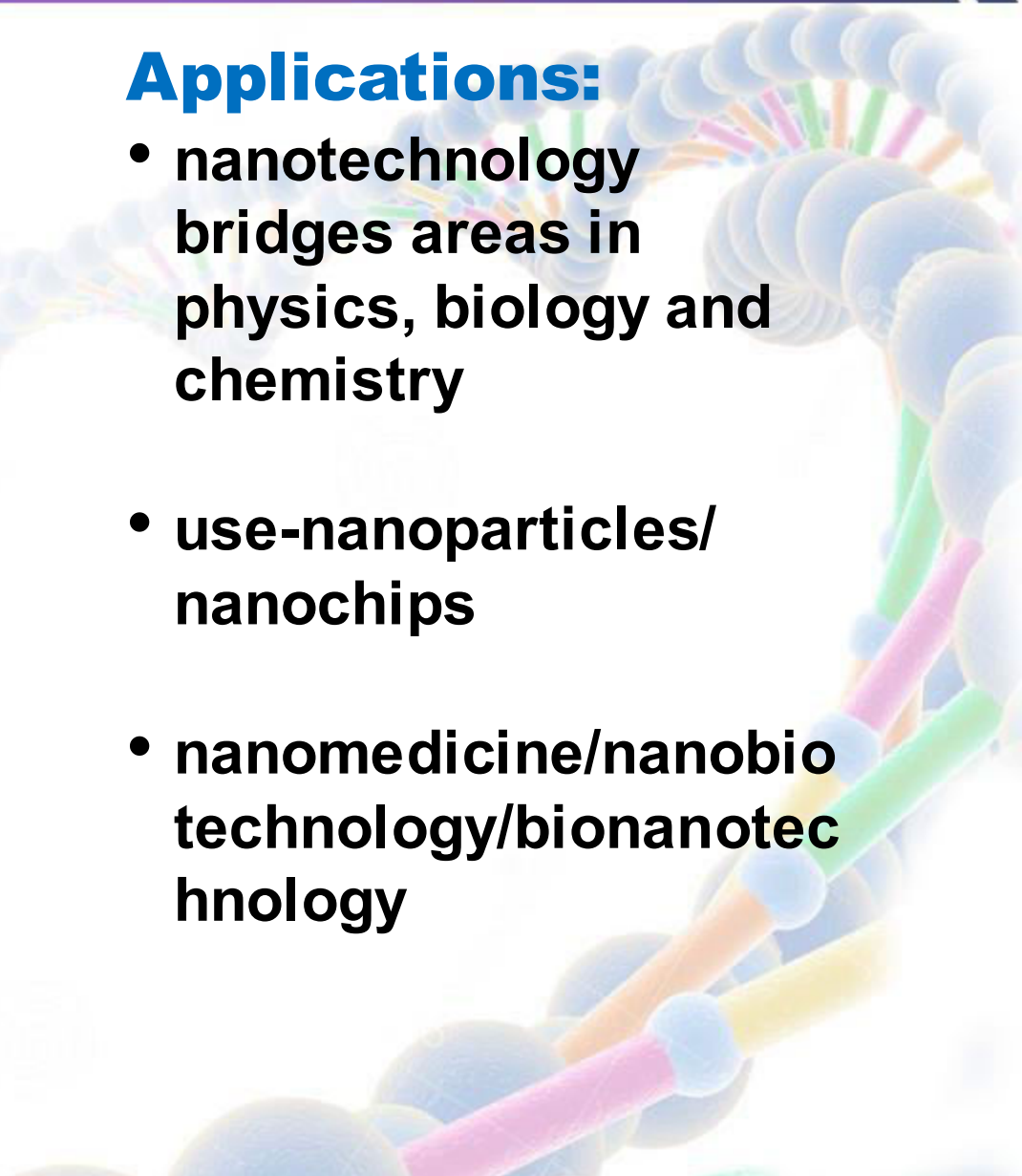
Nanotechnology



Nanotechnology

Applications:

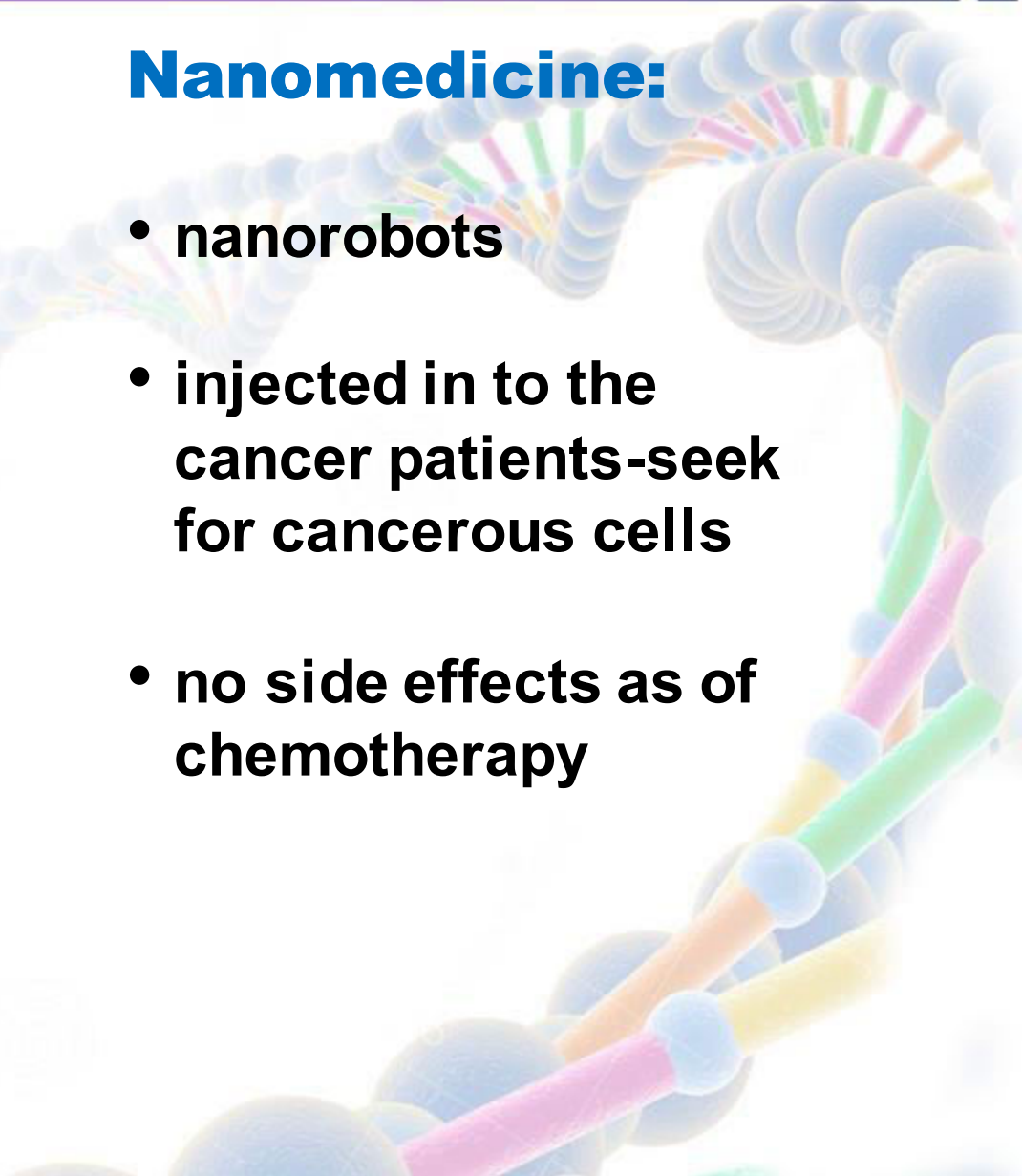
- nanotechnology bridges areas in physics, biology and chemistry
- use-nanoparticles/nanochips
- nanomedicine/nanobiotechnology/bionanotechnology



Nanotechnology

Nanomedicine:

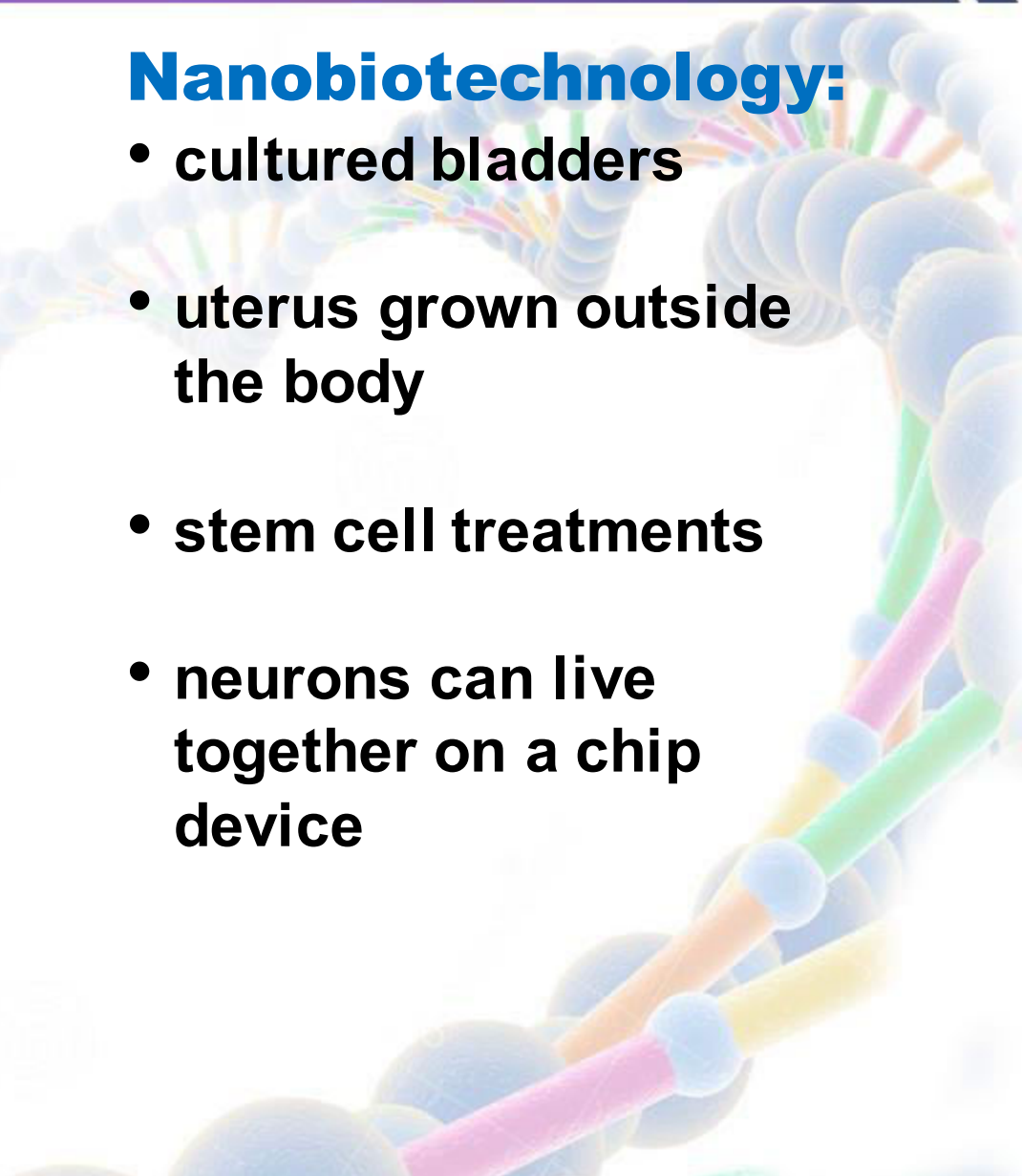
- nanorobots
- injected in to the cancer patients-
seek for cancerous cells
- no side effects as of chemotherapy



Nanotechnology

Nanobiotechnology:

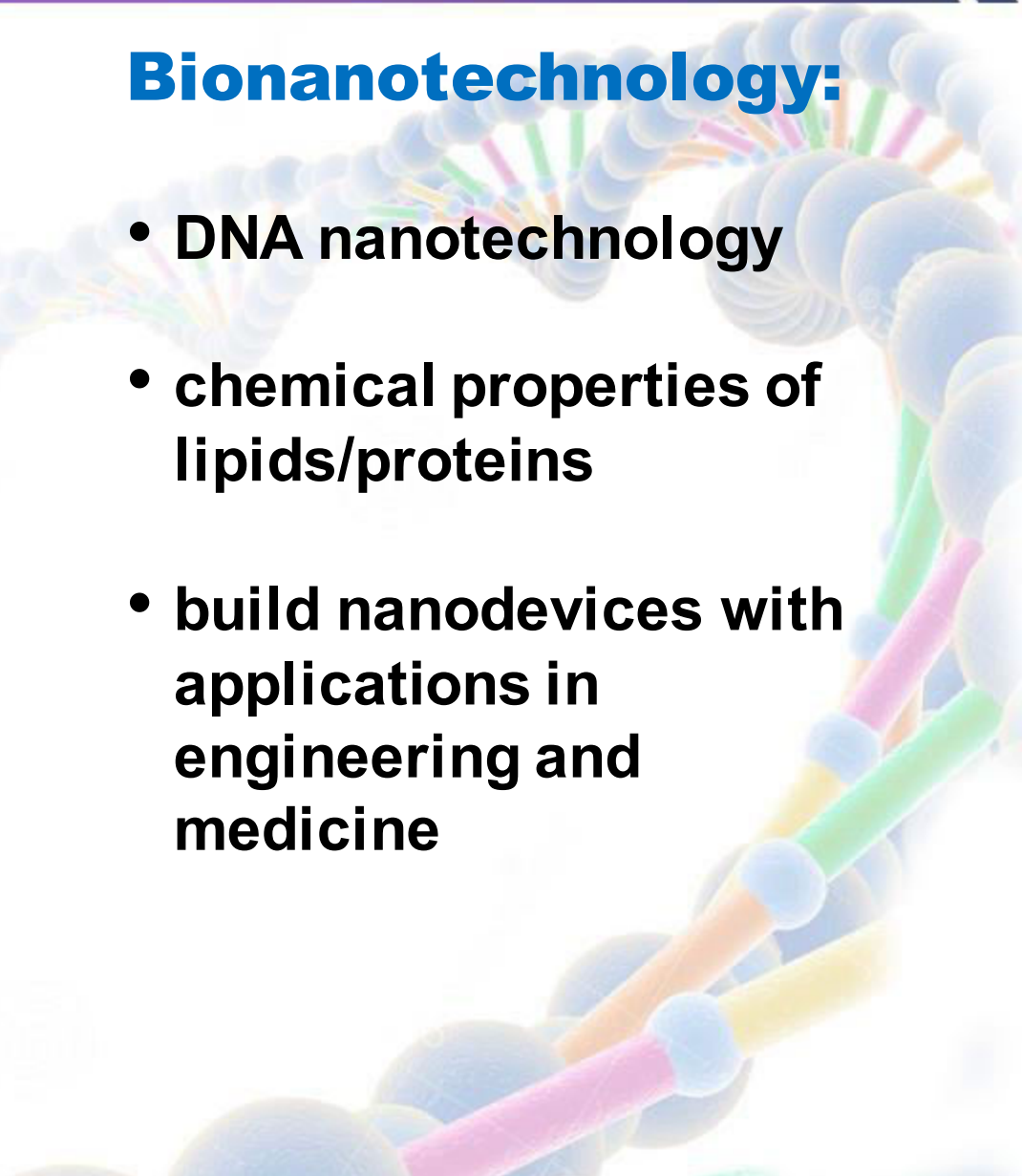
- cultured bladders
- uterus grown outside the body
- stem cell treatments
- neurons can live together on a chip device



Nanotechnology

Bionanotechnology:

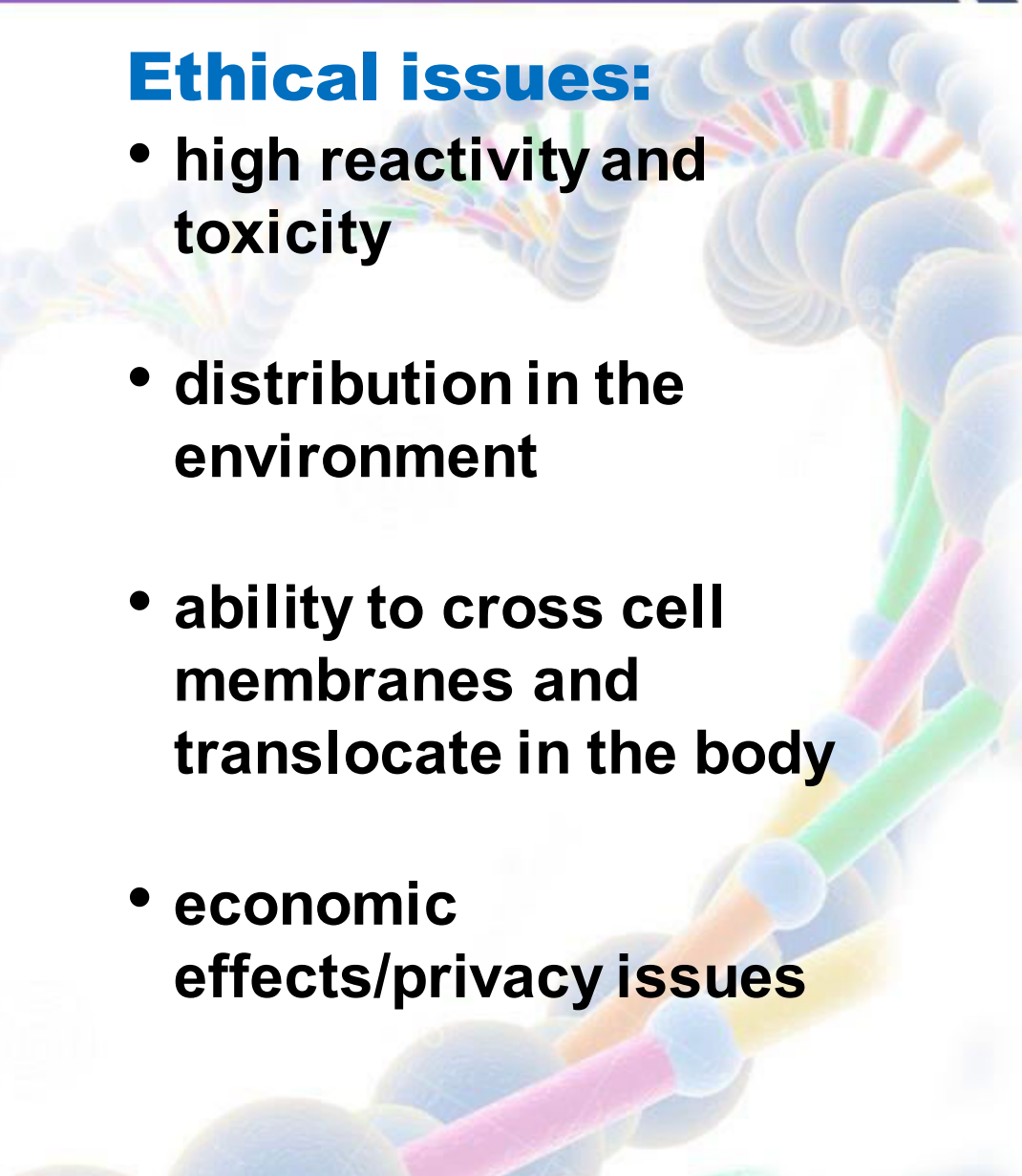
- DNA nanotechnology
- chemical properties of lipids/proteins
- build nanodevices with applications in engineering and medicine



Nanotechnology

Ethical issues:

- **high reactivity and toxicity**
- **distribution in the environment**
- **ability to cross cell membranes and translocate in the body**
- **economic effects/privacy issues**



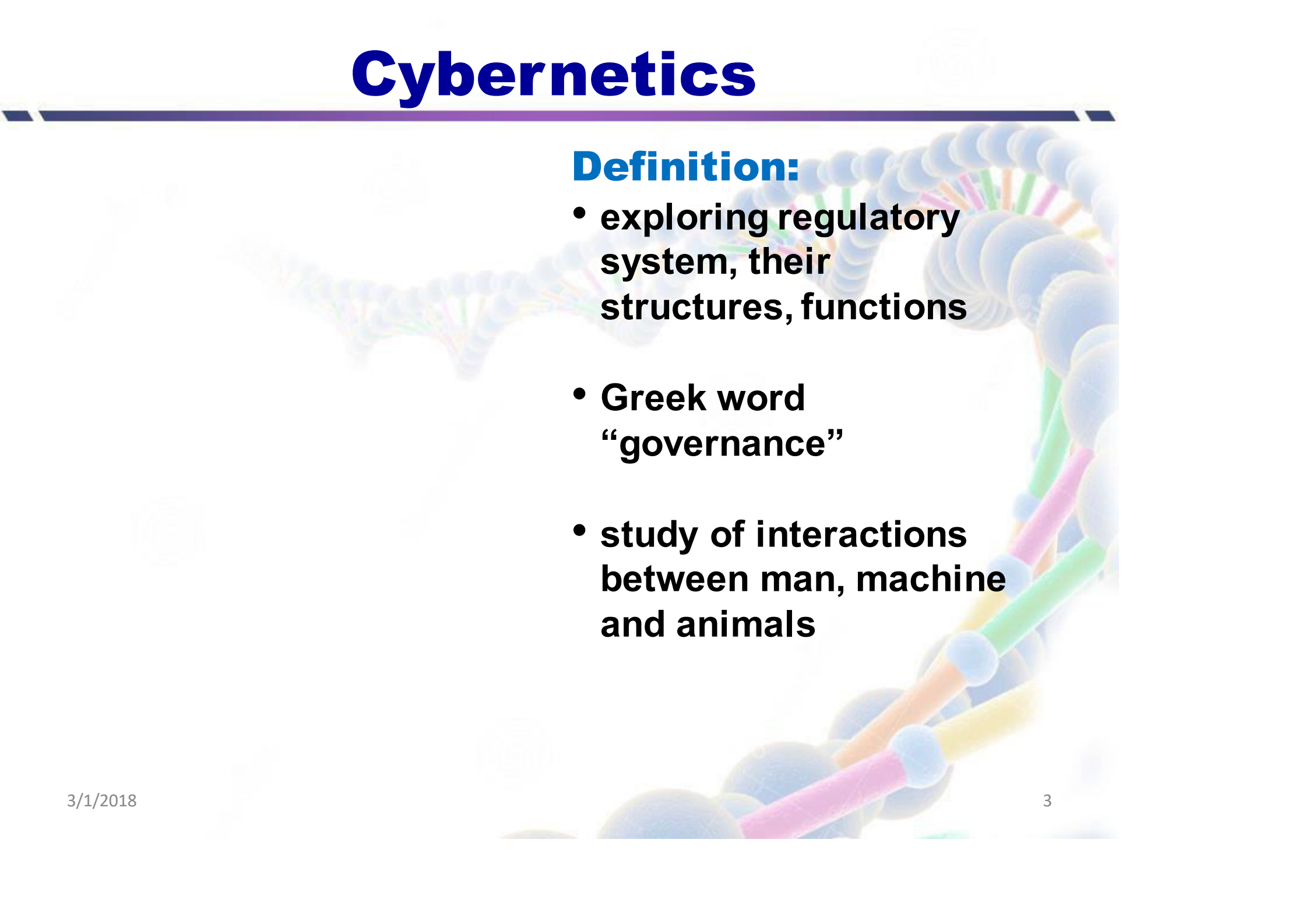
Bioethics

Lesson # 33

Bioethics

**Cybernetic
s**

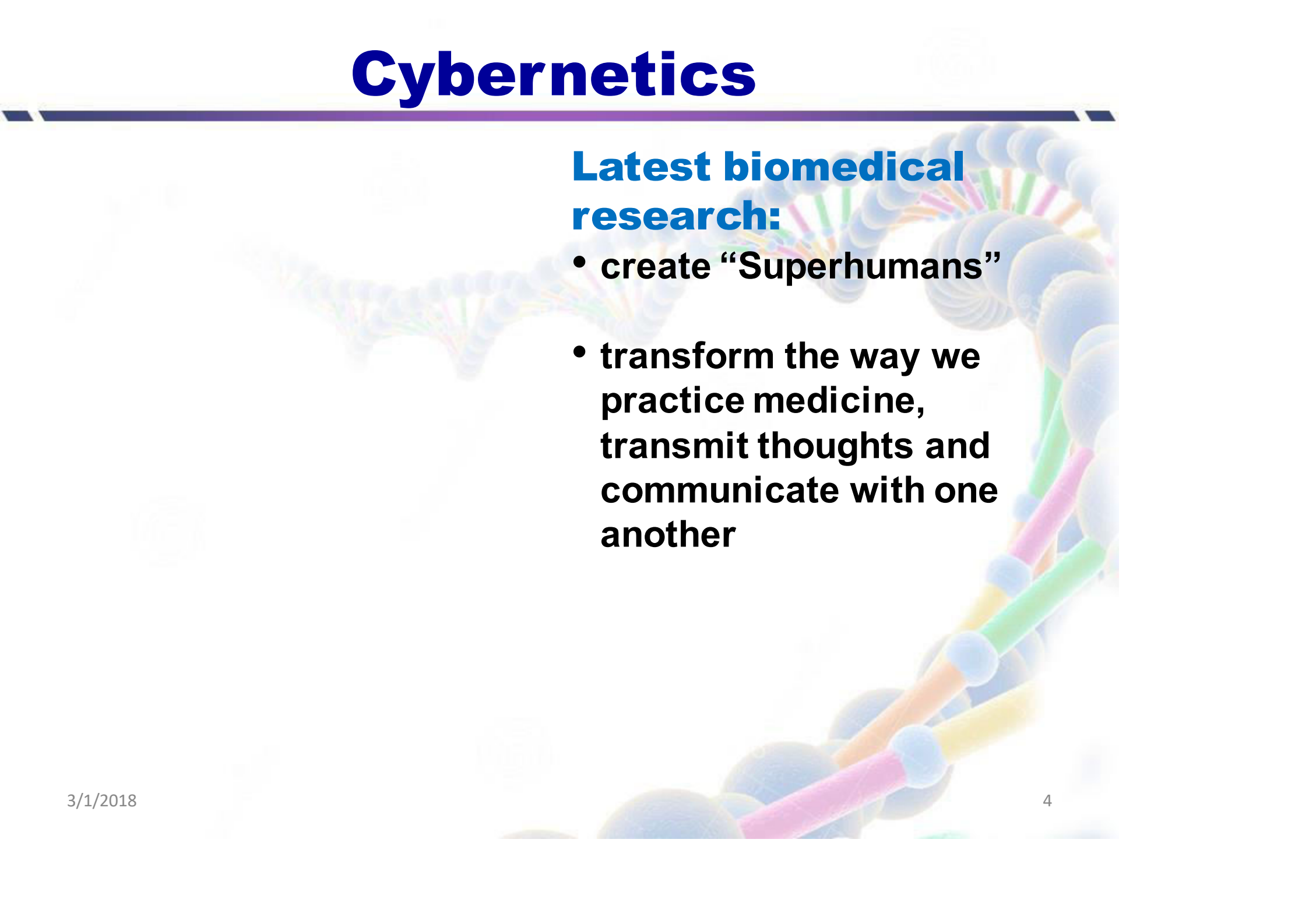
Cybernetics



Definition:

- exploring regulatory system, their structures, functions
- Greek word “governance”
- study of interactions between man, machine and animals

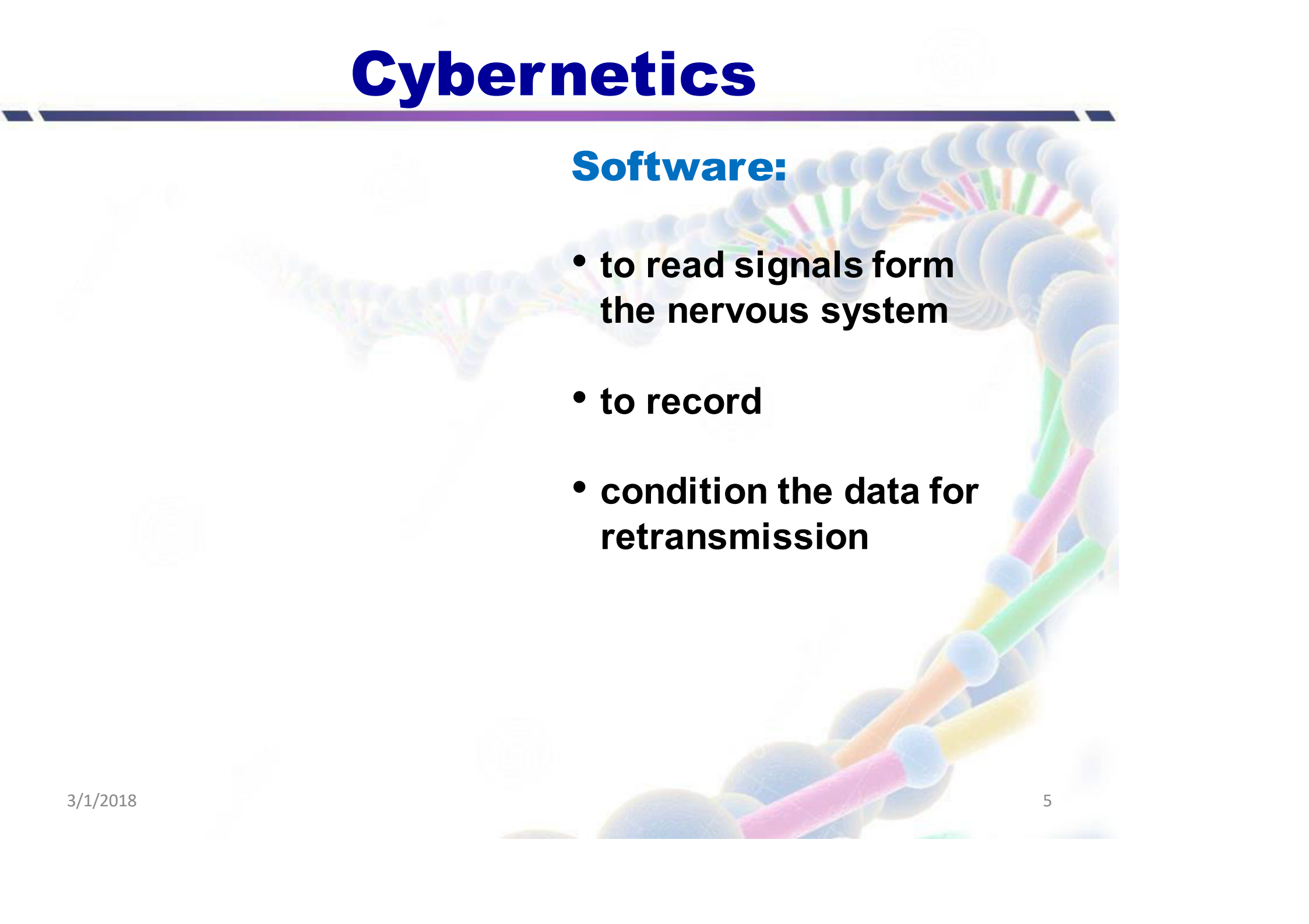
Cybernetics



Latest biomedical research:

- create “Superhumans”
- transform the way we practice medicine, transmit thoughts and communicate with one another

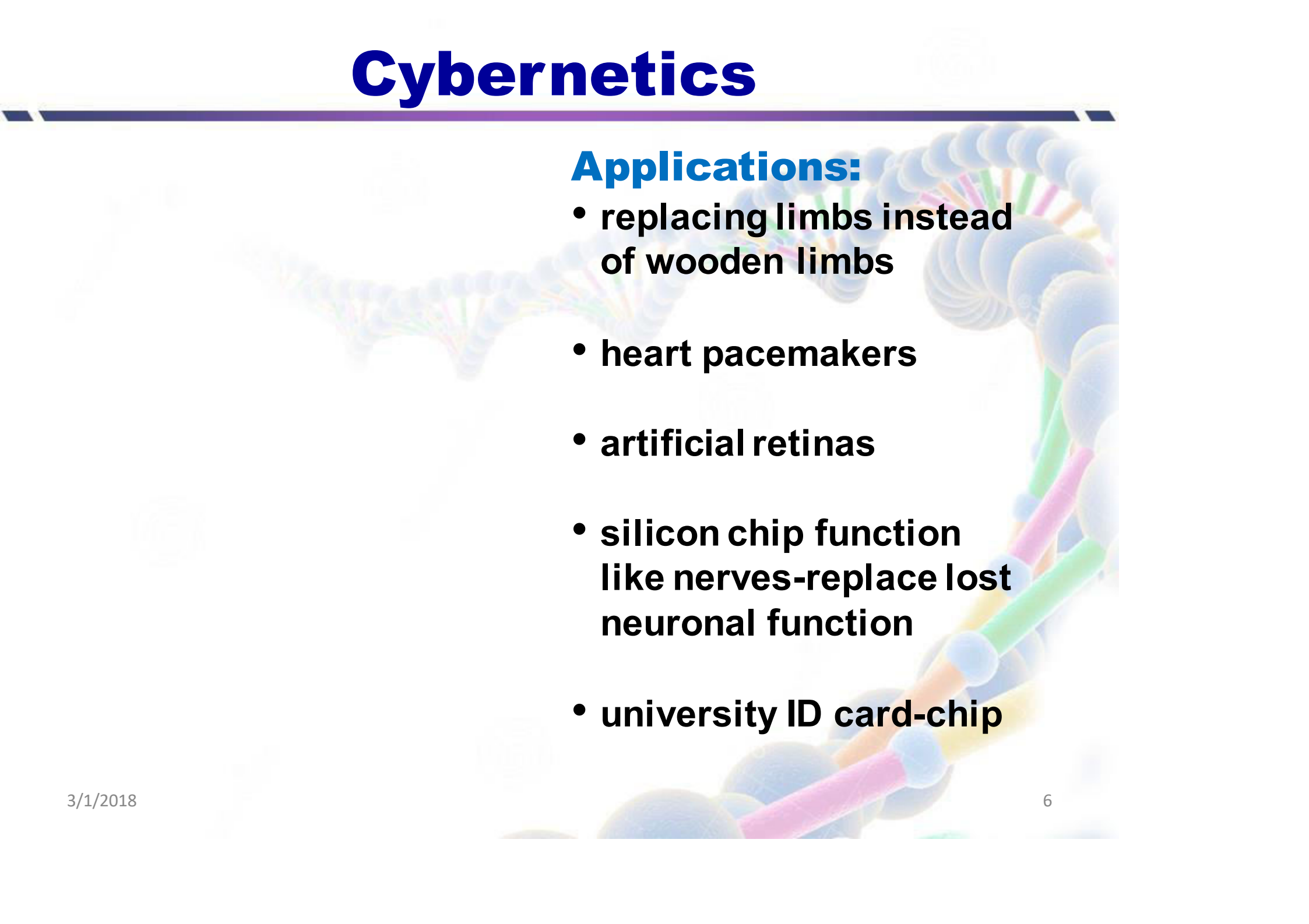
Cybernetics



Software:

- to read signals from the nervous system
- to record
- condition the data for retransmission

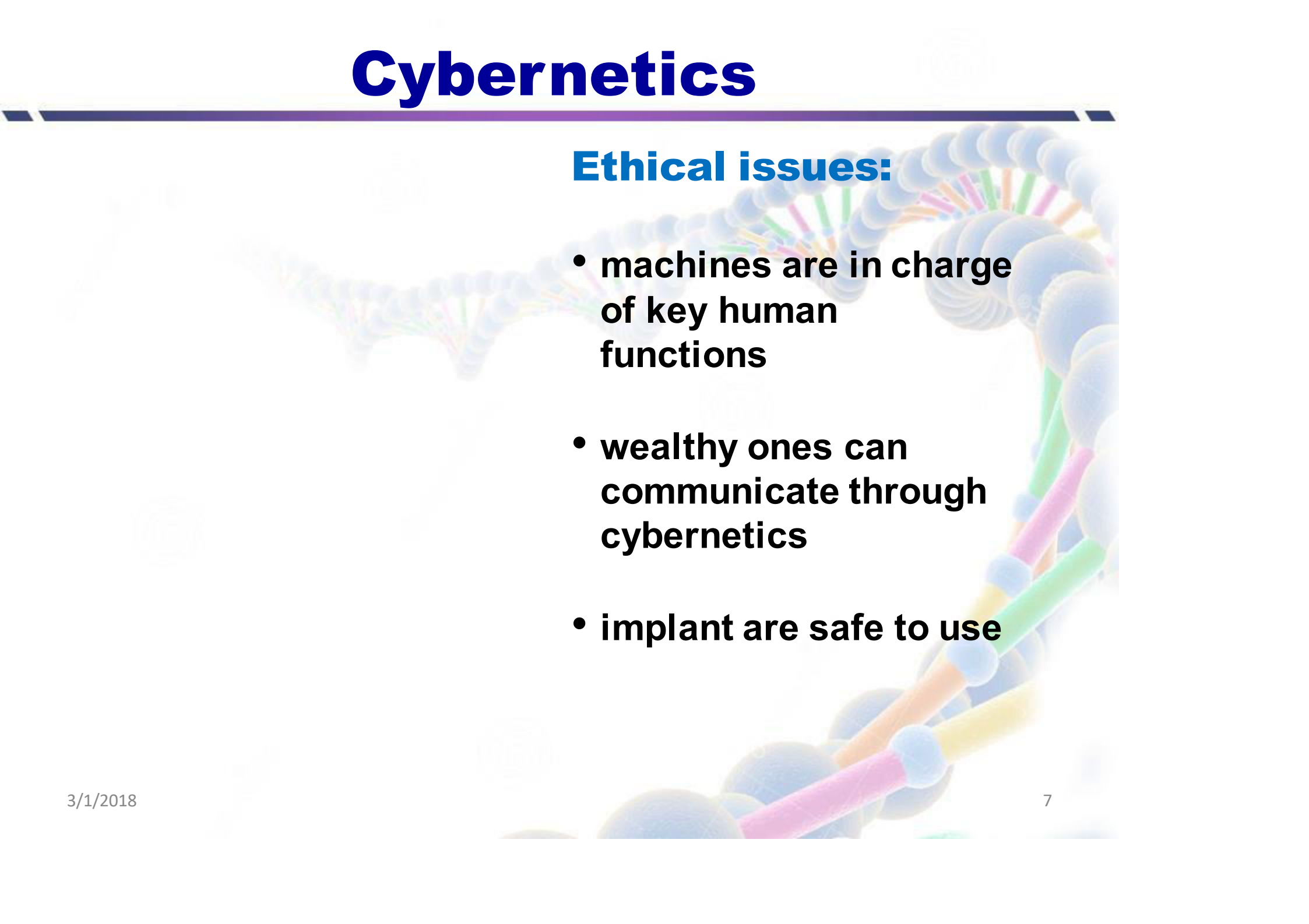
Cybernetics



Applications:

- replacing limbs instead of wooden limbs
- heart pacemakers
- artificial retinas
- silicon chip function like nerves-replace lost neuronal function
- university ID card-chip

Cybernetics



Ethical issues:

- machines are in charge of key human functions
- wealthy ones can communicate through cybernetics
- implant are safe to use

Cybernetics

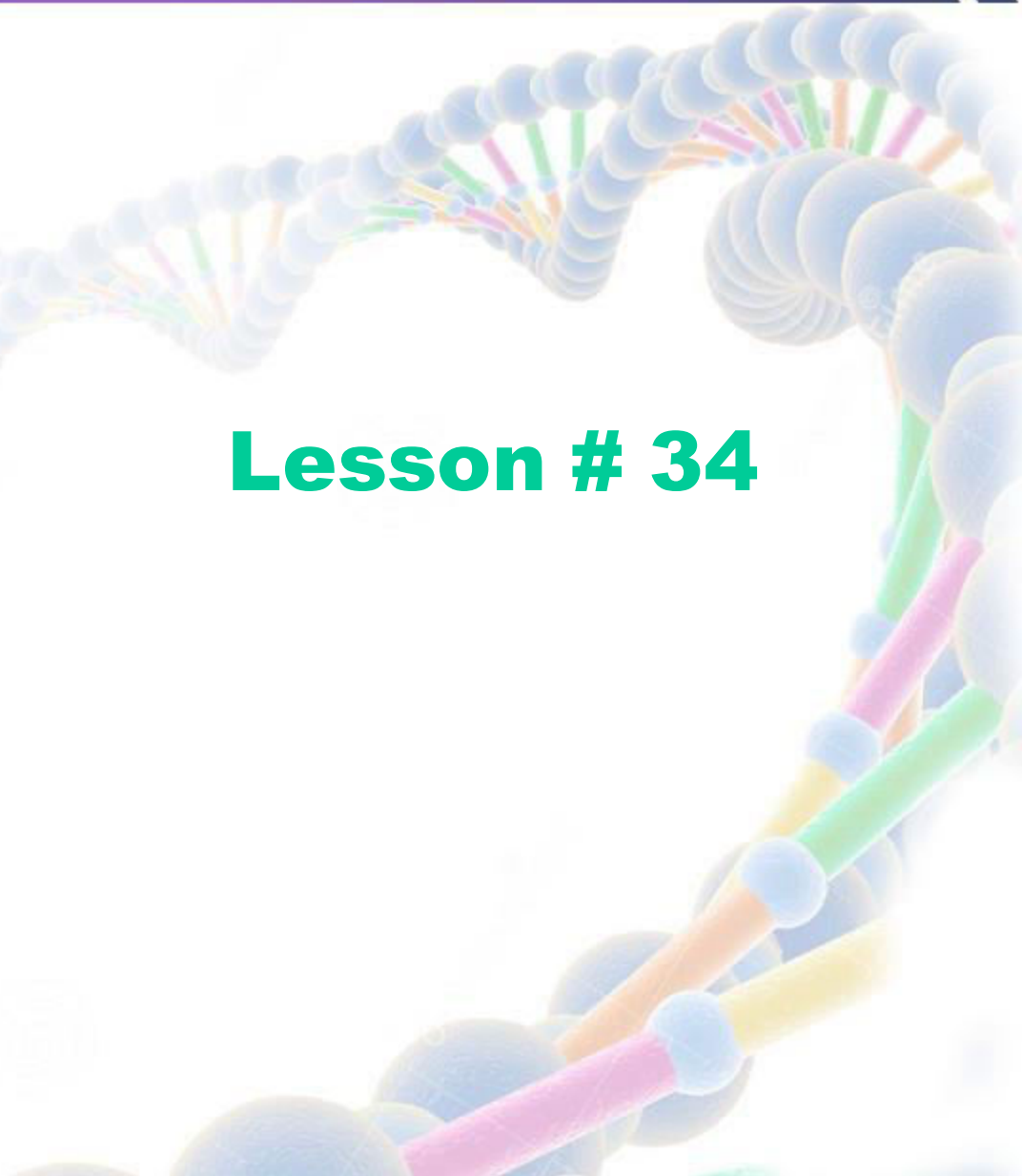


Ethical issues:

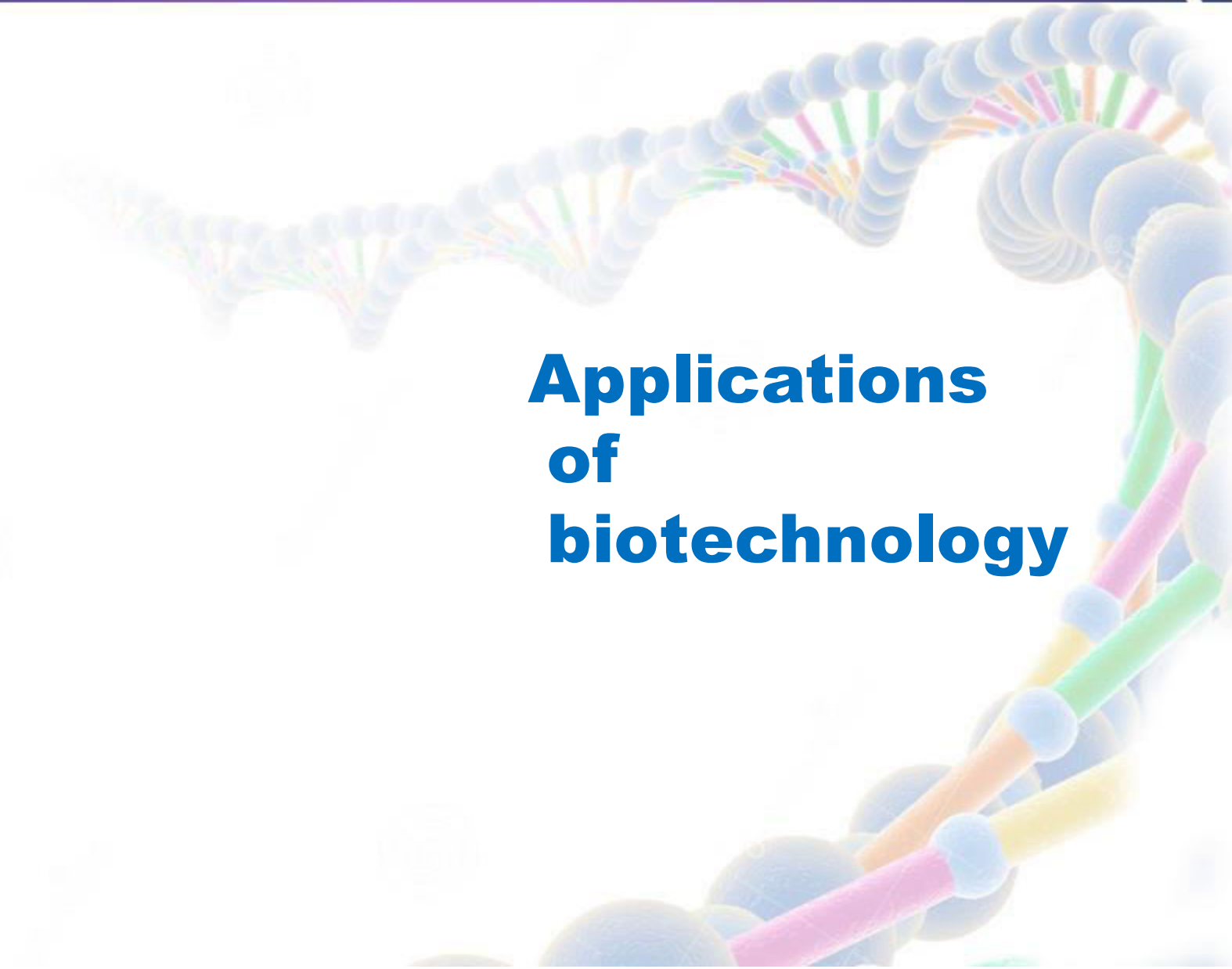
- **senses and impulses – transmitted in a harmful way**
- **can the senses be patented**
- **who regulates?**

Bioethics

Lesson # 34



Bioethics



Applications of biotechnology

Applications of biotechnology

Applications:

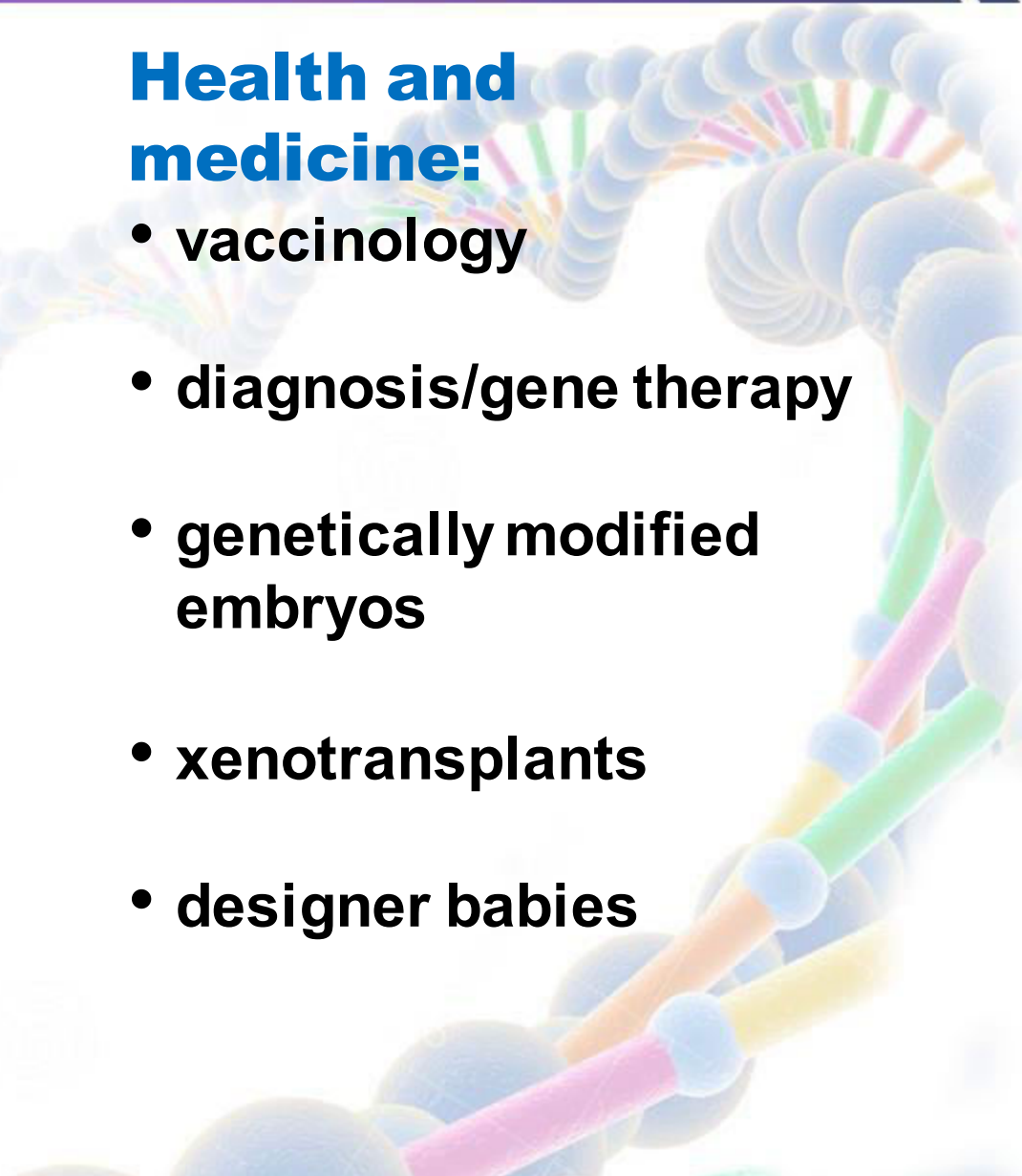
- health and medicine
- environmental use
- food and agriculture



Applications of biotechnology

Health and medicine:

- vaccinology
- diagnosis/gene therapy
- genetically modified embryos
- xenotransplants
- designer babies



Applications of biotechnology

Environmental use:

- oil spills-bioremediation
- pollution free environment
- remove algae
- use of fertilizers

Applications of biotechnology

Food and agriculture

:

- improved rice/potatoes/tomatoes
- pharmacrops
- improved sheep and cow milk

Bioethics

Lesson # 35



Bioethics



Ethical issues of GM food

Ethical issues of GM food

Ethical issues:

- **extrinsic concerns-**
how people view life,
nature
- **loss of biodiversity**
- **unfair to small farmers**
- **chances of transferring
antibiotic resistant
genes to bacteria**

Ethical issues of GM food

Ethical issues:

- gene flow and health issues
- intrinsic concerns- religion their personal emotions and values
- disrupts the beauty, integrity, balance of nature and might harm life

Bioethics

Lesson # 36



Bioethics



Risk factors of GM food

Risk factors of GM food

Risk factors:

- **create superbugs and superweeds**
- **kill bees and butterflies**
- **cross-pollination
contaminate regular
crops**
- **illegal to grow GM
plant accidentally**

Risk factors of GM food

Risk factors:

- harm biodiversity
- distract from healthy environmentally friendly technologies
- door between the government and biotechnology

Bioethics

Lesson # 37



Bioethics

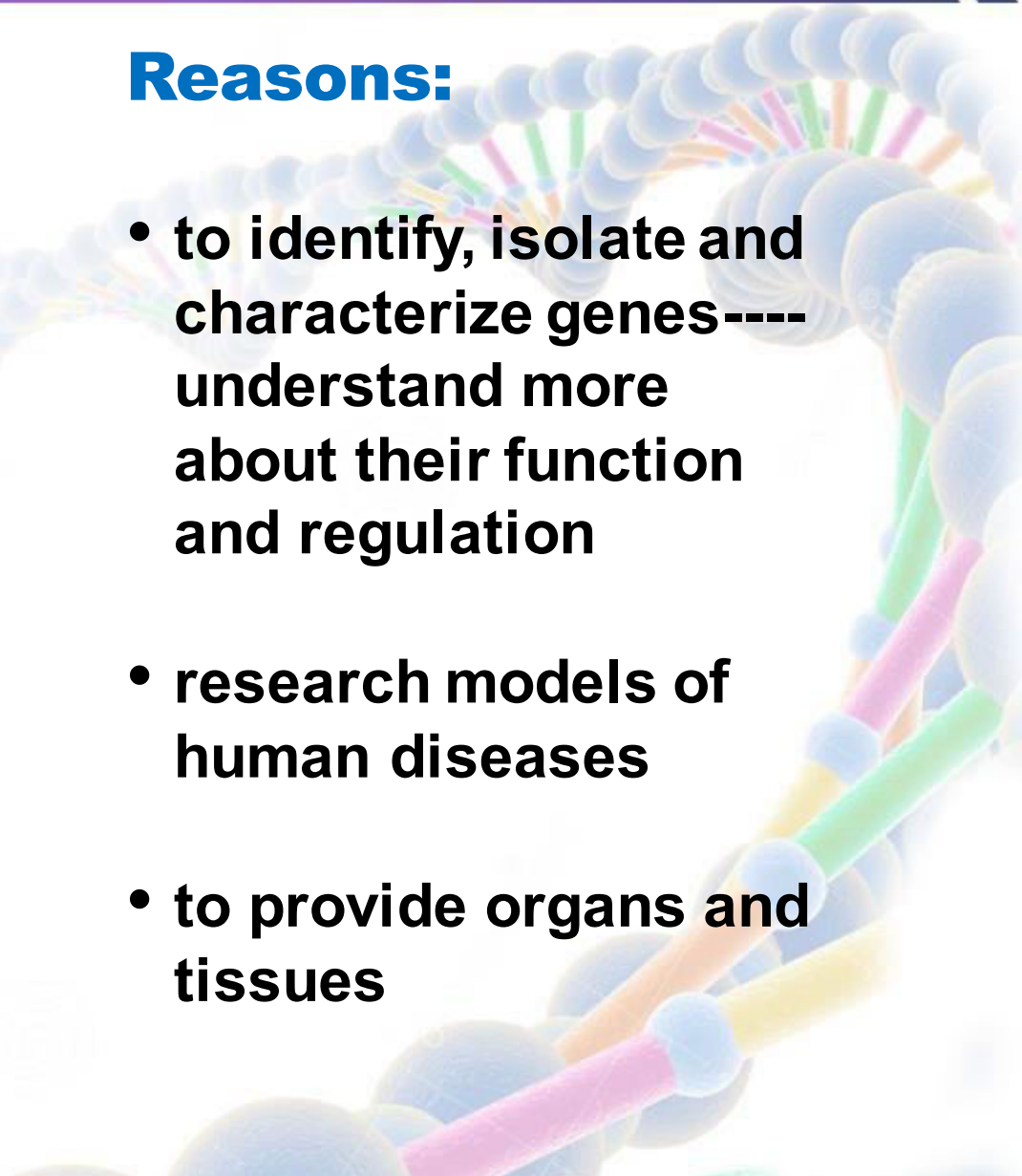


Ethics and animal biotechnology

Ethics and animal biotechnology

Reasons:

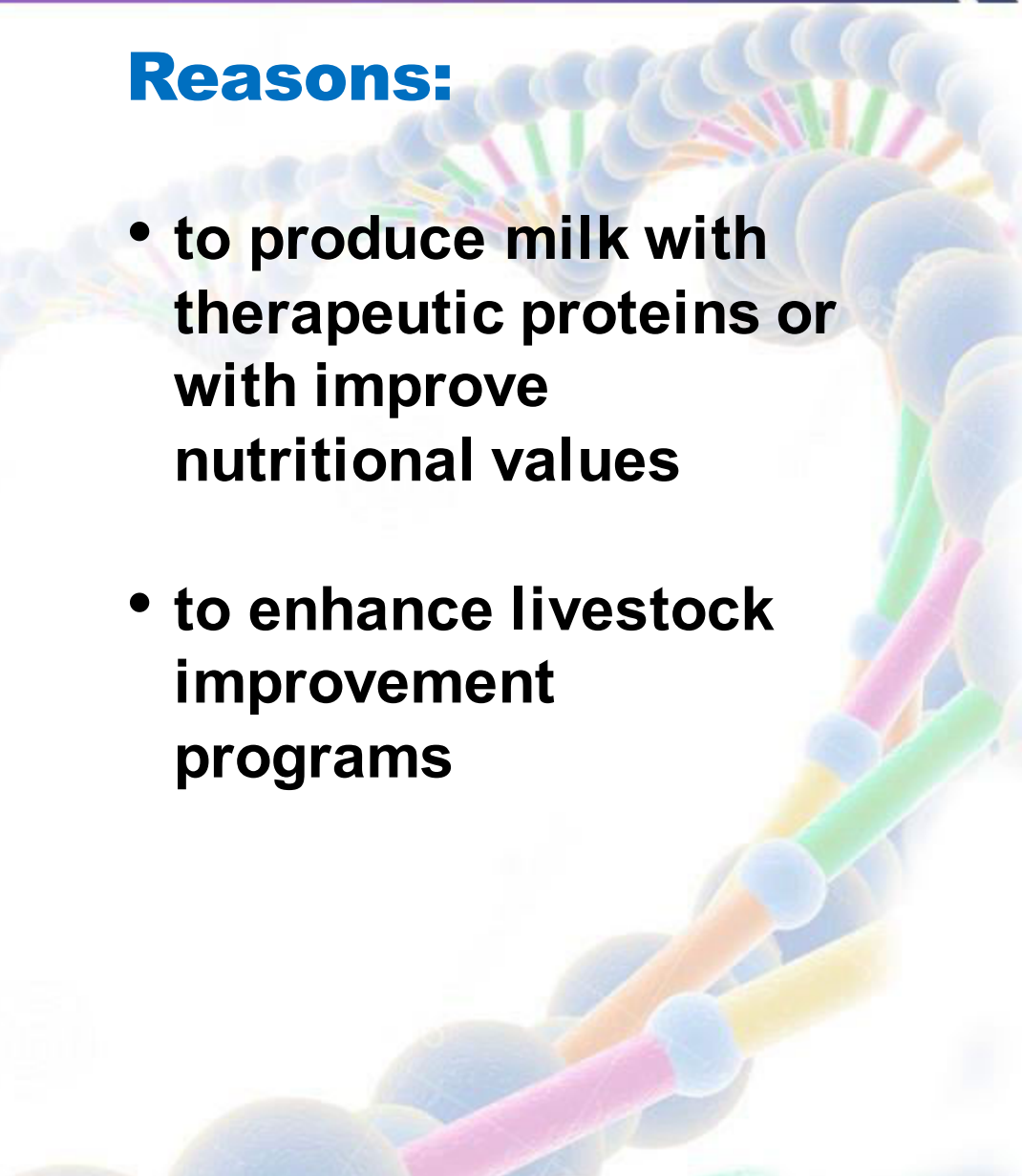
- to identify, isolate and characterize genes---- understand more about their function and regulation
- research models of human diseases
- to provide organs and tissues



Ethics and animal biotechnology

Reasons:

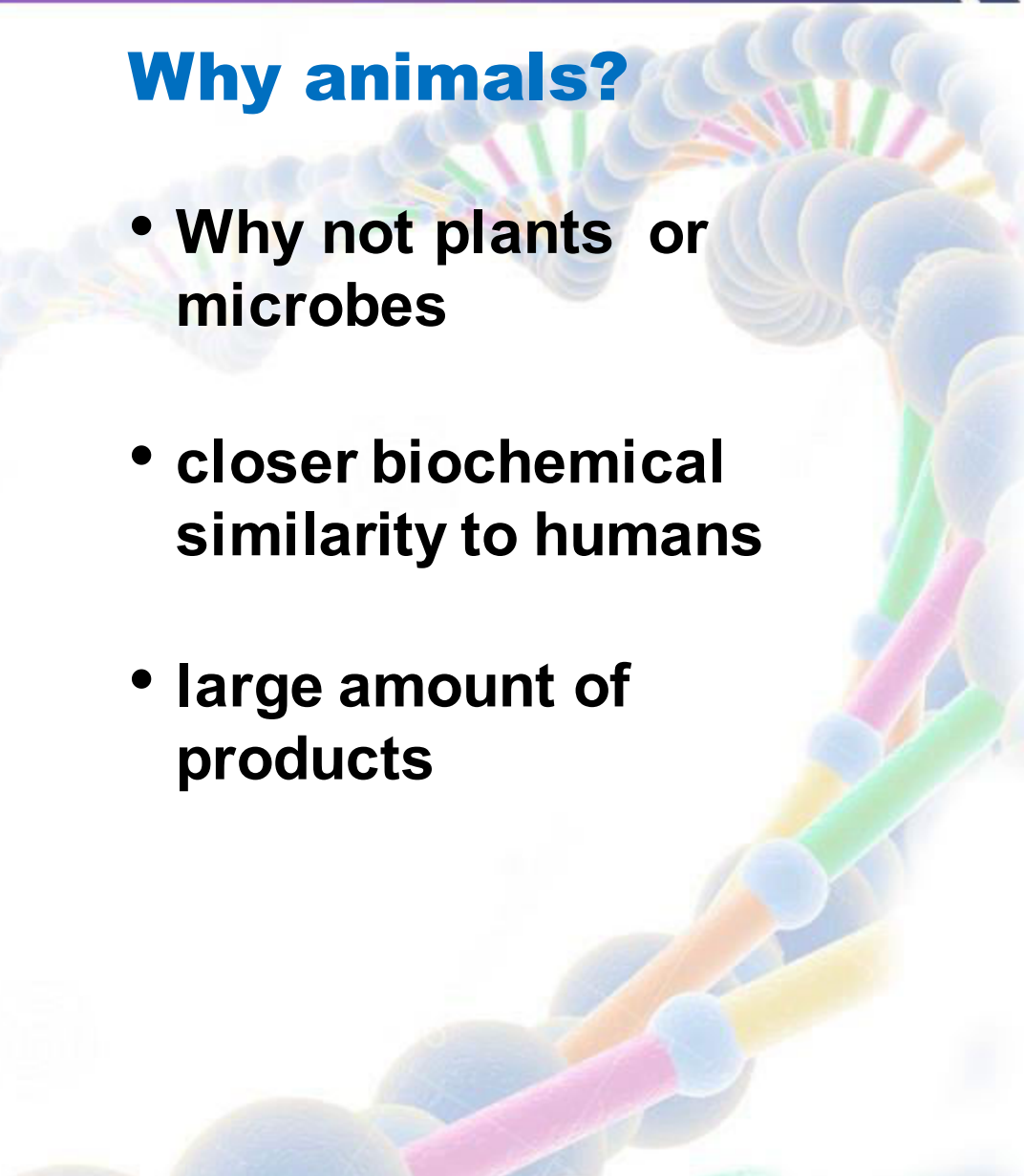
- to produce milk with therapeutic proteins or with improve nutritional values
- to enhance livestock improvement programs



Ethics and animal biotechnology

Why animals?

- Why not plants or microbes
- closer biochemical similarity to humans
- large amount of products



Ethics and animal biotechnology

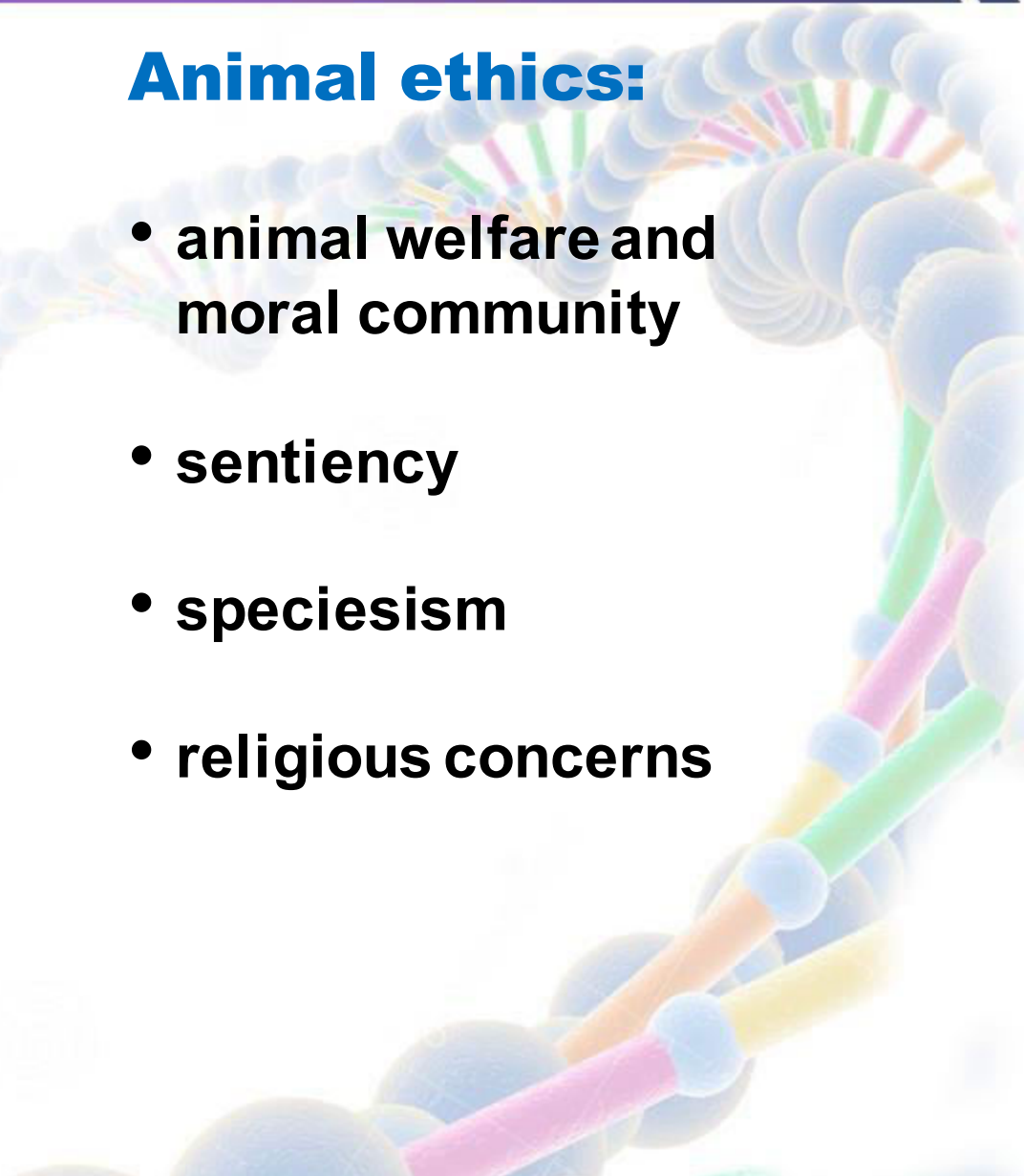
Nuclear transfer:

- Whole nuclei and the gene they carry are transferred---Dolly
- Providing cells as a source of replacement grafting
- Genetic conservation

Ethics and animal biotechnology

Animal ethics:

- **animal welfare and moral community**
- **sentientcy**
- **speciesism**
- **religious concerns**



Bioethics

Lesson # 38

Bioethics



Human genome project

Human genome project



HGP:

- **determine--sequence of chemical base pairs—make up human DNA**
- **identifying and mapping genes**
- **difficult—converting the idea into public policy**

Human genome project



State of Completion:

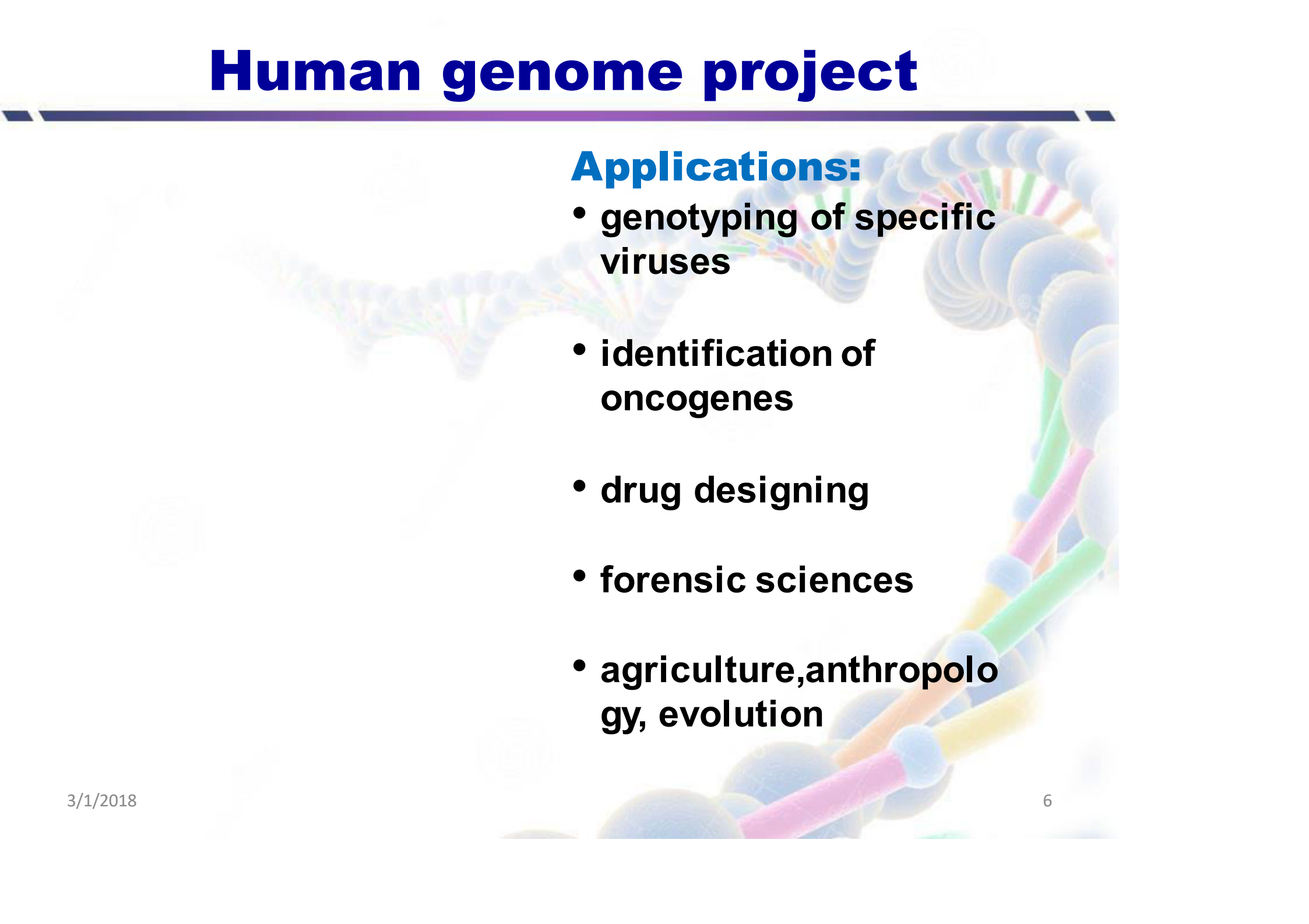
- April 2003
- 99% euchromatic human genome
- 99.99% accuracy
- heterochromatic regions are not sequenced

Human genome project

Techniques and analysis:

- genome annotation
- domain of bioinformatics
- 20,500 genes
- more segmental duplication

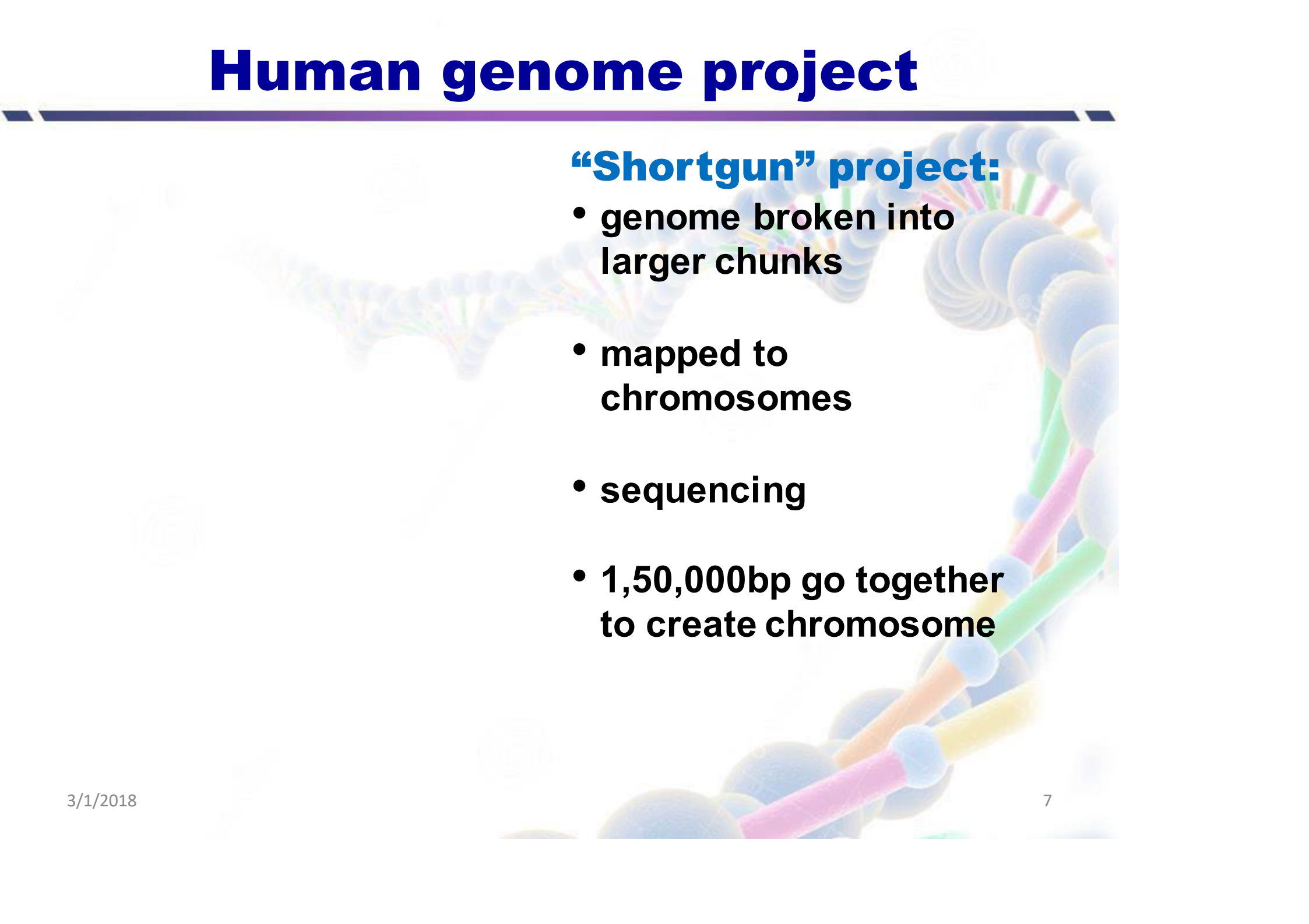
Human genome project



Applications:

- **genotyping of specific viruses**
- **identification of oncogenes**
- **drug designing**
- **forensic sciences**
- **agriculture, anthropology, evolution**

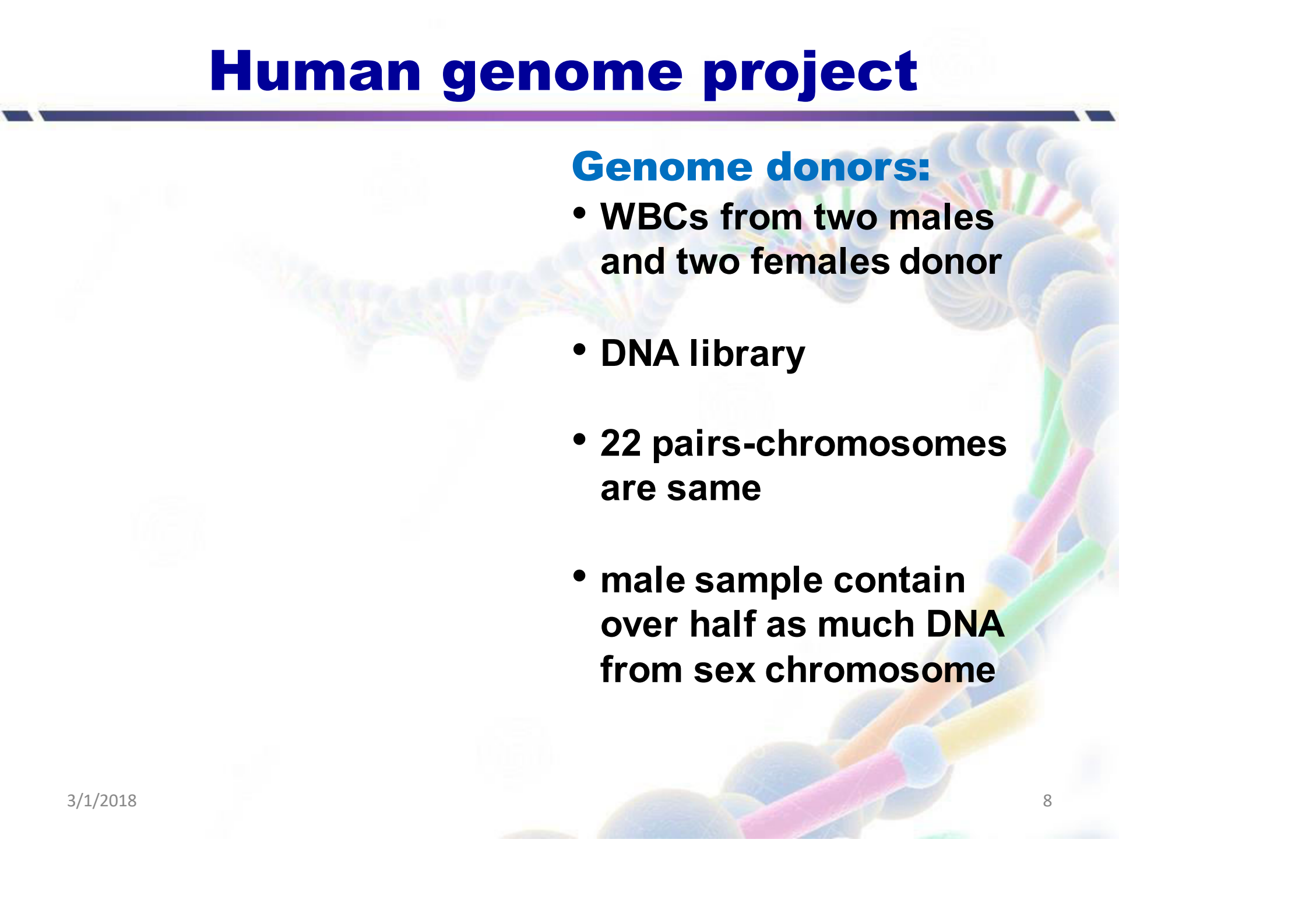
Human genome project



“Shotgun” project:

- genome broken into larger chunks
- mapped to chromosomes
- sequencing
- 1,50,000bp go together to create chromosome

Human genome project



Genome donors:

- WBCs from two males and two females donor
- DNA library
- 22 pairs-chromosomes are same
- male sample contain over half as much DNA from sex chromosome

Human genome project

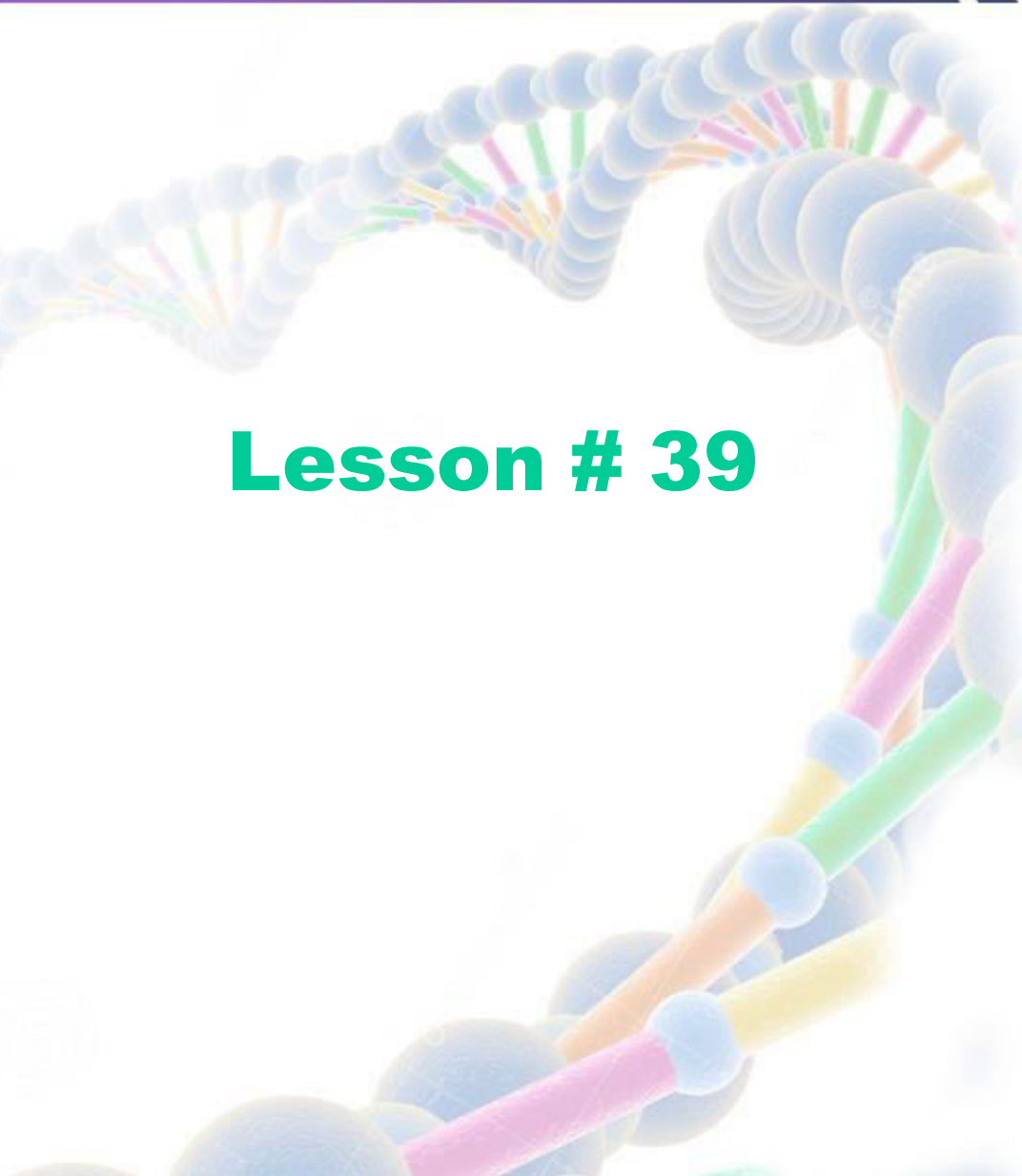


Ethical issues:

- used to discriminate against people
- refuse to provide insurance
- ethical, legal and social implications program (1990)

Bioethics

Lesson # 39



Bioethics



Thoughts on eugenics

Thoughts on eugenics

eugenics:

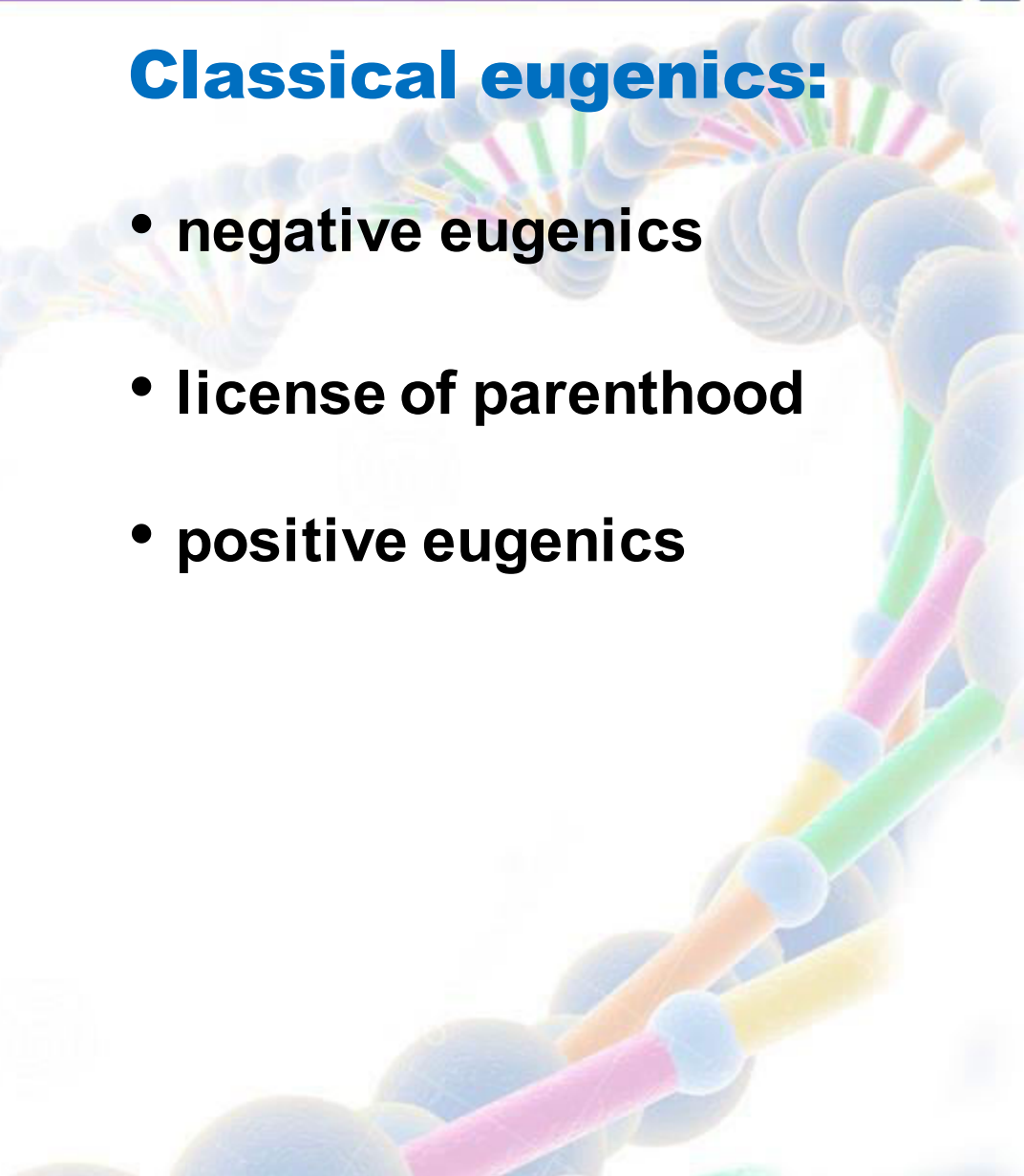
- **Greek “well-born”**
- **belief and practice –
improve the quality of
human population**
- **began early in 20
century**



Thoughts on eugenics

Classical eugenics:

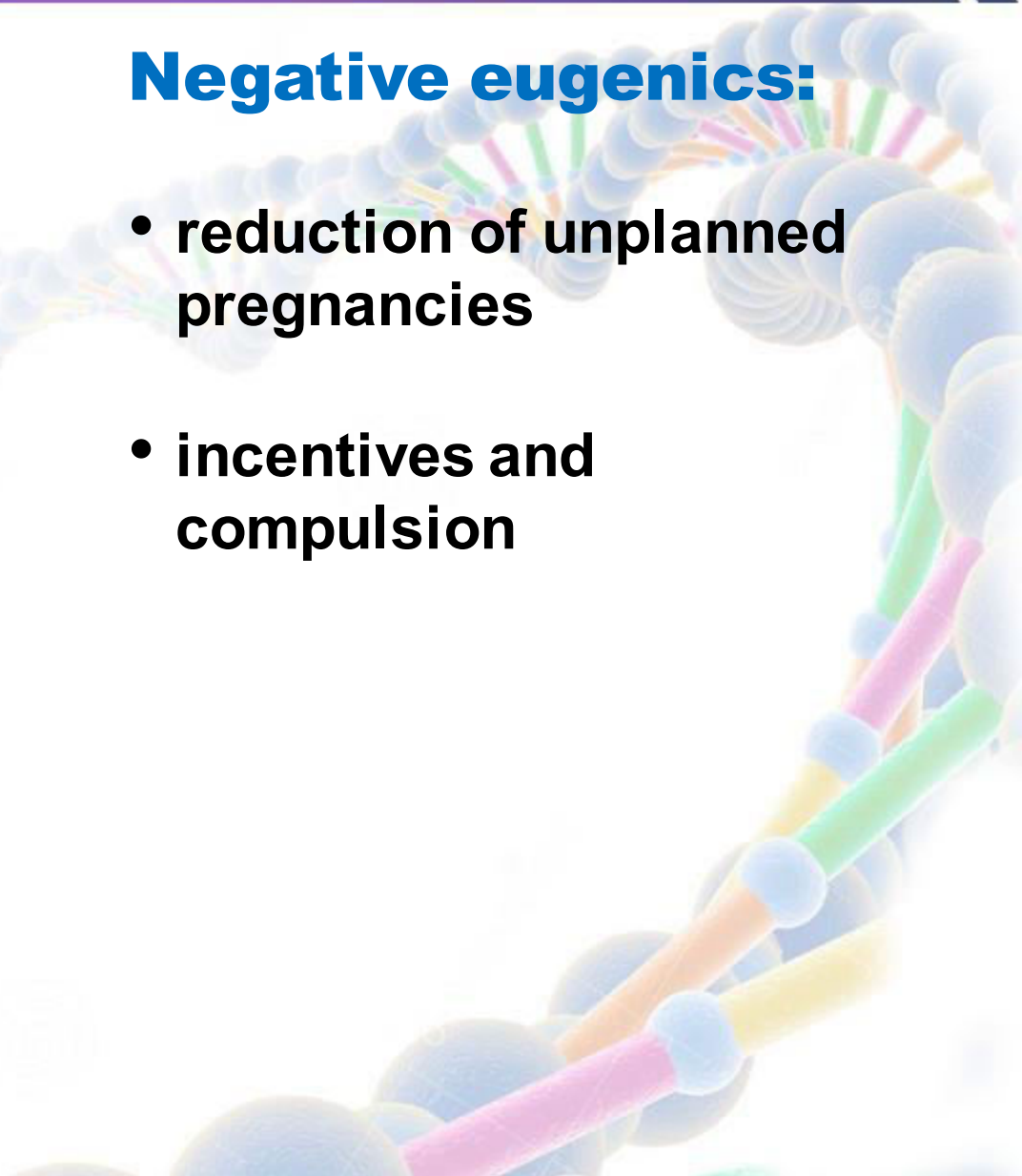
- negative eugenics
- license of parenthood
- positive eugenics



Thoughts on eugenics

Negative eugenics:

- reduction of unplanned pregnancies
- incentives and compulsion



Thoughts on eugenics

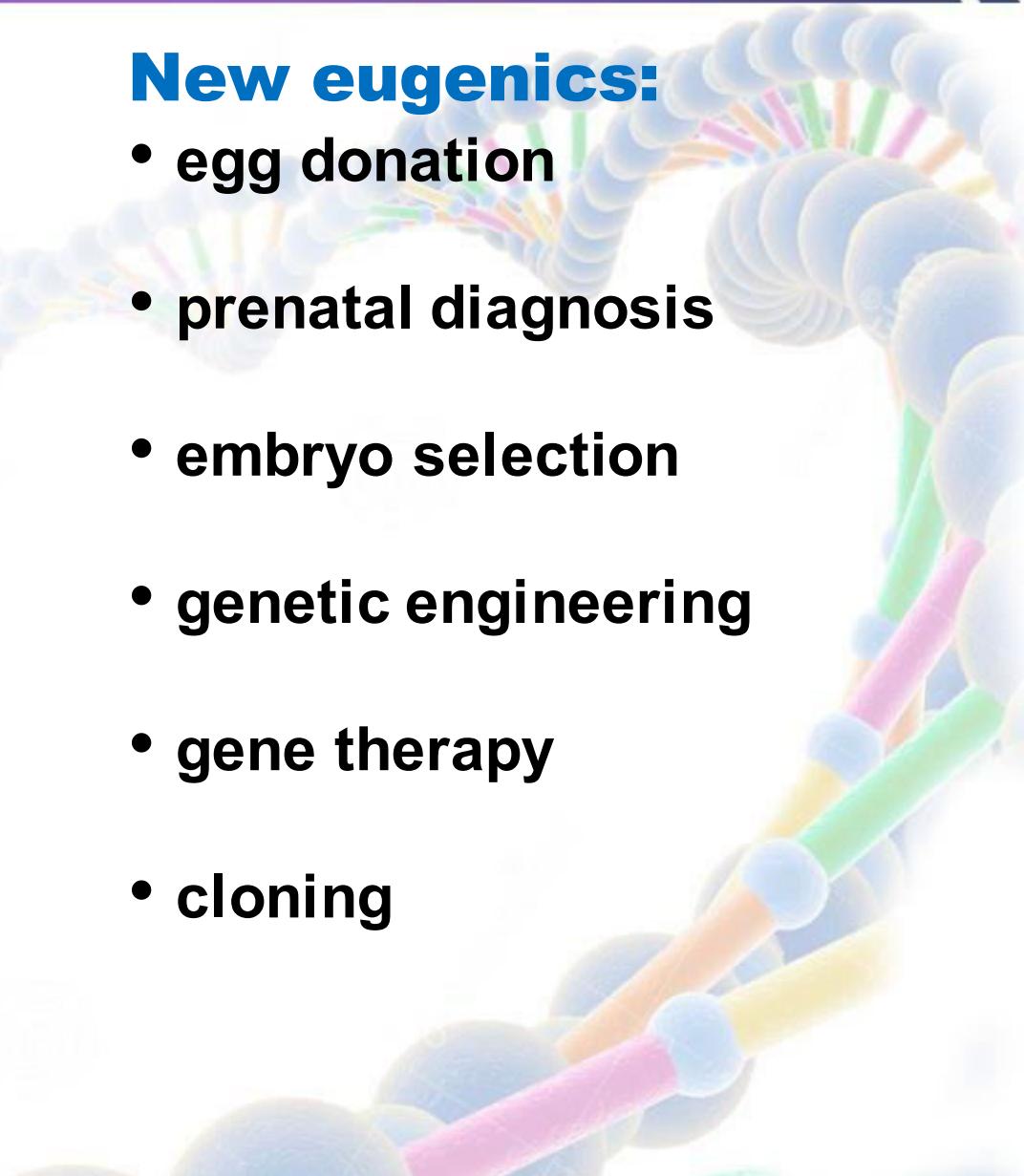
Positive eugenics:

- financial/selective incentives to have children
- taxation of the childless
- ethical obligations of the elite
- eugenic immigration

Thoughts on eugenics

New eugenics:

- egg donation
- prenatal diagnosis
- embryo selection
- genetic engineering
- gene therapy
- cloning



Thoughts on eugenics

Ethics:

- “It inevitably leads to measure that are unethical”
- no longer ex post facto regulation of the living
- preemptive action on the unborn
- unborn fetus lack the voice of consent

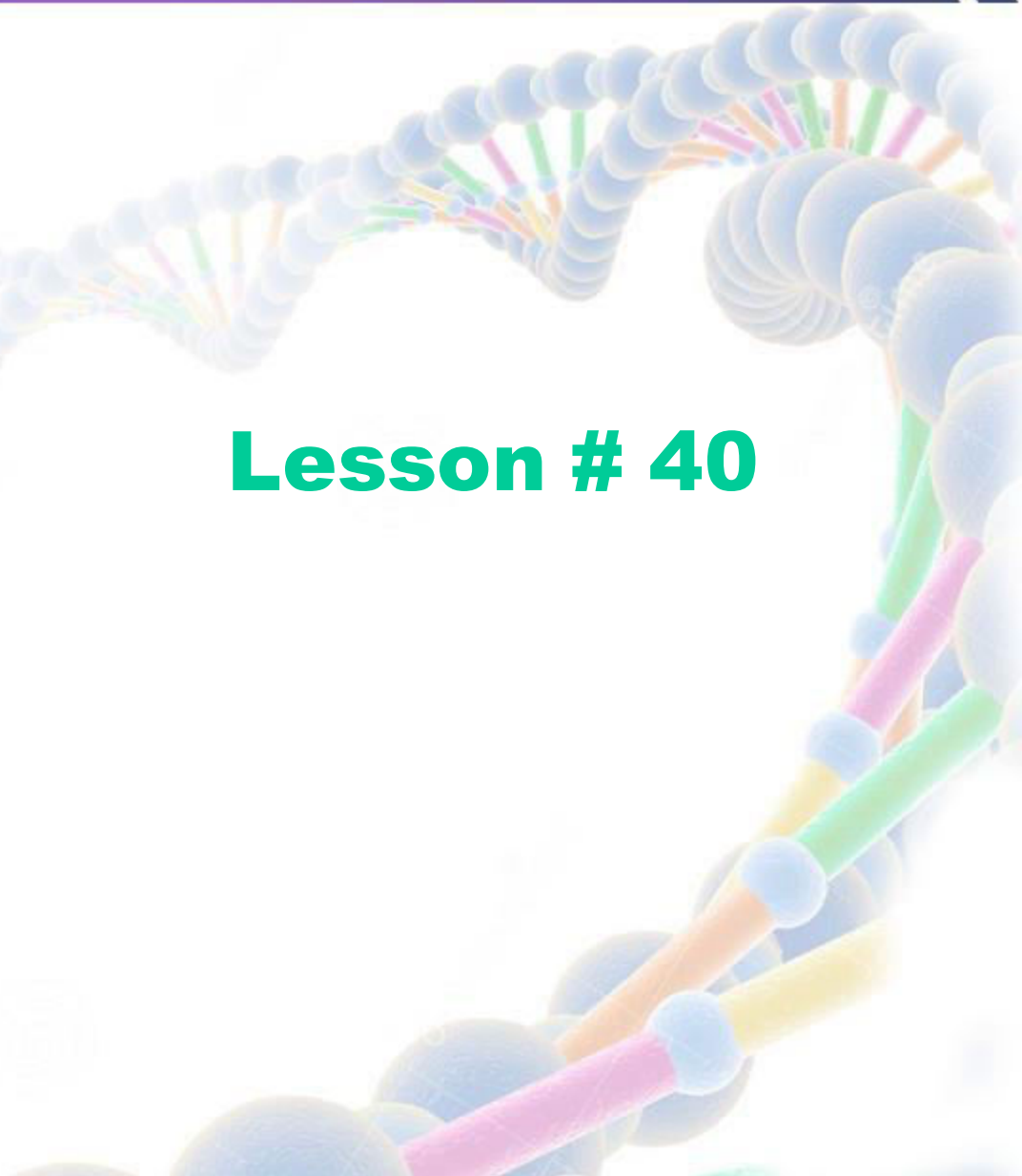
Thoughts on eugenics

Ethics:

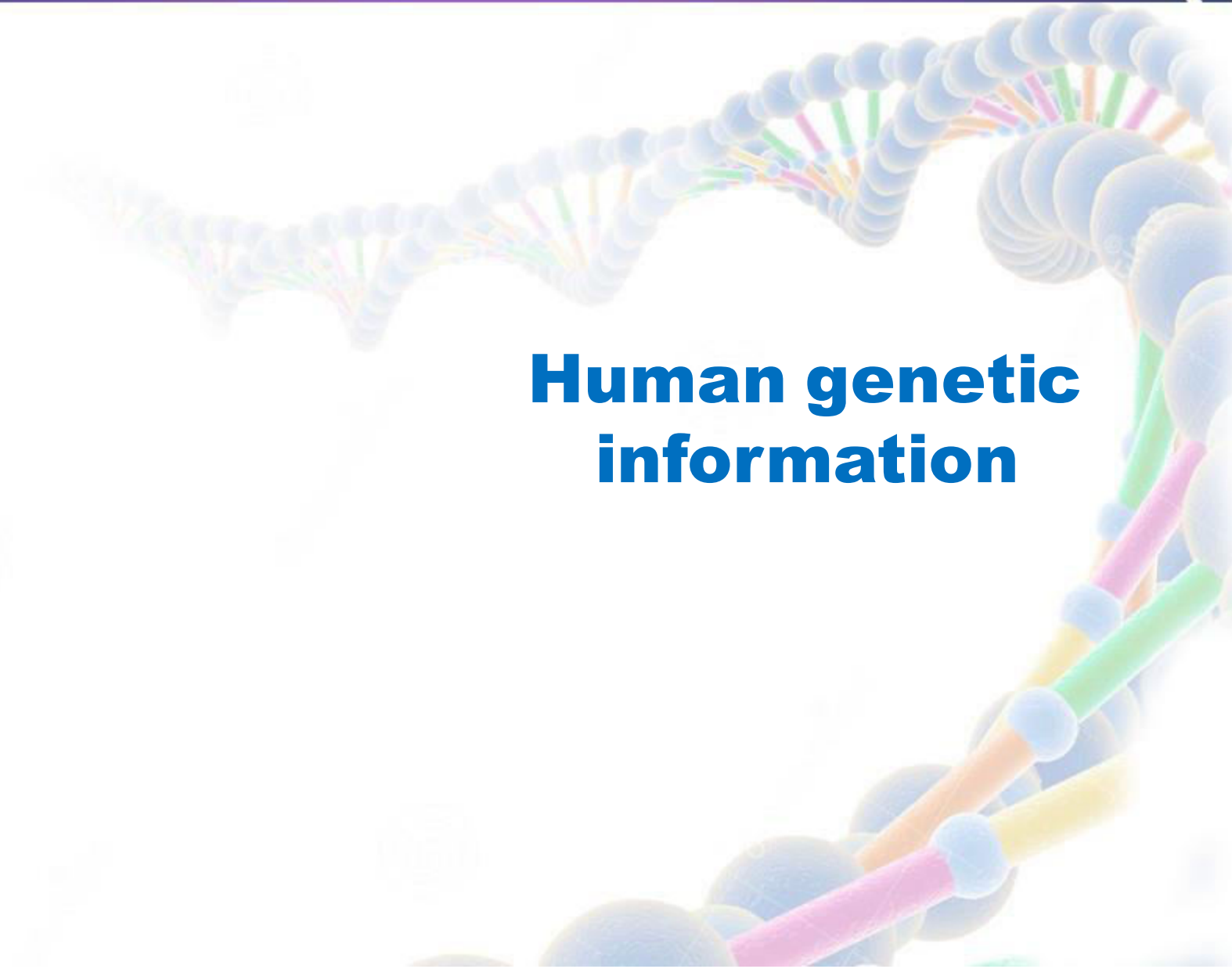
- public policy issues on sex and race
- political aspects of eugenics
- issues of morality and power
- loss of genetic diversity-pleiotropic genes-heterozygous recessive traits

Bioethics

Lesson # 40



Bioethics

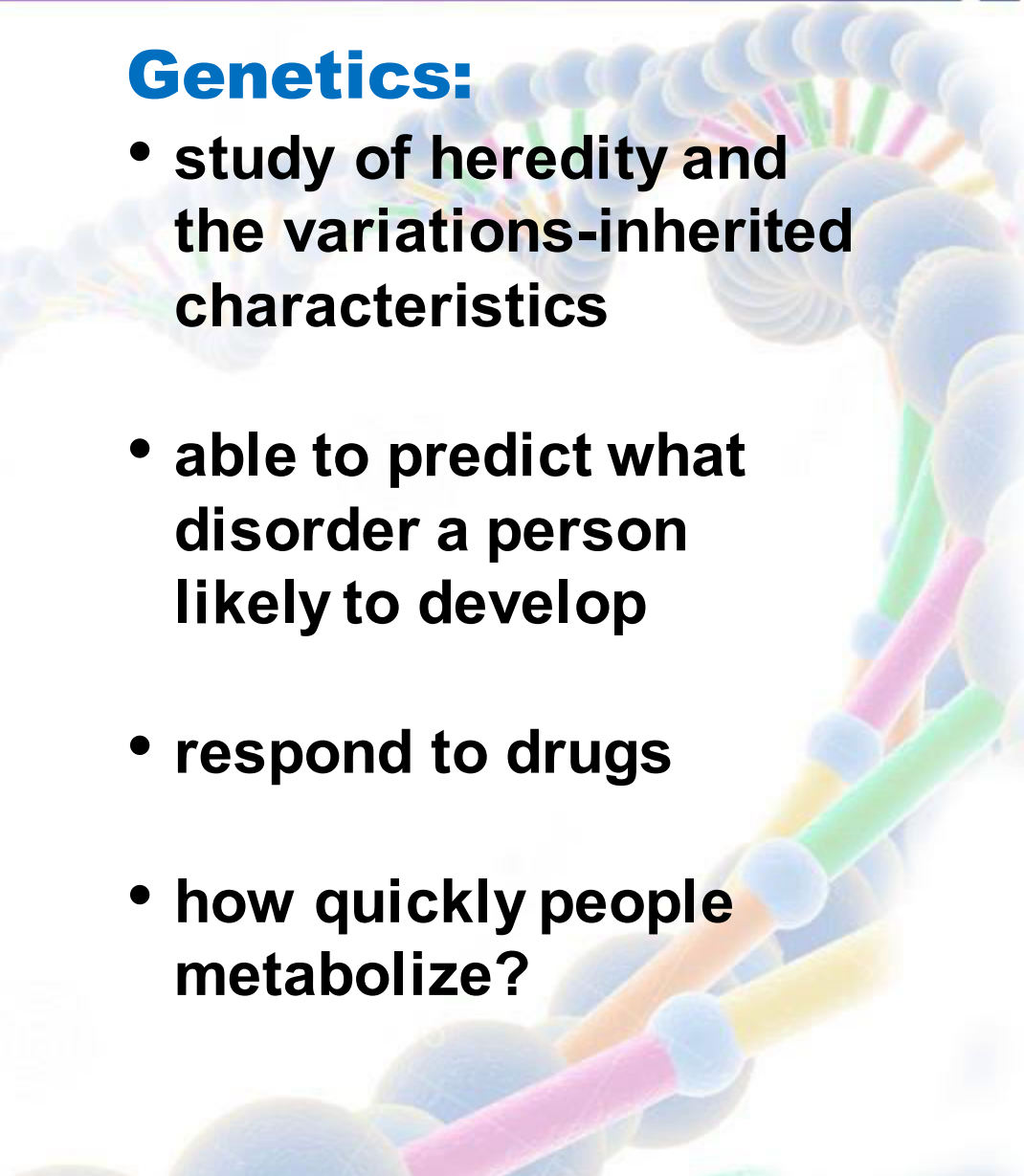


**Human genetic
information**

Human genetic information

Genetics:

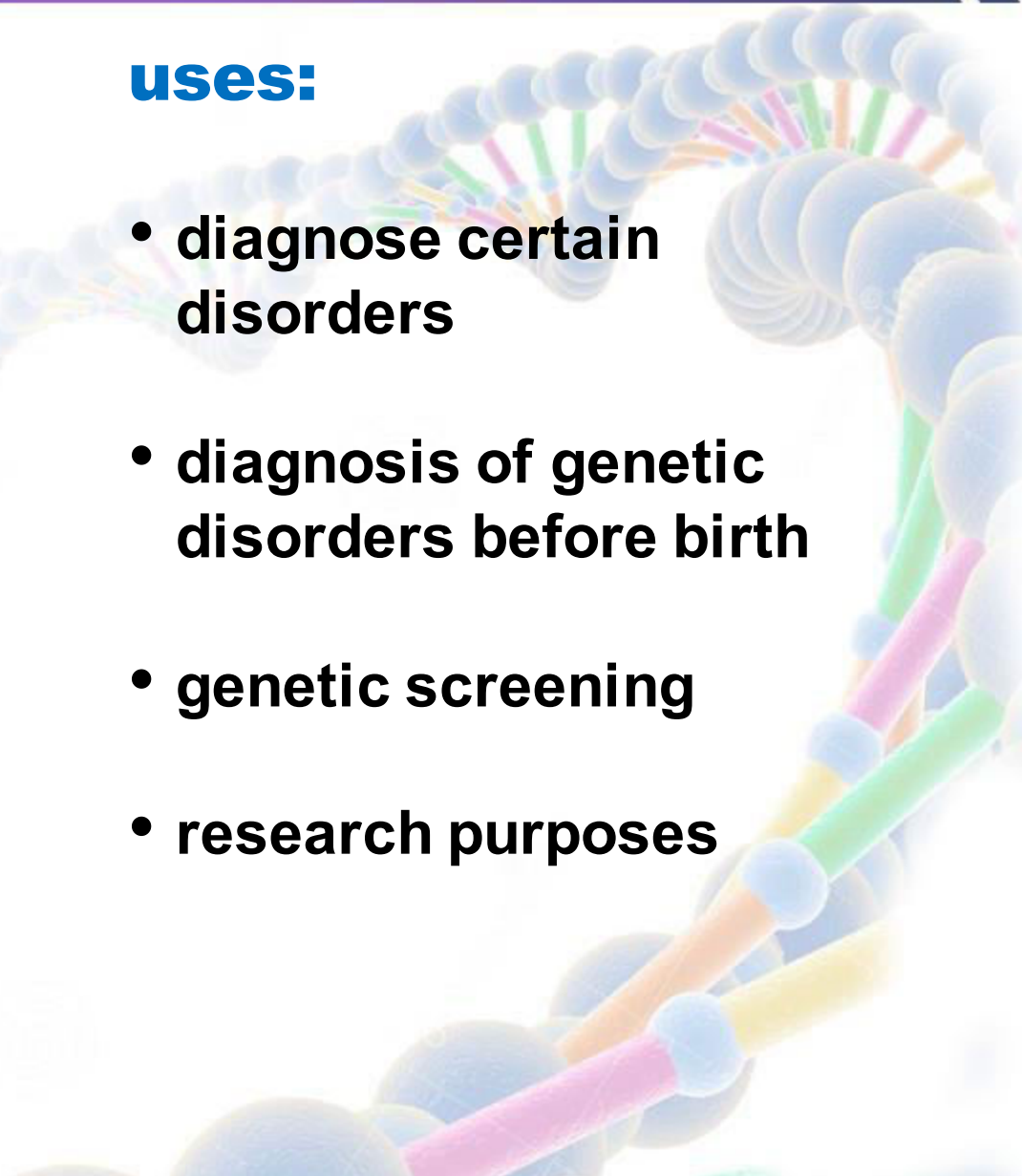
- study of heredity and the variations-inherited characteristics
- able to predict what disorder a person likely to develop
- respond to drugs
- how quickly people metabolize?



Human genetic information

uses:

- **diagnose certain disorders**
- **diagnosis of genetic disorders before birth**
- **genetic screening**
- **research purposes**



Bioethics

Lesson # 41

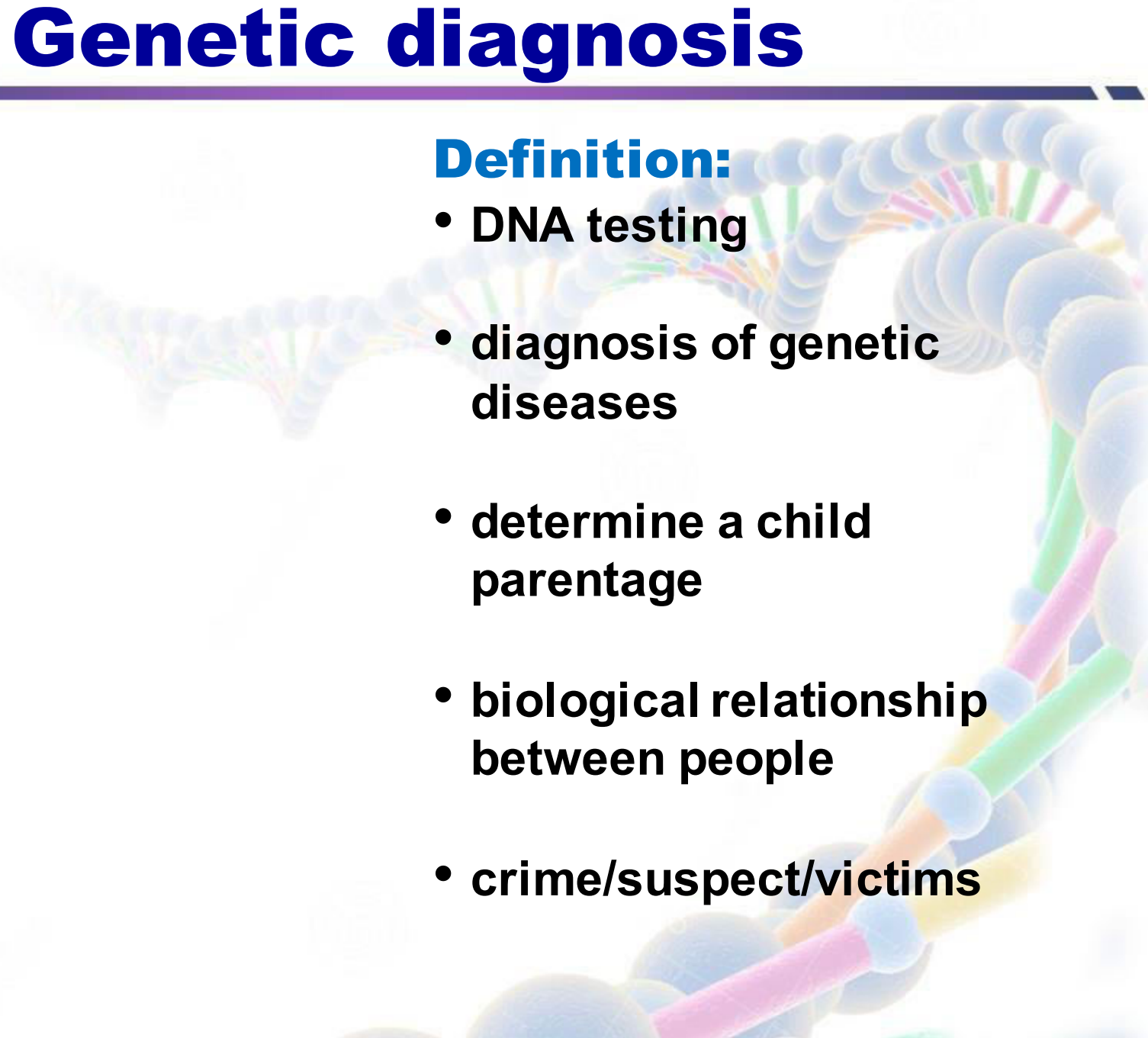


Bioethics



**Genetic
diagnosis**

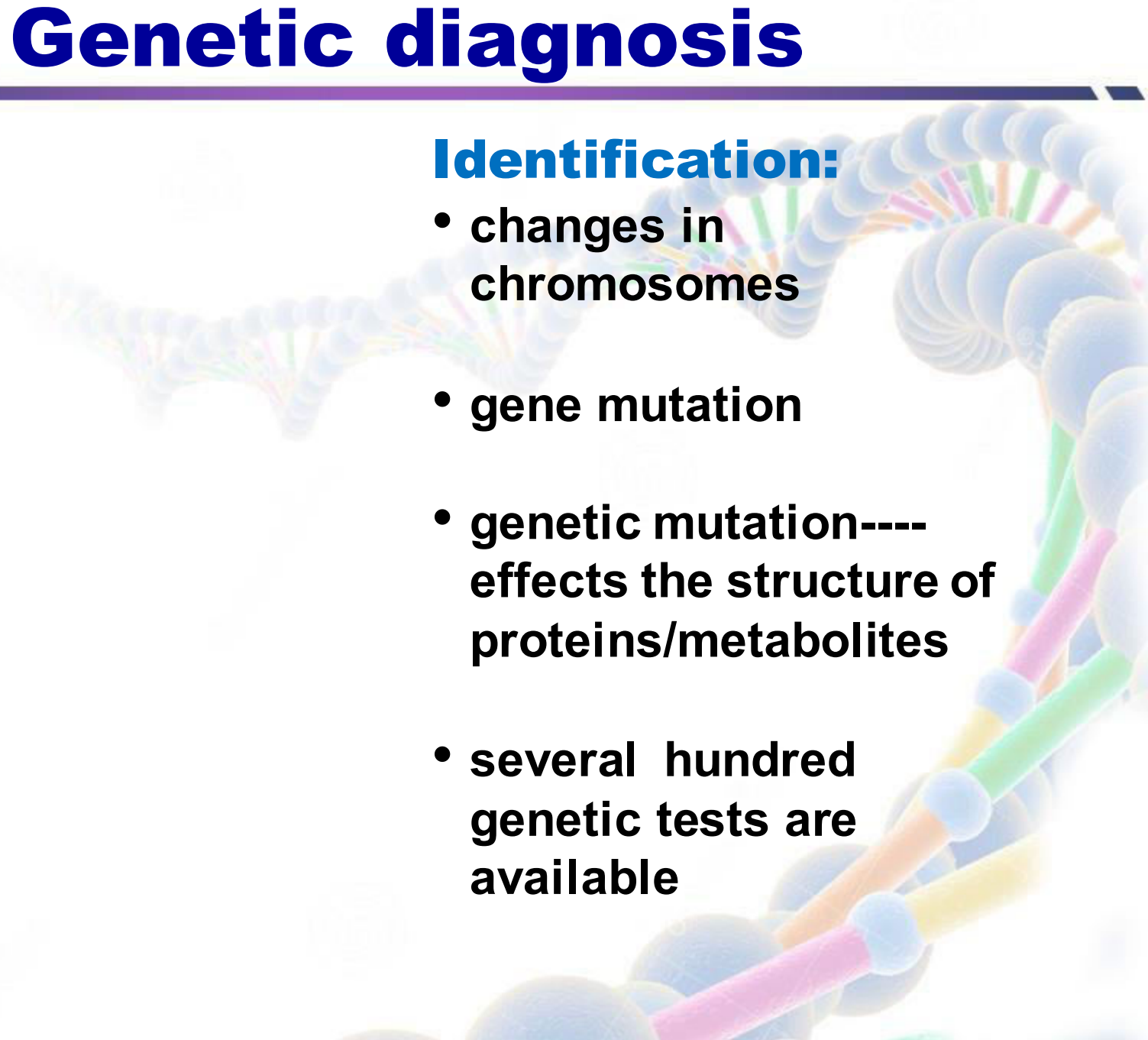
Genetic diagnosis



Definition:

- DNA testing
- diagnosis of genetic diseases
- determine a child parentage
- biological relationship between people
- crime/suspect/victims

Genetic diagnosis



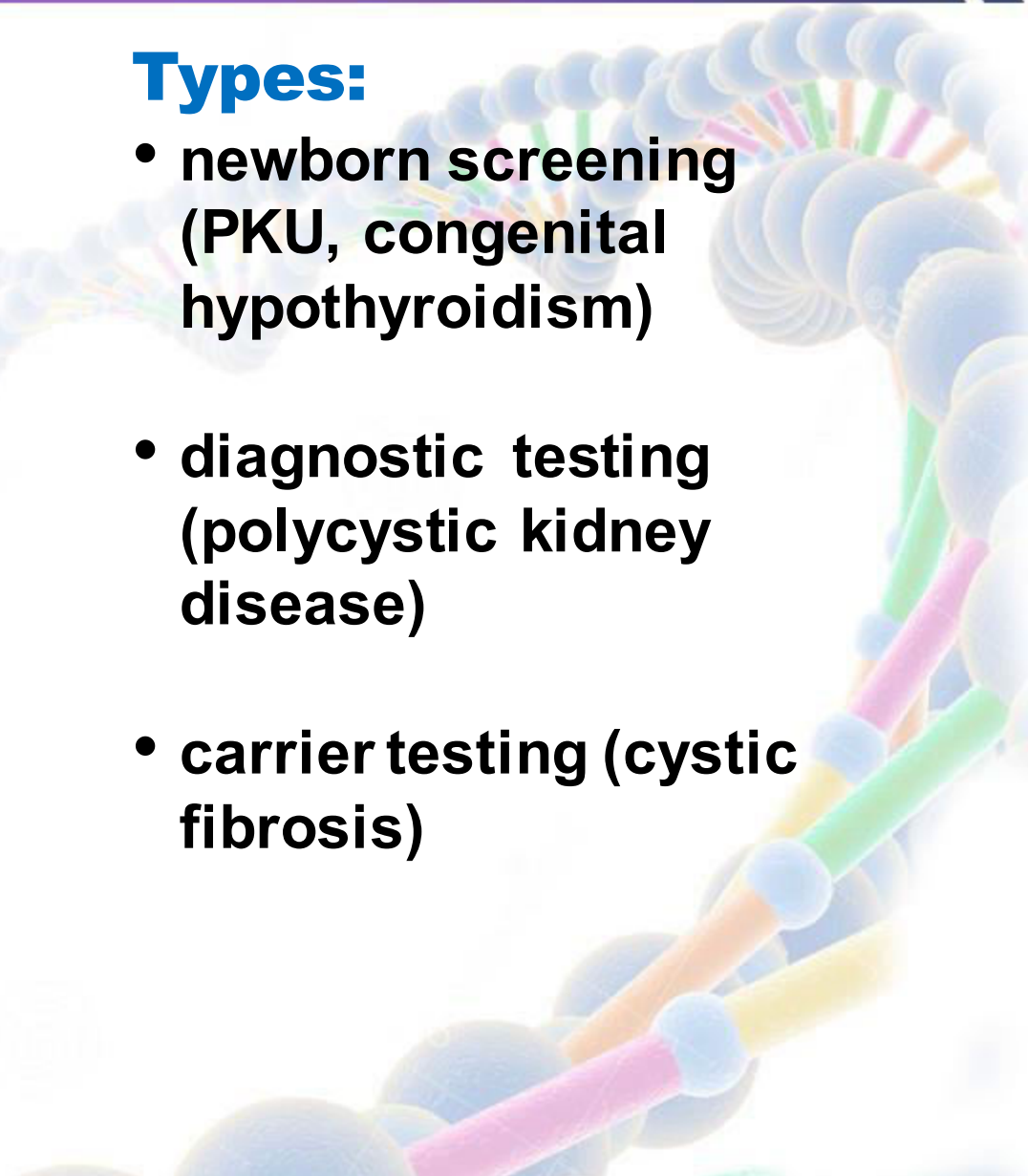
Identification:

- changes in chromosomes
- gene mutation
- genetic mutation----
effects the structure of
proteins/metabolites
- several hundred
genetic tests are
available

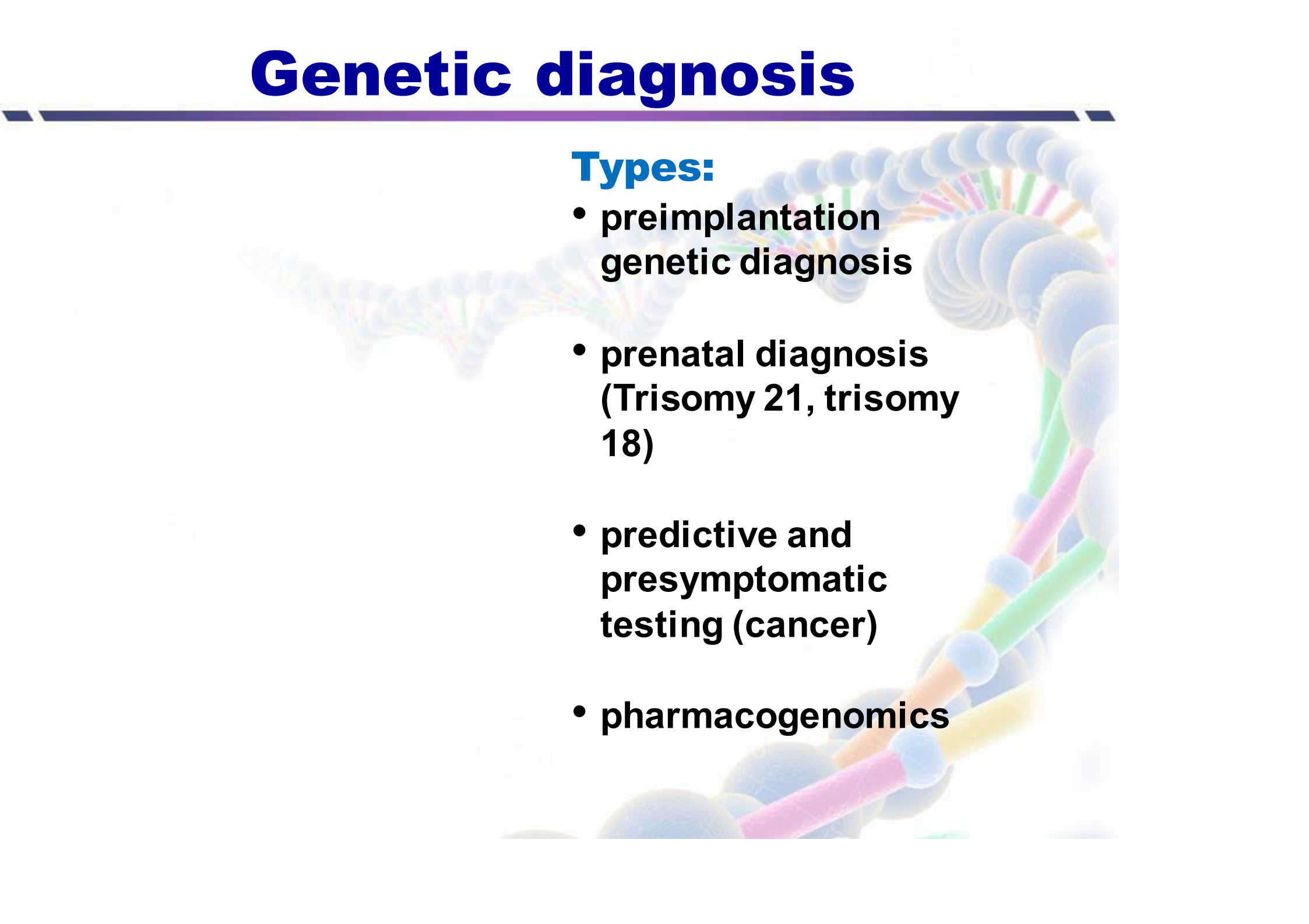
Genetic diagnosis

Types:

- newborn screening (PKU, congenital hypothyroidism)
- diagnostic testing (polycystic kidney disease)
- carrier testing (cystic fibrosis)



Genetic diagnosis



Types:

- preimplantation genetic diagnosis
- prenatal diagnosis (Trisomy 21, trisomy 18)
- predictive and presymptomatic testing (cancer)
- pharmacogenomics

Genetic diagnosis

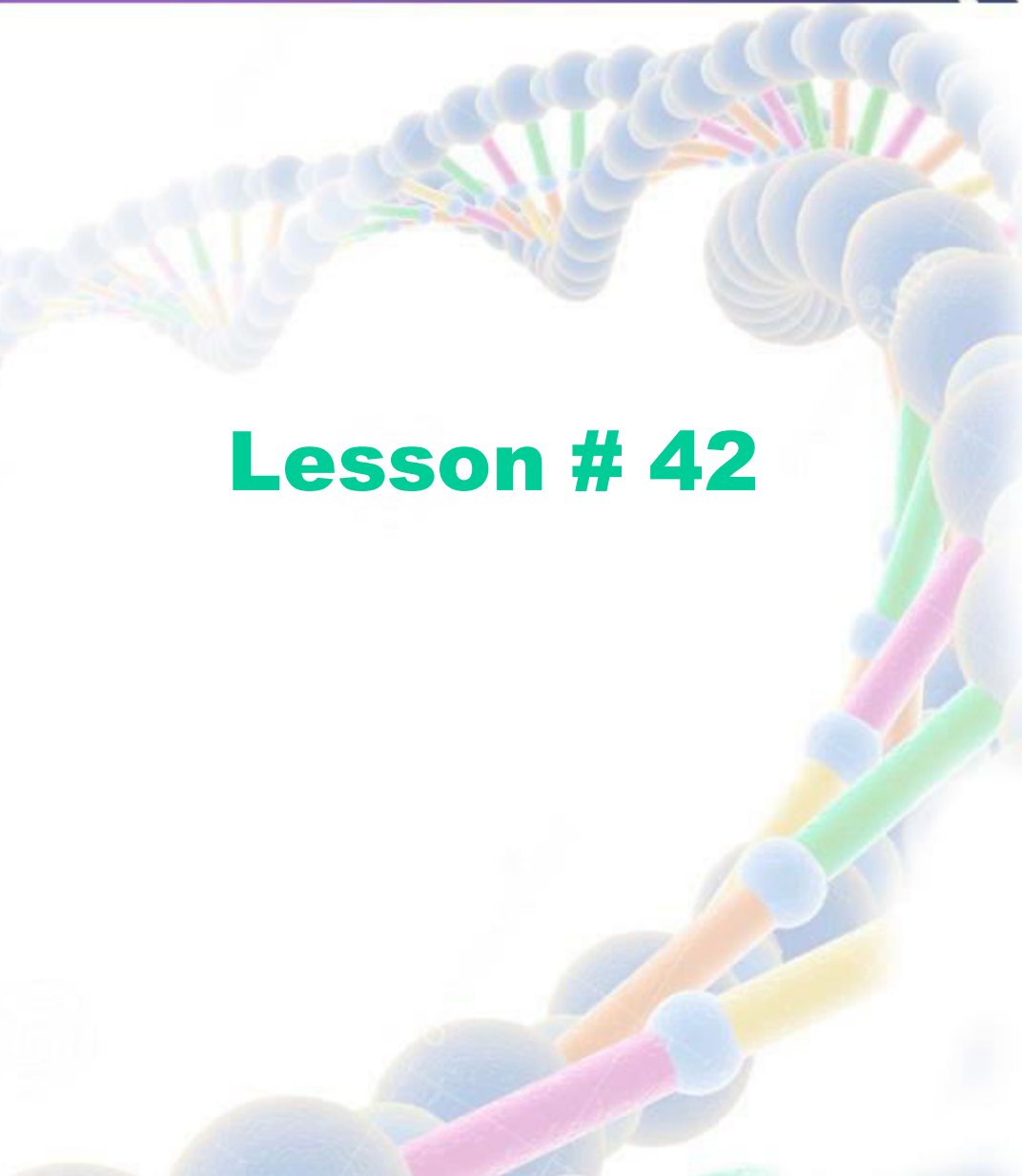
Risks:

- **risk of losing the pregnancy**
- **emotional consequences**
- **social issues**
- **financial issues**

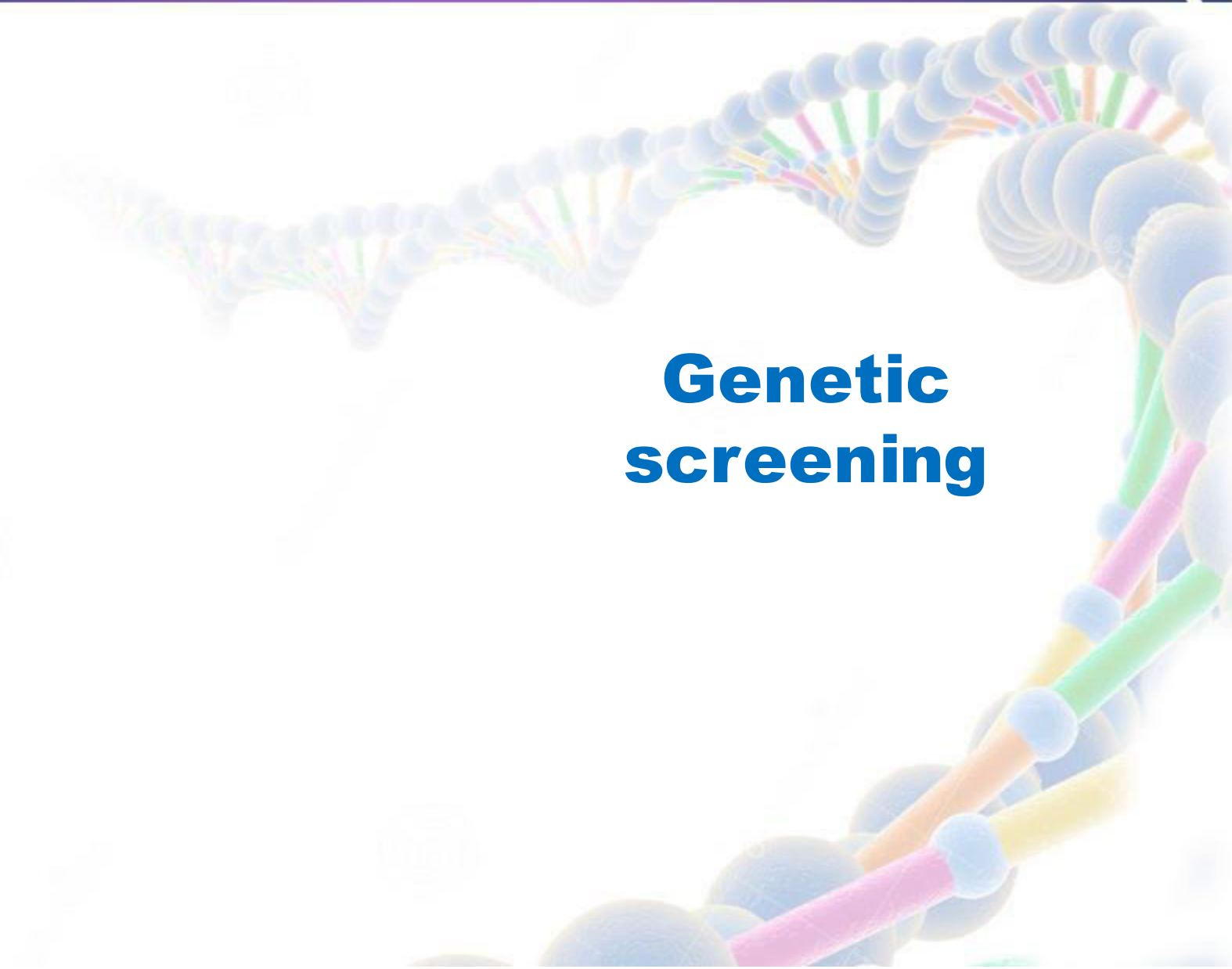


Bioethics

Lesson # 42



Bioethics

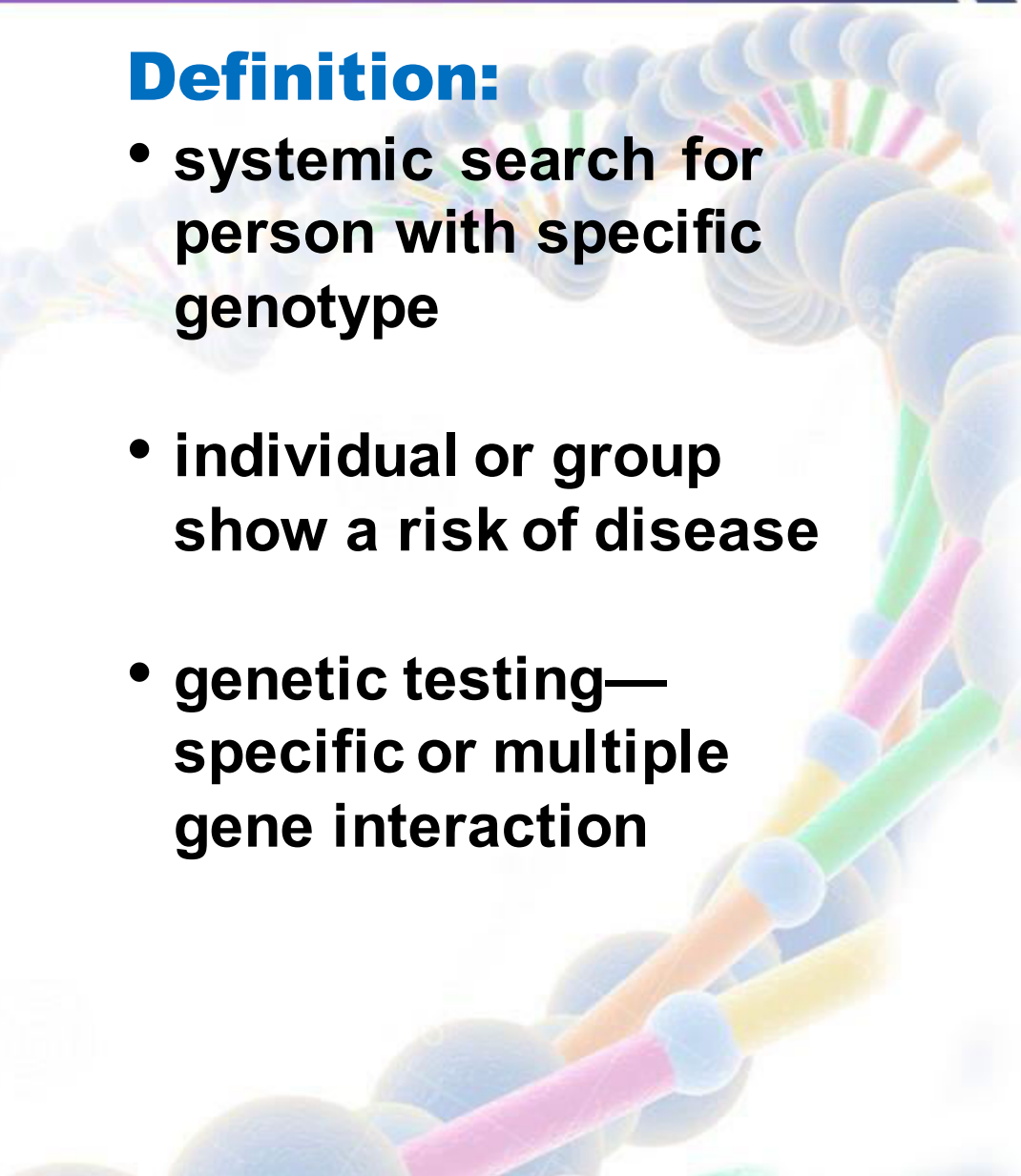


**Genetic
screening**

Genetic screening

Definition:

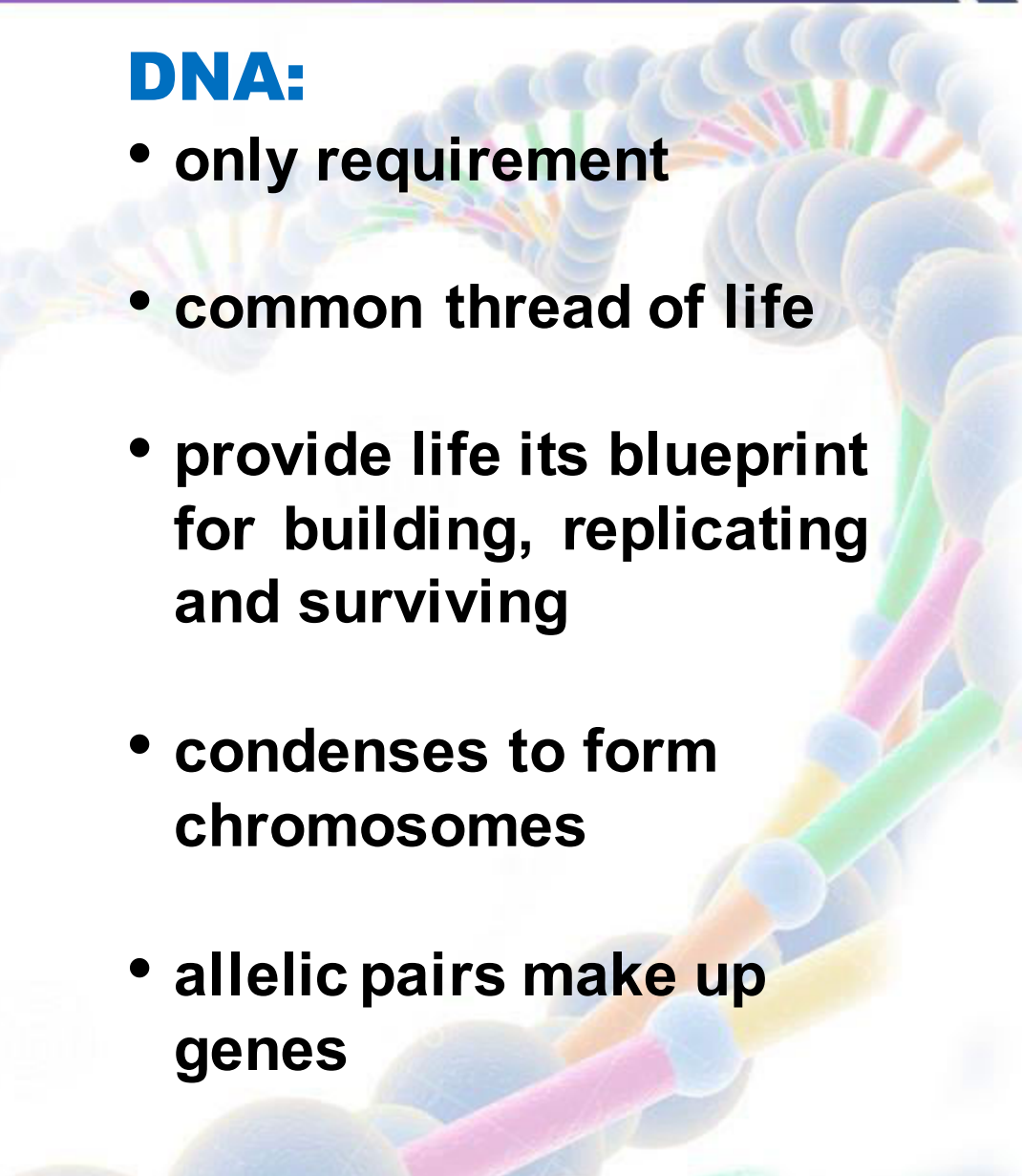
- systemic search for person with specific genotype
- individual or group show a risk of disease
- genetic testing—specific or multiple gene interaction



Genetic screening

DNA:

- only requirement
- common thread of life
- provide life its blueprint for building, replicating and surviving
- condenses to form chromosomes
- allelic pairs make up genes



Genetic screening

Ethics:

- religious groups
- expensive/emotional distress
- done for common diseases/part of medical record???
- deny employment, social services and insurance benefits



Bioethics

Lesson # 43



Bioethics



**Genetic
discrimination**

Genetic discrimination

Definition:

- when people are treated differently
- based on the individual genotype rather than their individual merits
- Genism----distinctive human characteristics and capacities are determined by genes

Genetic discrimination

Conditions:

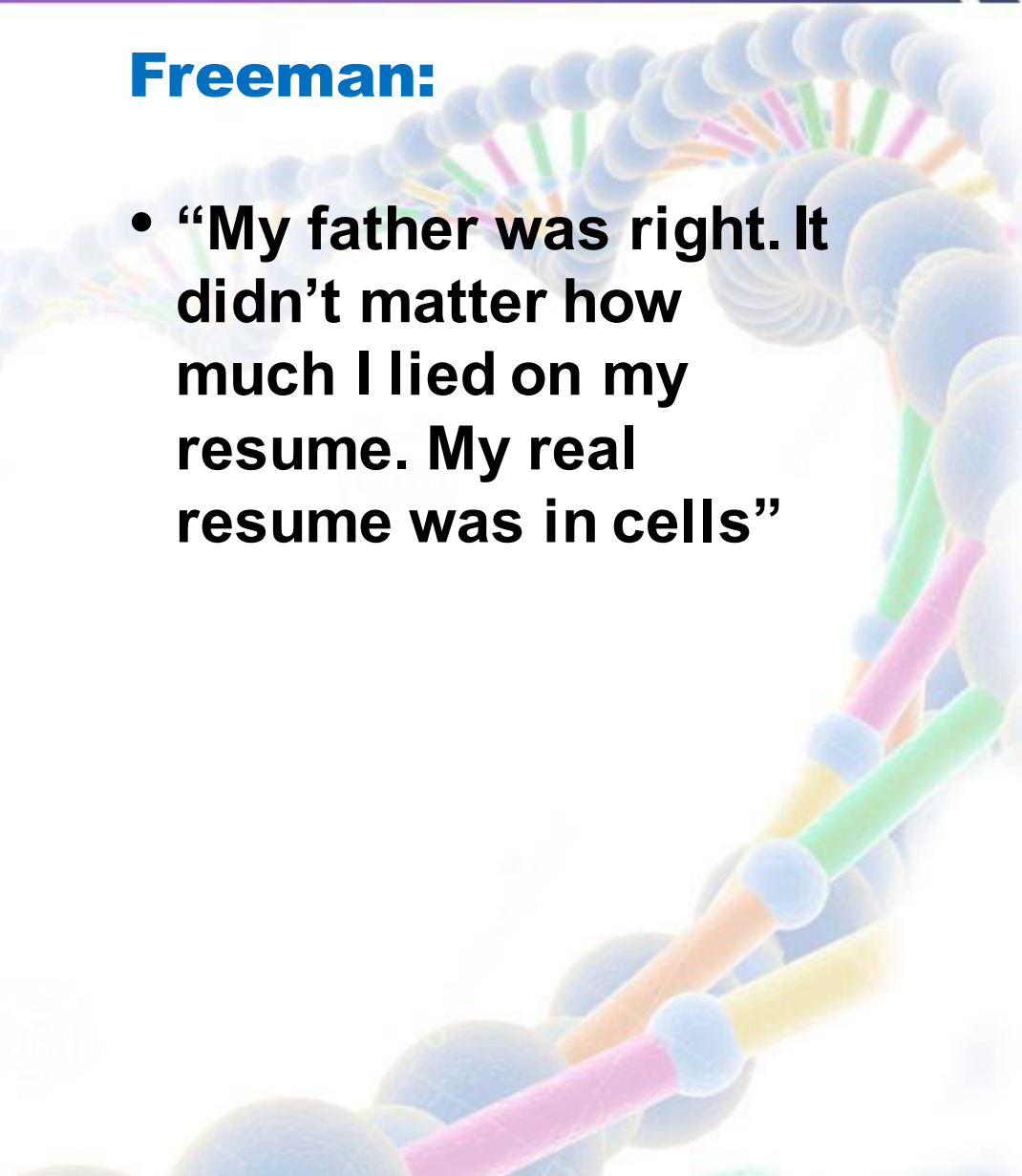
- genetic testing in the work place
- health insurance discrimination
- popular culture—
genoism—unethical
and illegal genetic
discrimination



Genetic discrimination

Freeman:

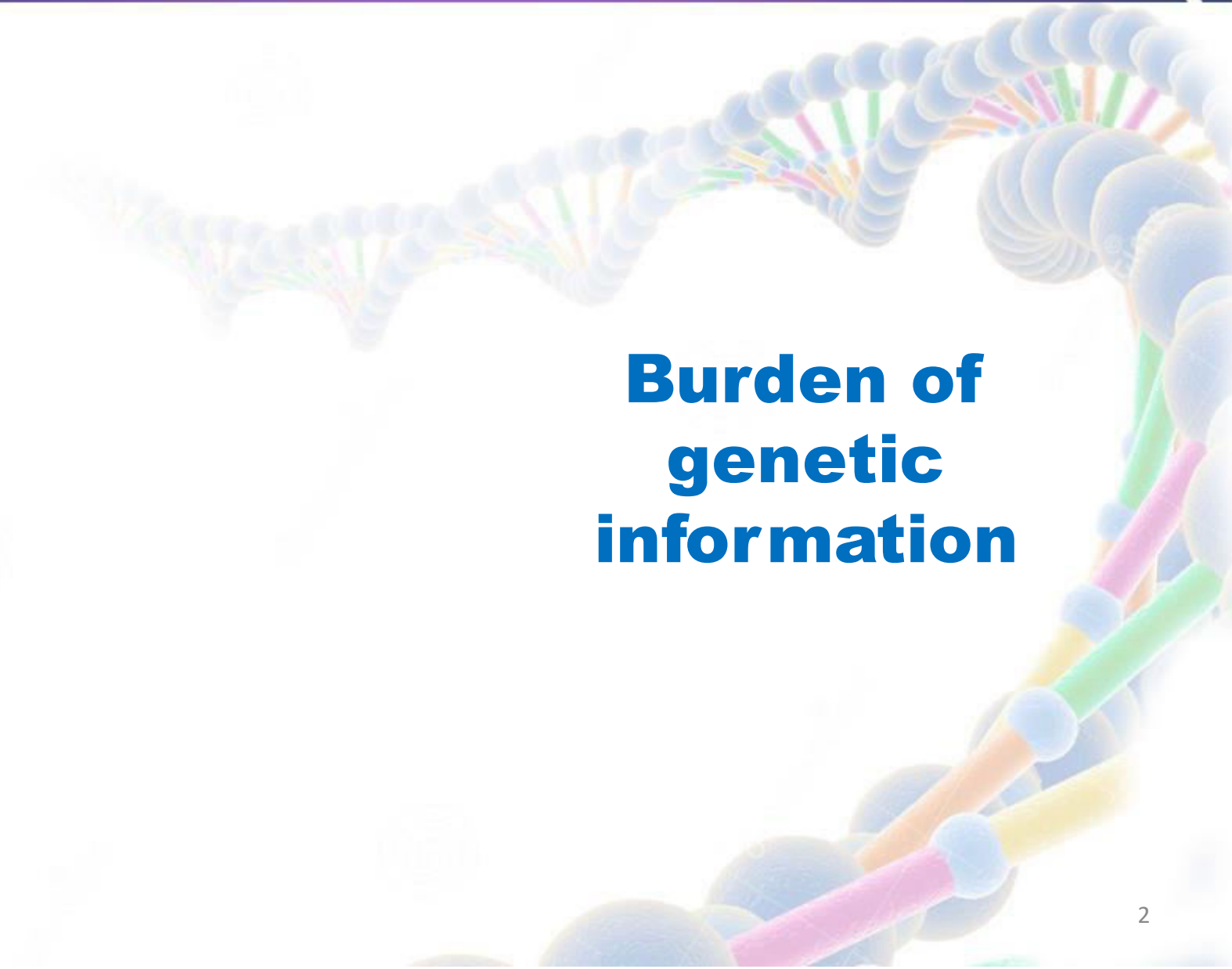
- “My father was right. It didn’t matter how much I lied on my resume. My real resume was in cells”



Bioethics

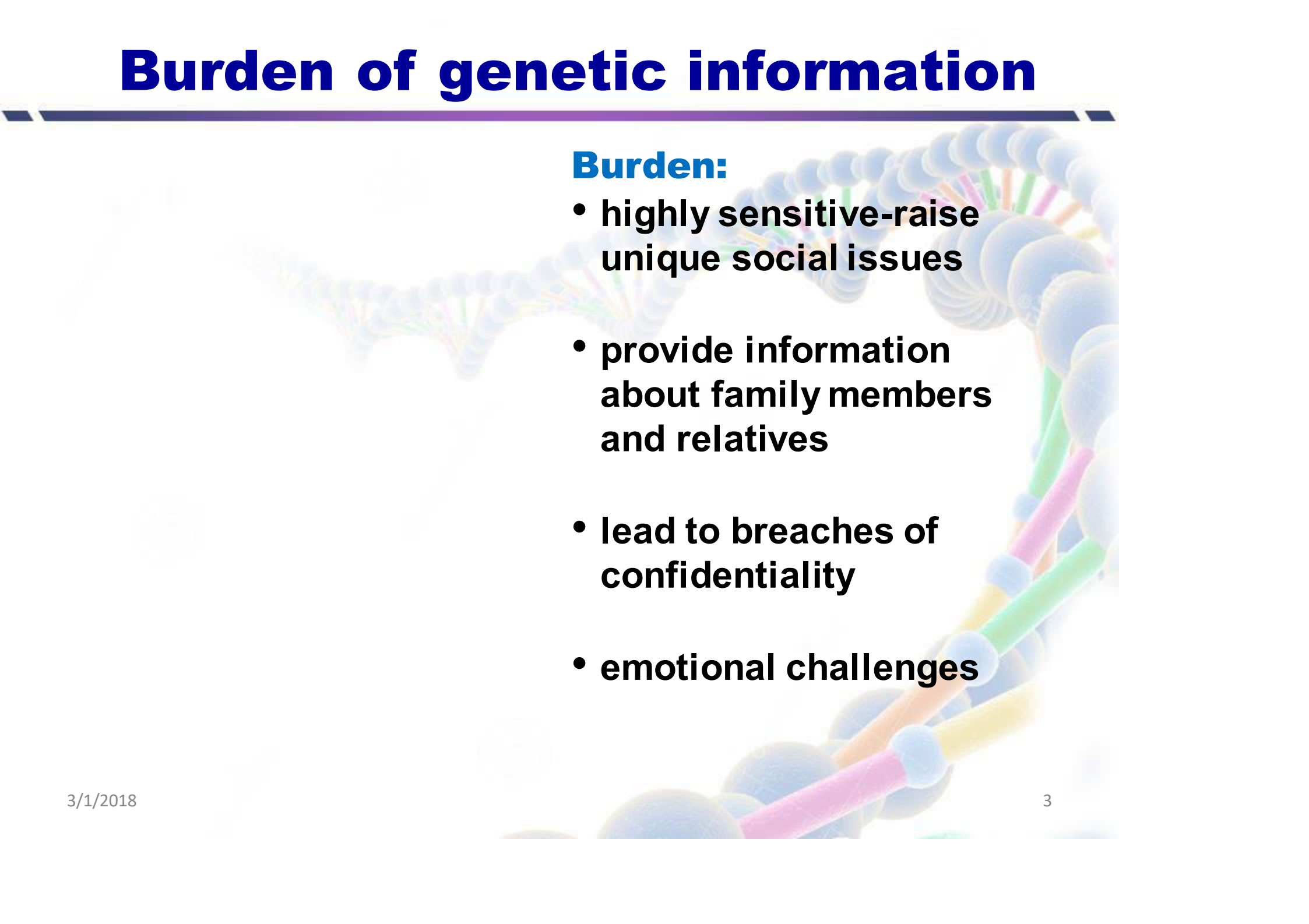
Lesson # 44

Bioethics



Burden of genetic information

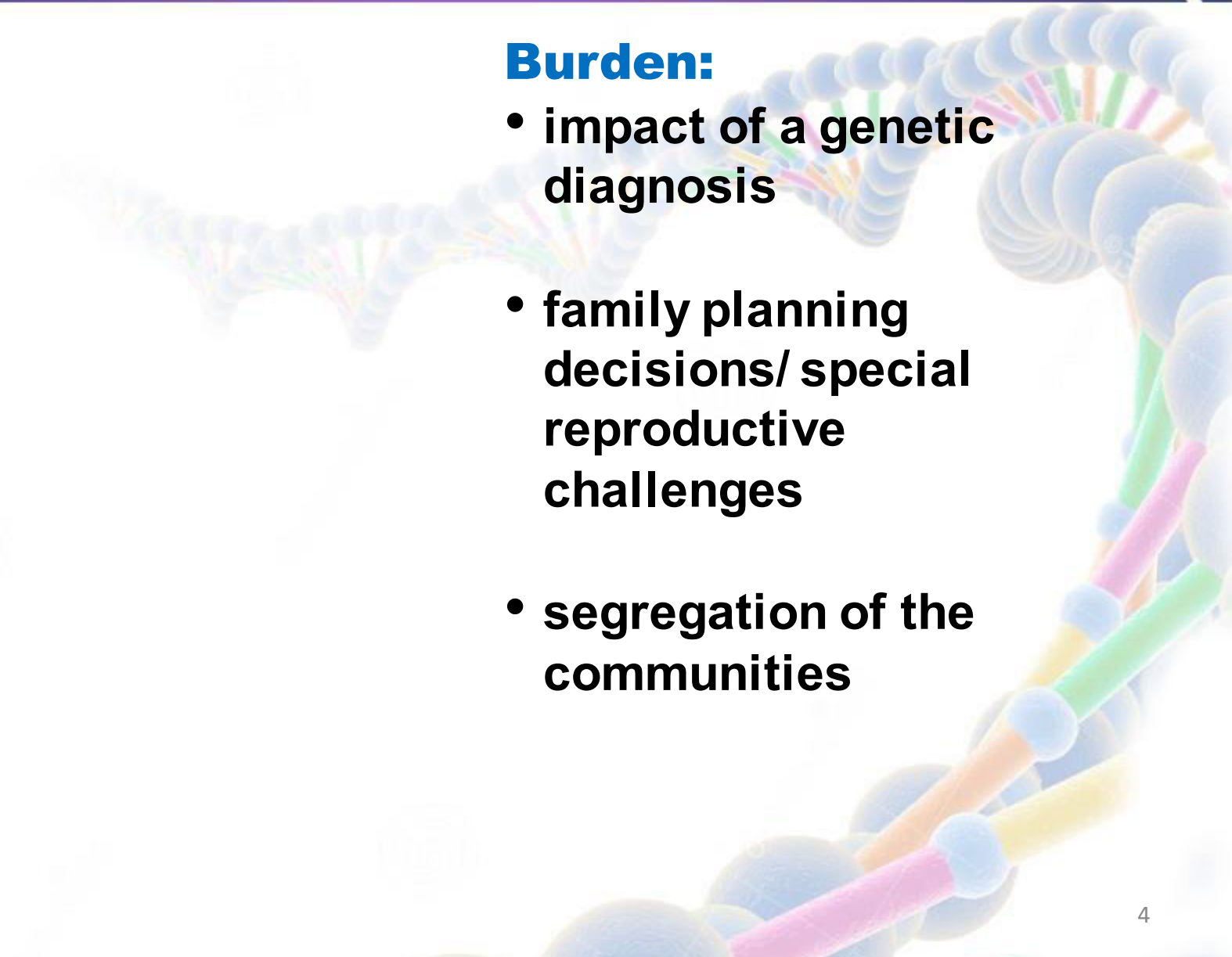
Burden of genetic information



Burden:

- highly sensitive-raise unique social issues
- provide information about family members and relatives
- lead to breaches of confidentiality
- emotional challenges

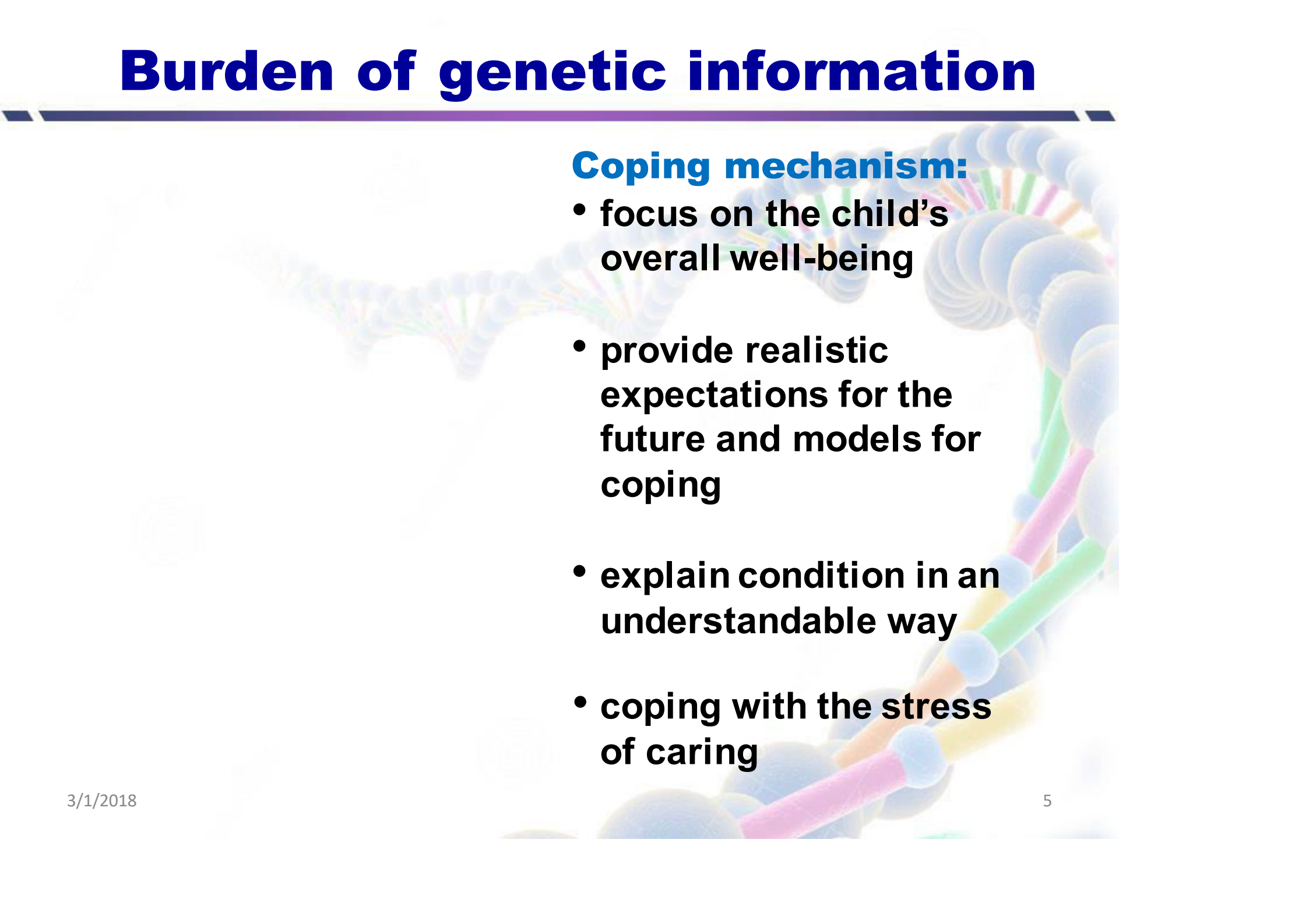
Burden of genetic information



Burden:

- impact of a genetic diagnosis
- family planning decisions/ special reproductive challenges
- segregation of the communities

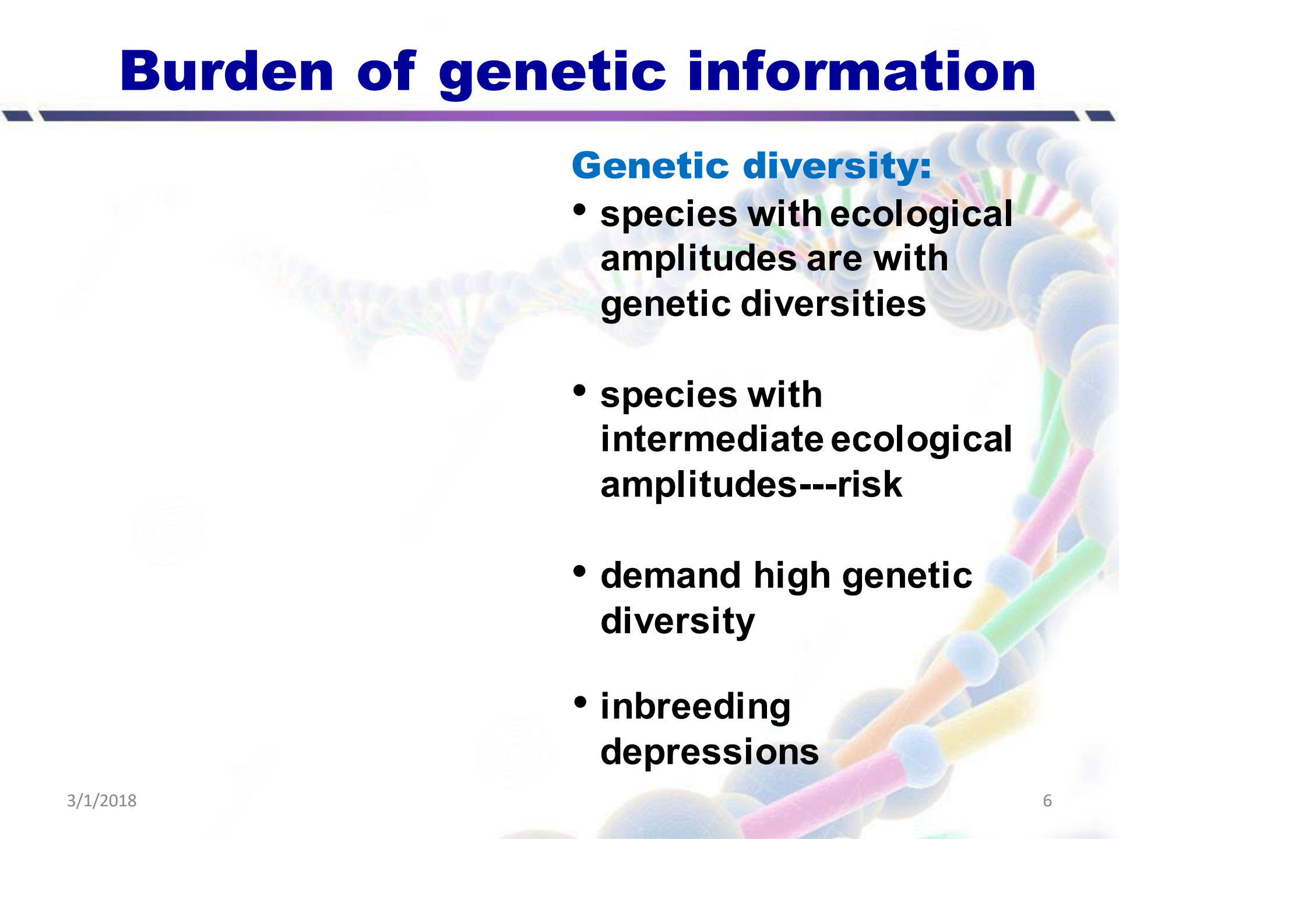
Burden of genetic information



Coping mechanism:

- focus on the child's overall well-being
- provide realistic expectations for the future and models for coping
- explain condition in an understandable way
- coping with the stress of caring

Burden of genetic information



Genetic diversity:

- species with ecological amplitudes are with genetic diversities
- species with intermediate ecological amplitudes---risk
- demand high genetic diversity
- inbreeding depressions

Bioethics

Lesson # 45



Bioethics

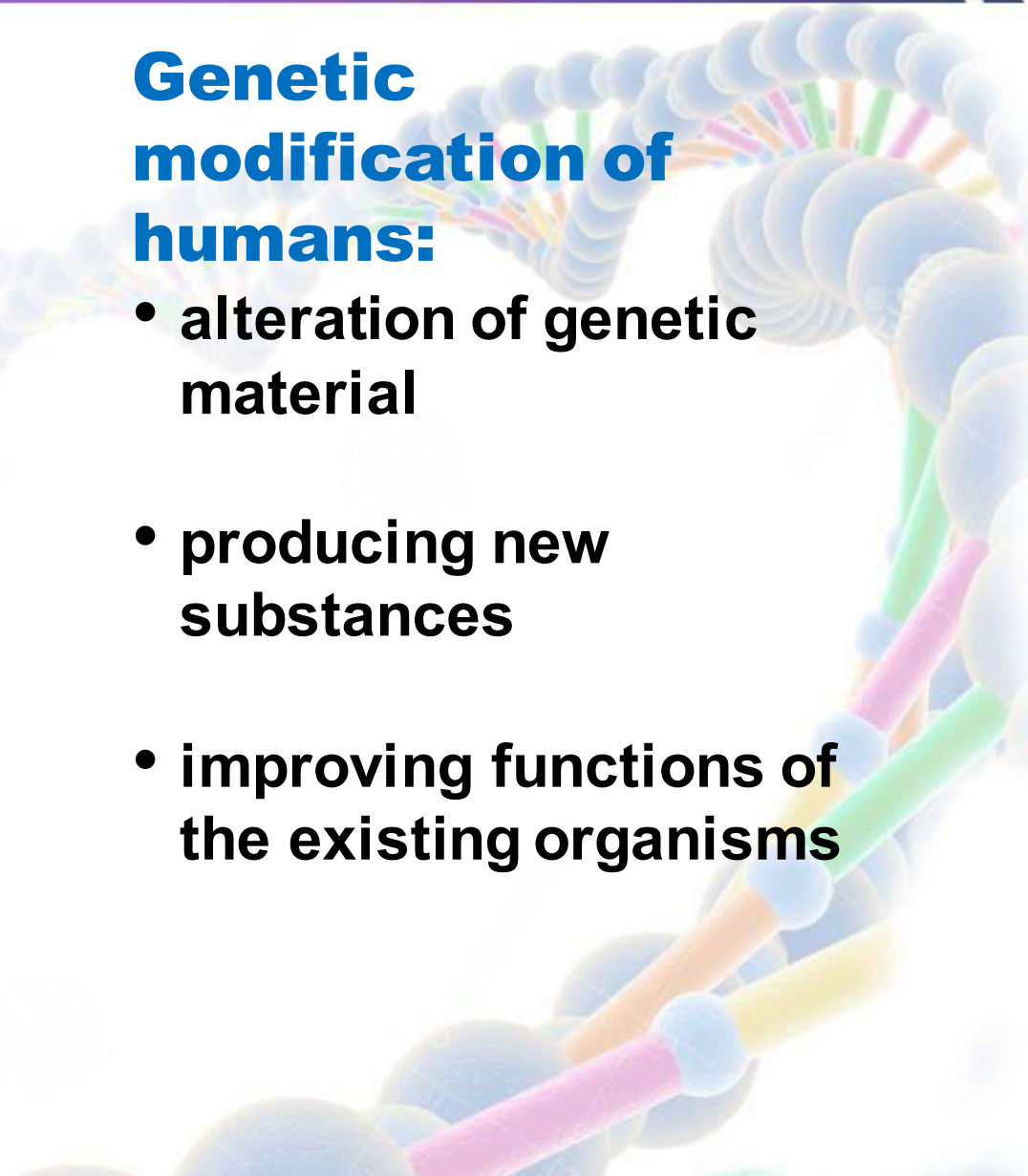
Fact or fiction



Fact or fiction

Genetic modification of humans:

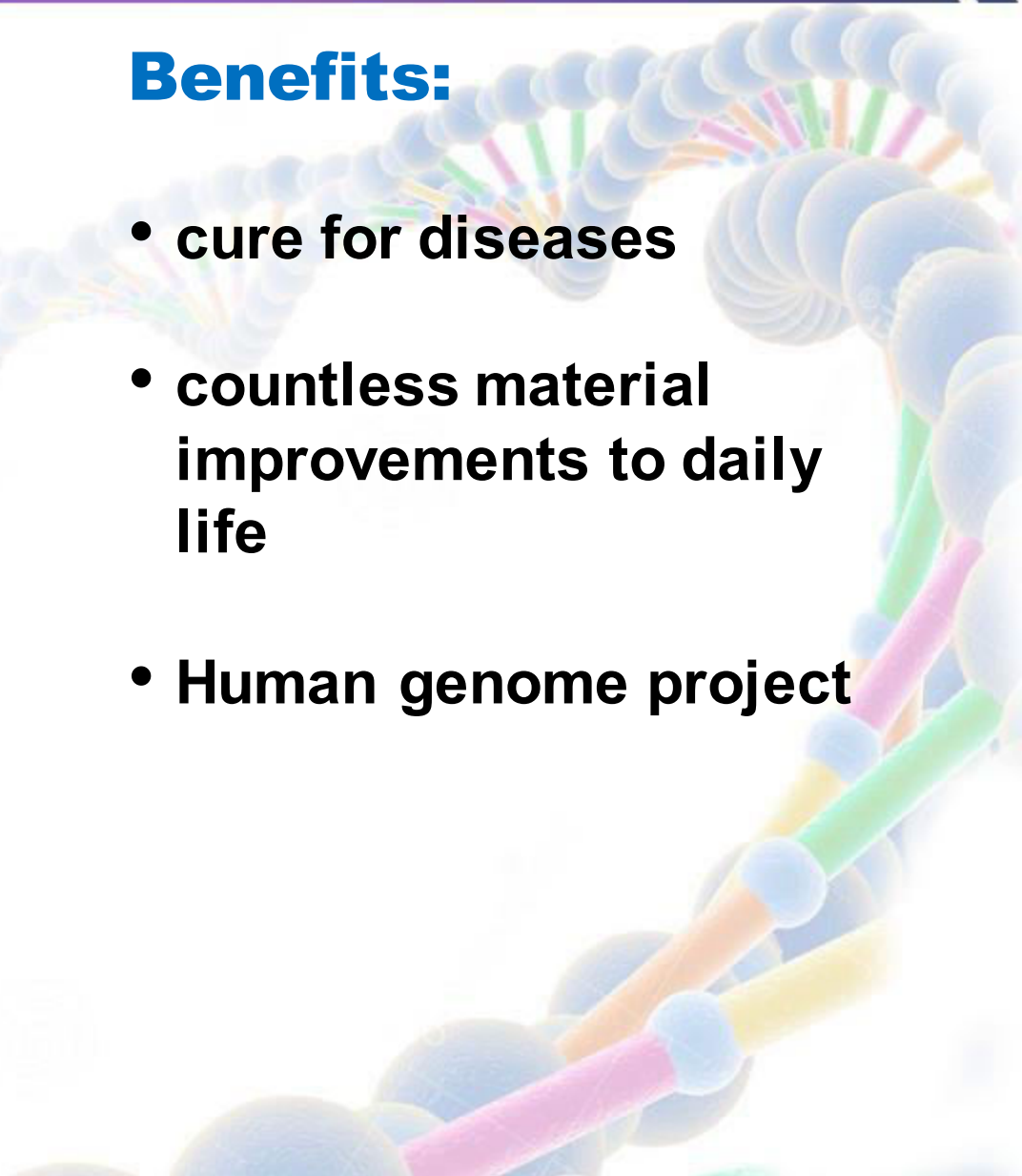
- alteration of genetic material
- producing new substances
- improving functions of the existing organisms



Fact or fiction

Benefits:

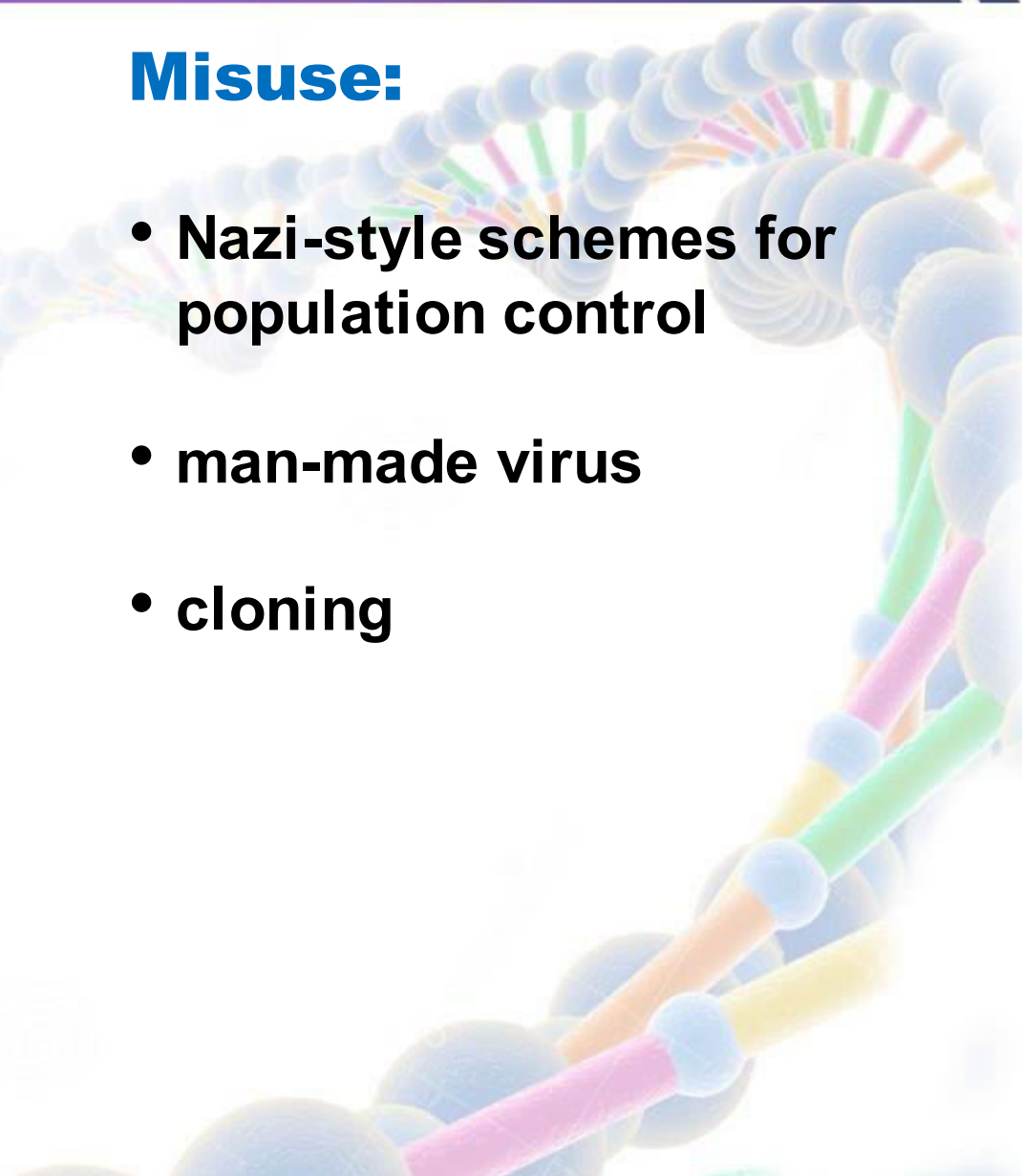
- cure for diseases
- countless material improvements to daily life
- Human genome project



Fact or fiction

Misuse:

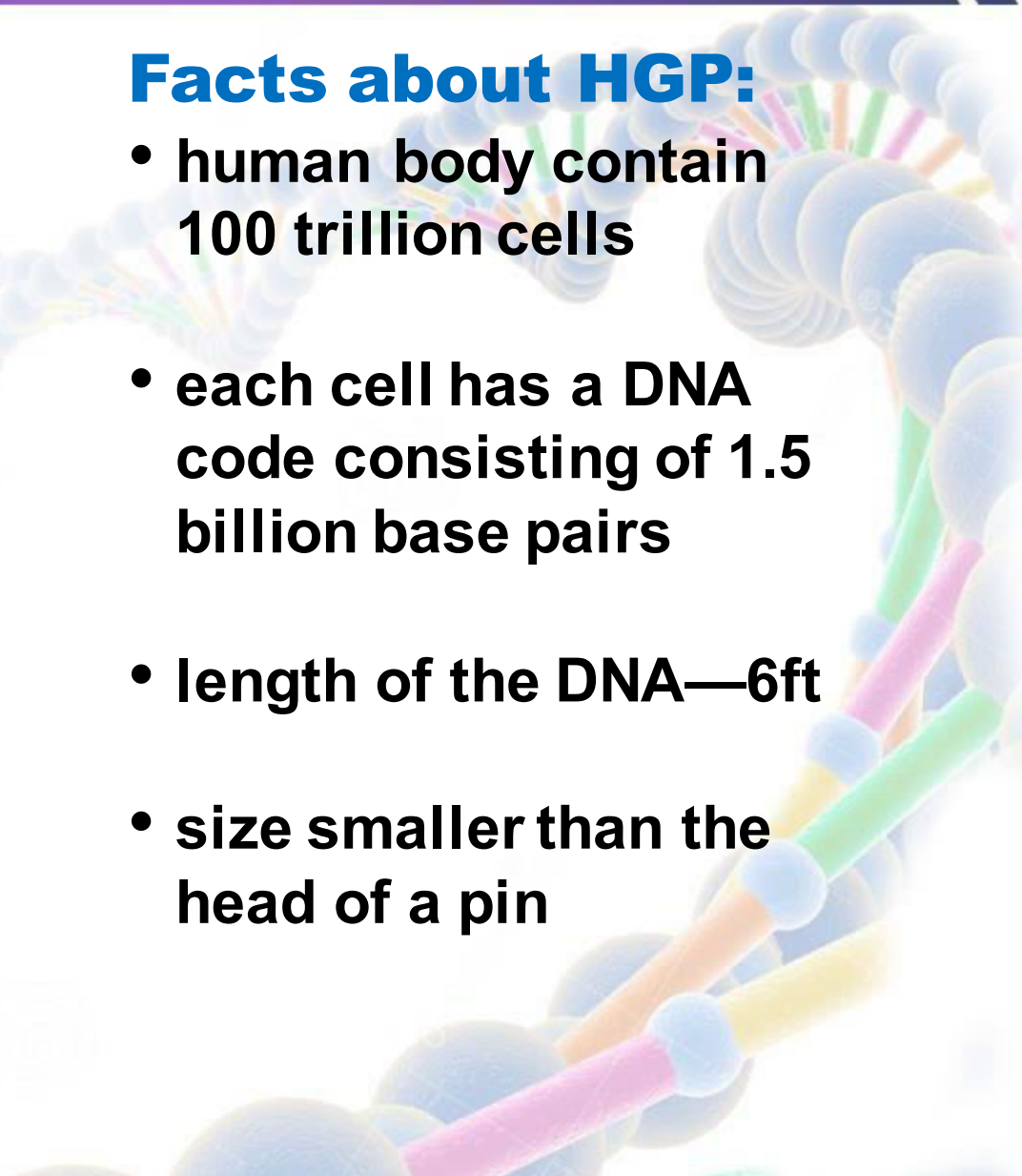
- **Nazi-style schemes for population control**
- **man-made virus**
- **cloning**



Fact or fiction

Facts about HGP:

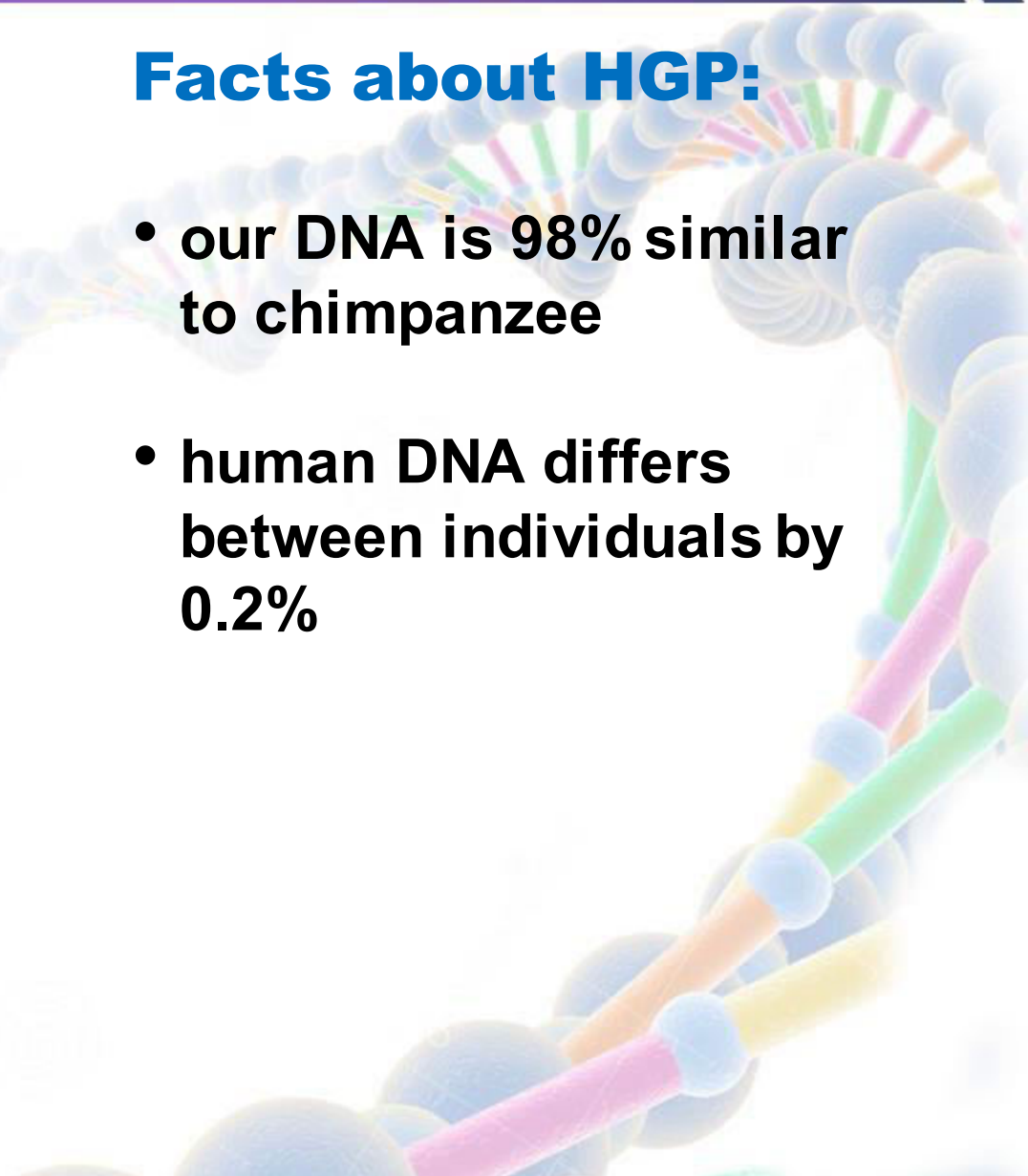
- human body contain 100 trillion cells
- each cell has a DNA code consisting of 1.5 billion base pairs
- length of the DNA—6ft
- size smaller than the head of a pin



Fact or fiction

Facts about HGP:

- our DNA is 98% similar to chimpanzee
- human DNA differs between individuals by 0.2%



Fact or fiction

Principle:

- **rDNA is the genetically altered DNA---process is known as gene splicing—sale of insulin**
- **gene therapy- genetically altered genes to cells**
- **use of restriction enzymes**

Bioethics

Lesson # 46

Bioethics



Genes-the wider issues

Genes-the wider issues

Genes:

- chromosomes are the physical entities that carry the genes
- replicate faithfully
- direct the synthesis of RNA and proteins
- accumulate mutation -- evolution

Genes-the wider issues



Future aspects:

- increasing knowledge of genetics---affect all future generations
- generation charged with--task of setting foundations and building the first few stages
- encourage scientists, technologists-look at what they are creating

Genes-the wider issues



Future aspects:

- encourage policy makers and public to stop moaning about the technology and problems
- effort and understand well-----enlightened decisions

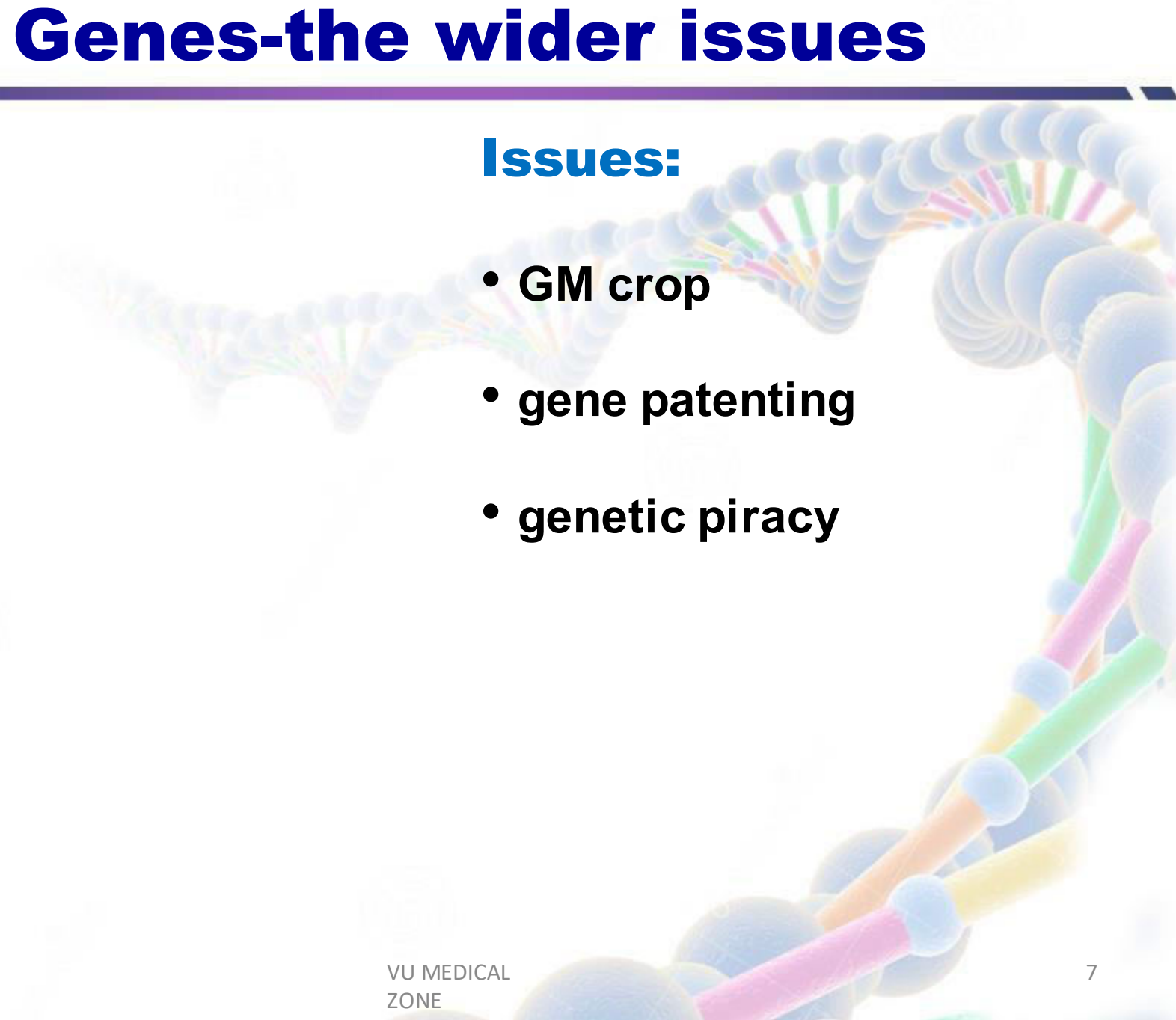
Genes-the wider issues



Socio-economic arguments:

- inequalities in ownership of the technology
- exploitation of the poor by rich
- act as lightning conductor for attracting opposition

Genes-the wider issues



Issues:

- GM crop
- gene patenting
- genetic piracy

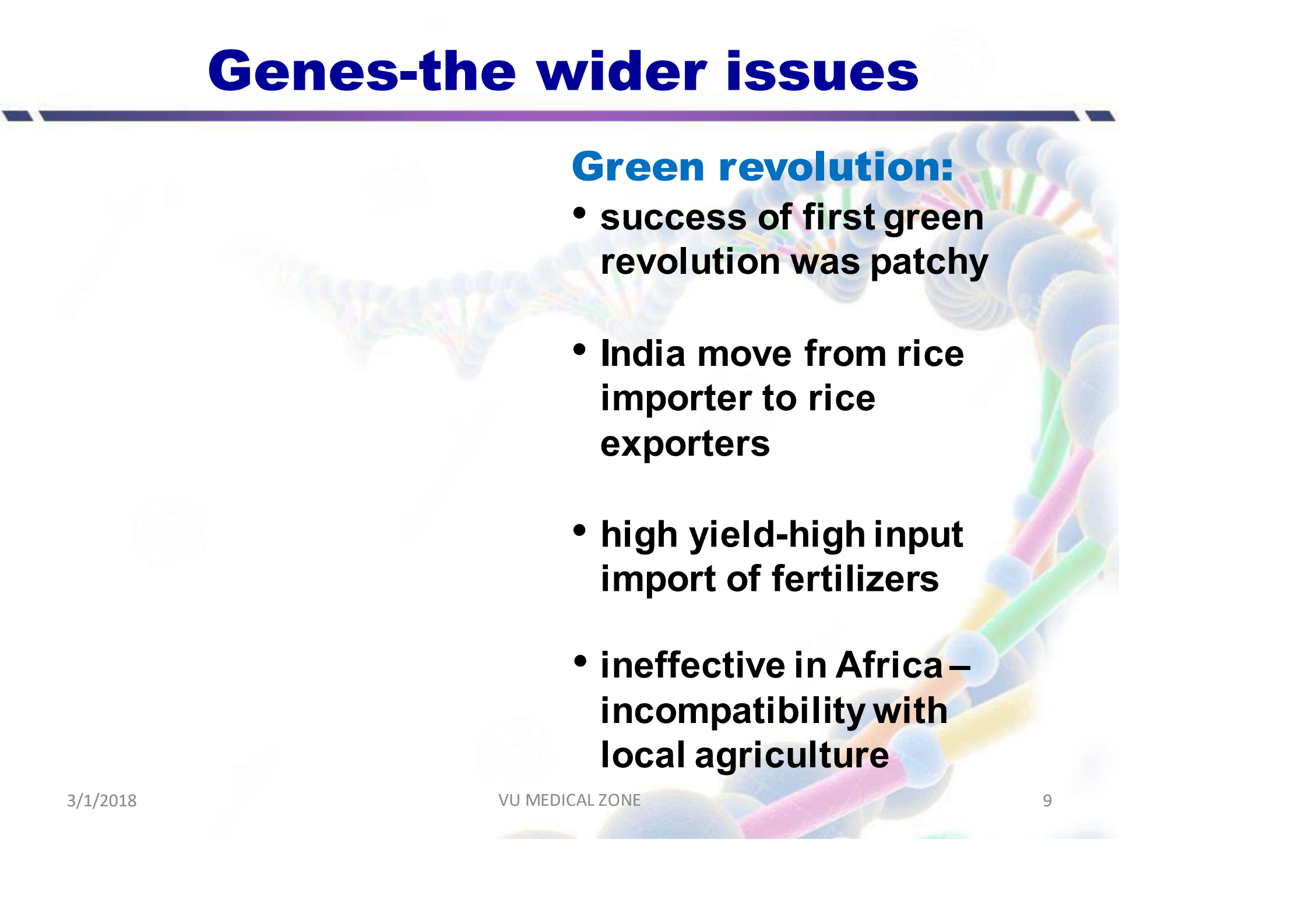
Genes-the wider issues



GM crop:

- world population is growing fast than the agriculture production
- million of hectares devoted world-wide to GM crops
- tool for the plant breeder-fight against food shortages

Genes-the wider issues



Green revolution:

- success of first green revolution was patchy
- India move from rice importer to rice exporters
- high yield-high input import of fertilizers
- ineffective in Africa – incompatibility with local agriculture

Genes-the wider issues

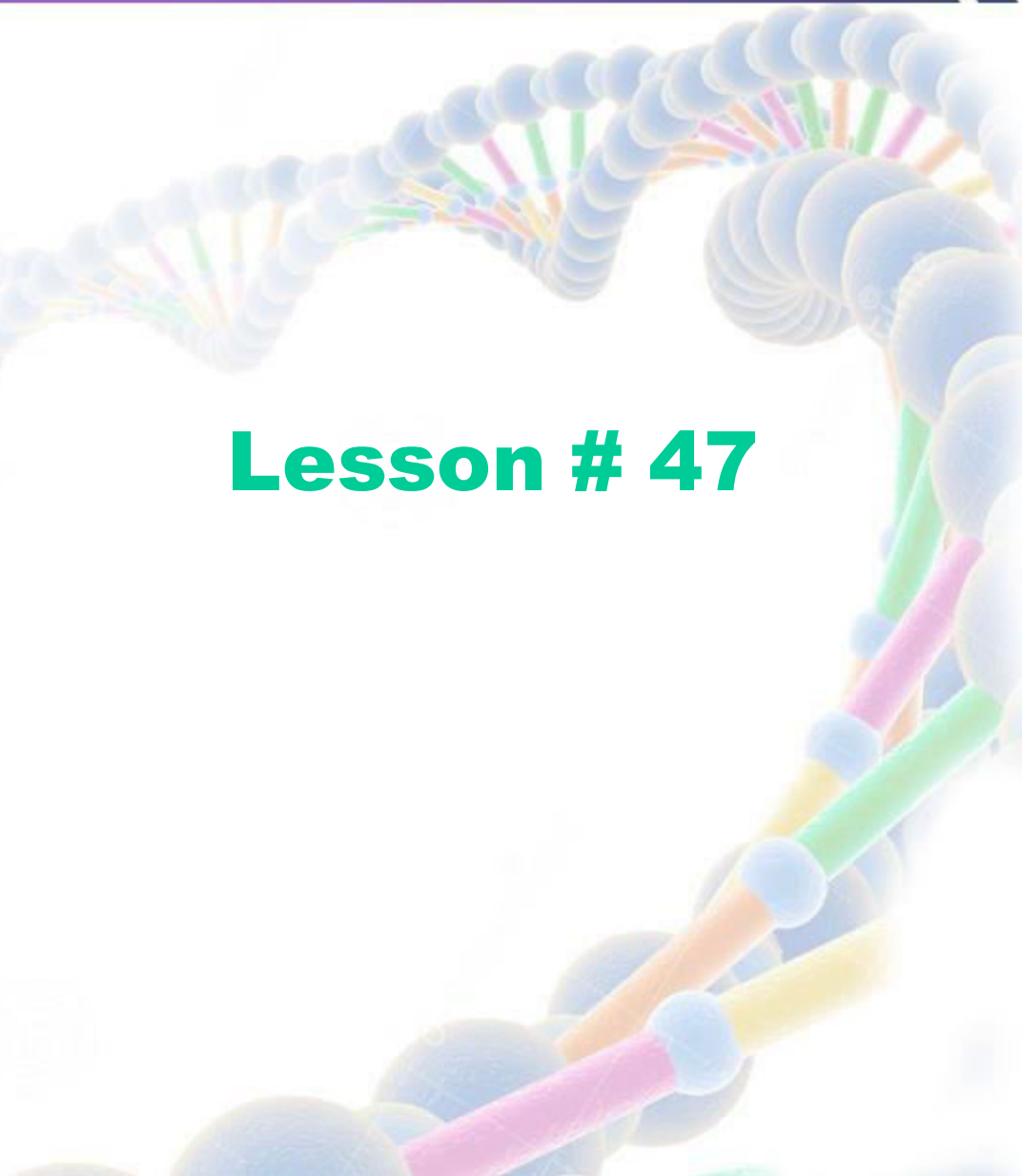


Green revolution:

- one-sixth of the world population is hungry
- poverty
- political factors
- based on research and government funds

Bioethics

Lesson # 47

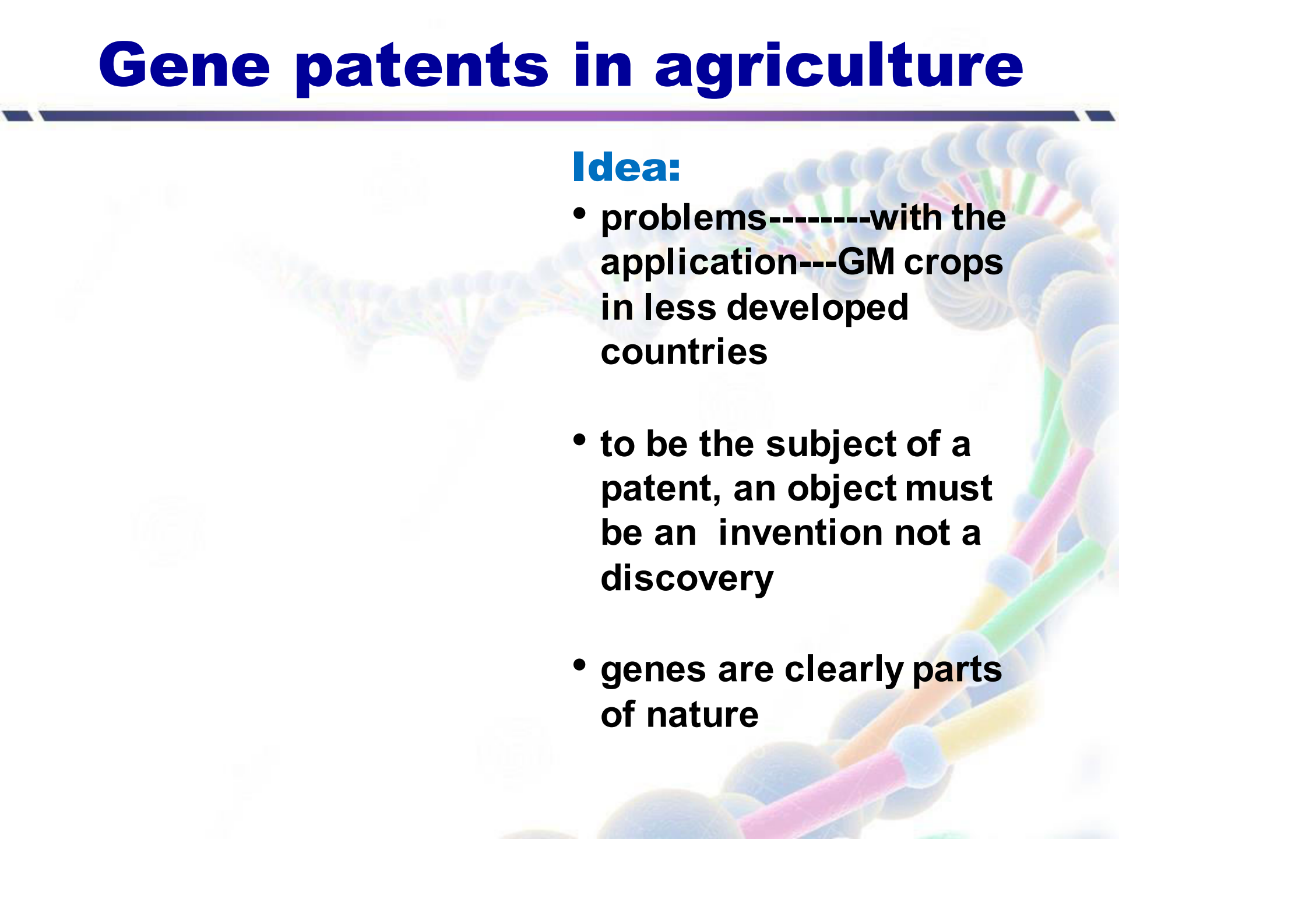


Bioethics



Gene patents in agriculture

Gene patents in agriculture



Idea:

- **problems-----with the application---GM crops in less developed countries**
- **to be the subject of a patent, an object must be an invention not a discovery**
- **genes are clearly parts of nature**

Gene patents in agriculture

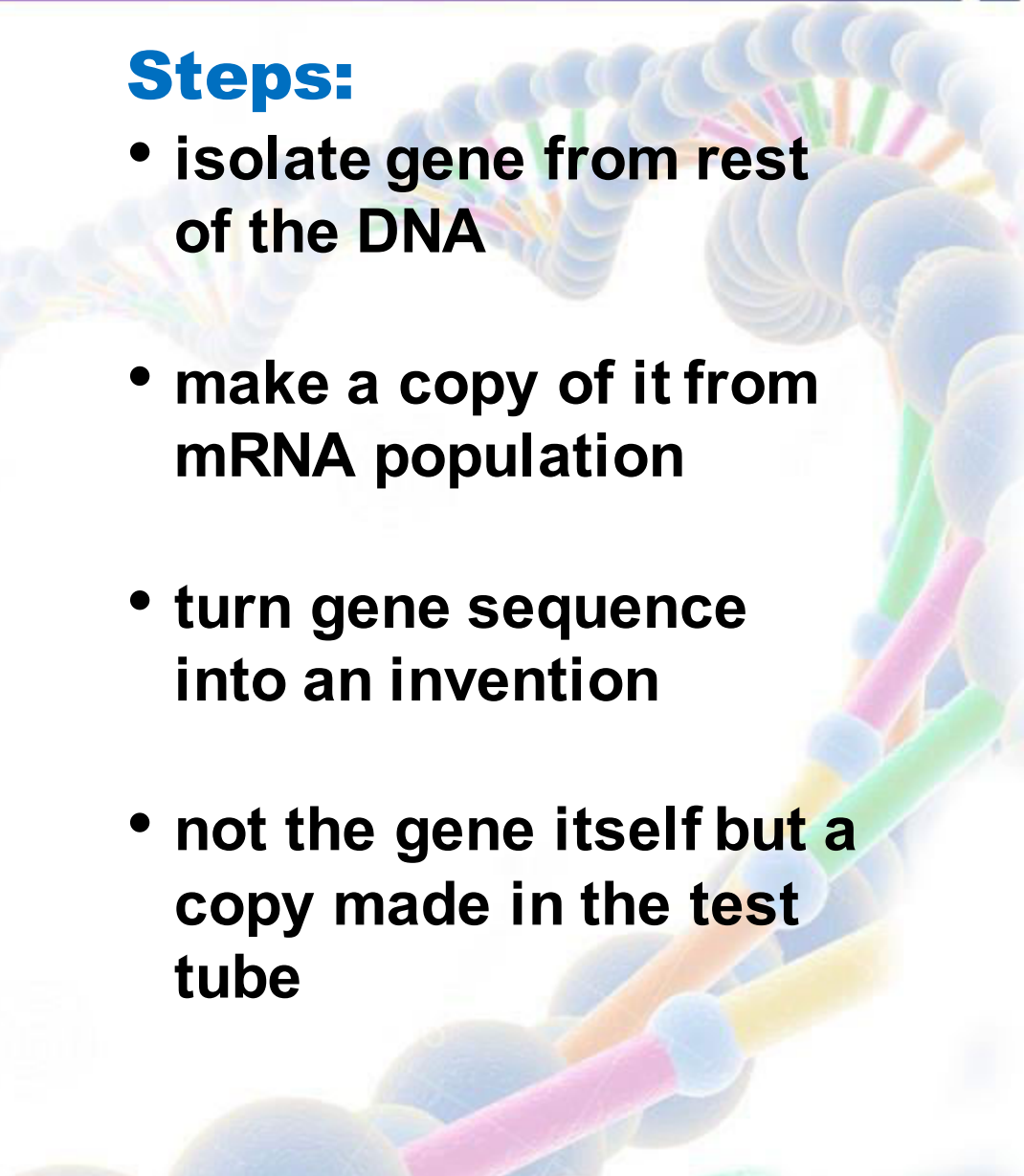
Granted:

- crop genes have been patented in USA
- groups who support patents—there is an inventive step
- argument not accepted “patenting a gene copy is not the same as patenting the gene”

Gene patents in agriculture

Steps:

- isolate gene from rest of the DNA
- make a copy of it from mRNA population
- turn gene sequence into an invention
- not the gene itself but a copy made in the test tube



Gene patents in agriculture

Advantages:

- gene sequence is legitimate intellectual property
- companies ensure an appropriate return on research/development investment
- Vitamin A enhanced Golden rice

Bioethics

Lesson # 48

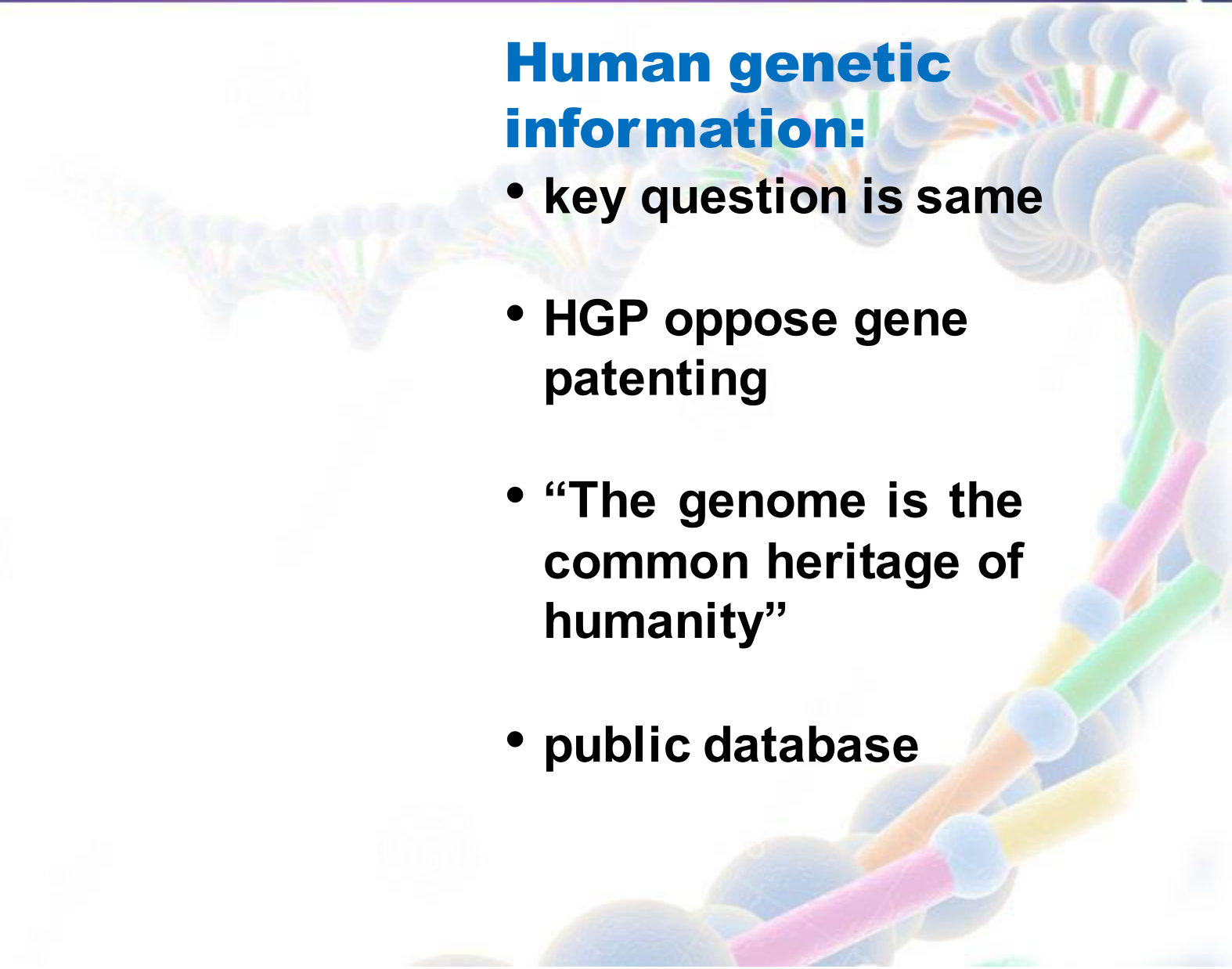


Bioethics



**Gene patents-
medical
genetics**

Gene patents-medical genetics



Human genetic information:

- key question is same
- HGP oppose gene patenting
- “The genome is the common heritage of humanity”
- public database

Gene patents-medical genetics

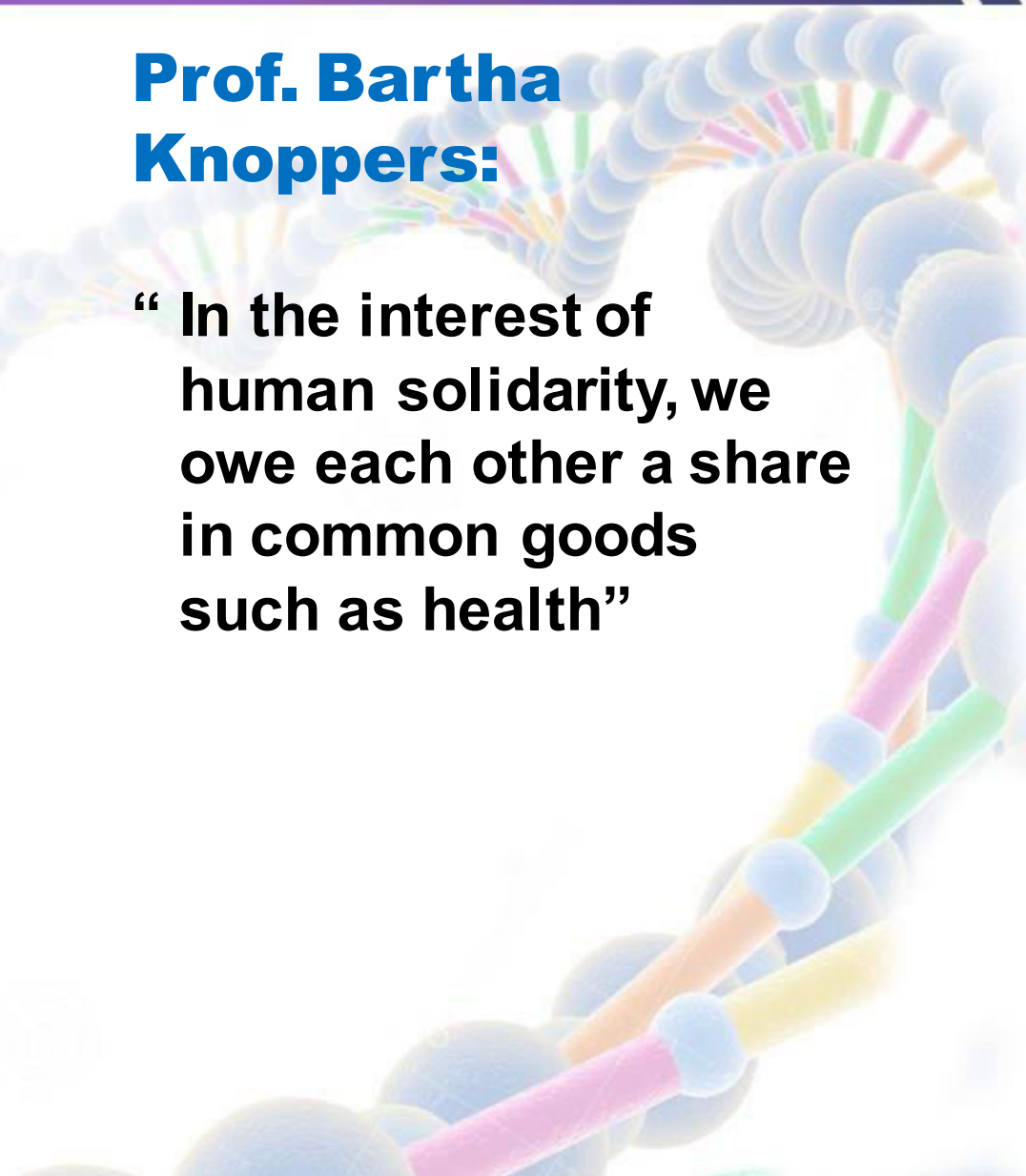
Celera Genomics:

- not a part of public or charity funded HGP
- purchase 300 DNA sequencing machines
- sequence most of the human genes
- commercial potential in the use of human gene sequences

Gene patents-medical genetics

**Prof. Bartha
Knoppers:**

**“ In the interest of
human solidarity, we
owe each other a share
in common goods
such as health”**

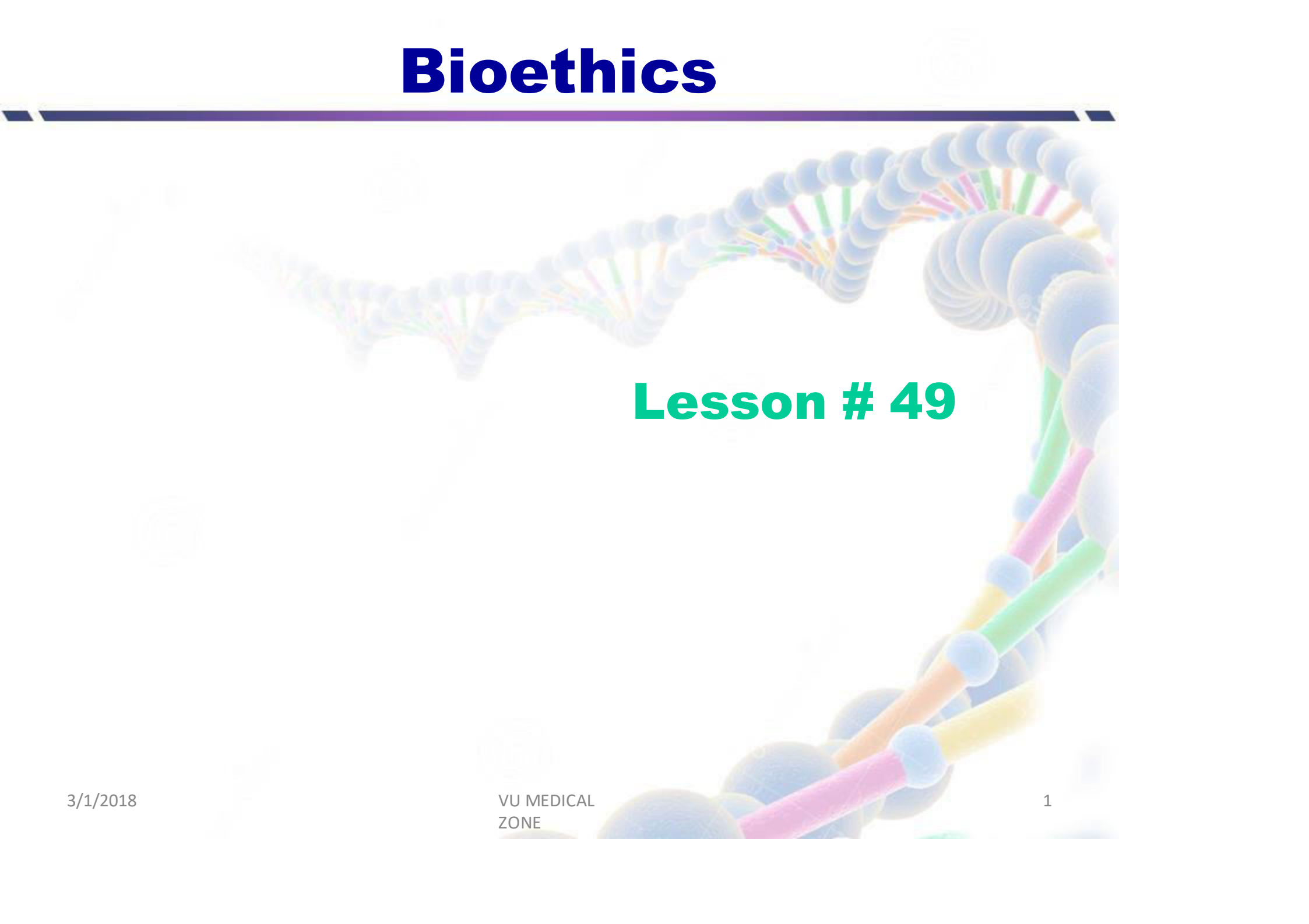


Gene patents-medical genetics

Advantages:

- commercial interest
- synthesis of generic drugs--AIDs
- genetic based treatments
- Africa– increased cost
–gene patenting

Bioethics



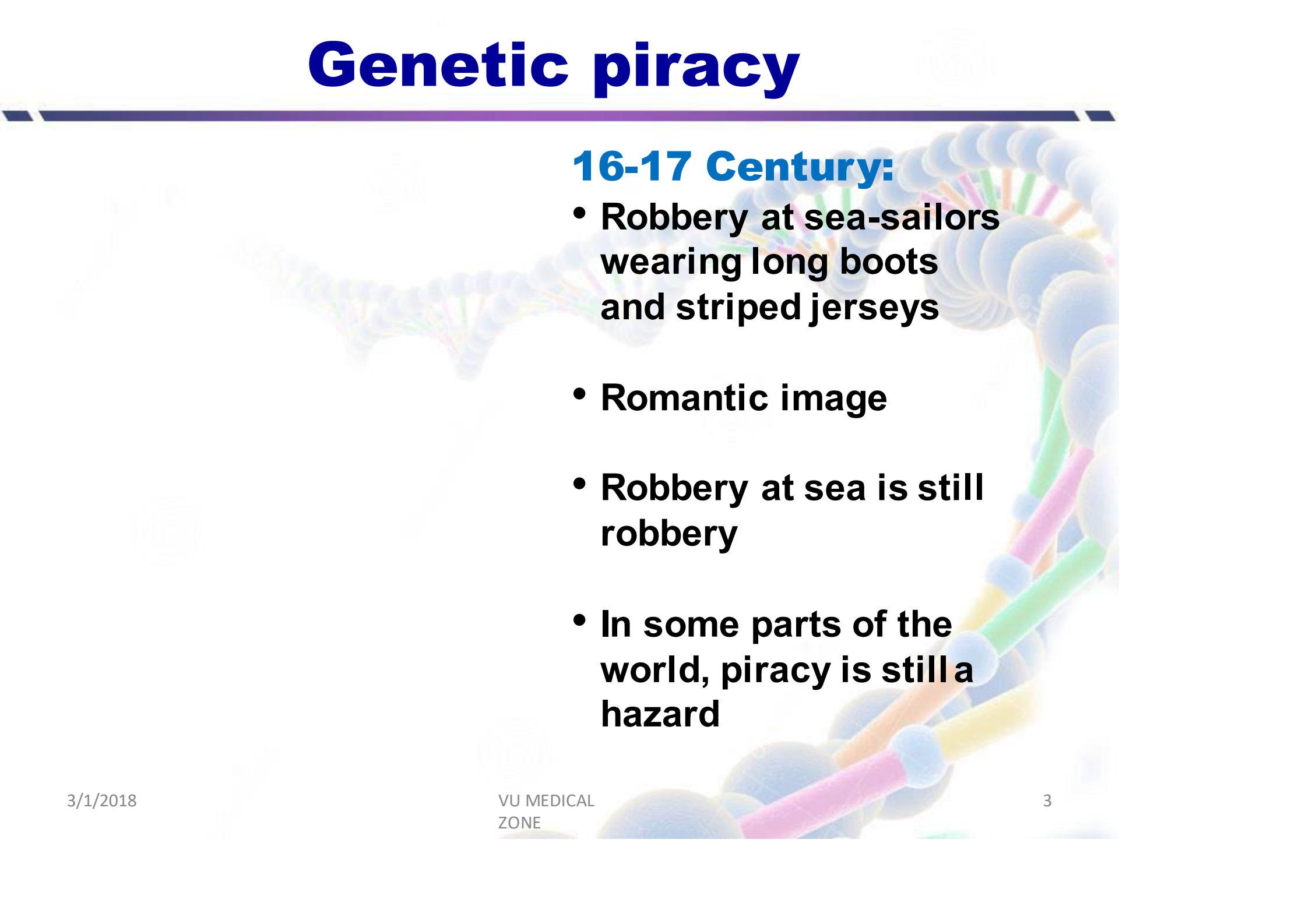
Lesson # 49

Bioethics



Genetic piracy

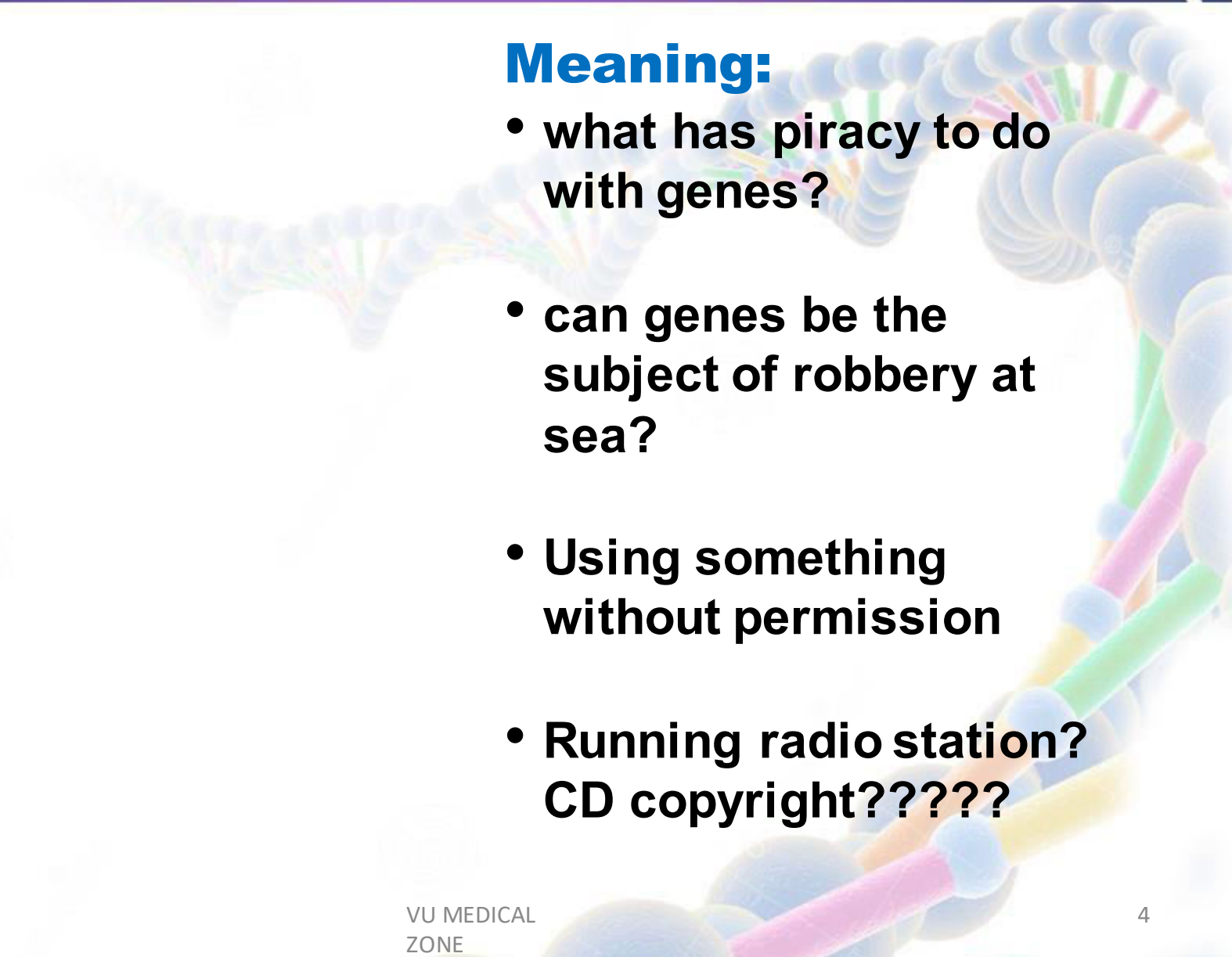
Genetic piracy



16-17 Century:

- Robbery at sea-sailors wearing long boots and striped jerseys
- Romantic image
- Robbery at sea is still robbery
- In some parts of the world, piracy is still a hazard

Genetic piracy



Meaning:

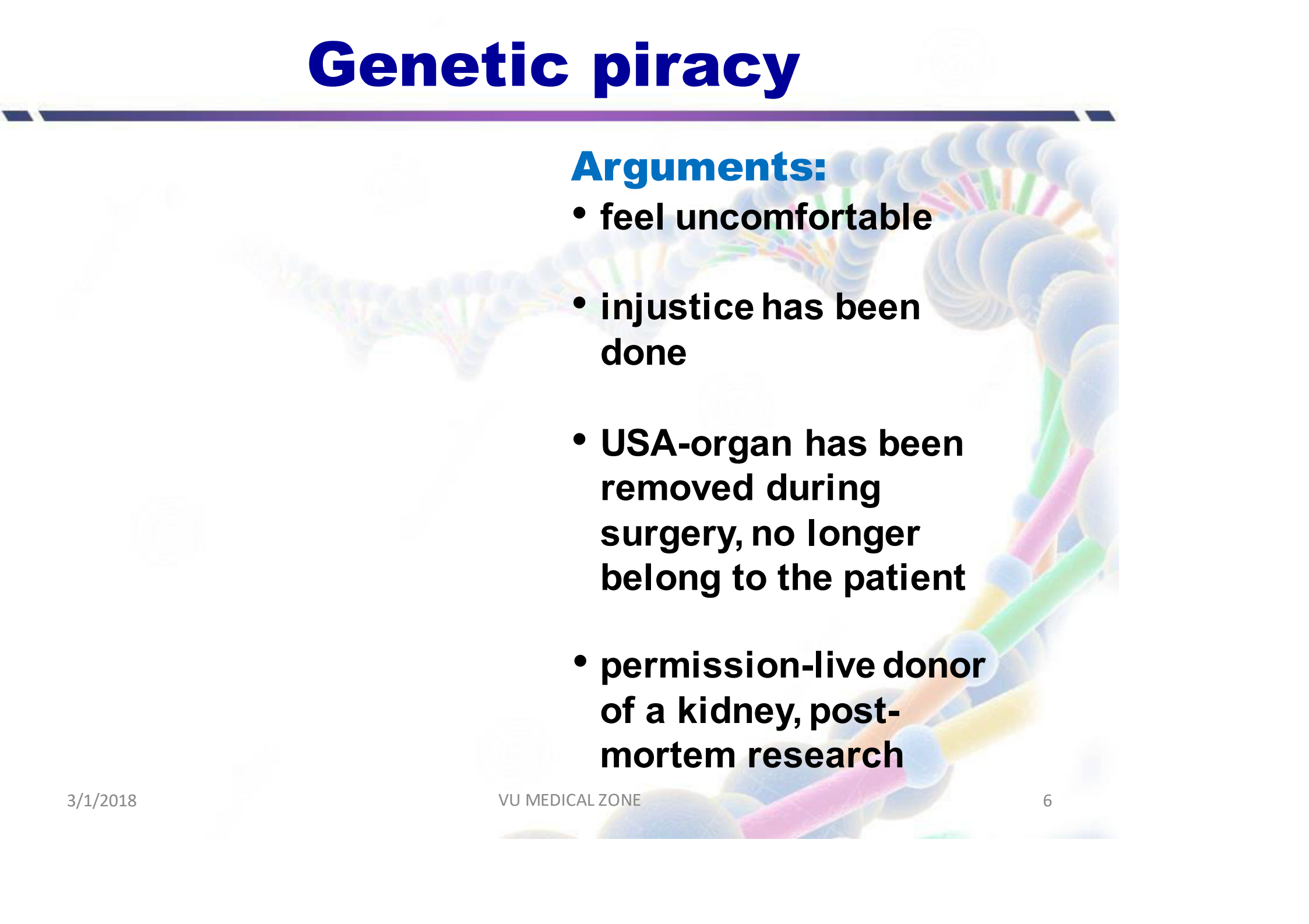
- what has piracy to do with genes?
- can genes be the subject of robbery at sea?
- Using something without permission
- Running radio station?
CD copyright?????

Genetic piracy

Is it genetic piracy?

- the patient's cells, for the sake of lesions they exhibit-----used without permission, brought gain to the user
- spleenectomy for the sake of patient's health
- no ownership rights

Genetic piracy



Arguments:

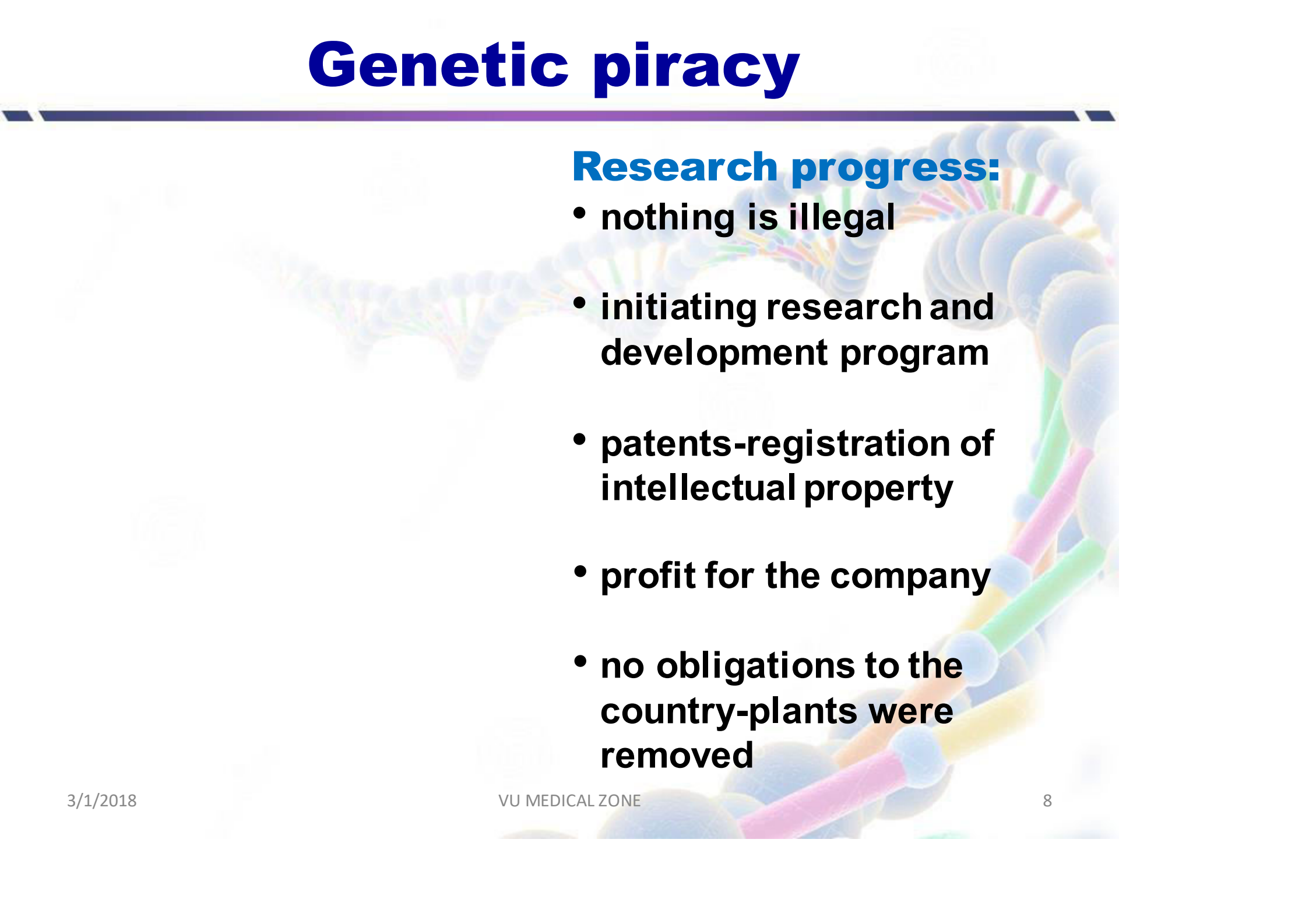
- feel uncomfortable
- injustice has been done
- USA-organ has been removed during surgery, no longer belong to the patient
- permission-live donor of a kidney, post-mortem research

Genetic piracy

Plants:

- Plants-pain relief
- Is this intellectual property?
- Do wild plants belong to anyone????
- NO
- Laws-prevent removal of plants----- private owner land

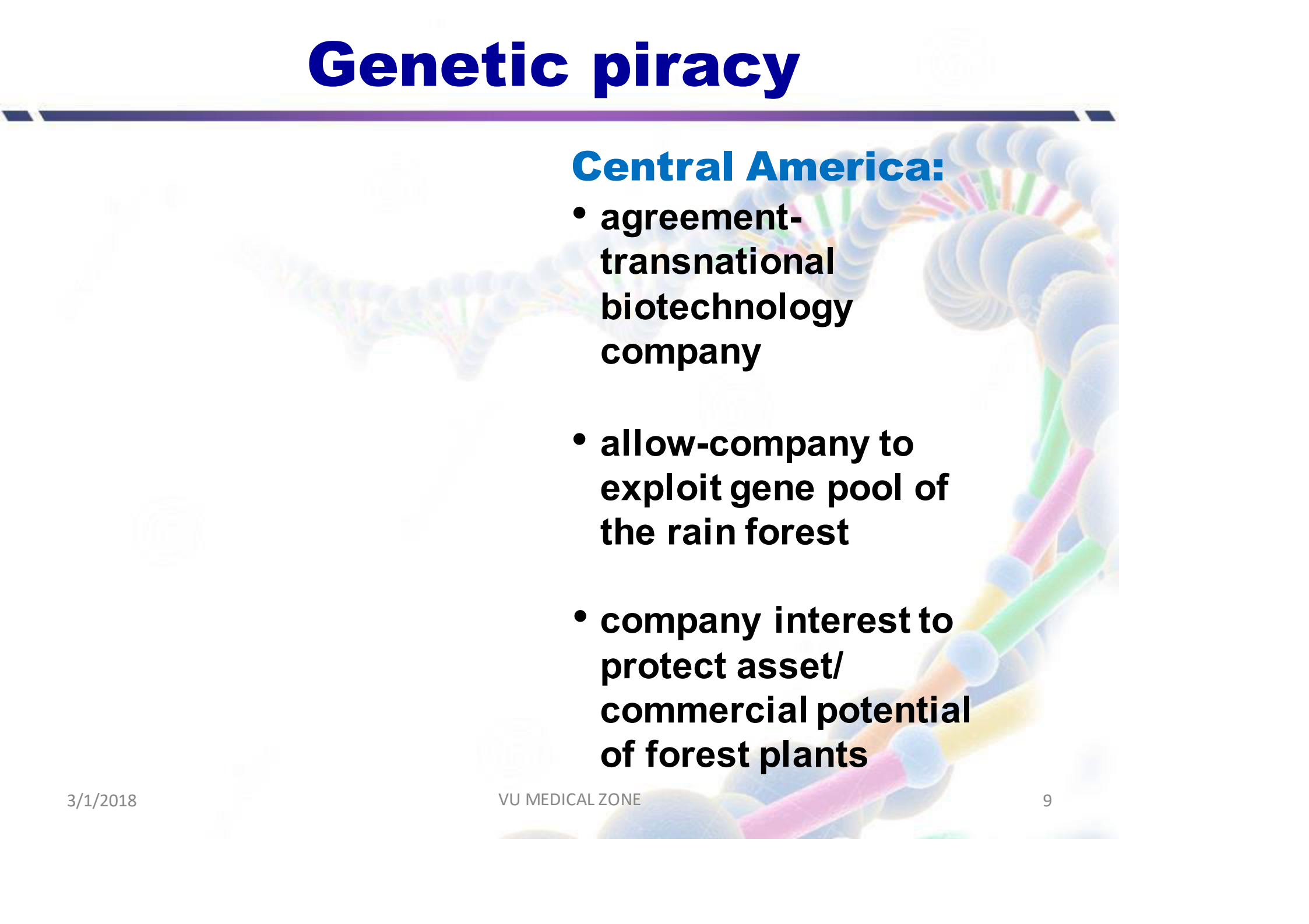
Genetic piracy



Research progress:

- nothing is illegal
- initiating research and development program
- patents-registration of intellectual property
- profit for the company
- no obligations to the country-plants were removed

Genetic piracy



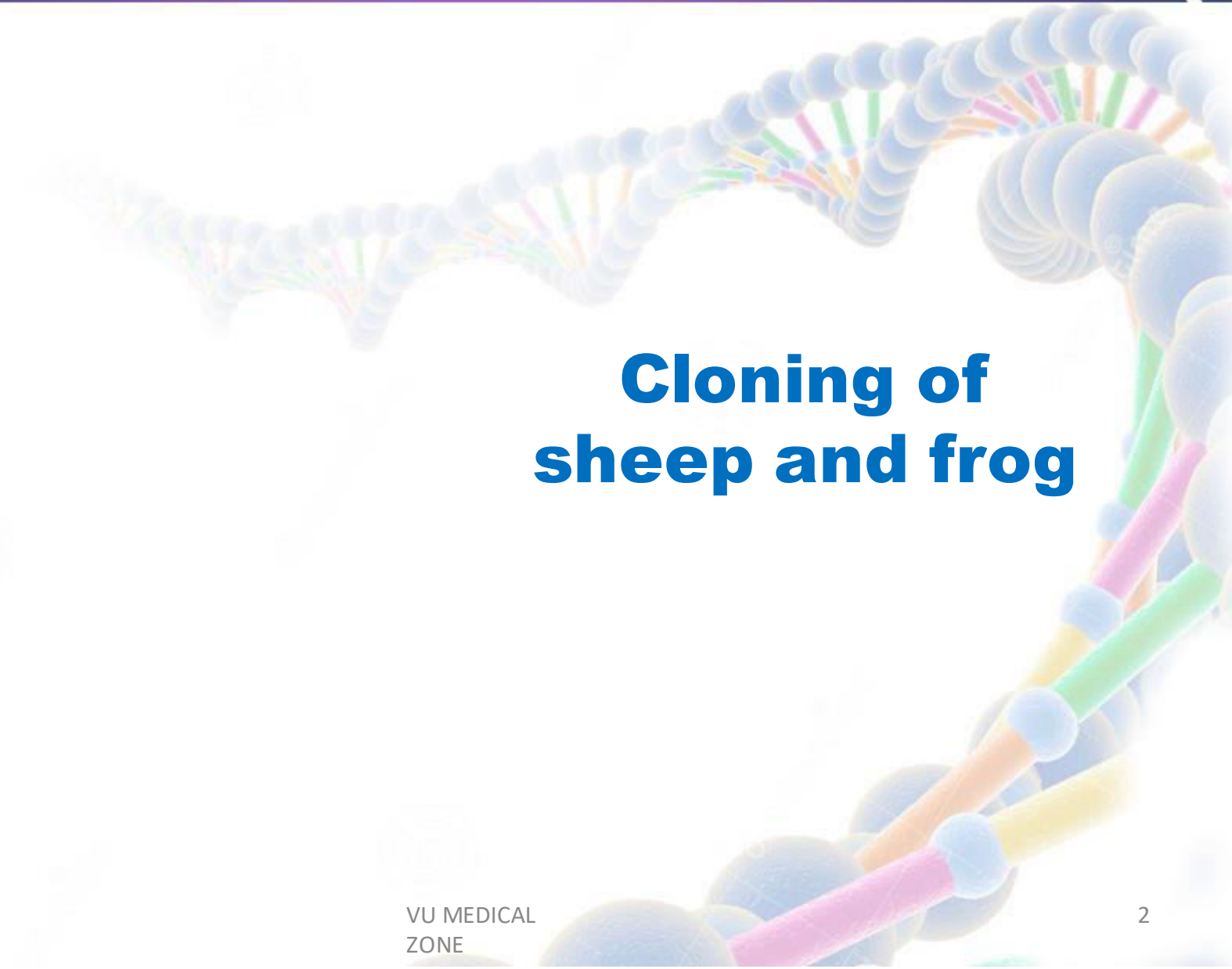
Central America:

- agreement-transnational biotechnology company
- allow-company to exploit gene pool of the rain forest
- company interest to protect asset/commercial potential of forest plants

Bioethics

Lesson # 50

Bioethics



Cloning of sheep and frog

Cloning of sheep and frog

Cloning:

- processes used to create copies of DNA fragments, cells or organisms
- Briggs and Kings-----normal tadpole clones using nuclei from early embryos
- nuclear transfer—viable technique

Cloning of sheep and frog

Nuclear transfer:

- nucleus directs cell growth
- embryonic cells early in development-better
- ultimately organism development

Cloning of sheep and frog

John Gurdon:

- transplanted the nucleus of a tadpole intestinal cell into an enucleated frog egg
- tadpoles- genetically identical to the one from which intestinal cells were taken
- cells retain genetic material as they divide and differentiate

Cloning of sheep and frog

Steen Willadsen:

- chemical process-----
separate one cell from
8-cell lamb embryo
- electric shock to fuse
in to an enucleated egg
cell
- lamb embryos--womb
of surrogate lamb
- three live lambs

Cloning of sheep and frog

Wilmut and Keith:

- transfer the nuclei from cultured cells into enucleated sheep egg cells
- lambs born “Megan and Morag”
- transgenic sheep----- Polly that produce Factor IX in her milk

Cloning of sheep and frog

Dolly:

- **adult somatic cells**
- **every cell's nucleus has a complete set of genetic information**
- **embryonic cells activate any gene**
- **differentiated adult cells shut down the genes they don't need**

Cloning of sheep and frog

Dolly:

- Of 277 attempts, one embryo was produced
- carried in to surrogate mother
- famous lamb-Dolly
- controversies arises

Cloning of sheep and frog

Ethical issues:

- **Dolly-1996**
- **adult cells can reprogram themselves into a new being**
- **cloning makes humans God**
- **health risk in cloned animals/all animals are created equally**

Bioethics

Lesson # 51



Bioethics

**Ethics-early
human embryo**



Ethics-early human embryo

Ethical issues:

- two moral principles
duty-prevent suffering,
duty-respect the value
of human life
- harvesting of human
embryo violate the
second duty
- aim of stem cell
research is good-what
about the moral
principles

Ethics-early human embryo

Ethical issues:

- fertilized eggs should be protected as they are human
- even unconscious individuals are treated as persons
- fertilized human egg before implantation doesn't satisfy the criteria of personhood

Ethics-early human embryo

Ethical issues:

- don't remember—not worthy of respect-early stage of development
- embryos don't have emotional, intellectual or psychological properties
- degrees of respect
- before implantation lesser degree

Ethics-early human embryo

Ethical issues:

- natural loss of embryos same as it occurs in stem cell research
- nervous system of early embryos is not developed fully
- In Jewish religion, human fetus < 40 days old-doesn't have the full human status

Ethics-early human embryo

Ethical issues:

- soul is “breathed in” to the human embryo on the 40 day after fertilization--Islam
- stem cell research is acceptable due to therapeutic benefits
- embryos cannot be donated to other couples

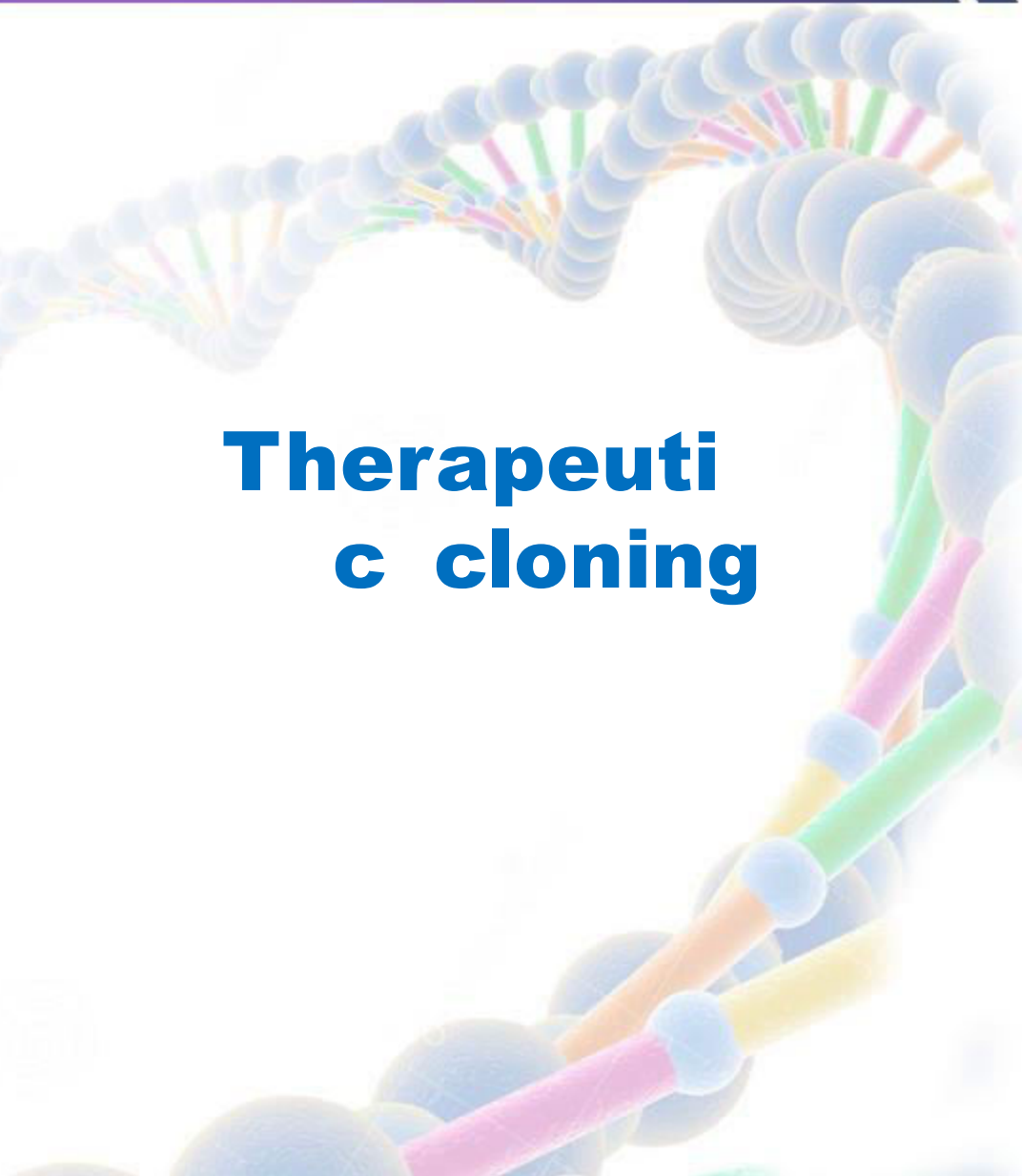
Bioethics

Lesson # 52



Bioethics

Therapeutic cloning



Therapeutic cloning

Human embryonic stem cells (hESCs):

- derived from embryo-5 to 7 days old-before implantation
- proliferate and differentiate
- hESCs generated organs
- face rejection by the immune system

Therapeutic cloning

Solution:

- somatic cell nuclear transfer
- somatic cell is taken from the patient own body
- nucleus from this cell is placed into an enucleated egg
- same genome as that of the donor cell

Therapeutic cloning

Ethical issues:

- moral status of the embryo ---- destruction
- patient has the right to live
- morally right for in vitro fertilization but morally wrong to save a child's life
- potential donor exploitation

Therapeutic cloning

Ethical issues:

- **slippery slope -----
reproductive cloning
different from
therapeutic cloning**
- **no access and benefit
to poor communities**
- **raises issues of social
justice and healthcare
disparities**

Bioethics

Lesson # 53



Bioethics

Designer babies



Designer babies

Designer babies:

- children-genetically engineered in the womb to have desired qualities
- made through in vitro fertilization
- embryo is removed-manipulated for desired qualities-placed in the womb

Designer babies

Disadvantages:

- expensive---not 100% save
- better looking---create gap in society
- affect the gene pool
- genes can have more than one use
- infants cannot give the consent

Designer babies

Advantages:

- increases human life span up to 30 years
- prevent genetic disorders
- infertile women can have children
- parents set their own limits for genetically engineered babies

Designer babies

Ethical issues:

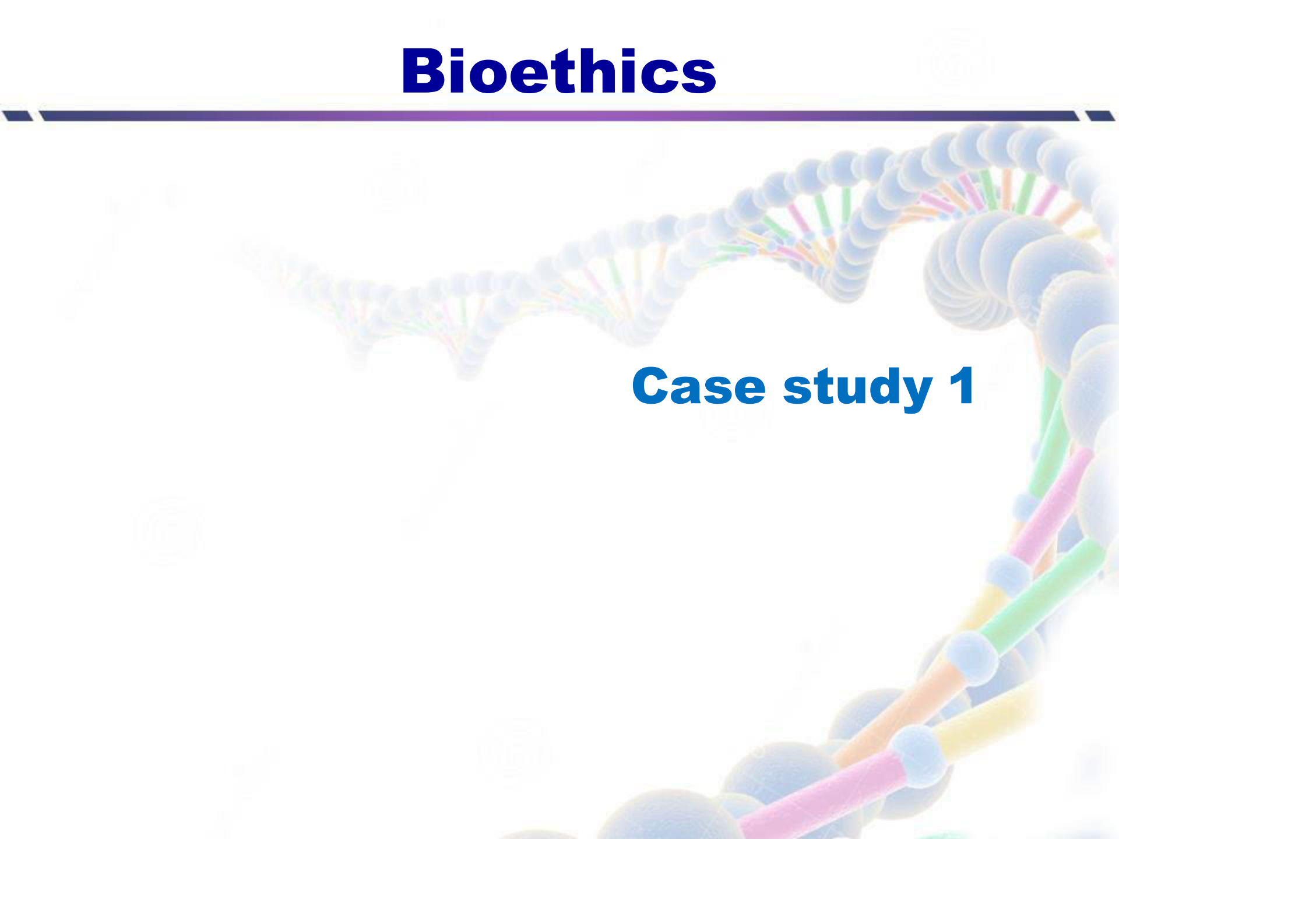
- unethical and unnatural
- morally wrong
- parents get upset when trait didn't pay off
- problems in the child/parent relationships

Bioethics

Lesson # 54



Bioethics



Case study 1

Case study 1

Case study:

- donated gametes- sperm and ova- are used in fertility treatments for patients who are unable to produce their own
- It is much easier to donate sperm than ova
- donated ova are very scarce.

Case study 1

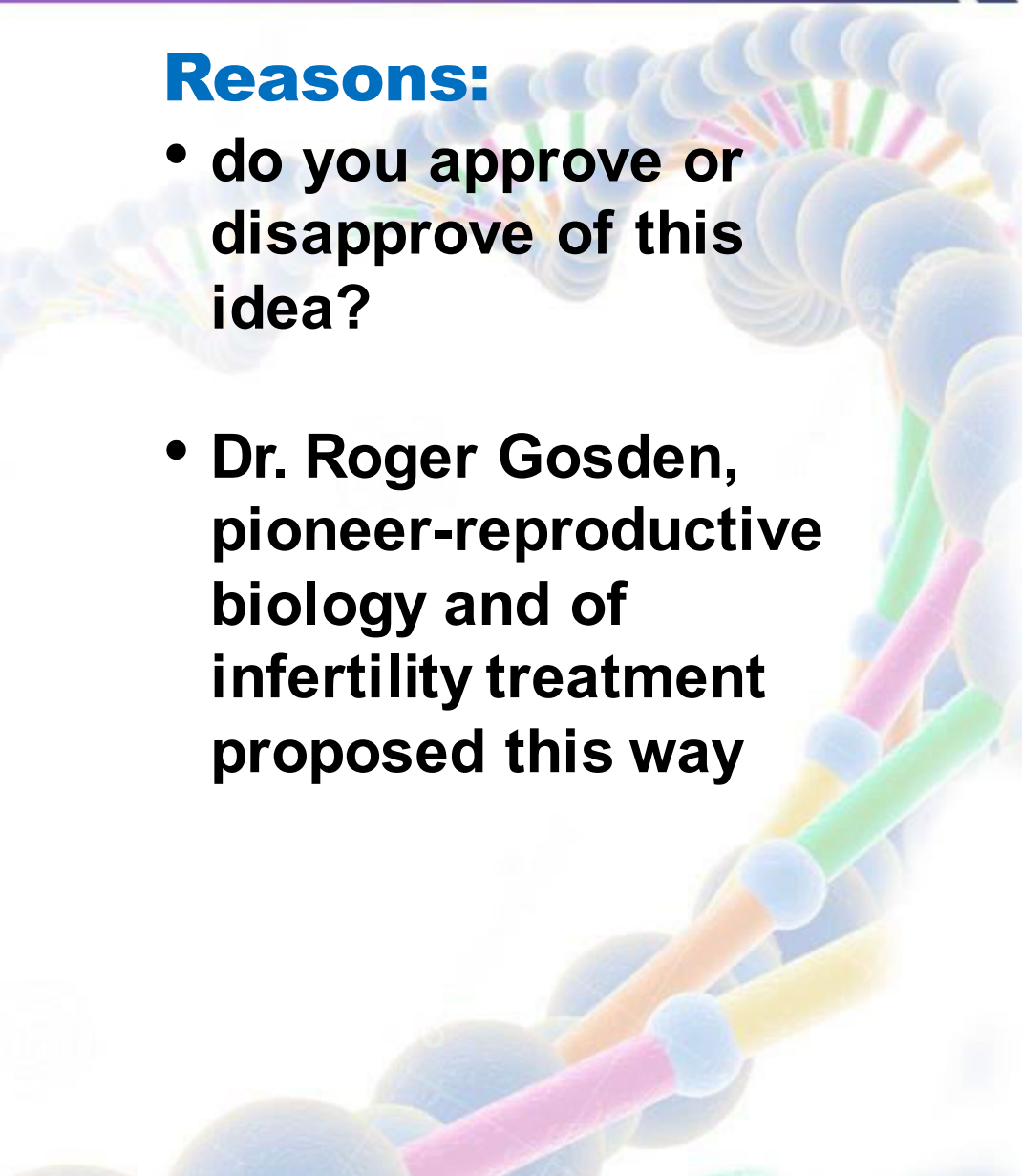
Case study:

- during fetal development, females lay down a lifetime's supply of oocytes
- It is therefore suggested that aborted female fetus may be used to supply oocytes for fertility treatments

Case study 1

Reasons:

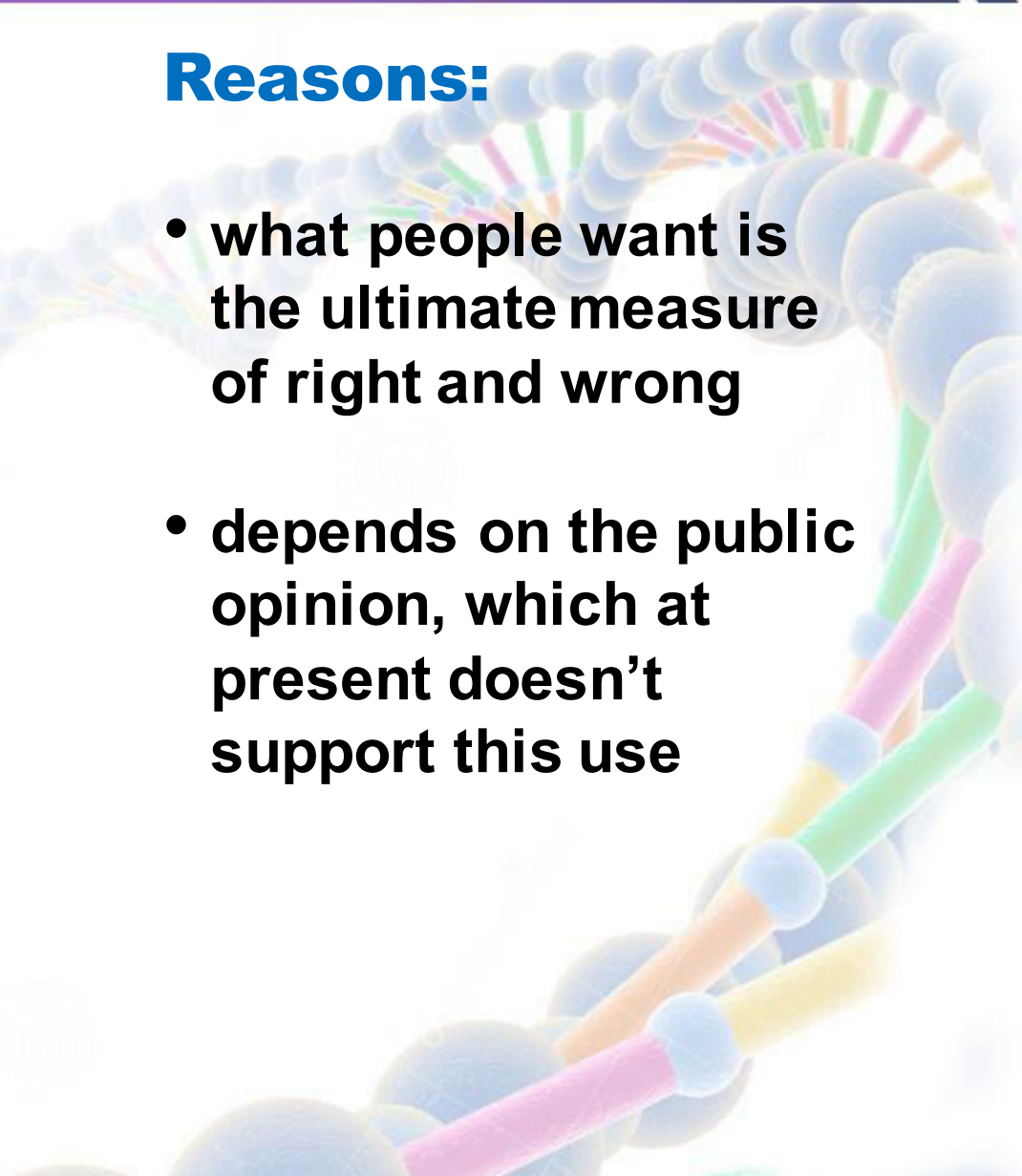
- do you approve or disapprove of this idea?
- Dr. Roger Gosden, pioneer-reproductive biology and of infertility treatment proposed this way



Case study 1

Reasons:

- what people want is the ultimate measure of right and wrong
- depends on the public opinion, which at present doesn't support this use



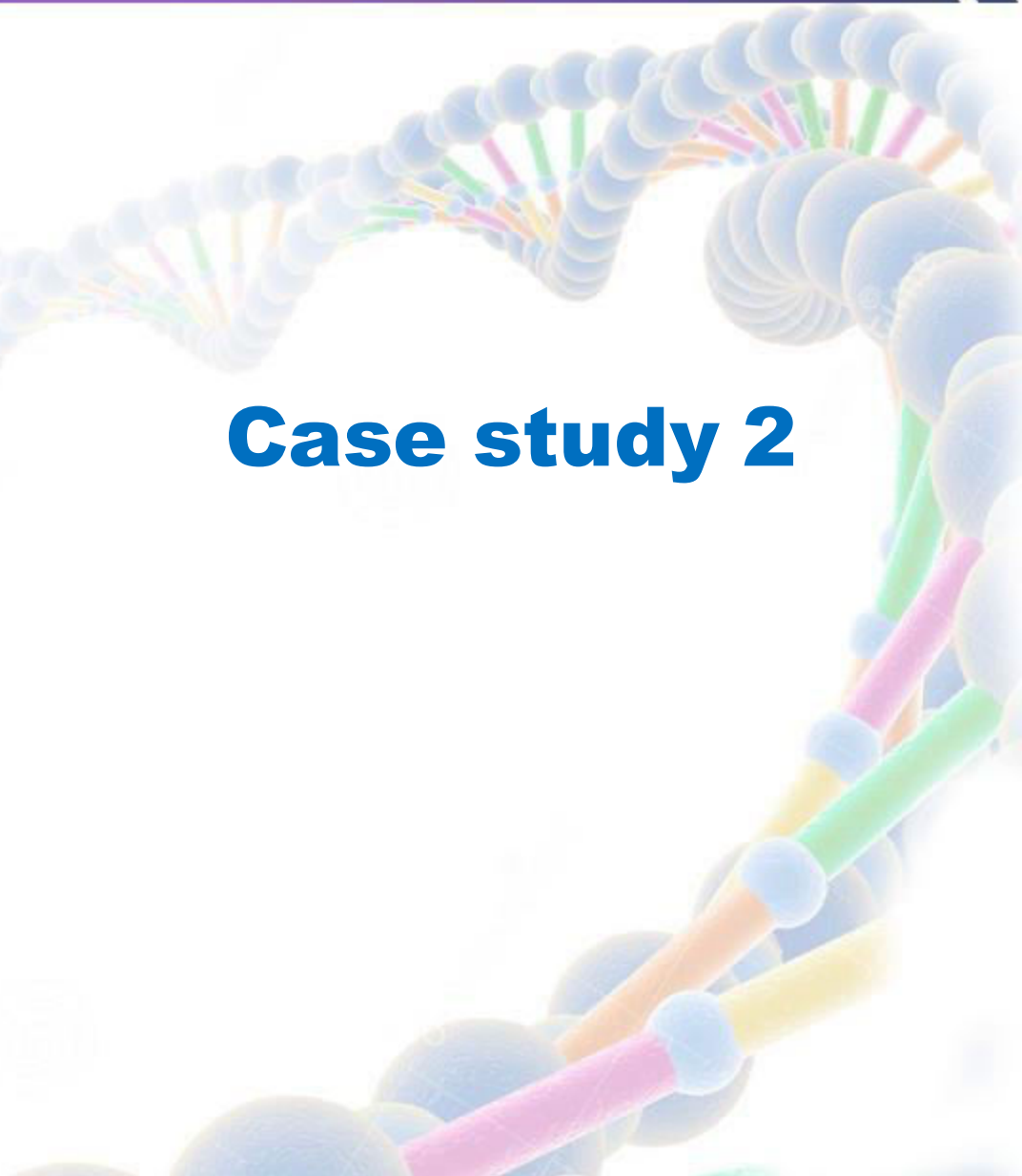
Bioethics



Lesson # 55

Bioethics

Case study 2



Case study 2

Case study 2:

- A small less developed country in South America is deep in debt
- Its main source is its rain forest

Case study 2

Solution:

- **What parent want this is an ultimate measure**
- **The land has been cleared used for cattle ranching to raise beef in the US market**

Case study 2

Case study 2:

- The government has also granted a license to transnational biotechnology company to exploit the forest's gene pool
- The company has agreed to pay royalties on income generated from discoveries based on rain forest gene pool

Case study 2

Issues:

- what are the issues in dealing with this situation?
- deleterious effects on biodiversity
- right- to exploit any living organism or any ecological community

Case study 2

Issues:

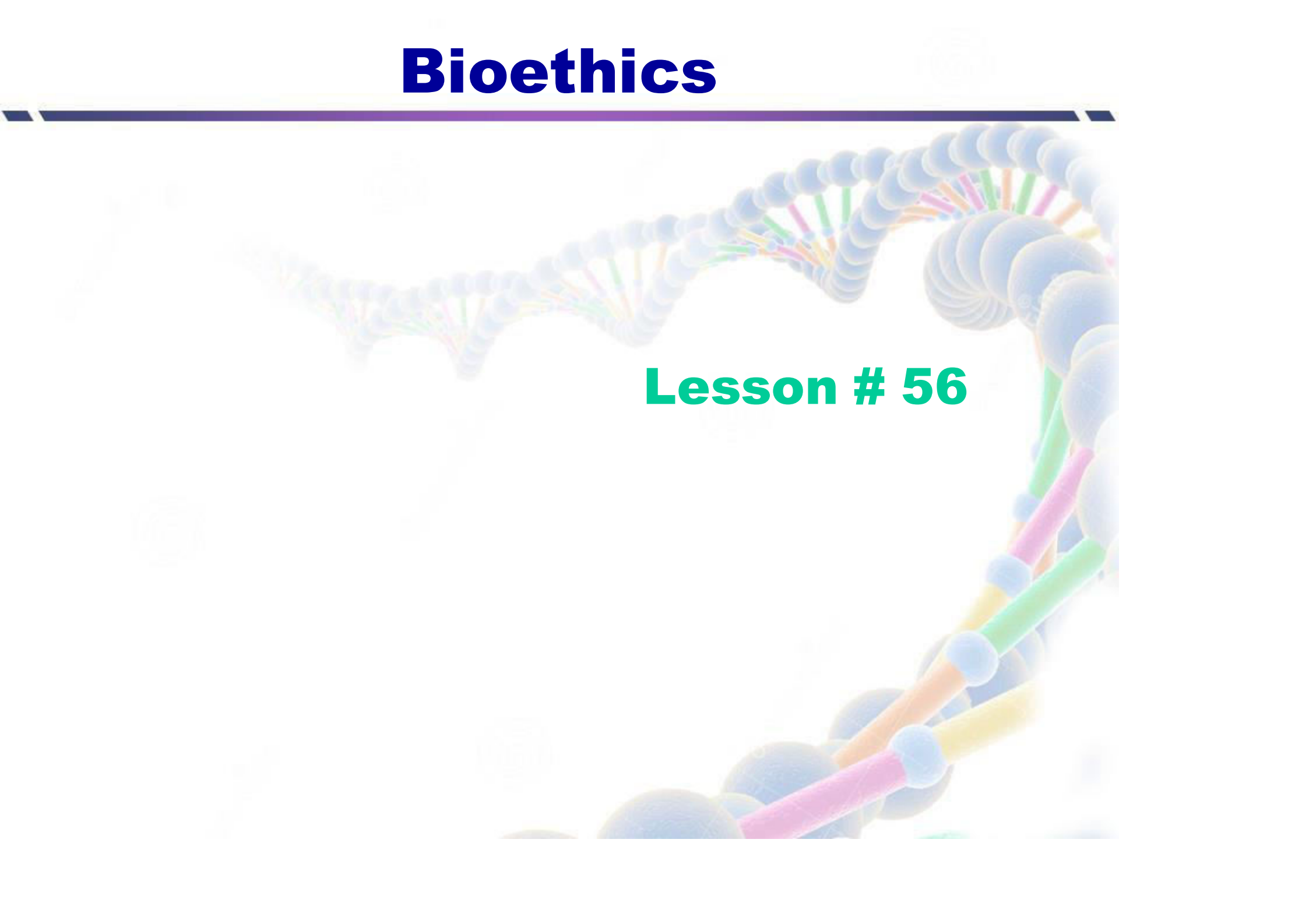
- agreement might create a genuine commercial flow of money from the richer to some of the poorer nations
- wealth of local knowledge on biodiversity

Case study 2

Issues:

- working to bring traditional knowledge under an extended intellectual property umbrella
- it appears that an imbalance of power is being corrected within this general area of exploiting exotic gene pools.

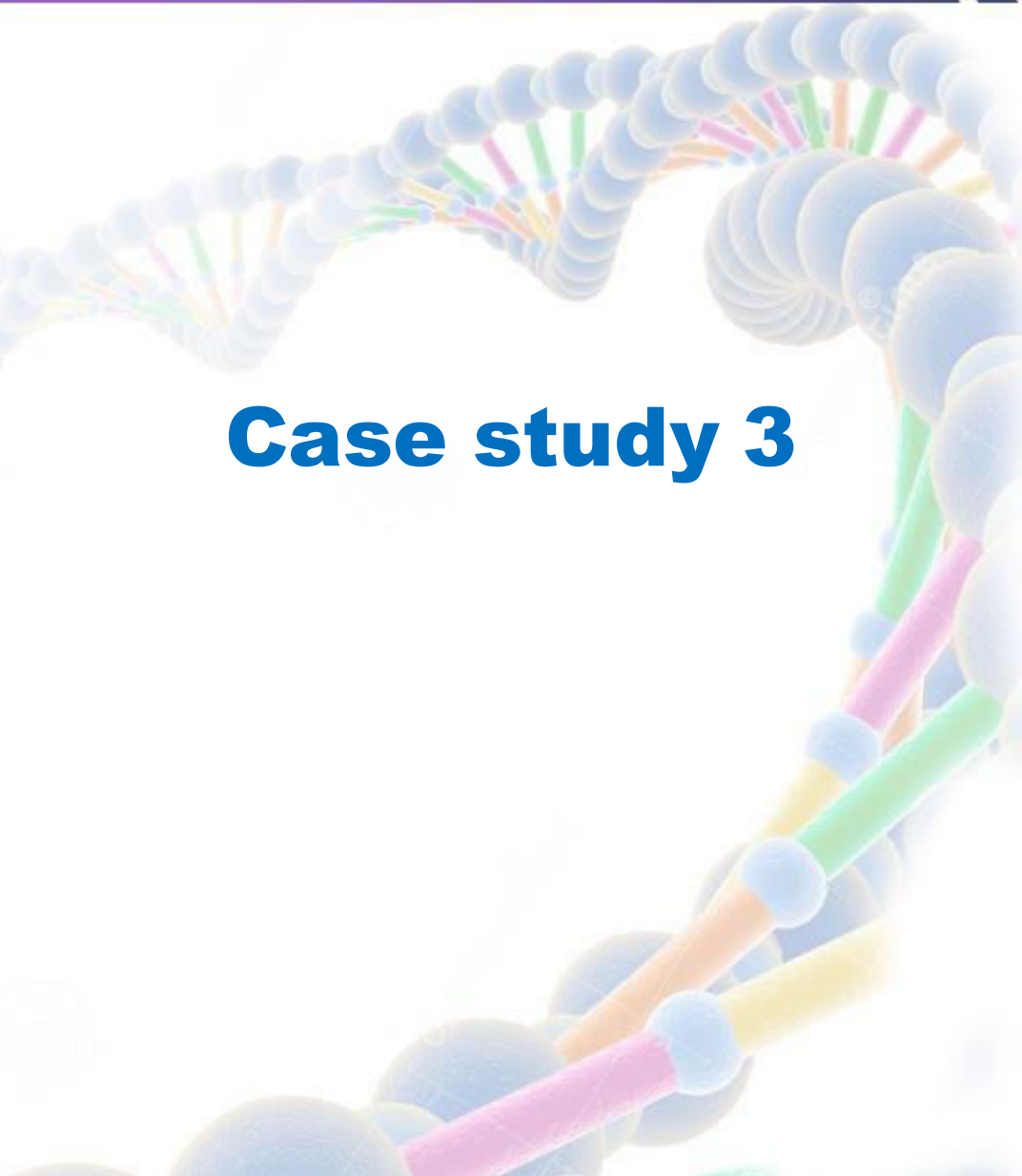
Bioethics



Lesson # 56

Bioethics

Case study 3



Case study 3

Case study:

- in which of the following cases, would you grant permission?
- normal fertile couples undergo in vitro fertilization in order to produce a baby that can be a stem cell donor for an older sibling

Case study 3

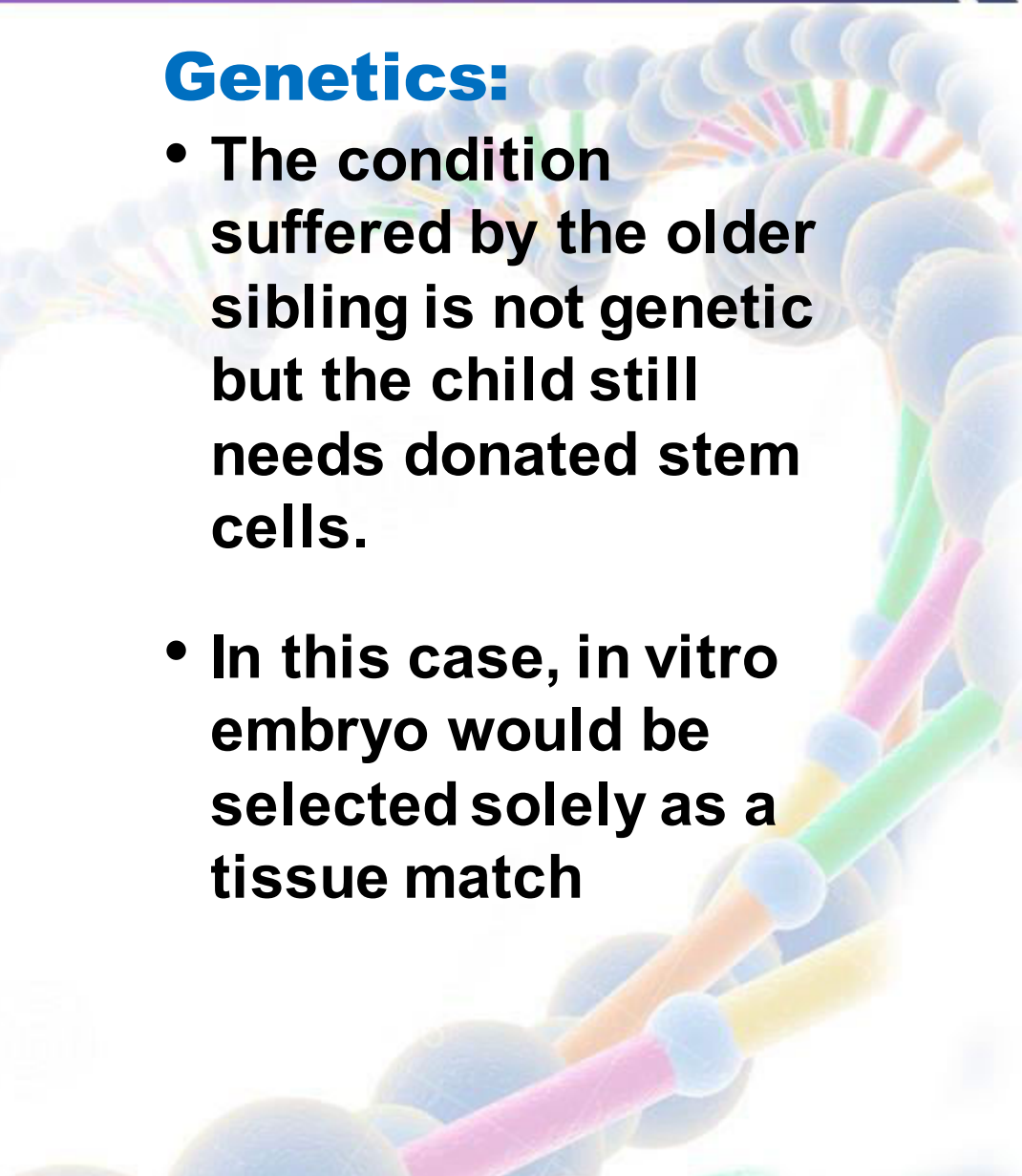
Genetics:

- The older sibling suffers from genetic disorder and the embryo created in vitro would be tested for the absence of mutation and is the positive tissue match to the older sibling

Case study 3

Genetics:

- The condition suffered by the older sibling is not genetic but the child still needs donated stem cells.
- In this case, in vitro embryo would be selected solely as a tissue match



Case study 3

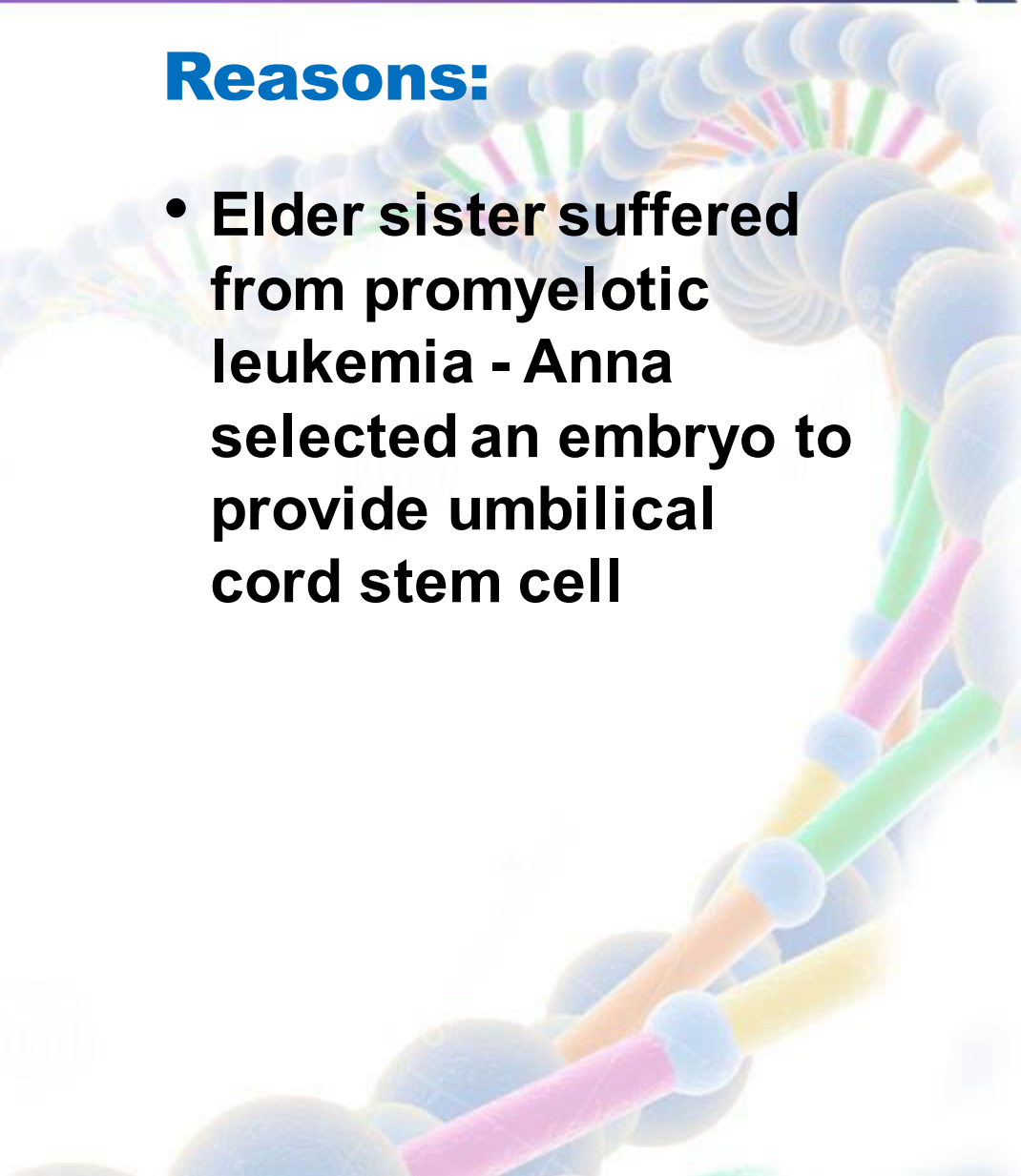
Reasons:

- There should be clear cut regulations surrounding these concepts
- HFE 1990 Act, creation of saviour sibling - enable the identification of a tissue match for an older sibling suffer from life-threatening disease

Case study 3

Reasons:

- **Elder sister suffered from promyelotic leukemia - Anna selected an embryo to provide umbilical cord stem cell**



Case study 3

Reasons:

- Nash family elder daughter- Fanconi's anemia
- In 2000, Adam was born a suitable match for her sister
- “If you use one of your children to save the life of another, are you being a good mother or a very bad one”

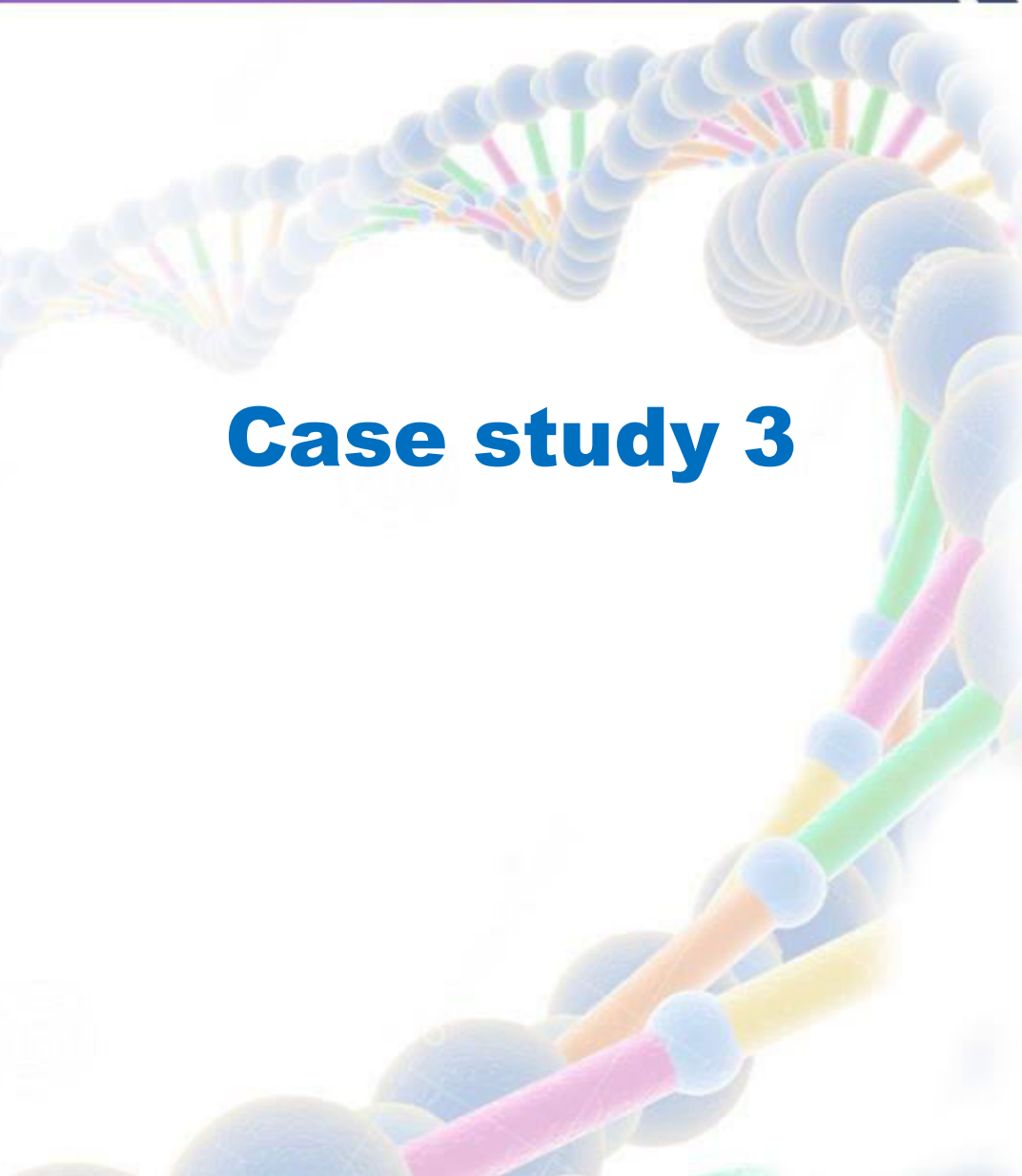
Bioethics

Lesson # 57



Bioethics

Case study 3



Case study 3

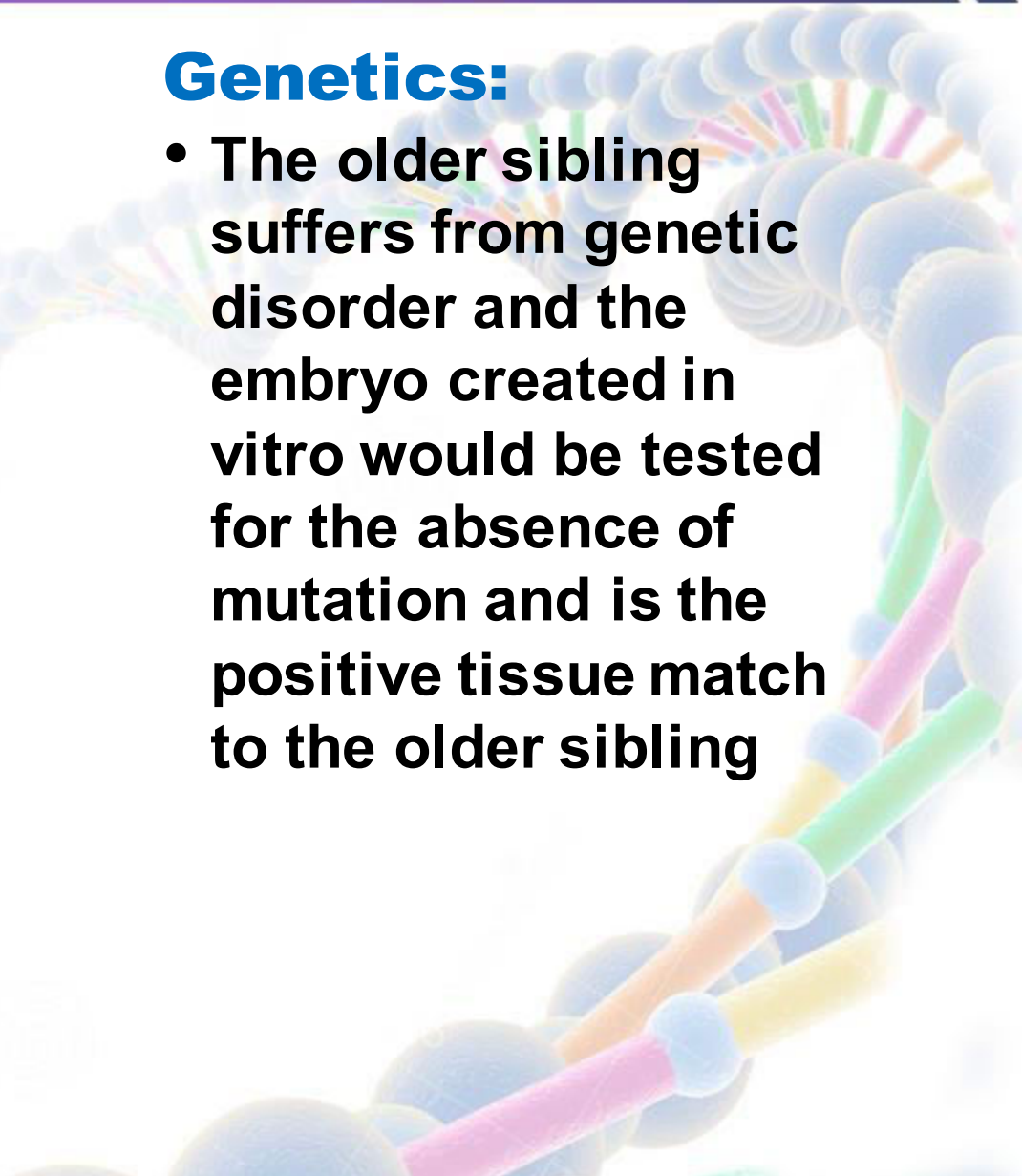
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Case study 3

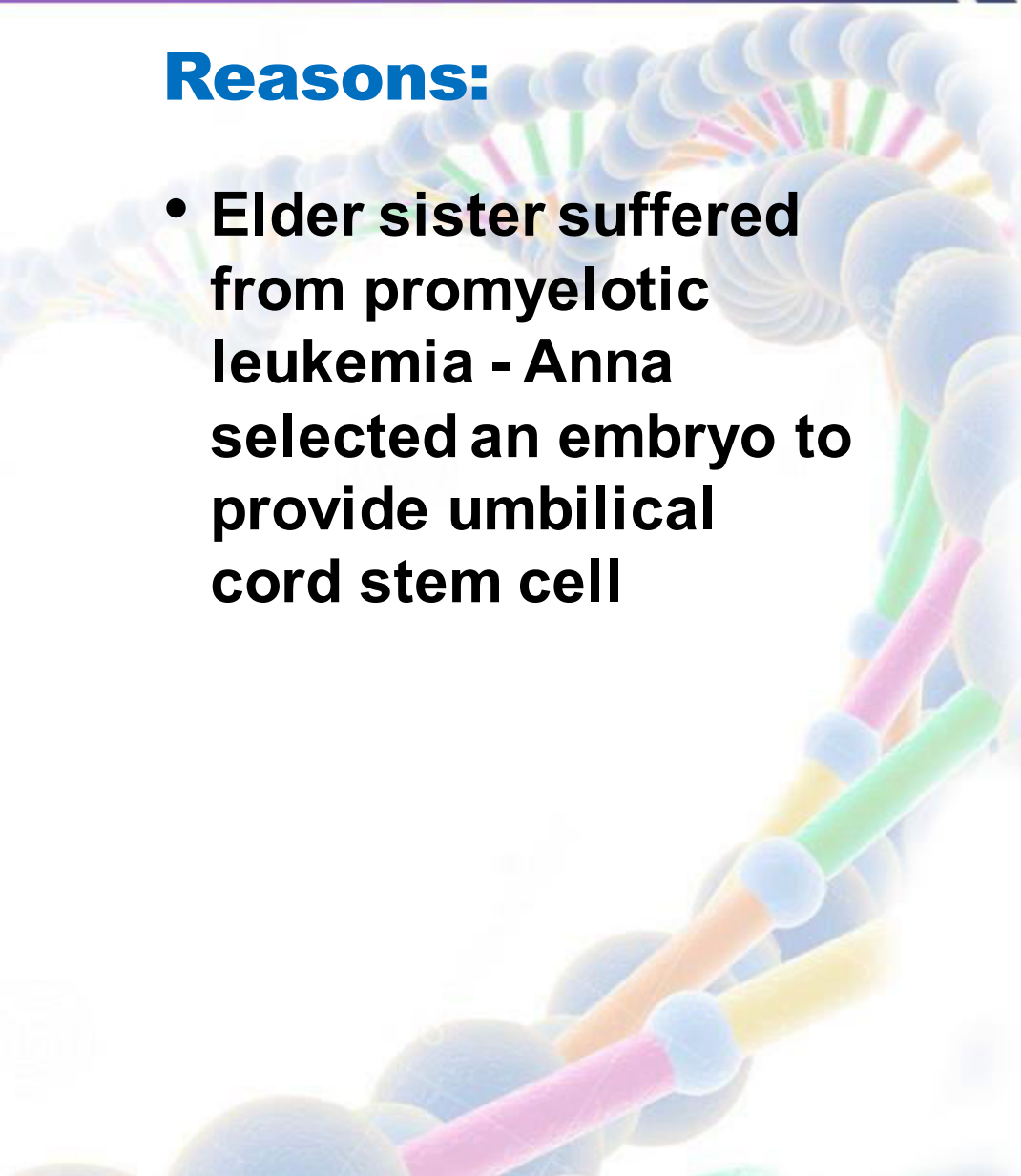
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Case study 3

Reasons:

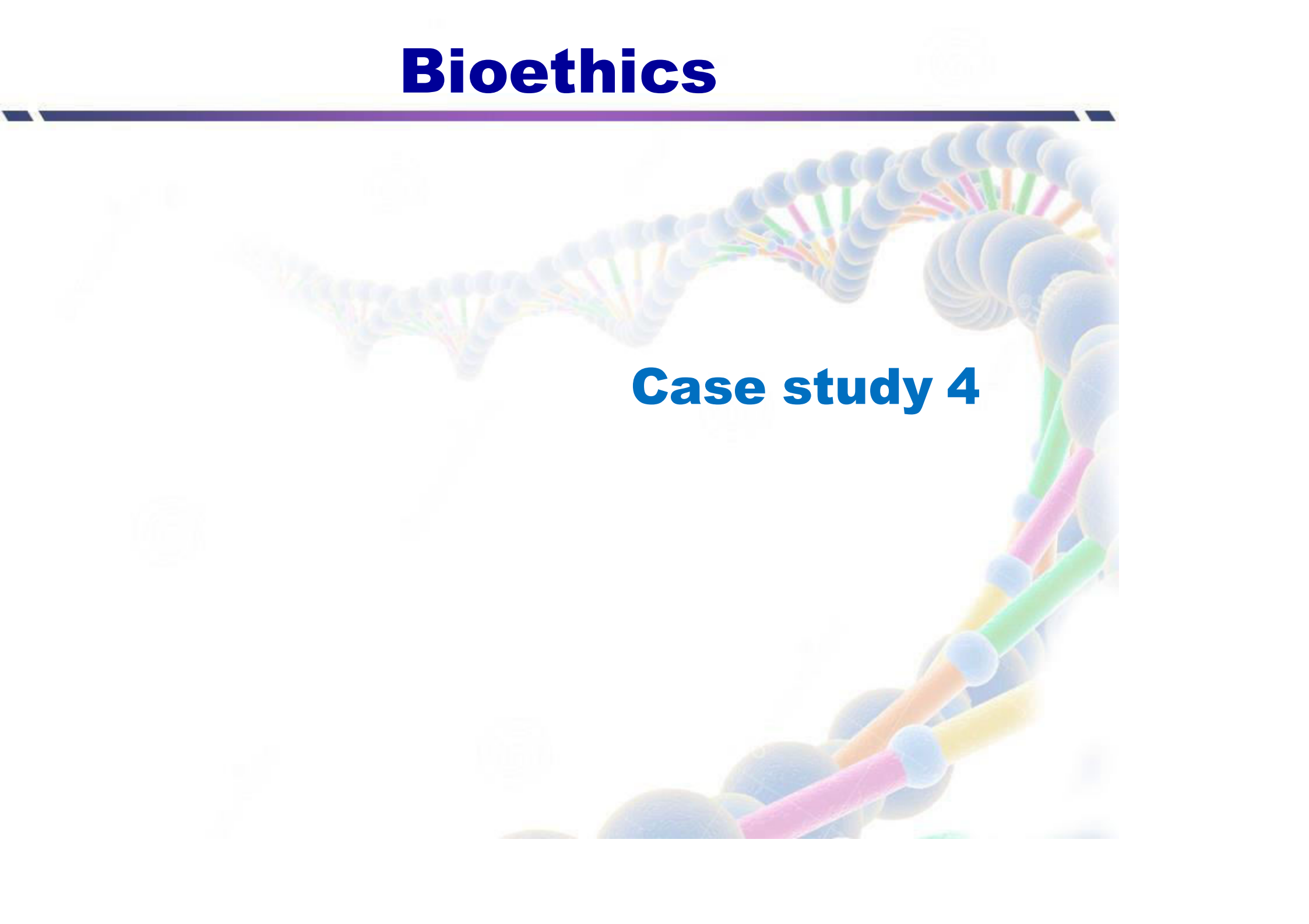
- Nash family elder daughter- Fanconi's anemia
- In 2000, Adam was born a suitable match for her sister
- “If you use one of your children to save the life of another, are you being a good mother or a very bad one”

Bioethics

Lesson # 58



Bioethics

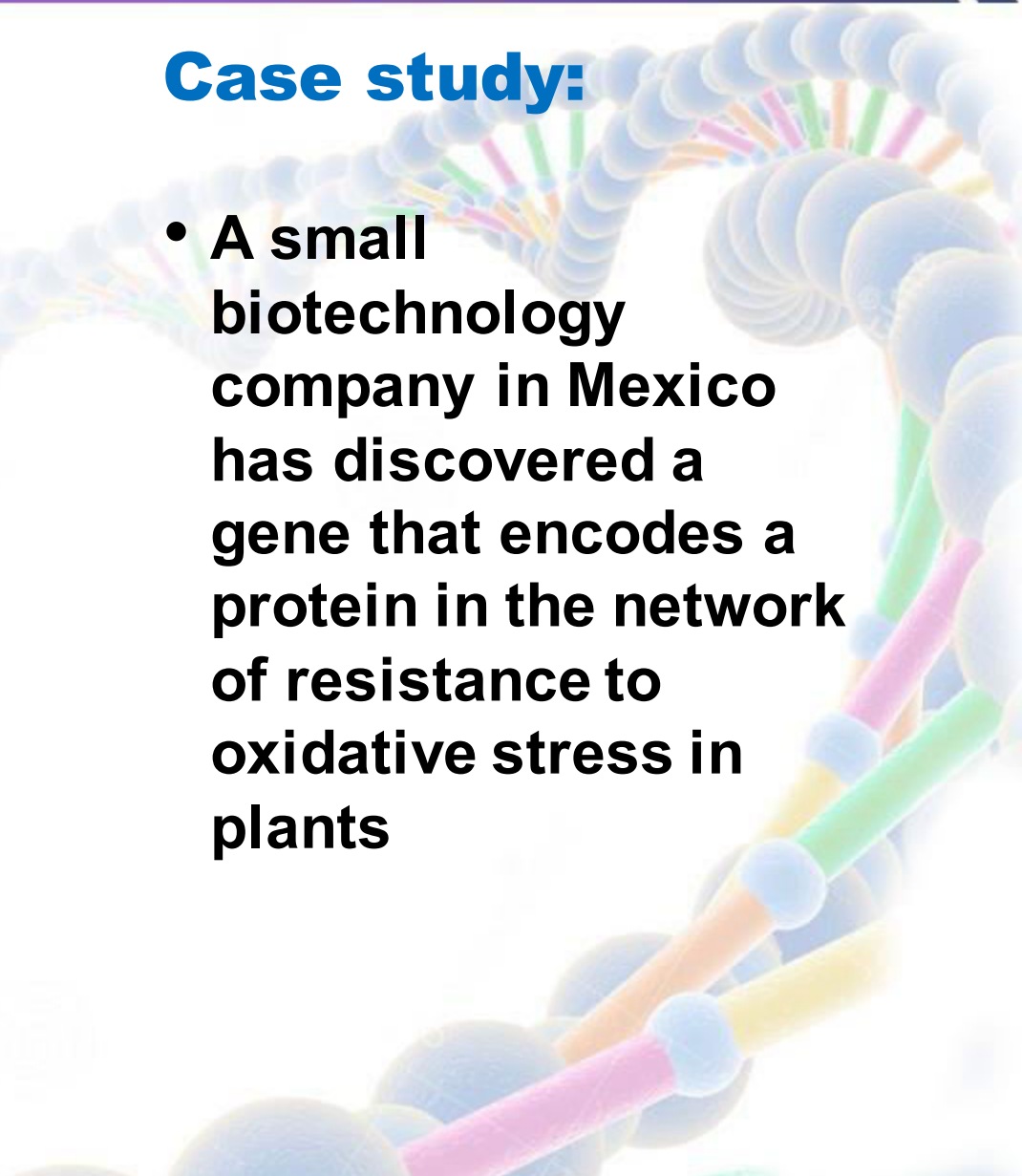


Case study 4

Case study 4

Case study:

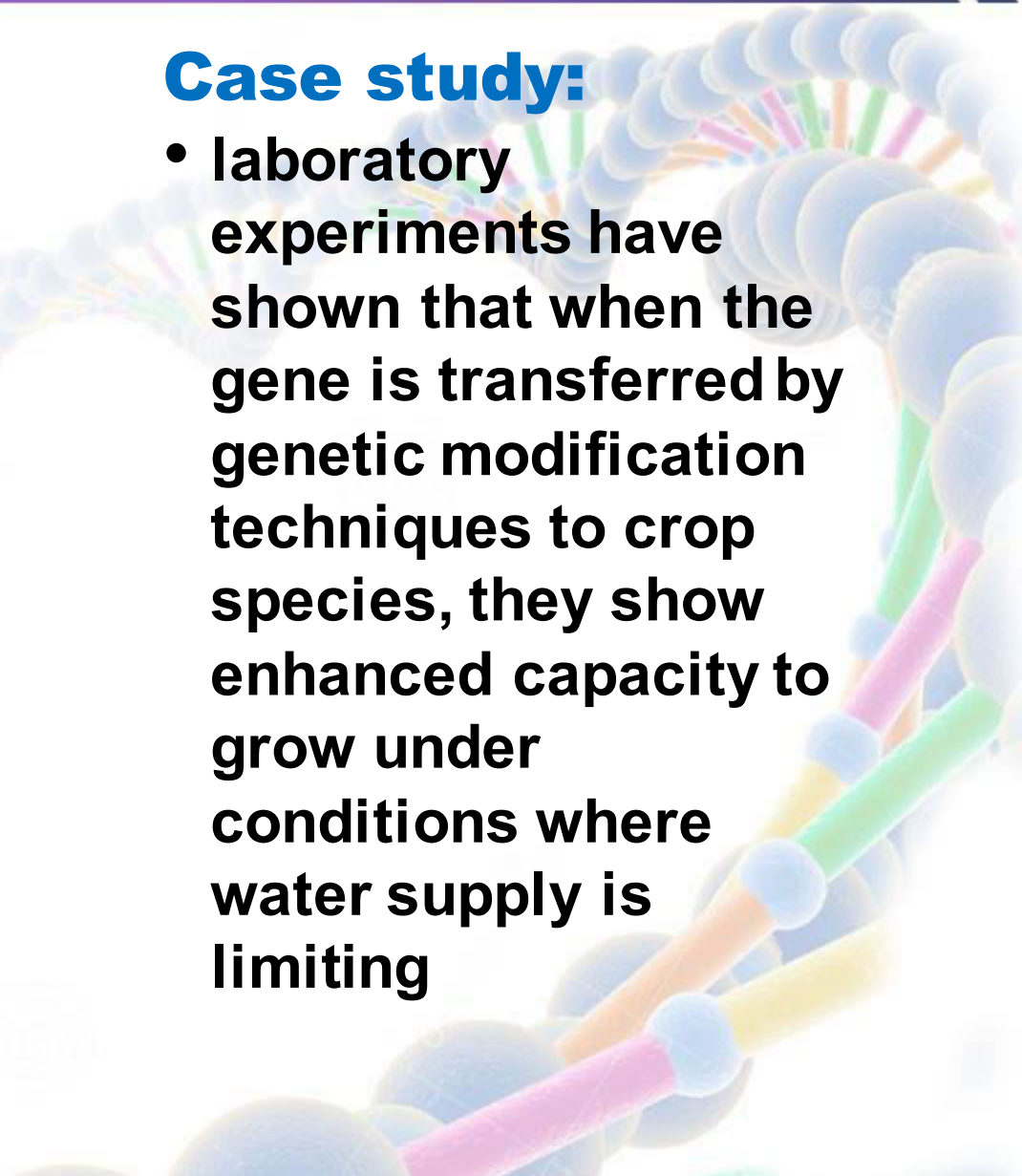
- A small biotechnology company in Mexico has discovered a gene that encodes a protein in the network of resistance to oxidative stress in plants



Case study 4

Case study:

- **laboratory experiments have shown that when the gene is transferred by genetic modification techniques to crop species, they show enhanced capacity to grow under conditions where water supply is limiting**



Case study 4

Case study:

- The company has not published its data because it is filing a patent on the gene
- If the patent is granted, the company plans to license it out to a major trans-national agri-chemical company

Case study 4

Reasons:

- should the patent be granted?
- yes, the term oxidative stress is used-comprising all kind of biotic and abiotic stress conditions
- helpful in reducing the damaging of crops caused by stress conditions

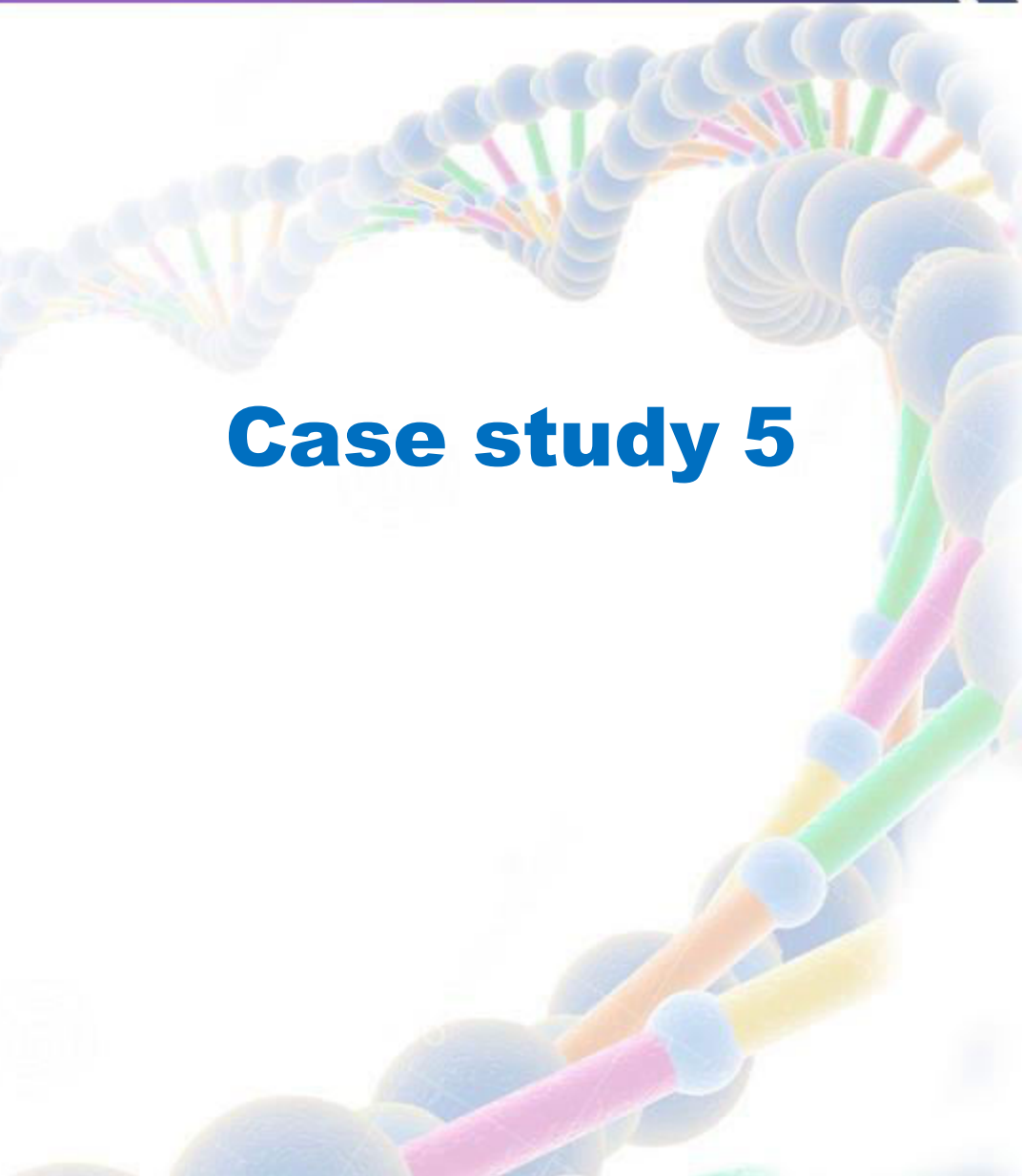
Bioethics

Lesson # 59



Bioethics

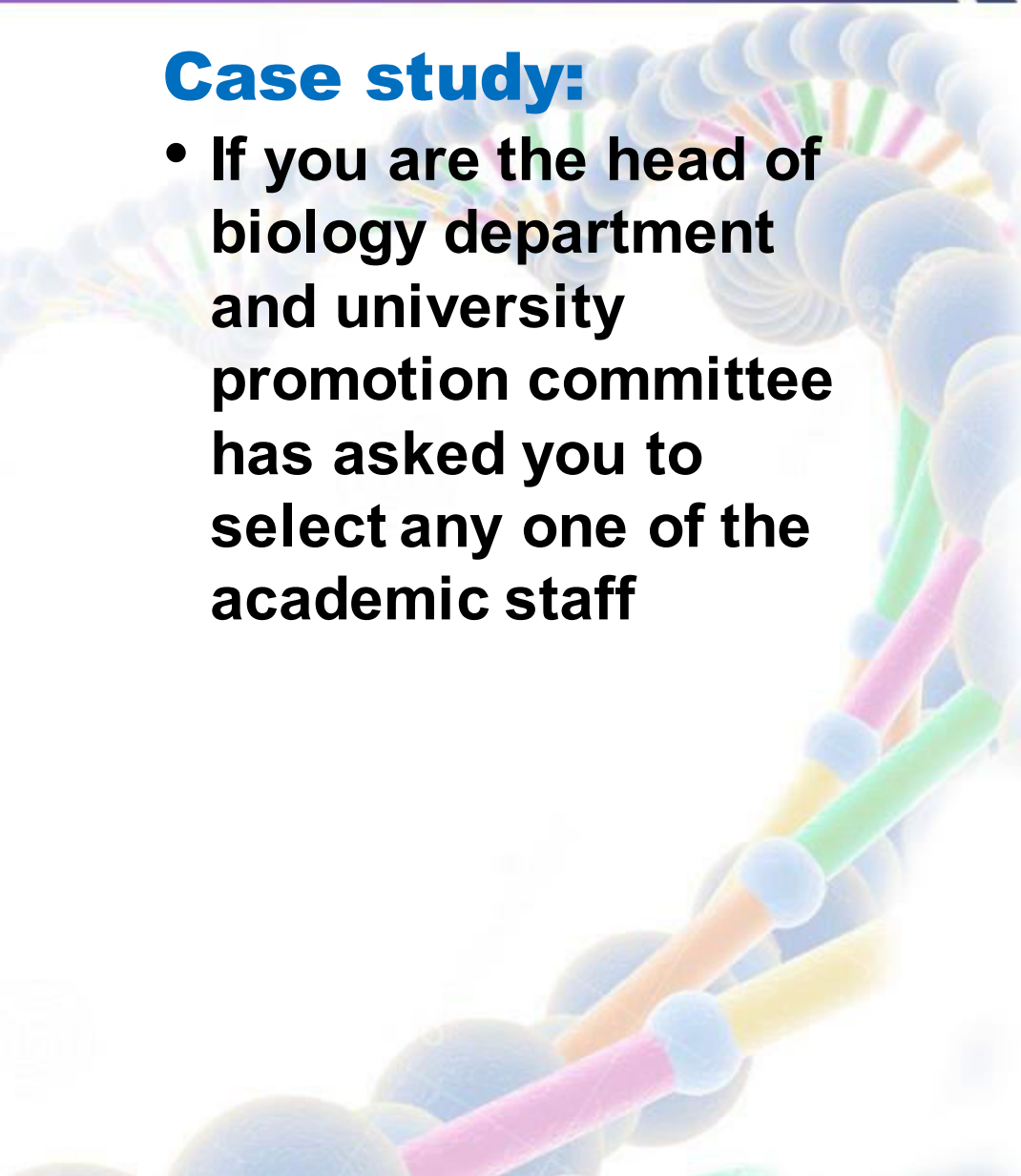
Case study 5



Case study 5

Case study:

- If you are the head of biology department and university promotion committee has asked you to select any one of the academic staff



Case study 5

Candidate A:

- **Candidate A is 37, working on the ecology of plant-insect relations. His research on the evolution of pollination mechanisms is widely respected. The research has steady flow of grant**

Case study 5

Candidate B:

- candidate B is 34, working on the regulation of gene expression in programmed cell death, especially in relation to cancer. This work is of great interest in the biomedical community. The work is supported by extensive funds

Case study 5

Reasons:

- What should be the criteria of selecting according to research?
- look at the particular research goals
- candidate aspirations and world-view
- beneficial for the public

Bioethics

Lesson # 60



Bioethics



Case study 6

Case study 6

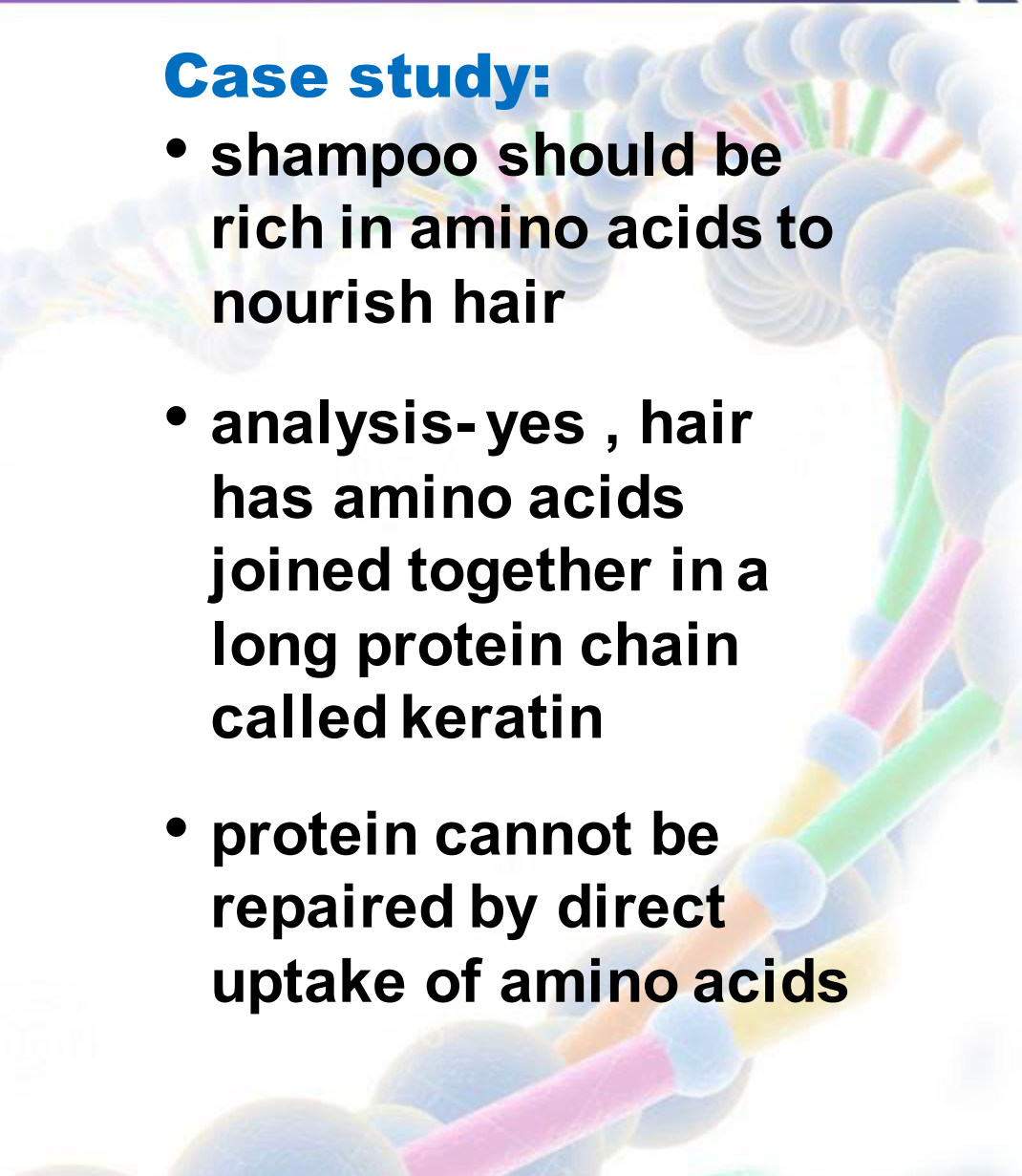
Case study:

- **employing science to sell a product, the modernist and post-modernist version**
- **a female actor told viewers about shampoo on UK TV in 2004**
- **she told hair is 96% amino acids**

Case study 6

Case study:

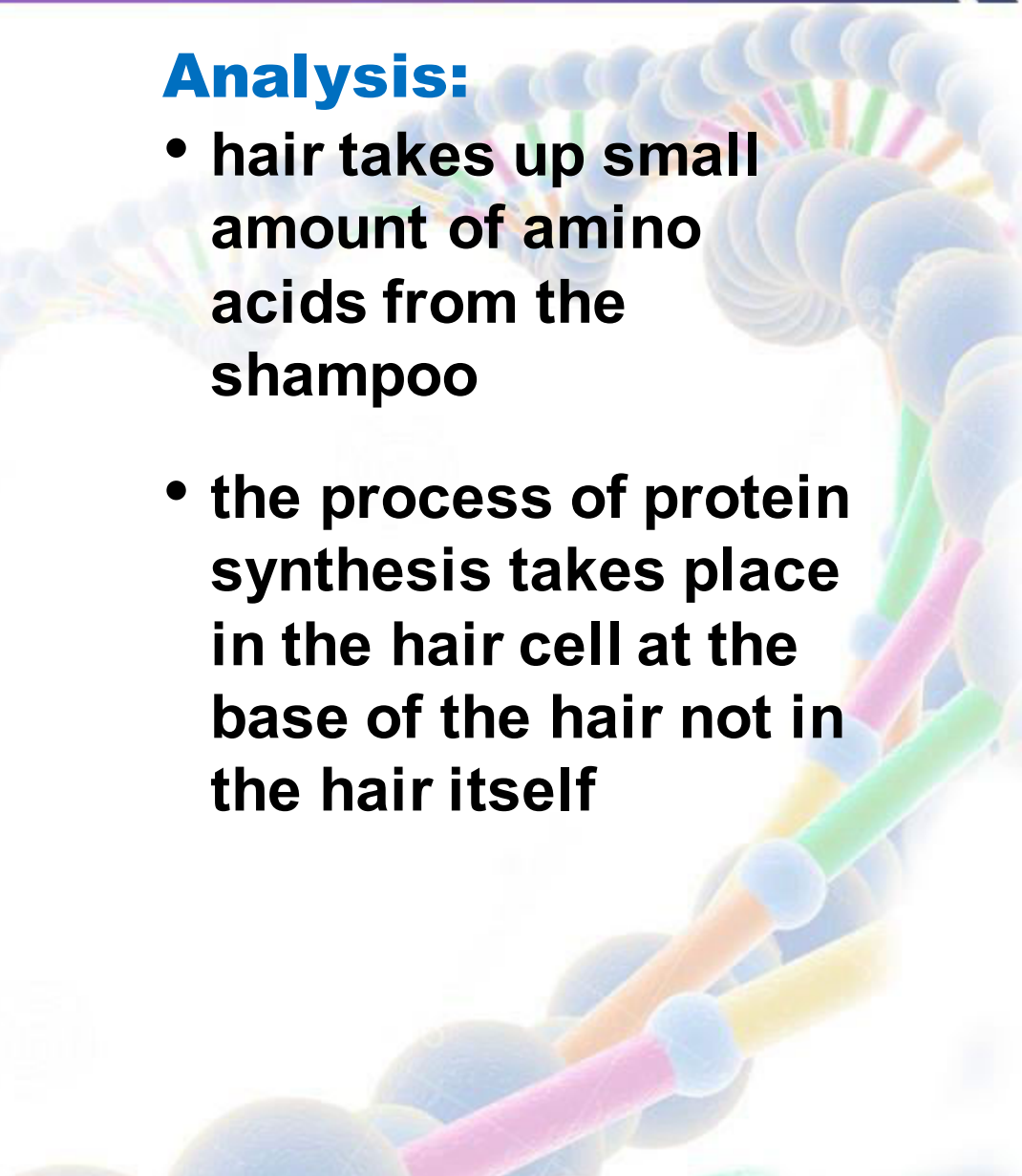
- shampoo should be rich in amino acids to nourish hair
- analysis-yes , hair has amino acids joined together in a long protein chain called keratin
- protein cannot be repaired by direct uptake of amino acids



Case study 6

Analysis:

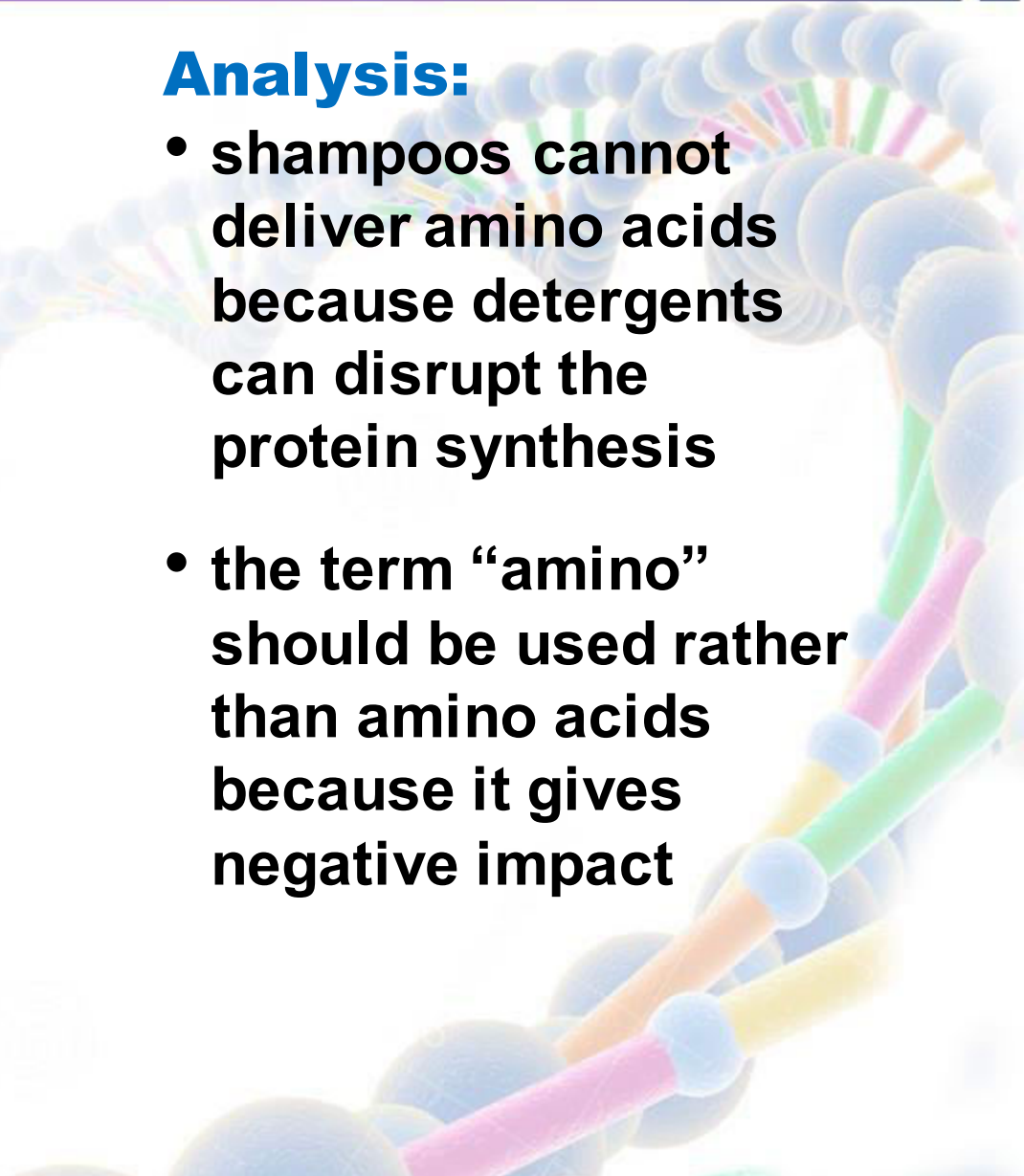
- hair takes up small amount of amino acids from the shampoo
- the process of protein synthesis takes place in the hair cell at the base of the hair not in the hair itself



Case study 6

Analysis:

- **shampoos cannot deliver amino acids because detergents can disrupt the protein synthesis**
- **the term “amino” should be used rather than amino acids because it gives negative impact**



Case study 6

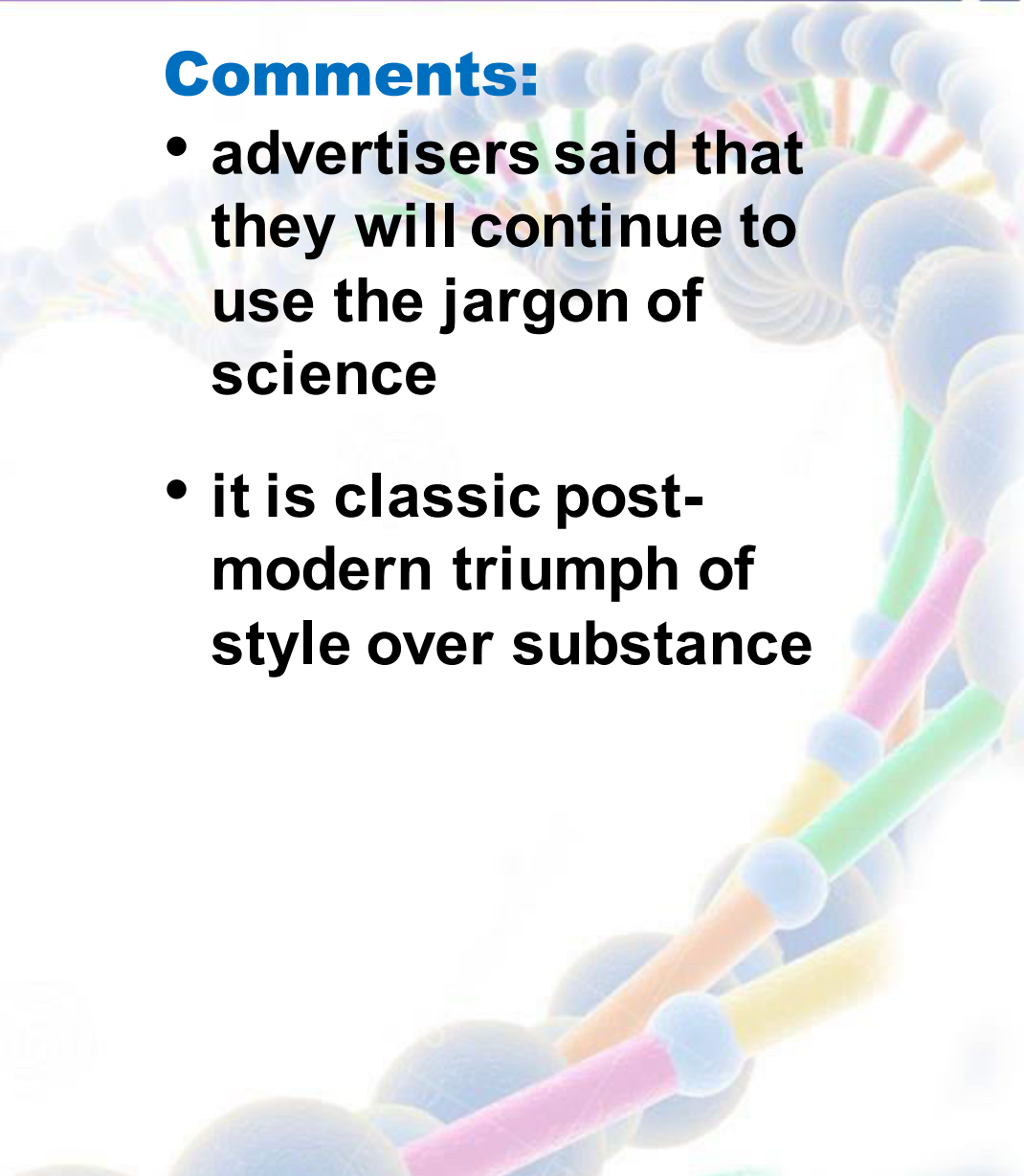
Comment:

- **this is not a comment on the shampoo efficacy; we are sure that modern shampoos clean the hair and scalp and leave the hair shiny and manageable**
- **comment on the dishonest use of scientific terminology to imply things that cannot happen**

Case study 6

Comments:

- **advertisers said that they will continue to use the jargon of science**
- **it is classic post-modern triumph of style over substance**



Bioethics



Lesson # 61

Bioethics



Case study 7

Case study 7

Case study:

- because of the family history I know I am likely to be an unaffected carrier of a gene that causes a serious and so far untreatable condition
- do I request a test for that gene? If the test is positive should I tell my spouse?

Case study 7

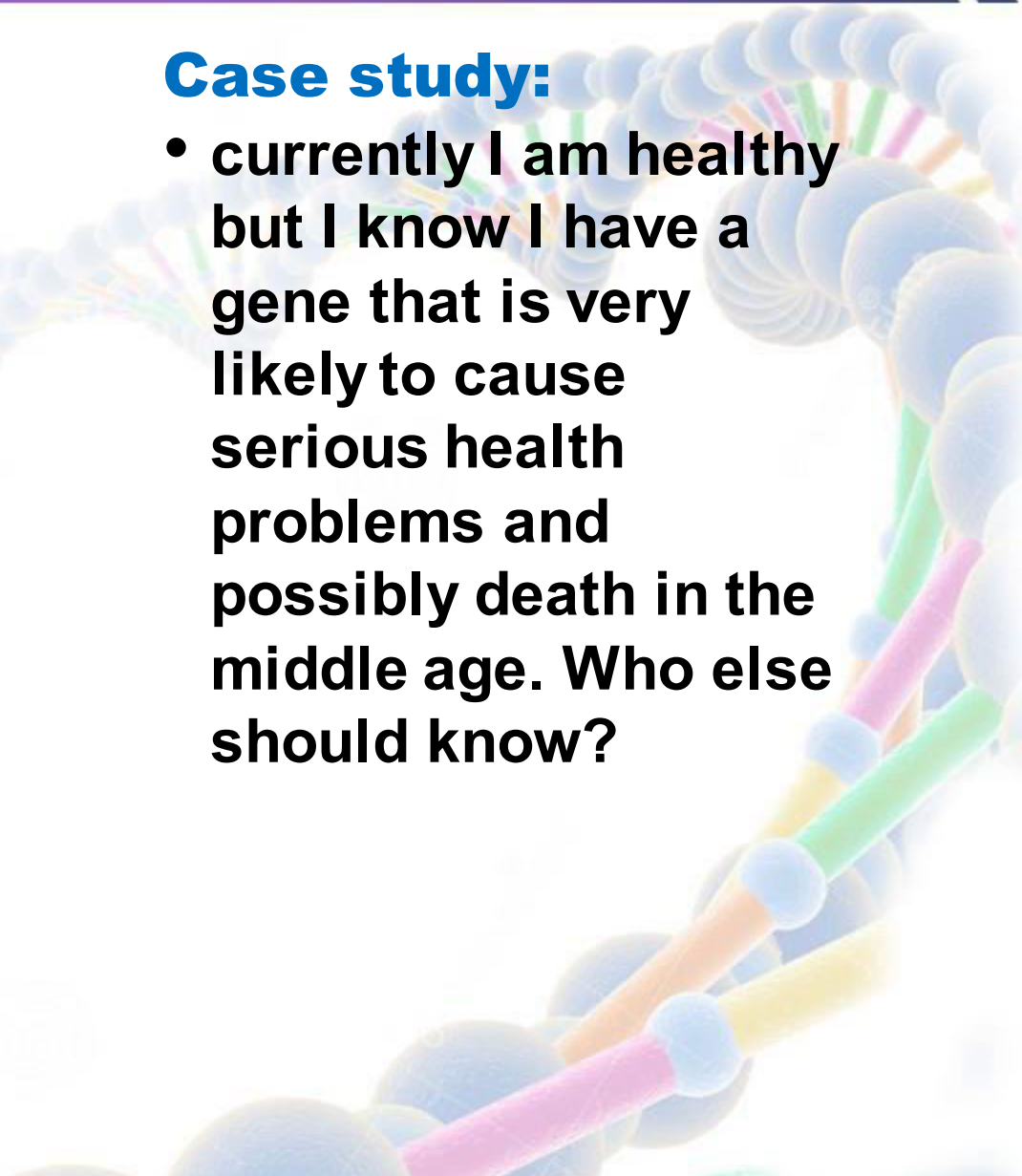
Case study:

- **family history informs me that I have 50-50 chance of possessing a gene that at the age of 40 cause serious neuro-degenerative disease for which there is no treatment**
- **do I want the test? if the test is positive should I tell my spouse or children**

Case study 7

Case study:

- currently I am healthy but I know I have a gene that is very likely to cause serious health problems and possibly death in the middle age. Who else should know?



Case study 7

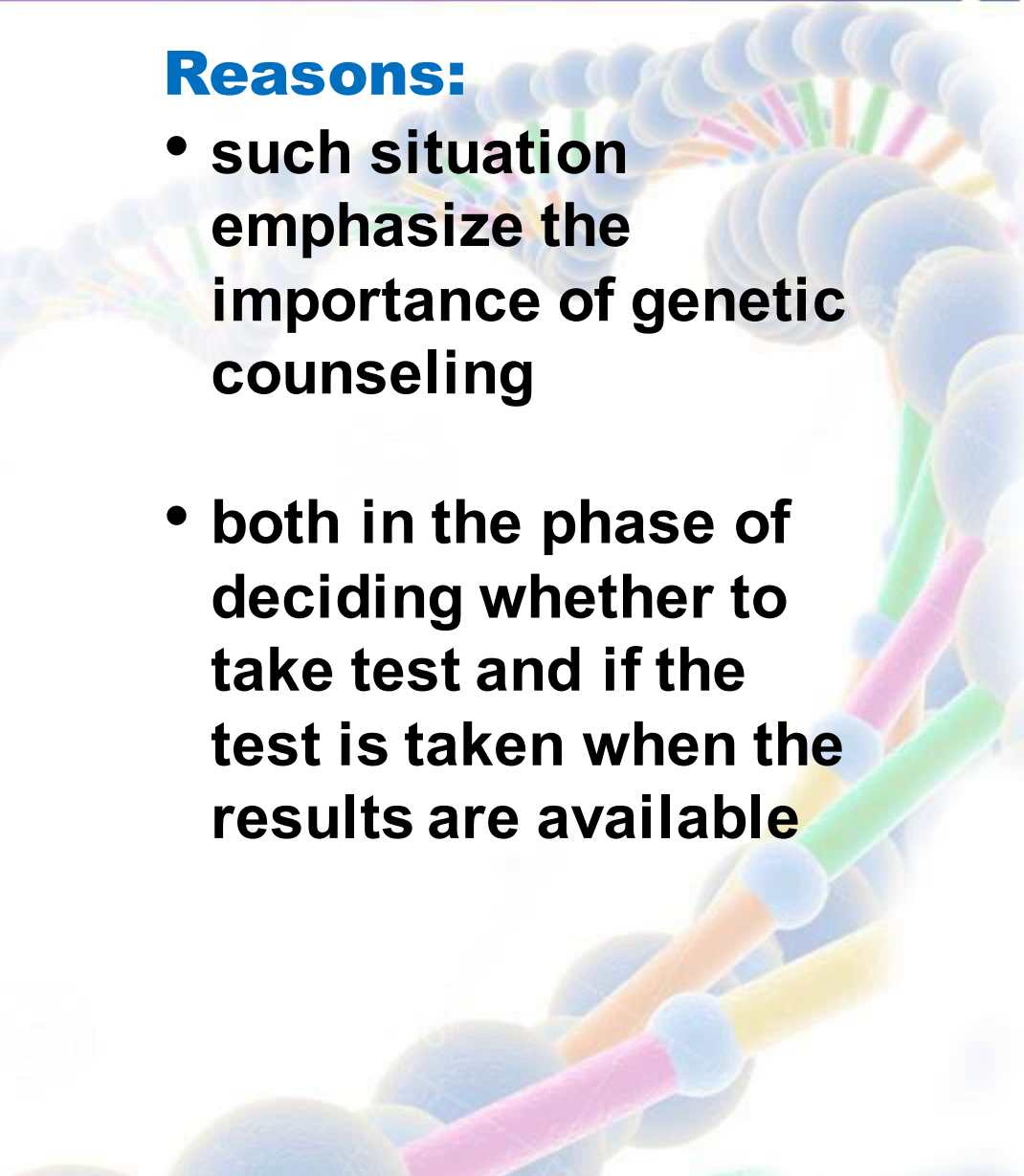
Reasons:

- sometimes the knowledge that one is certain to suffer a serious and distressing condition is a burden too heavy to bear
- thus ignorance is a bliss
- Social stigma

Case study 7

Reasons:

- such situation emphasize the importance of genetic counseling
- both in the phase of deciding whether to take test and if the test is taken when the results are available



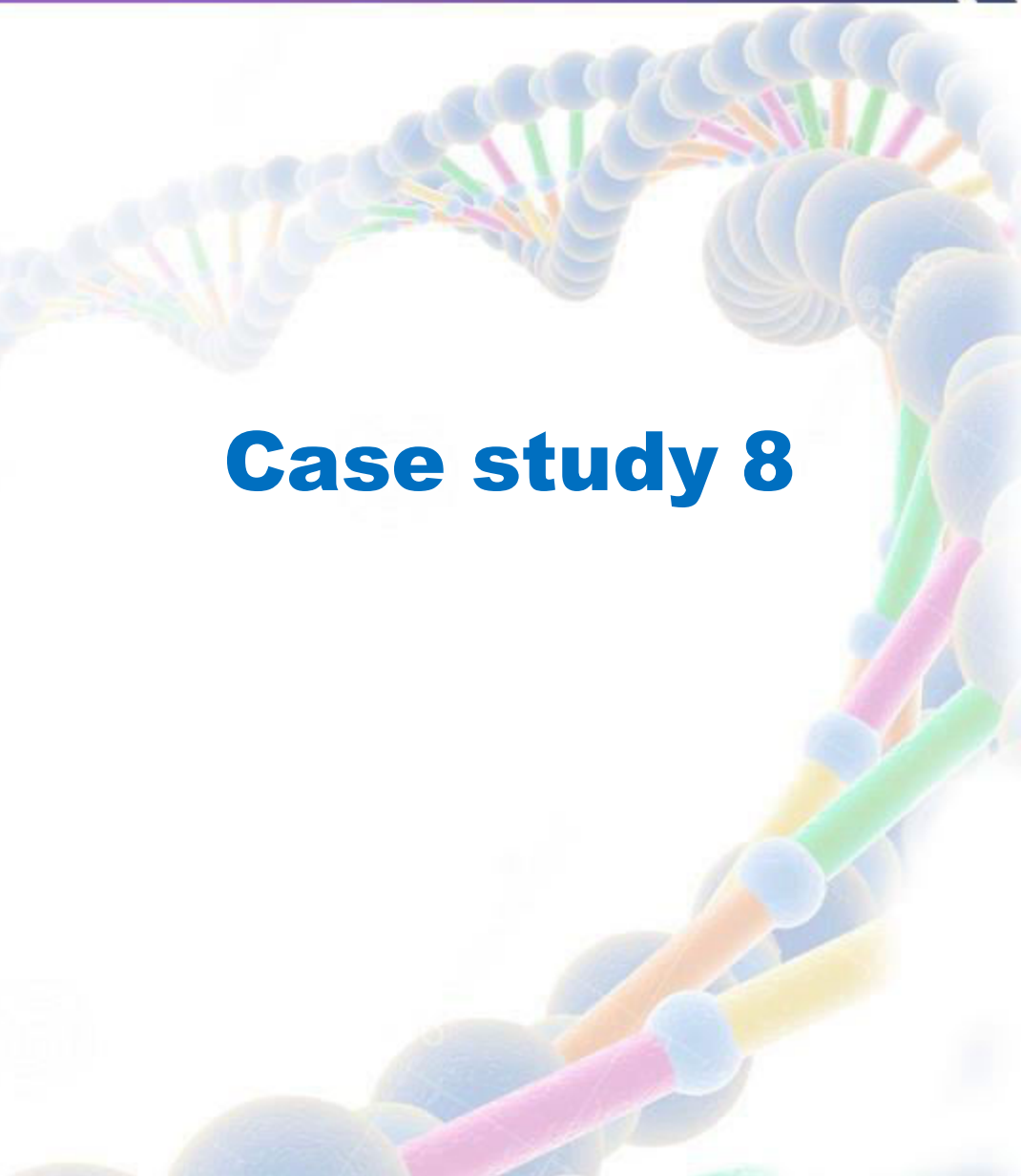
Bioethics

Lesson # 62



Bioethics

Case study 8



Case study 8

Case study:

- a man presents with symptoms representing cancer and as part of his treatment spleen should be removed.
- the pathology department use it to establish a cultured cell line in order to study the rare cancer

Case study 8

Case study:

- the cell line performs so well that the scientists collaborate with biotechnology company to patent it.
- they start to earn royalties from other laboratories and organizations that wish to use the cell line

Case study 8

Case study:

- when patient find all this he was amazed
- nobody has taken the consent from him nor he has been informed by anyone about these developments
- analyze the ethical issues

Case study 8

Reasons:

- in terms of medical ethics, removal of the spleen was an act of doing good-beneficence
- patient's personal autonomy had been respected
- it was for the sake of his health that spleen has been removed

Case study 8

Reasons:

- in UK, there is great sensitivity concerning the fate of organ removed
- under new legislation, if there is no pre-death consent then kin permission must be taken to retain any organs from dead bodies

Case study 8

Reasons:

- in USA, once the organ is removed during surgery, it is no longer belong to the patient
- what is the purpose of keeping their appendix or diseased kidney in a jar in their office?

Case study 8

Reasons:

- donor can not claim on the income gained as a result of research
- anyone who donates a kidney makes a gift not an investment in the recipient

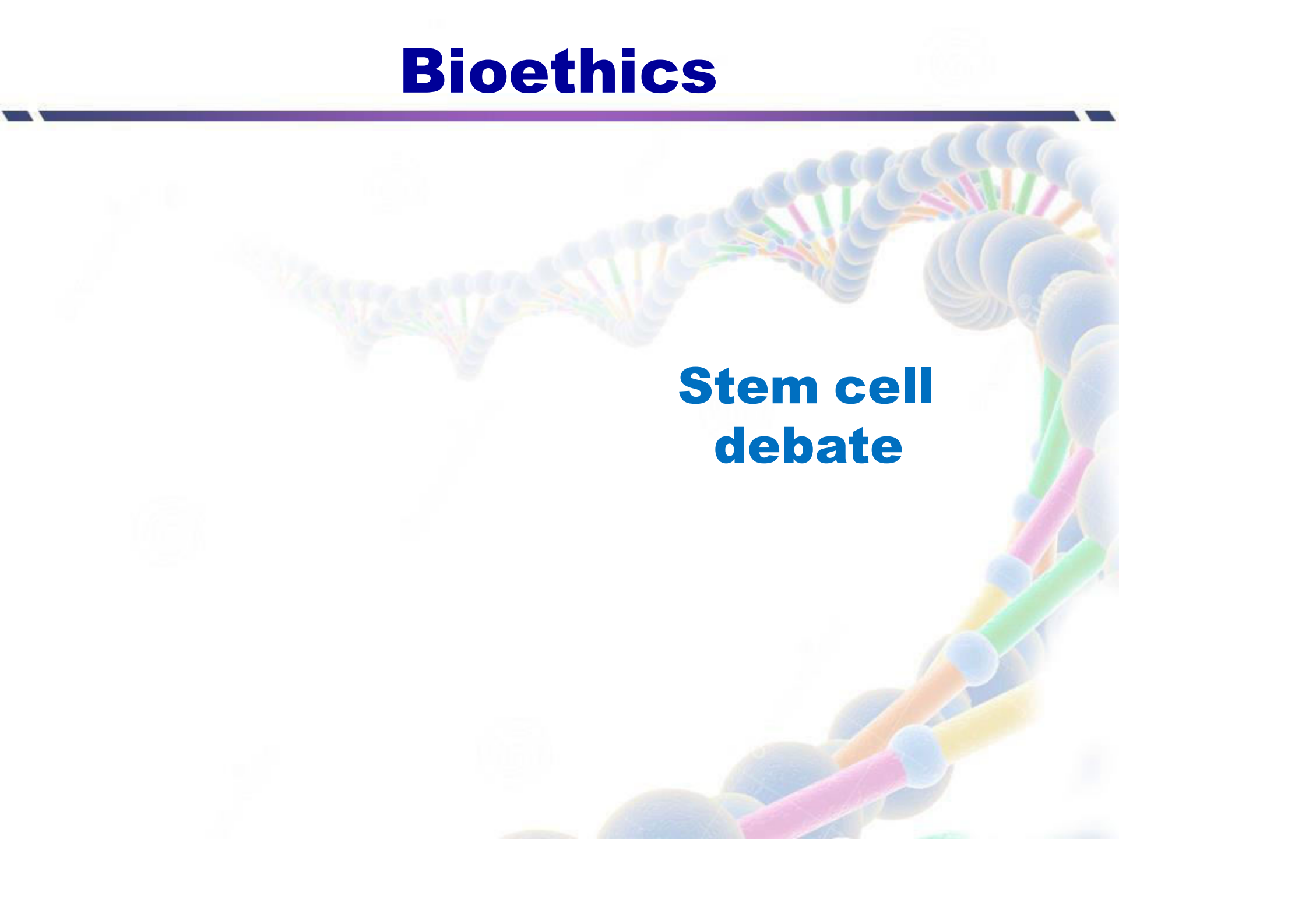
Bioethics

Lesson # 63



Bioethics

**Stem cell
debate**

A 3D illustration of a DNA double helix structure, rendered in blue, green, yellow, and pink, set against a white background. The helix is shown in a perspective view, curving from the top left towards the bottom right. The sugar-phosphate backbones are represented by blue spheres, and the nitrogenous base pairs are shown as colorful sticks connecting the two strands.

Stem cell debate

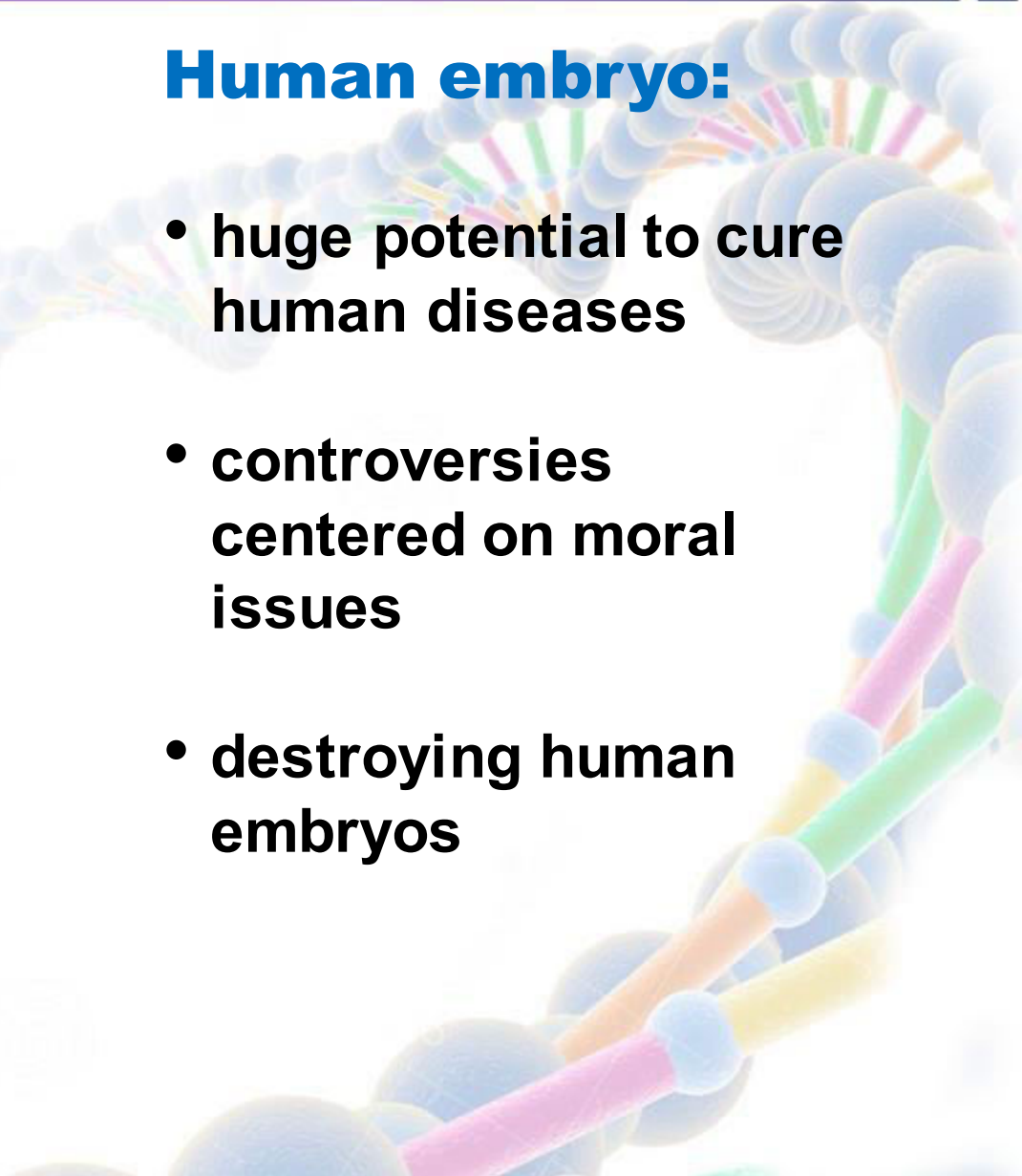
Stem cell:

- **stem cell therapies are not new**
- **bone marrow stem cell transplants**
- **removal of stem cells from human embryo**
- **excitement and controversies started**

Stem cell debate

Human embryo:

- huge potential to cure human diseases
- controversies centered on moral issues
- destroying human embryos



Stem cell debate

Difficult questions:

- does life begin at fertilization, in the womb or at birth?
- is a human embryo equivalent to a human child
- does human embryo have any rights?
- is the embryo really destroyed?

Stem cell debate

Legislations:

- to regulate stem cell research
- prohibit the creation of embryos-research
- should taxpayer money used-believe it to be unethical

Stem cell debate

Legislations:

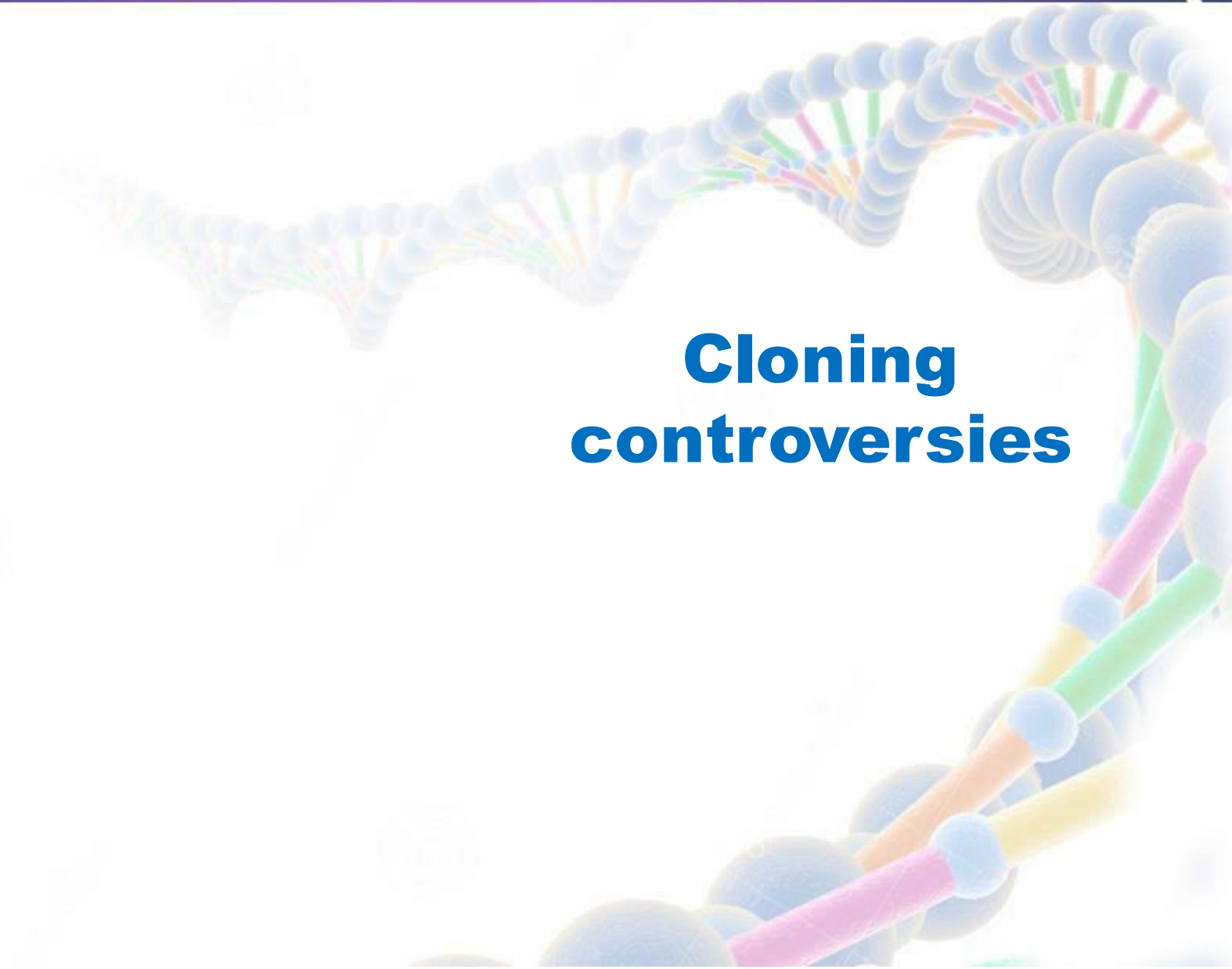
- Bush reduces the funds
- Obama expand the funds
- policy makers with new questions

Bioethics

Lesson # 64



Bioethics

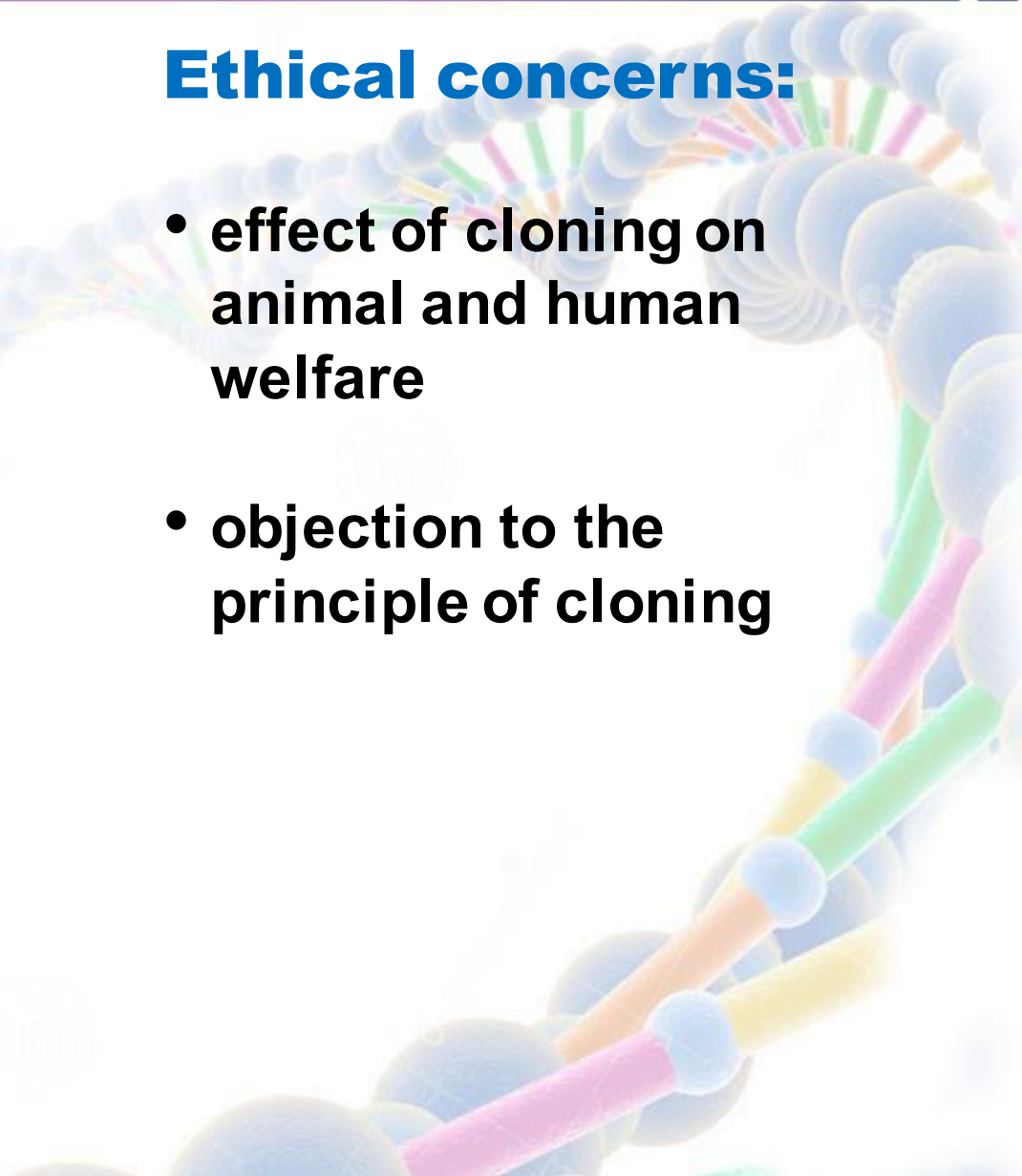


Cloning controversies

Cloning controversies

Ethical concerns:

- effect of cloning on animal and human welfare
- objection to the principle of cloning



Cloning controversies

Effect on animals:

- increase animal suffering-standard breeding methods
- surgeries performed to obtain oocytes
- animals produced as diseased models

Cloning controversies

Arguments:

- these findings are not unique to cloning
- associated with other procedures
- embryo transfer, oocyte transfer, in vitro fertilization

Cloning controversies

Human welfare:

- cloned animal species
--housed/slaughtered
and eaten
- potential benefits---
understanding life
processes and animal
diseases
- human health
- food production

Cloning controversies

objections:

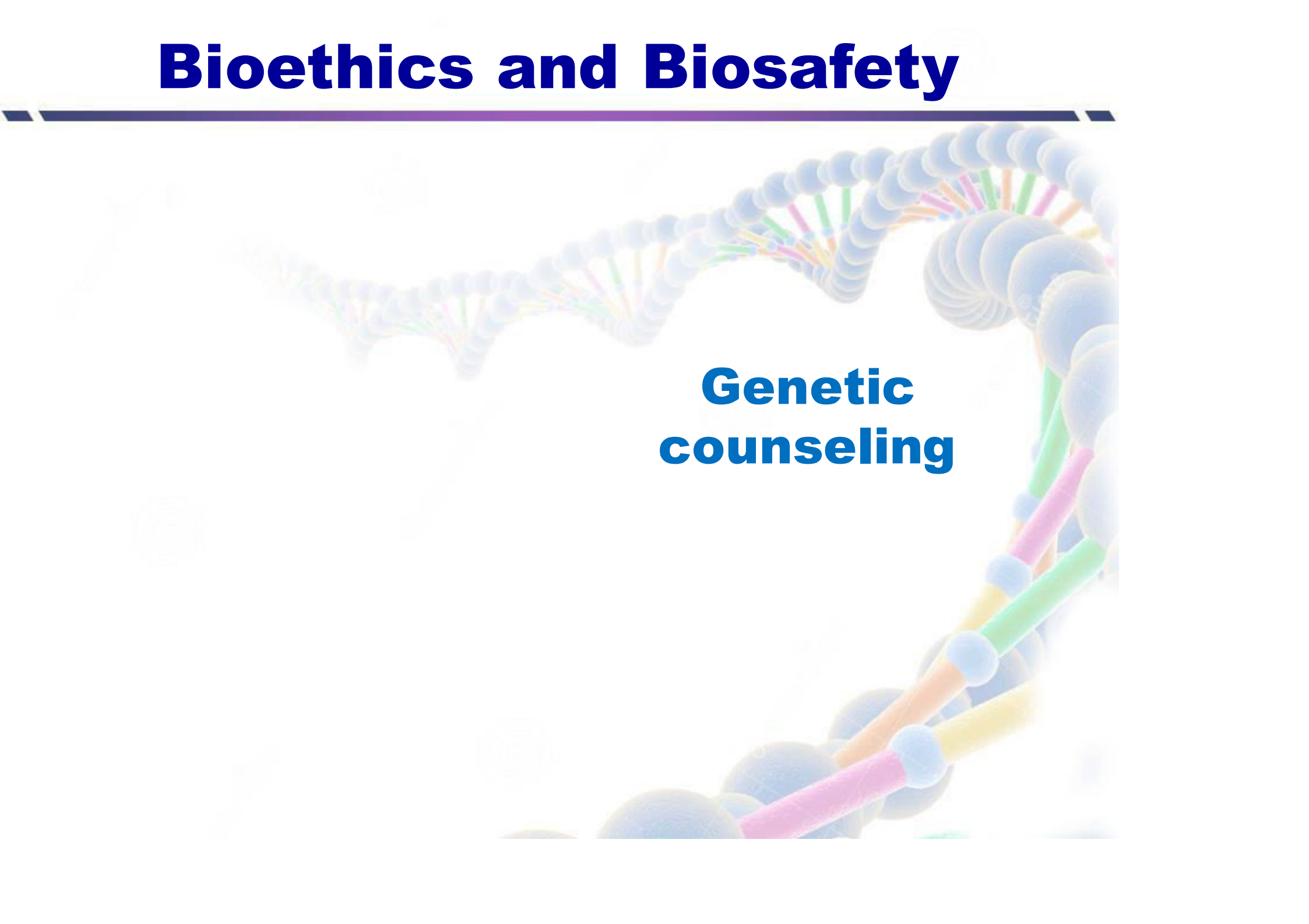
- **genetic variations of the species**
- **cloned animals poses no public health risk**
- **transgenic animals-unnatural means**
- **potential hazards to animals, humans and on the environment**

Bioethics and Biosafety

Lesson # 65



Bioethics and Biosafety



**Genetic
counseling**

Genetic counseling

Definition:

- patient-inherited disorder
- advised the consequences and nature of the disorder
- probability of developing and transmitting it
- options-management and family planning

Genetic counseling

Genetic counselors

- understanding and adaptation to the medical
- psychological
- familial implications-
genetic contributions
to disease



Genetic counseling

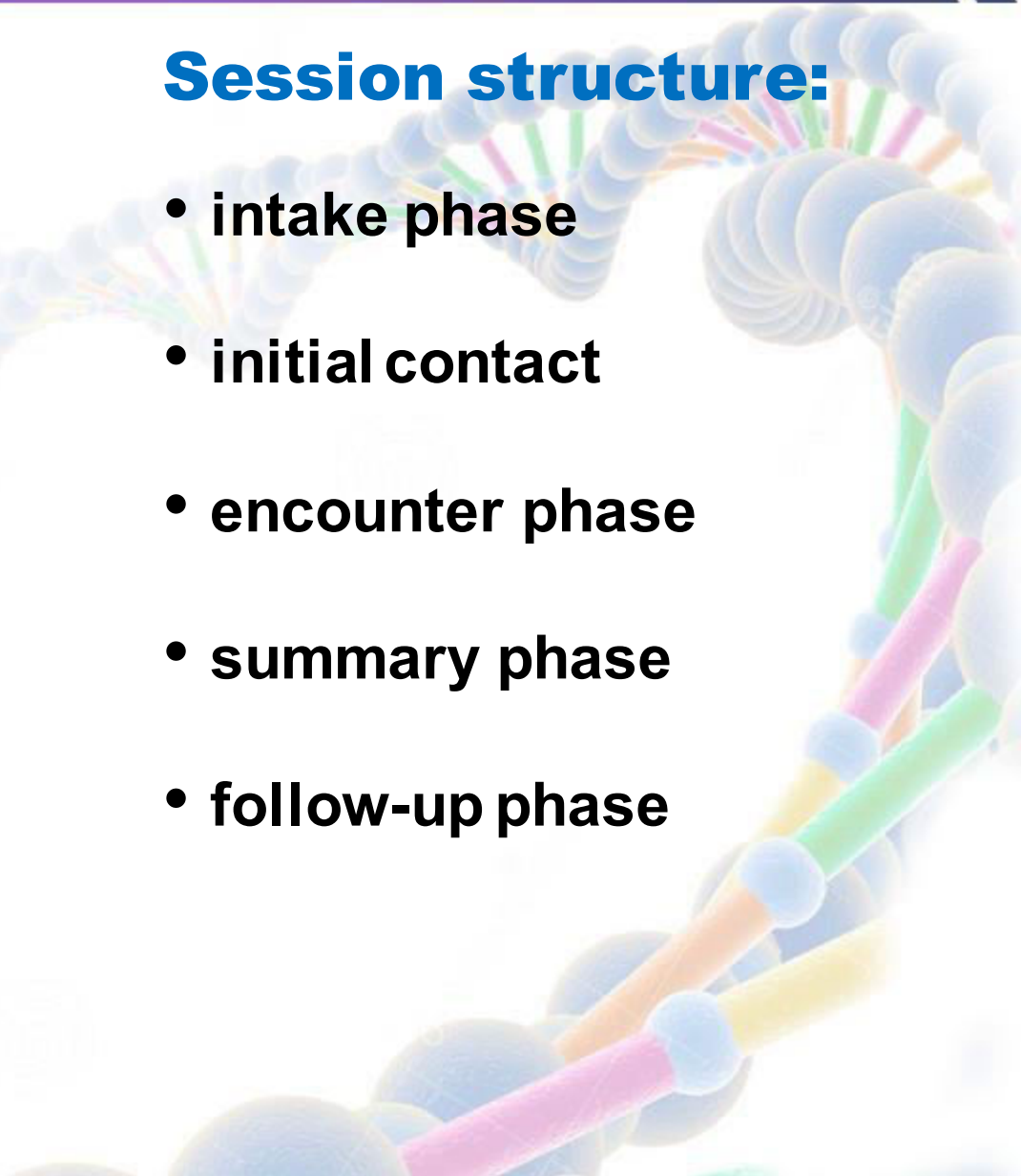
Role:

- **interpretation of family and medical histories**
 - **education - about inheritance, testing, management, prevention, resources**
 - **counseling to promote informed choices and adaptation to the risk**
- 

Genetic counseling

Session structure:

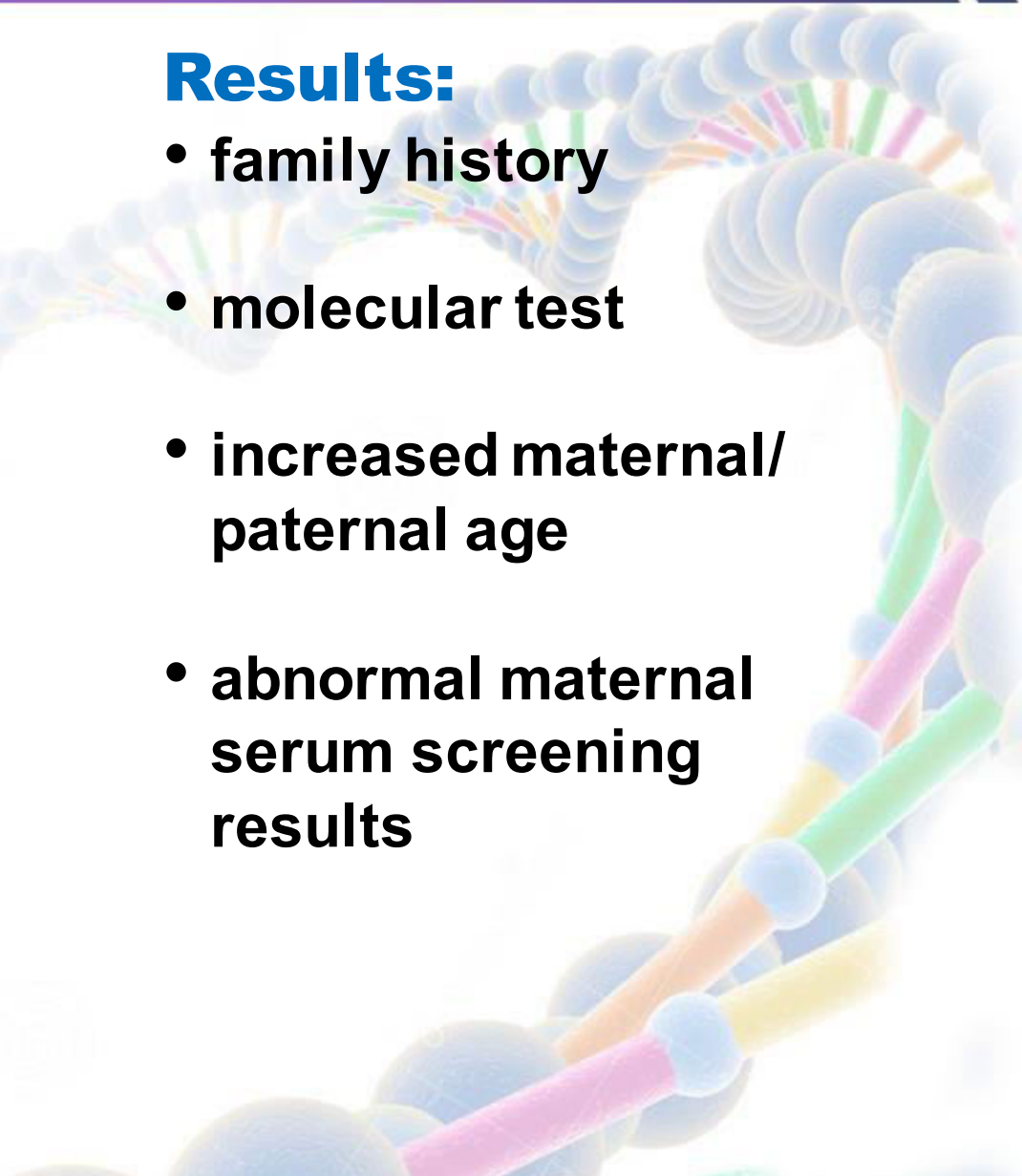
- intake phase
- initial contact
- encounter phase
- summary phase
- follow-up phase



Genetic counseling

Results:

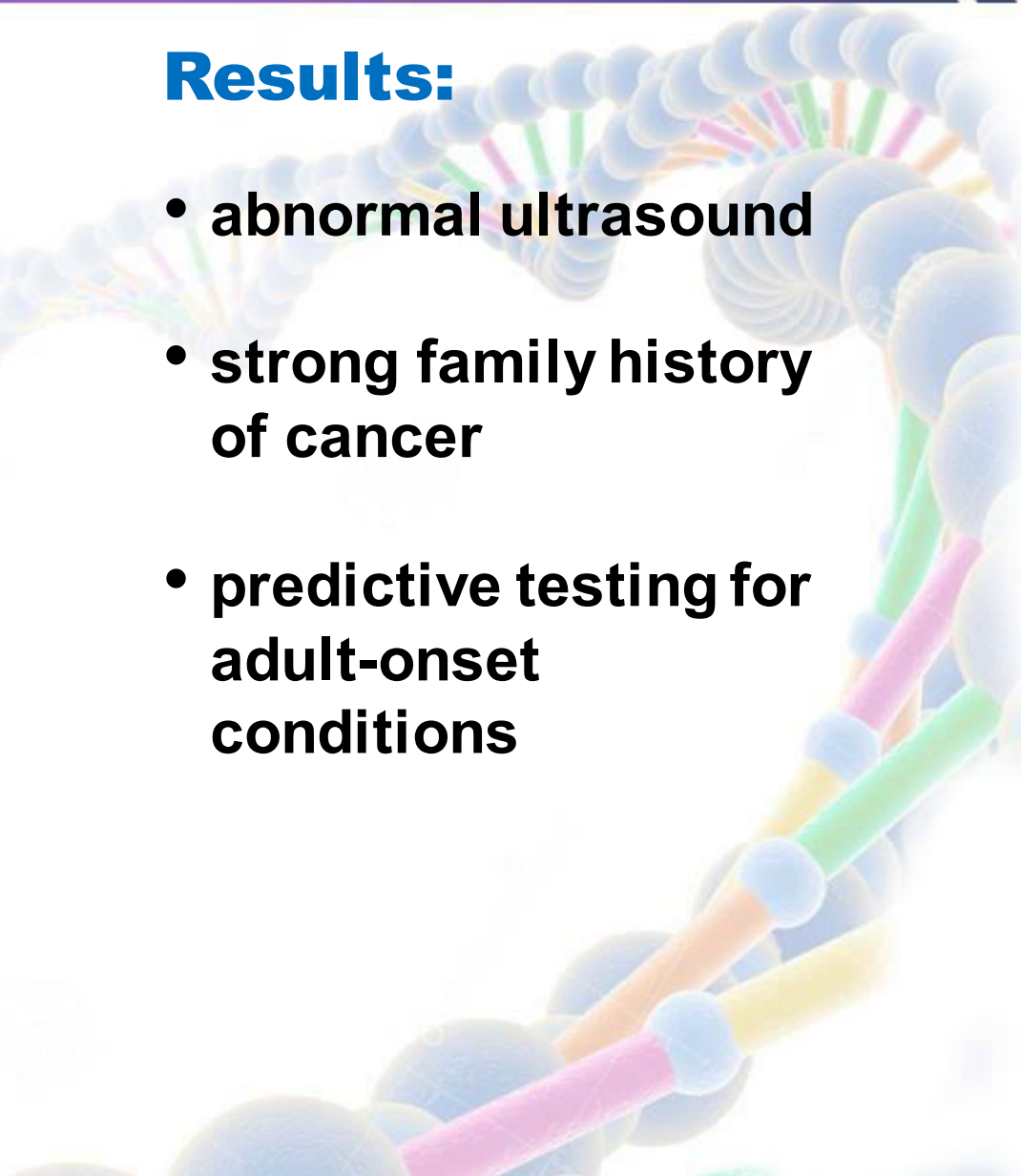
- family history
- molecular test
- increased maternal/
paternal age
- abnormal maternal
serum screening
results



Genetic counseling

Results:

- abnormal ultrasound
- strong family history of cancer
- predictive testing for adult-onset conditions



Bioethics and Biosafety

Lesson # 66



Bioethics and Biosafety

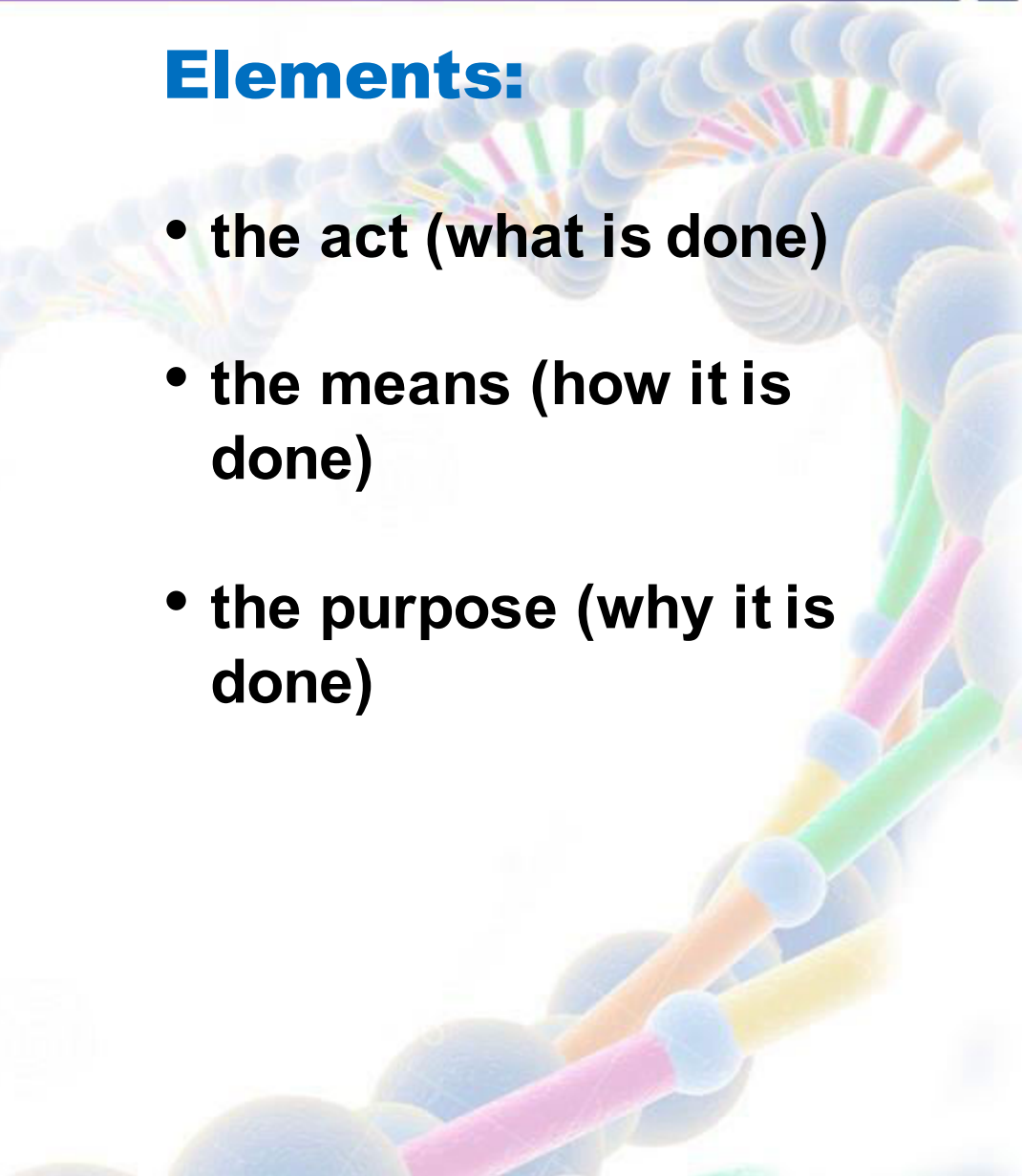


**Responding to
trafficking**

Responding to trafficking

Elements:

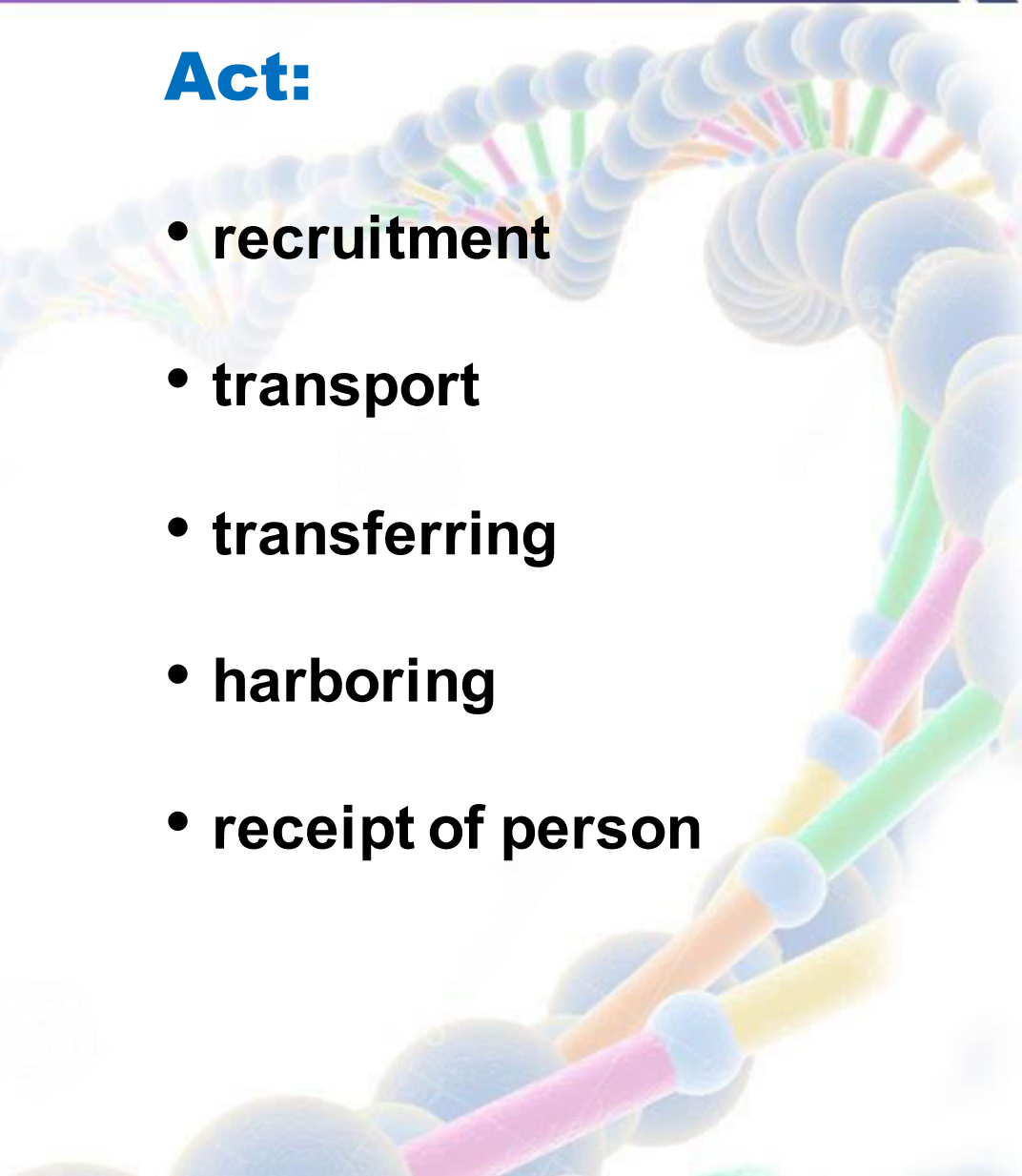
- the act (what is done)
- the means (how it is done)
- the purpose (why it is done)



Responding to trafficking

Act:

- **recruitment**
- **transport**
- **transferring**
- **harboring**
- **receipt of person**



Responding to trafficking

Means:

- threat
- use of force
- abduction
- fraud
- abuse of power
- payments/benefits

Responding to trafficking

Exploitation:

- prostitution
- sexual exploitation
- forced labor
- slavery
- removal of organs

Responding to trafficking

Response:

- within the country or across borders
- range of exploitative purposes
- victimizes children, men, women
- involve organized victim groups

Responding to trafficking

Prevention:

- trafficking in persons
- victims of human trafficking
- trafficking offenders

Responding to trafficking

Education:

- research and awareness raising
- promotion of protocols and capacity building
- strengthening of partnerships and coordination

Bioethics and Biosafety

Lesson # 67



Bioethics and Biosafety



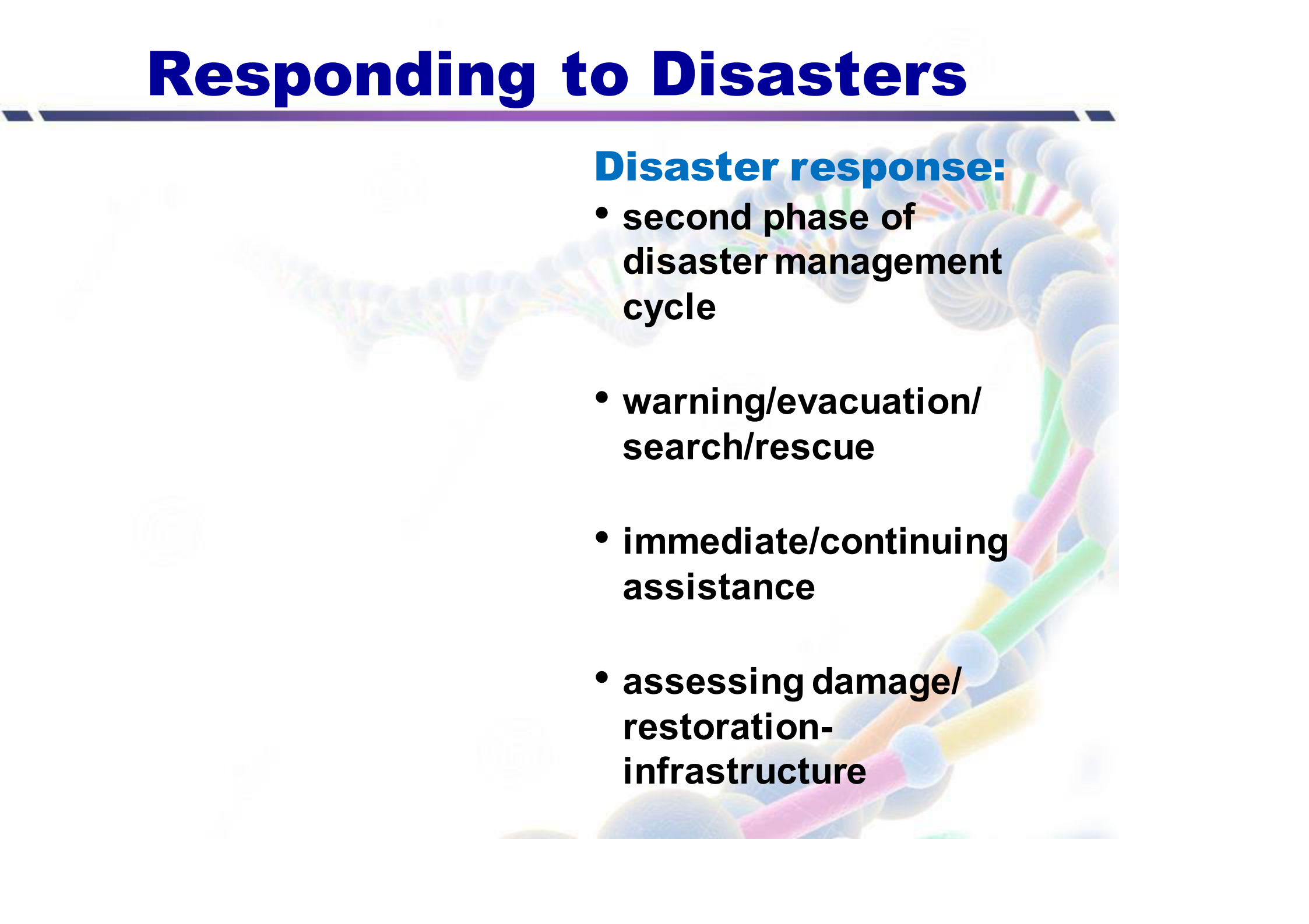
**Responding to
Disasters**

Responding to Disasters

Definition:

- event occurring suddenly-causing loss of life damage or hardship
- sudden overwhelming and unforeseen event

Responding to Disasters



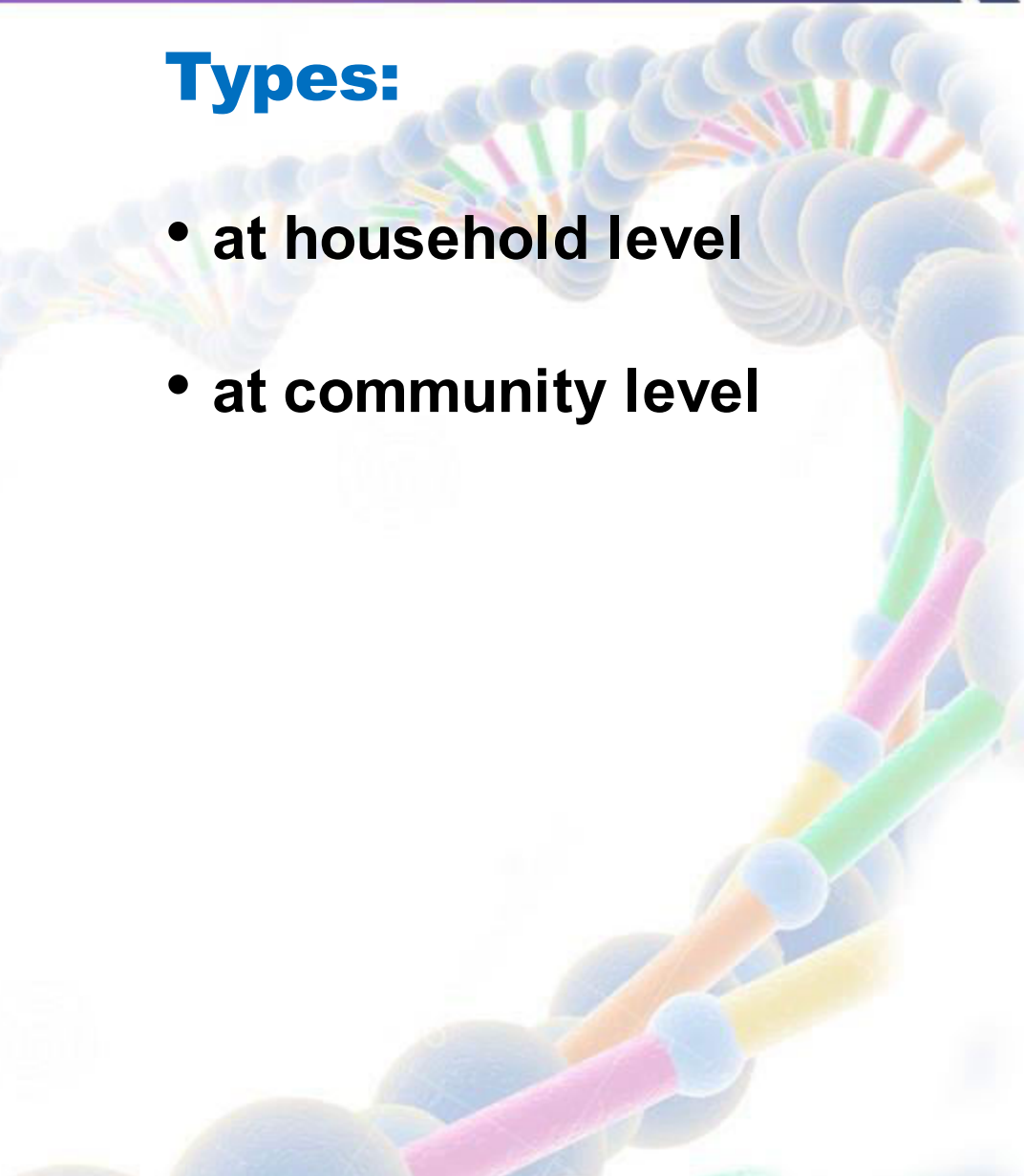
Disaster response:

- second phase of disaster management cycle
- warning/evacuation/search/rescue
- immediate/continuing assistance
- assessing damage/restoration-infrastructure

Responding to Disasters

Types:

- at household level
- at community level



Responding to Disasters

Aims:

- **assistance to maintain life**
- **improve health**
- **support the morale-affected population**
- **limited aid**



Biosafety

Lesson # 68

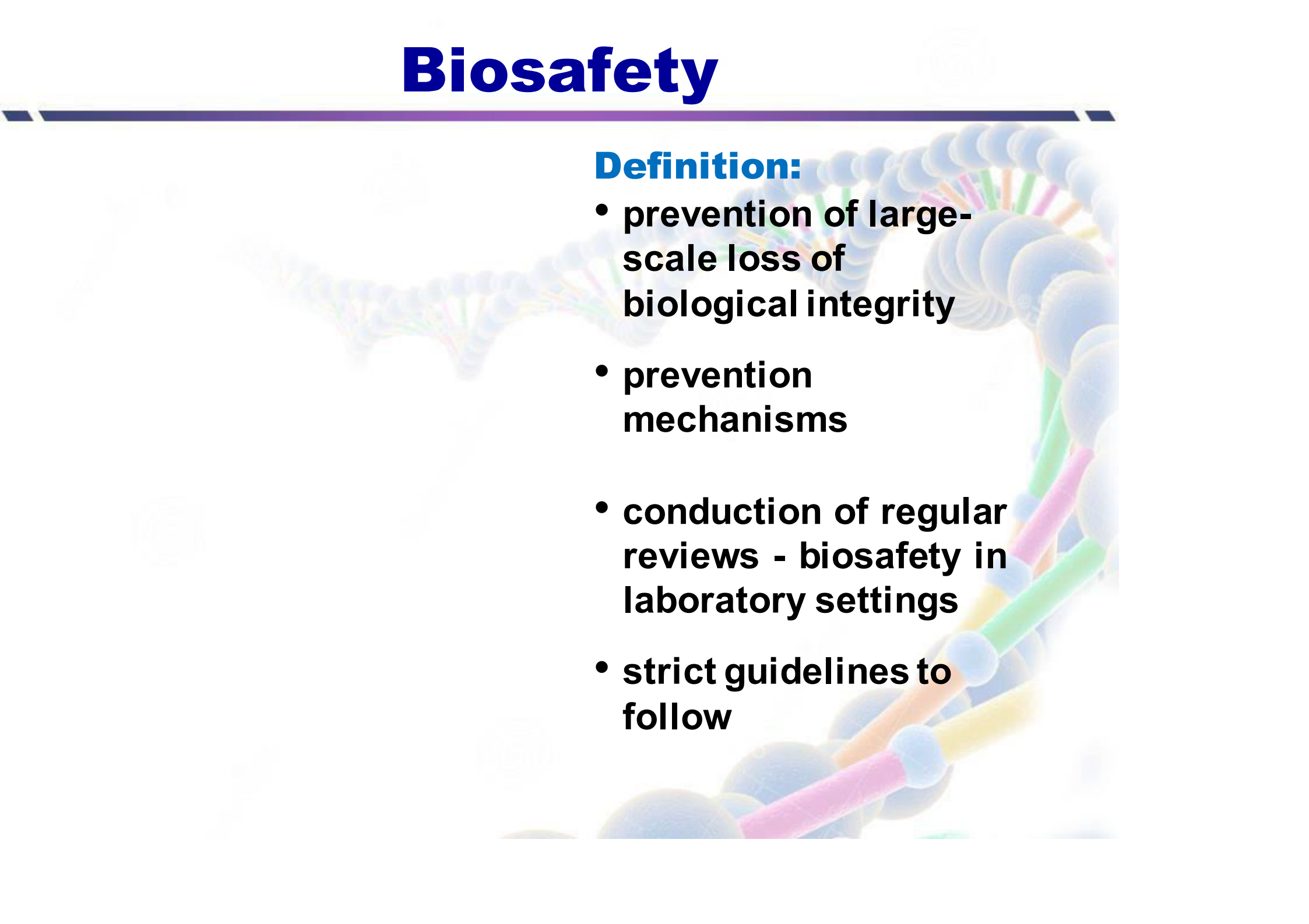


Biosafety



Biosafety

Biosafety



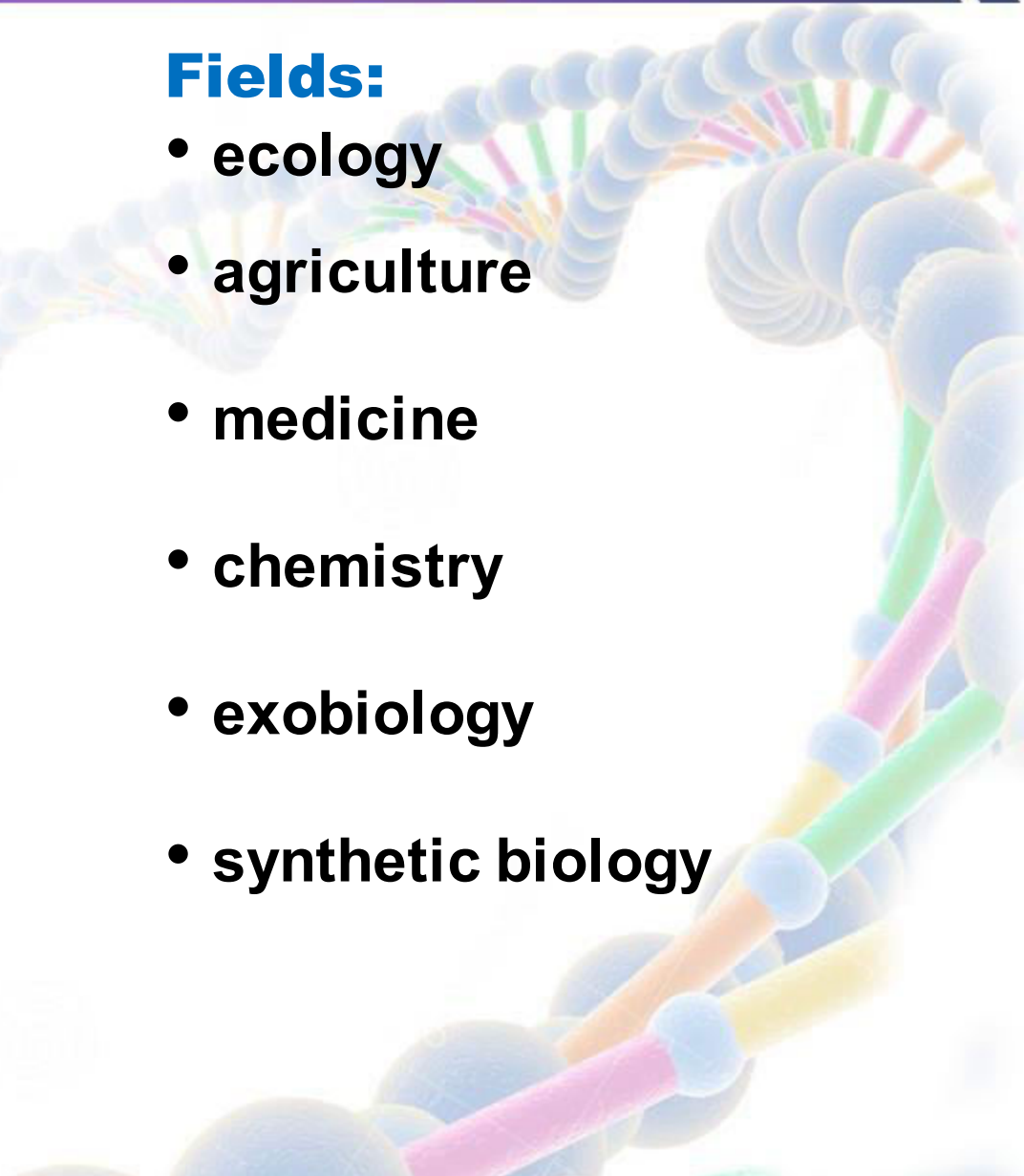
Definition:

- **prevention of large-scale loss of biological integrity**
- **prevention mechanisms**
- **conduction of regular reviews - biosafety in laboratory settings**
- **strict guidelines to follow**

Biosafety

Fields:

- ecology
- agriculture
- medicine
- chemistry
- exobiology
- synthetic biology



Biosecurity

Lesson # 69

3/1/2018

VU MEDICAL
ZONE

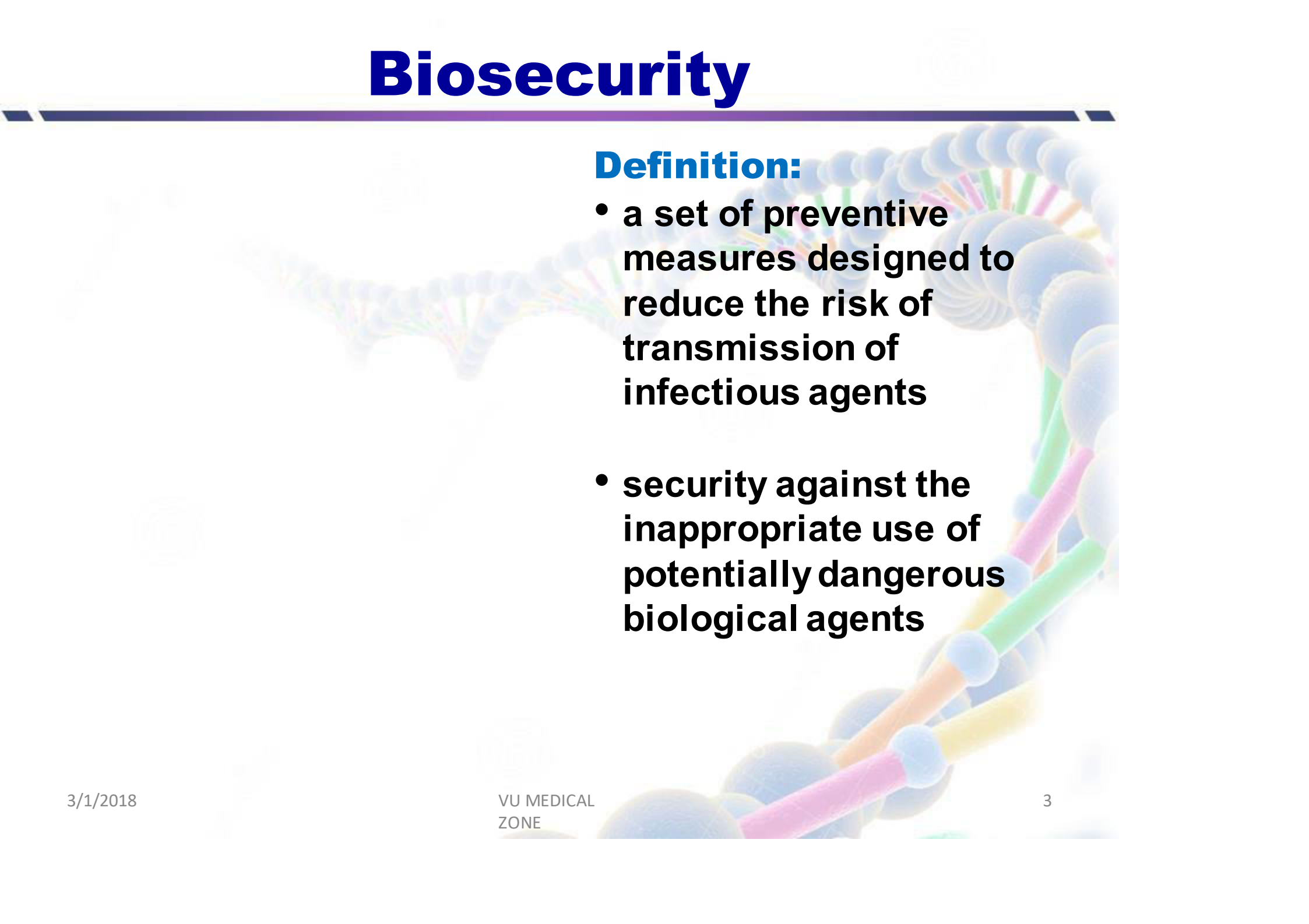
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Biosecurity



Biosecurity

Biosecurity



Definition:

- a set of preventive measures designed to reduce the risk of transmission of infectious agents
- security against the inappropriate use of potentially dangerous biological agents

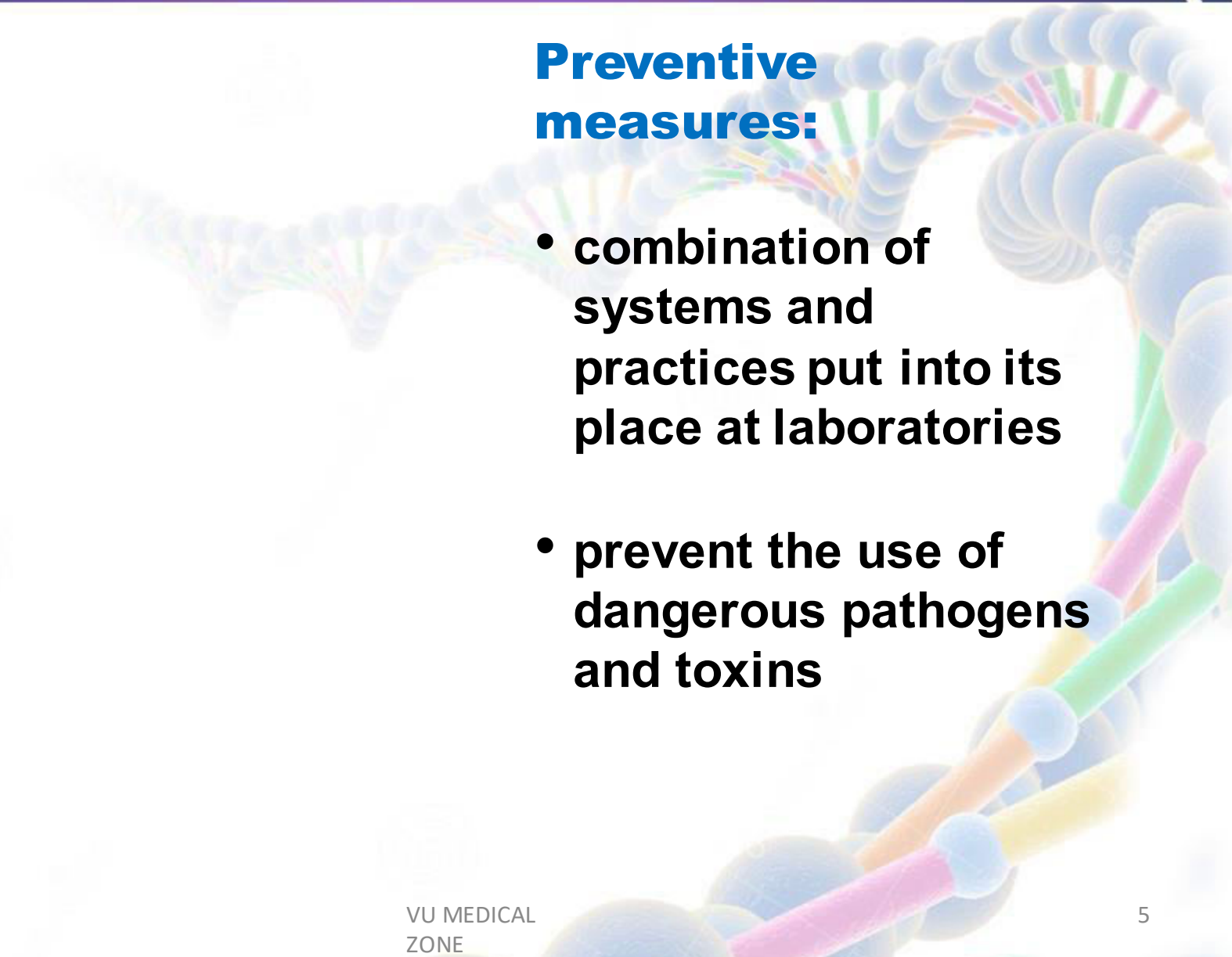
Biosecurity



Security issues:

- non-traditional security
- international security
- cooperation - of scientists, technicians, policy makers, security engineers and law enforcement officials

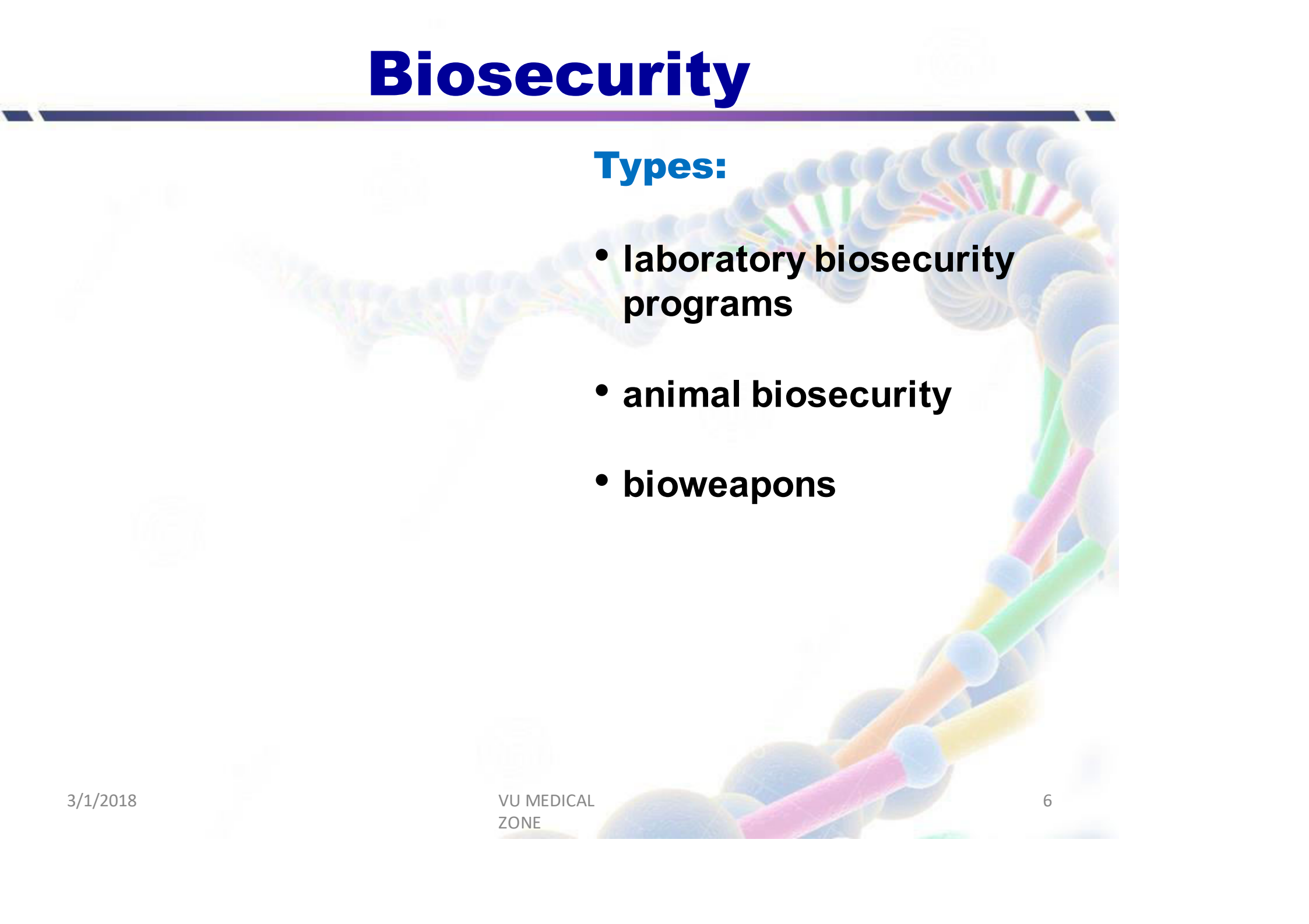
Biosecurity



Preventive measures:

- combination of systems and practices put into its place at laboratories
- prevent the use of dangerous pathogens and toxins

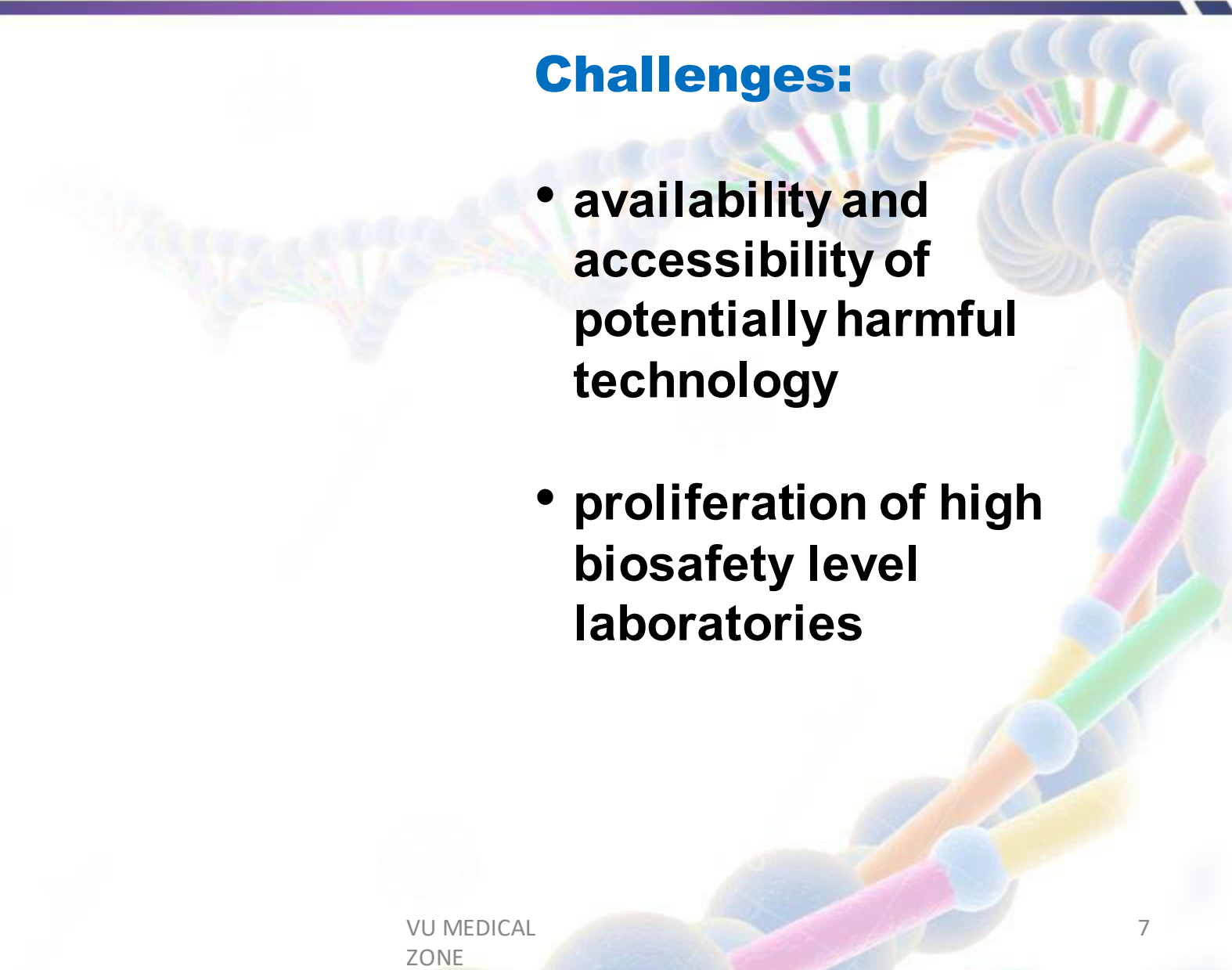
Biosecurity



Types:

- **laboratory biosecurity programs**
- **animal biosecurity**
- **bioweapons**

Biosecurity

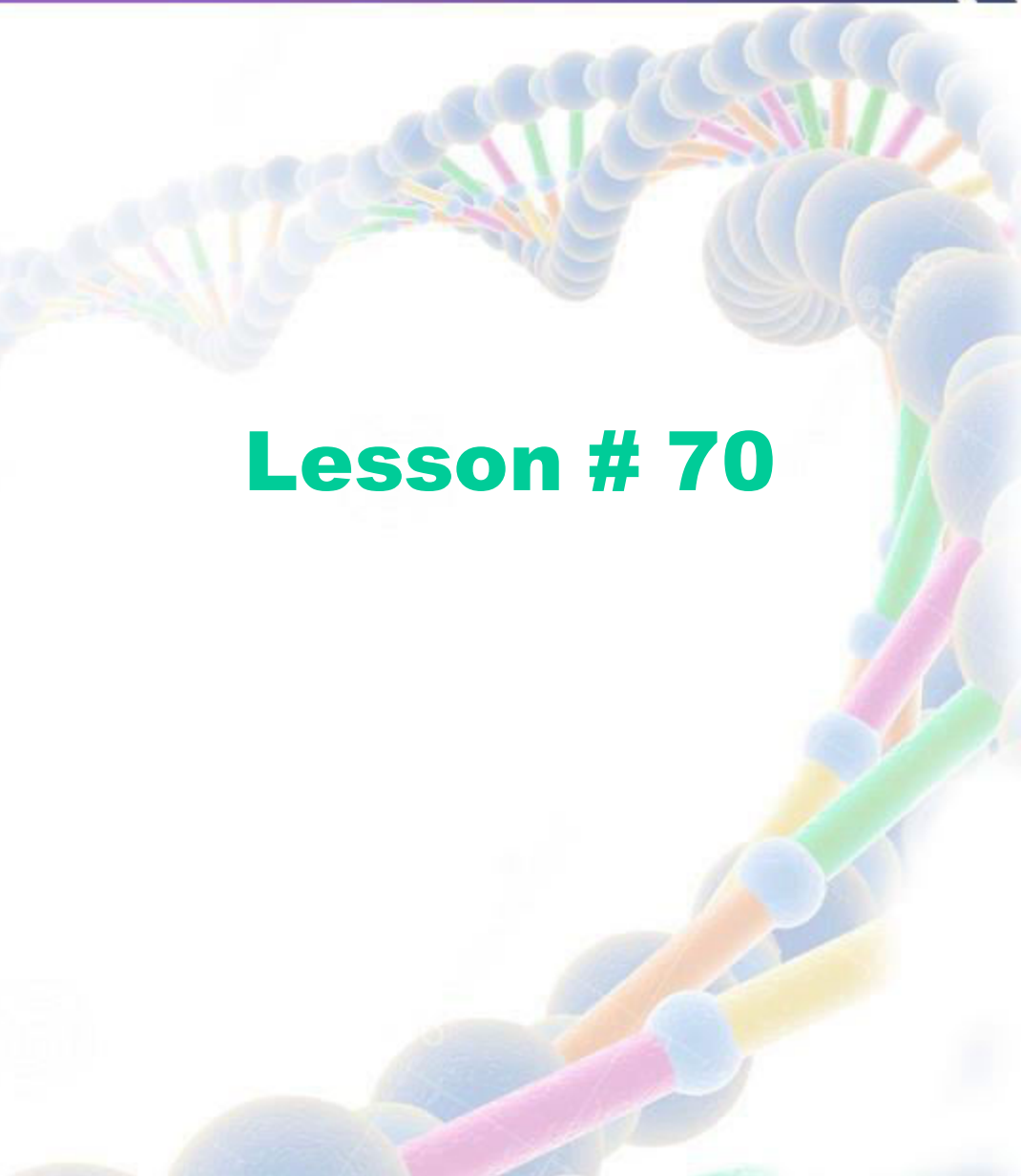


Challenges:

- availability and accessibility of potentially harmful technology
- proliferation of high biosafety level laboratories

Biosecurity

Lesson # 70

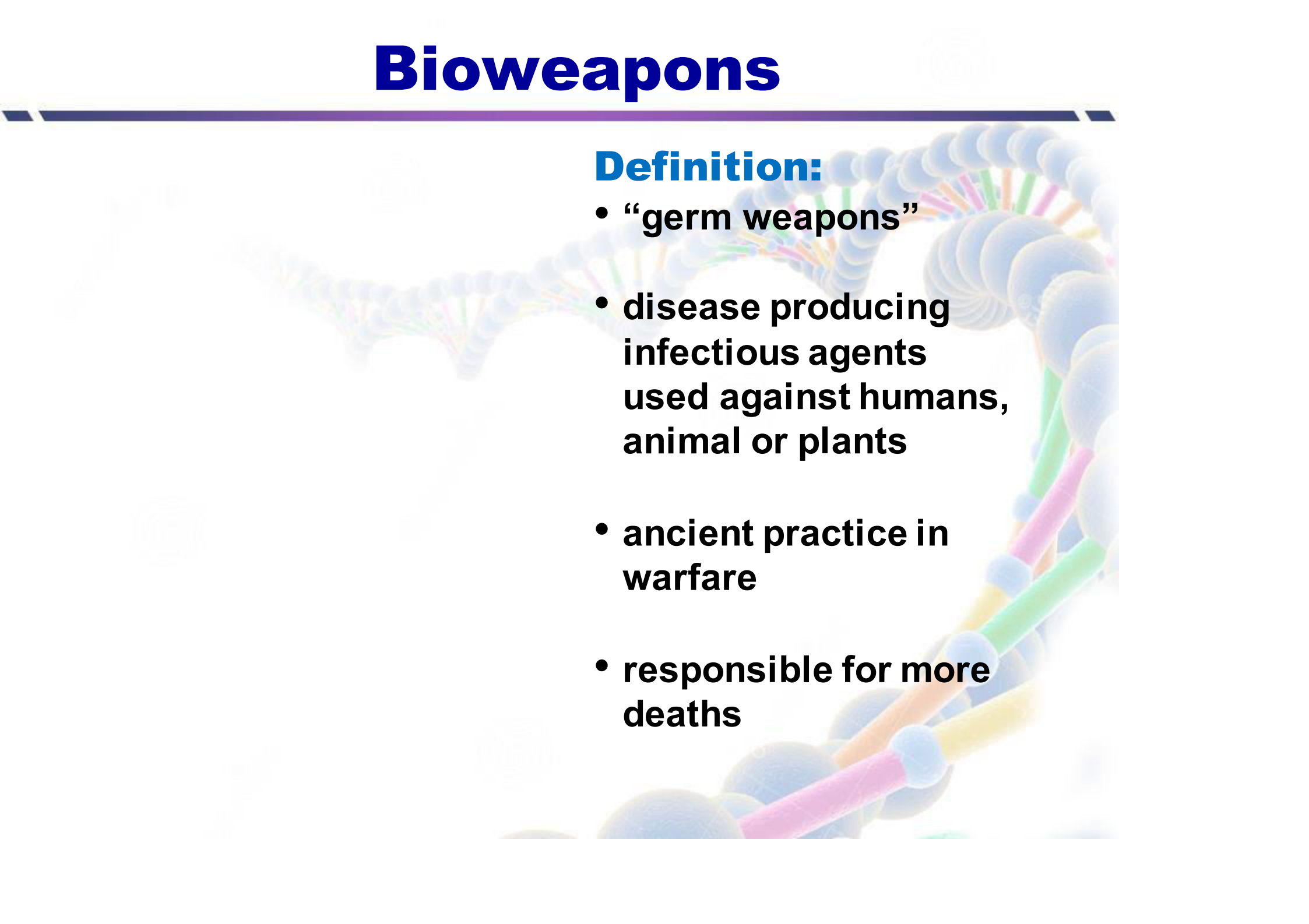


Biosecurity

Bioweapons



Bioweapons



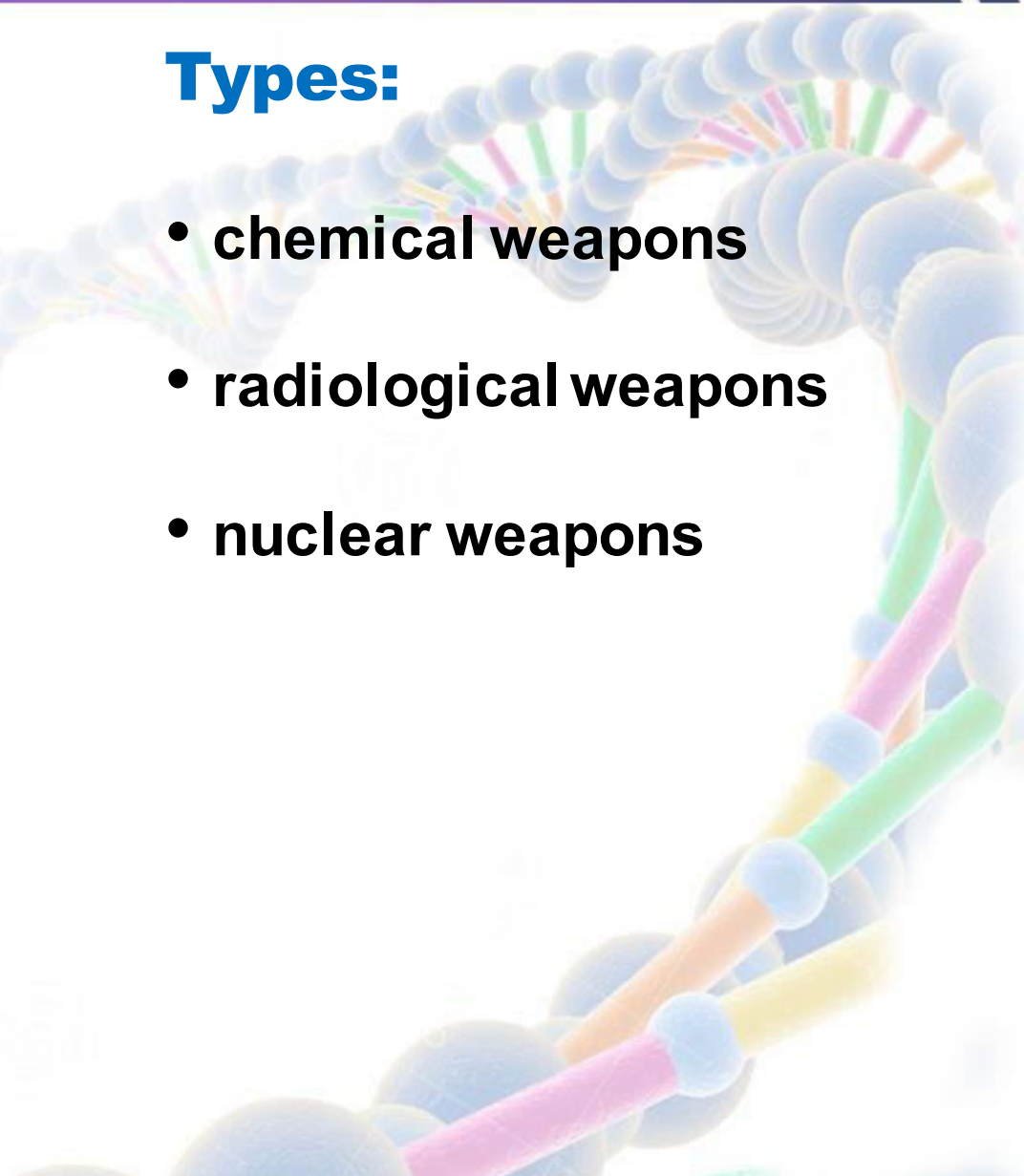
Definition:

- “germ weapons”
- disease producing infectious agents used against humans, animal or plants
- ancient practice in warfare
- responsible for more deaths

Bioweapons

Types:

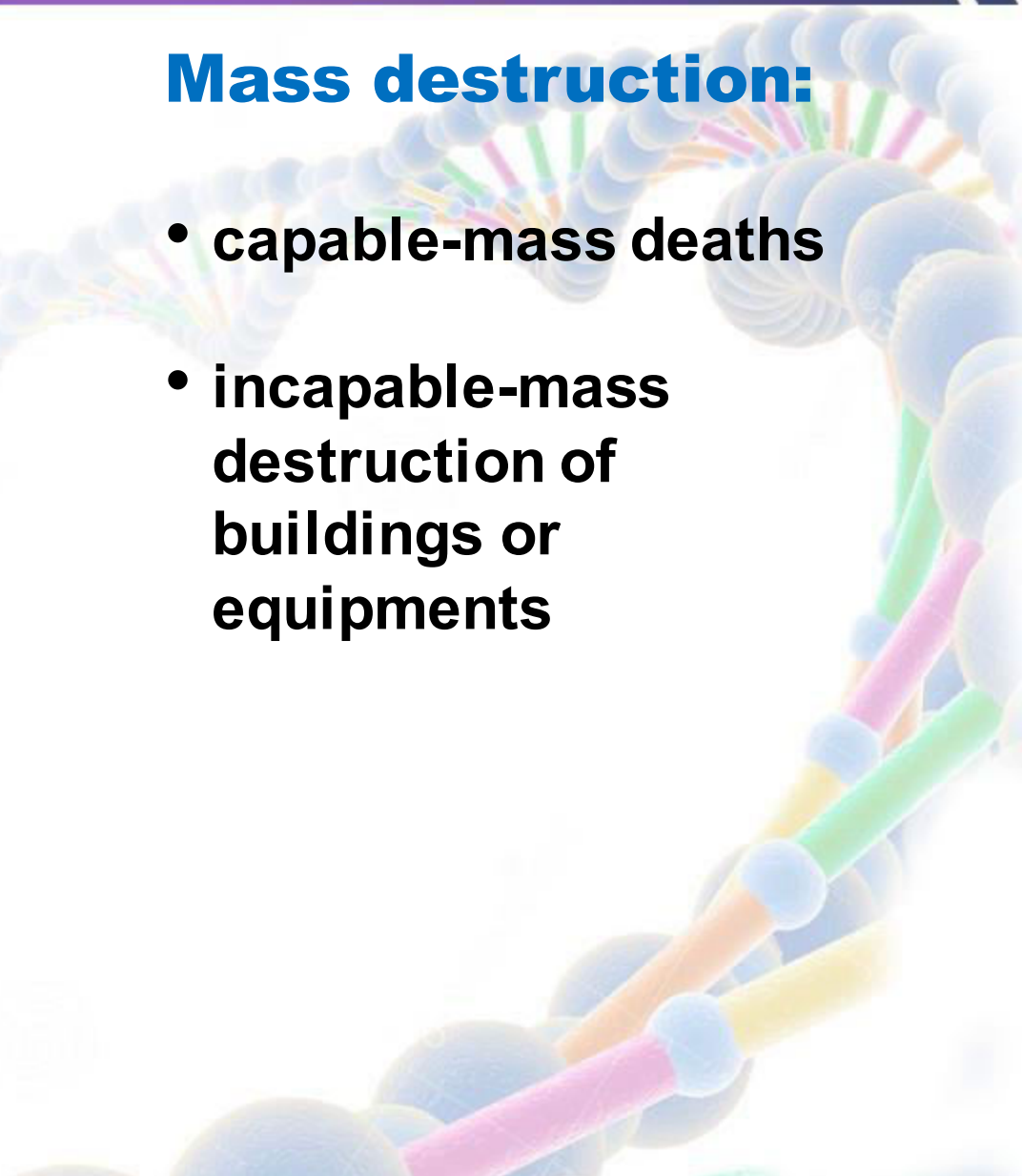
- chemical weapons
- radiological weapons
- nuclear weapons



Bioweapons

Mass destruction:

- capable-mass deaths
- incapable-mass destruction of buildings or equipments



Bioweapons

Agents:

- anthrax
- brucellosis
- small pox
- viral hemorrhagic fever
- Staphylococcal enterotoxin B
- botulinum toxins

Bioweapons

Precautions:

- mass equipped with filters
- boots/ gloves
- prevent the contacts with wounds
- biological weapon sensors

Biosafety

Lesson # 71

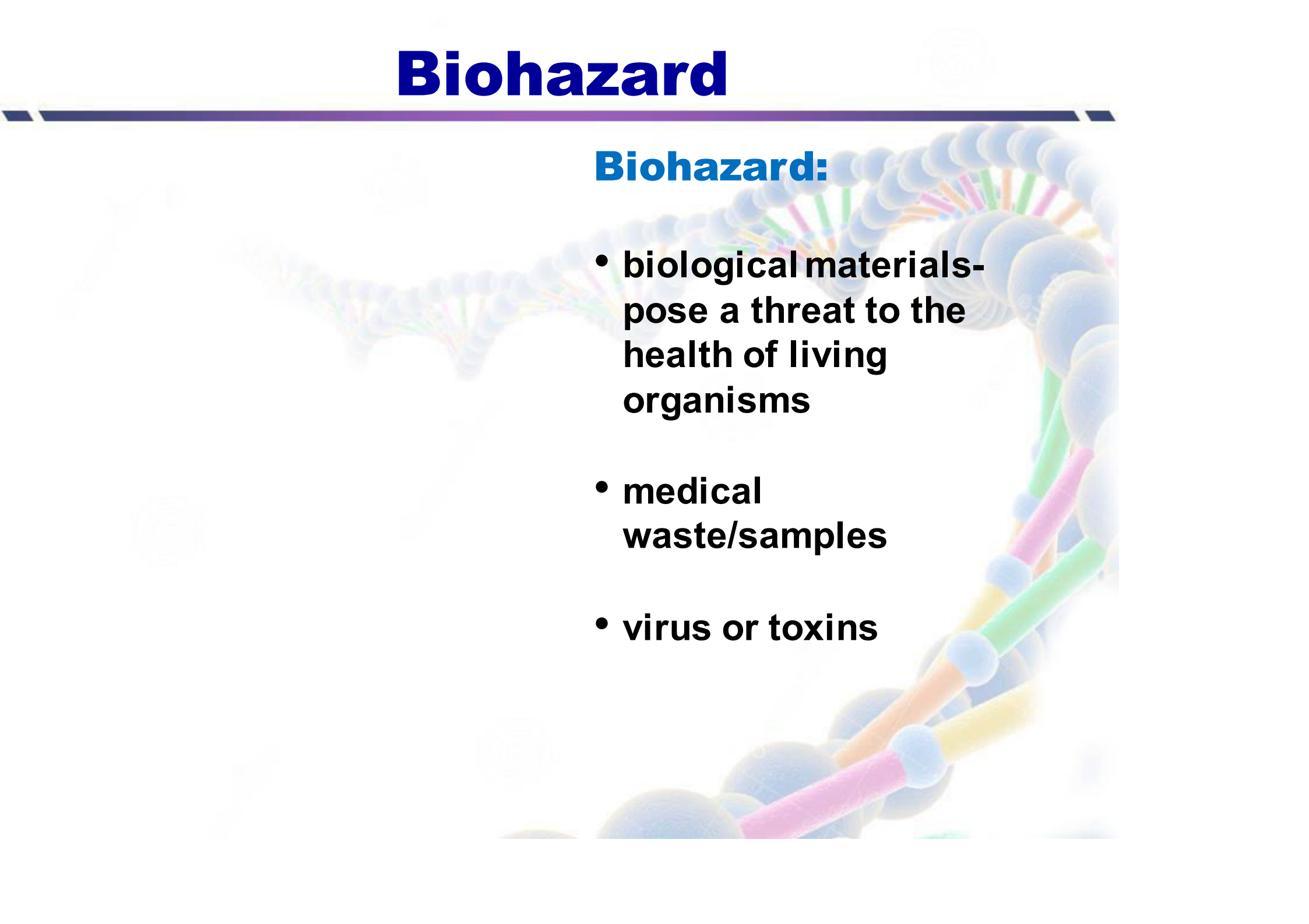


Biosafety



Biohazard

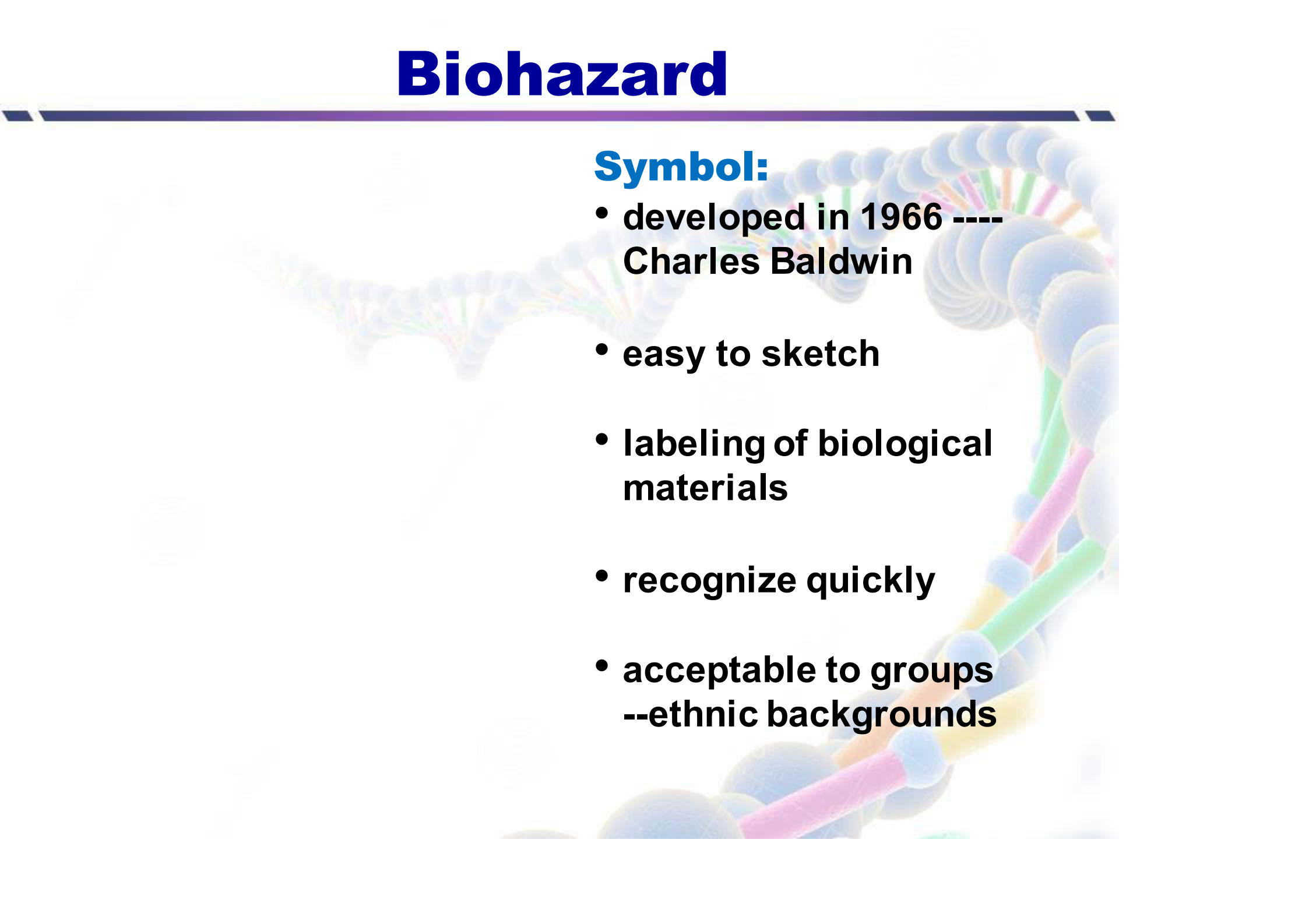
Biohazard



Biohazard:

- **biological materials-
pose a threat to the
health of living
organisms**
- **medical
waste/samples**
- **virus or toxins**

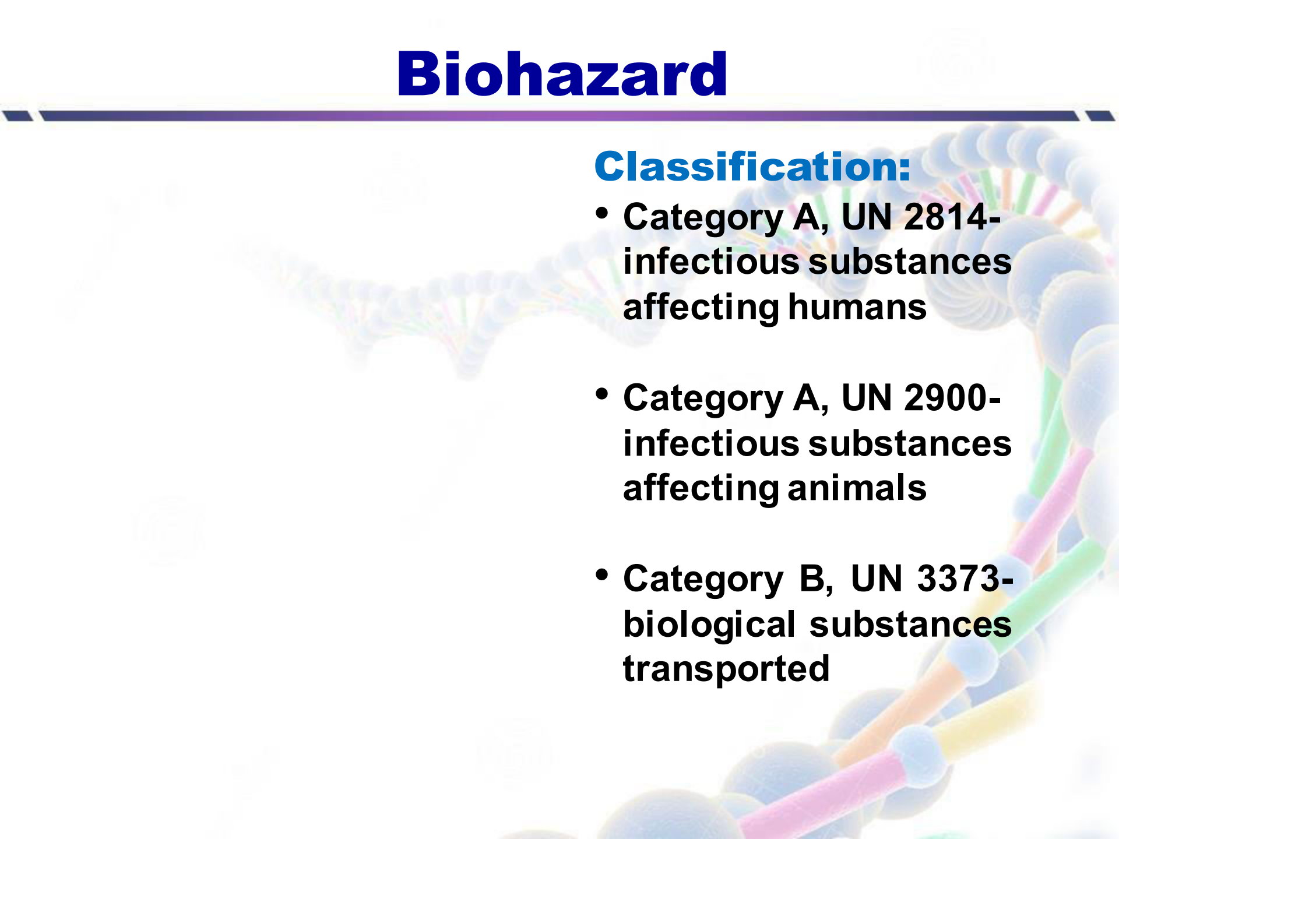
Biohazard



Symbol:

- developed in 1966 ----
Charles Baldwin
- easy to sketch
- labeling of biological materials
- recognize quickly
- acceptable to groups
--ethnic backgrounds

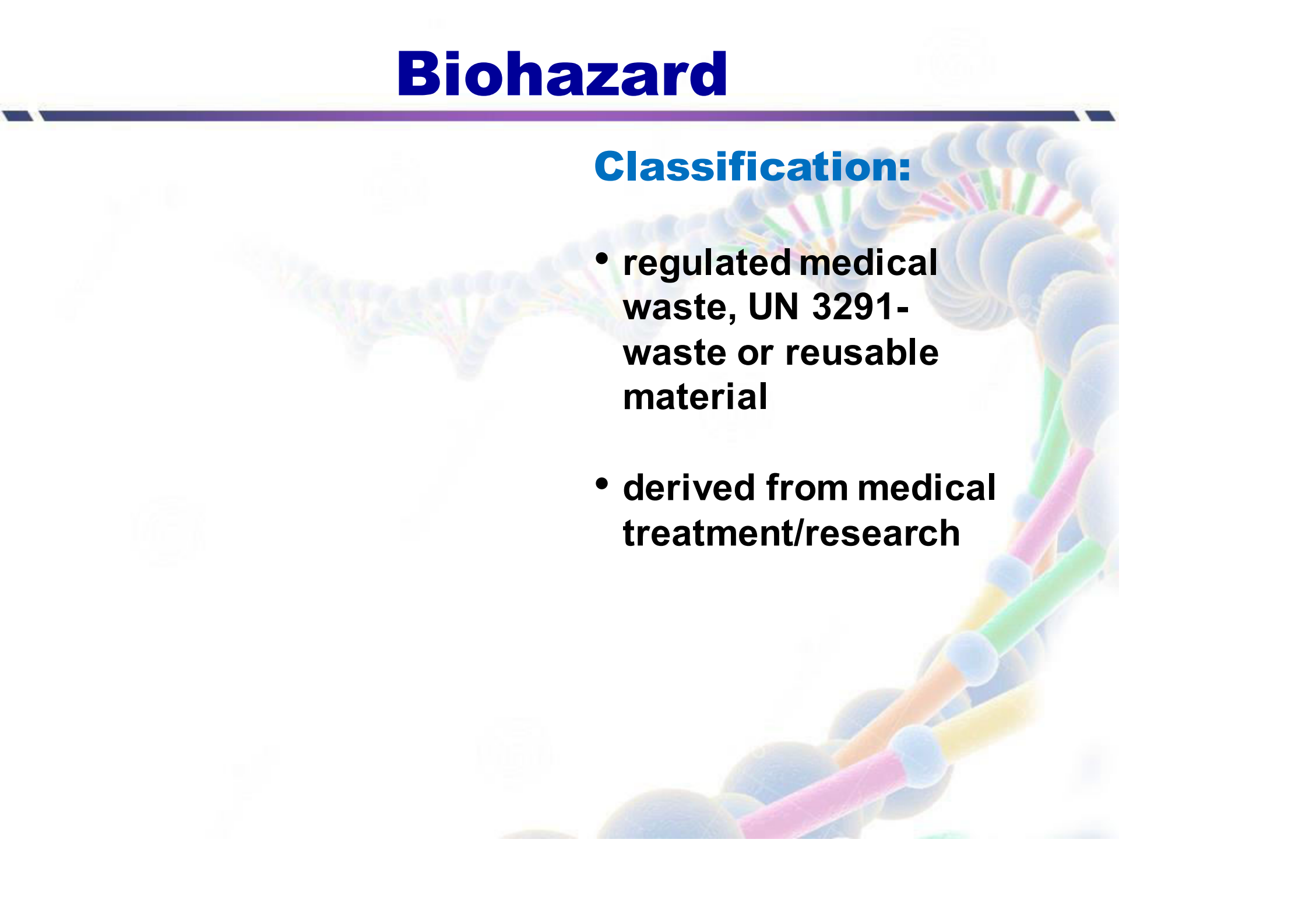
Biohazard



Classification:

- **Category A, UN 2814-**
infectious substances
affecting humans
- **Category A, UN 2900-**
infectious substances
affecting animals
- **Category B, UN 3373-**
biological substances
transported

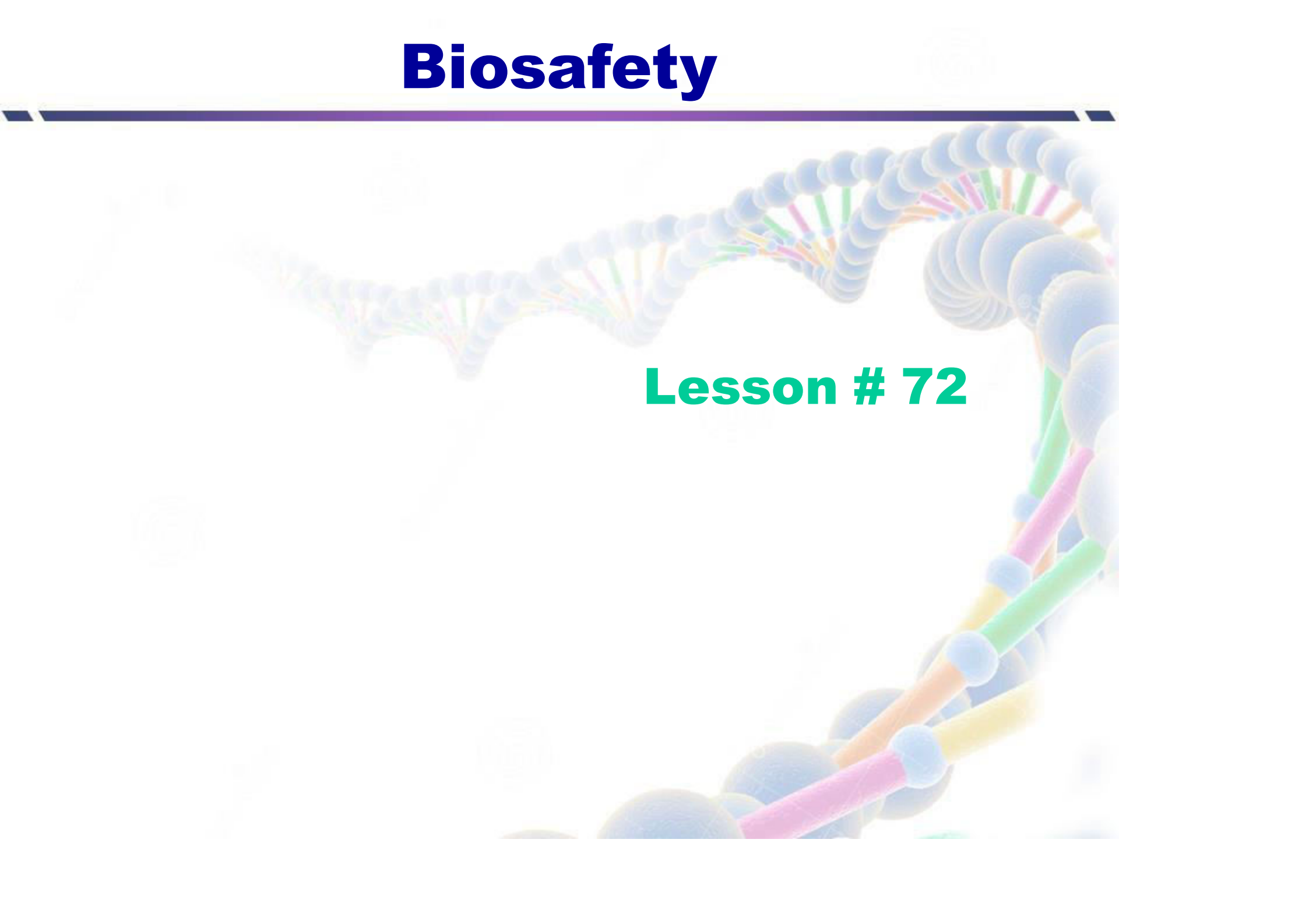
Biohazard



Classification:

- **regulated medical waste, UN 3291- waste or reusable material**
- **derived from medical treatment/research**

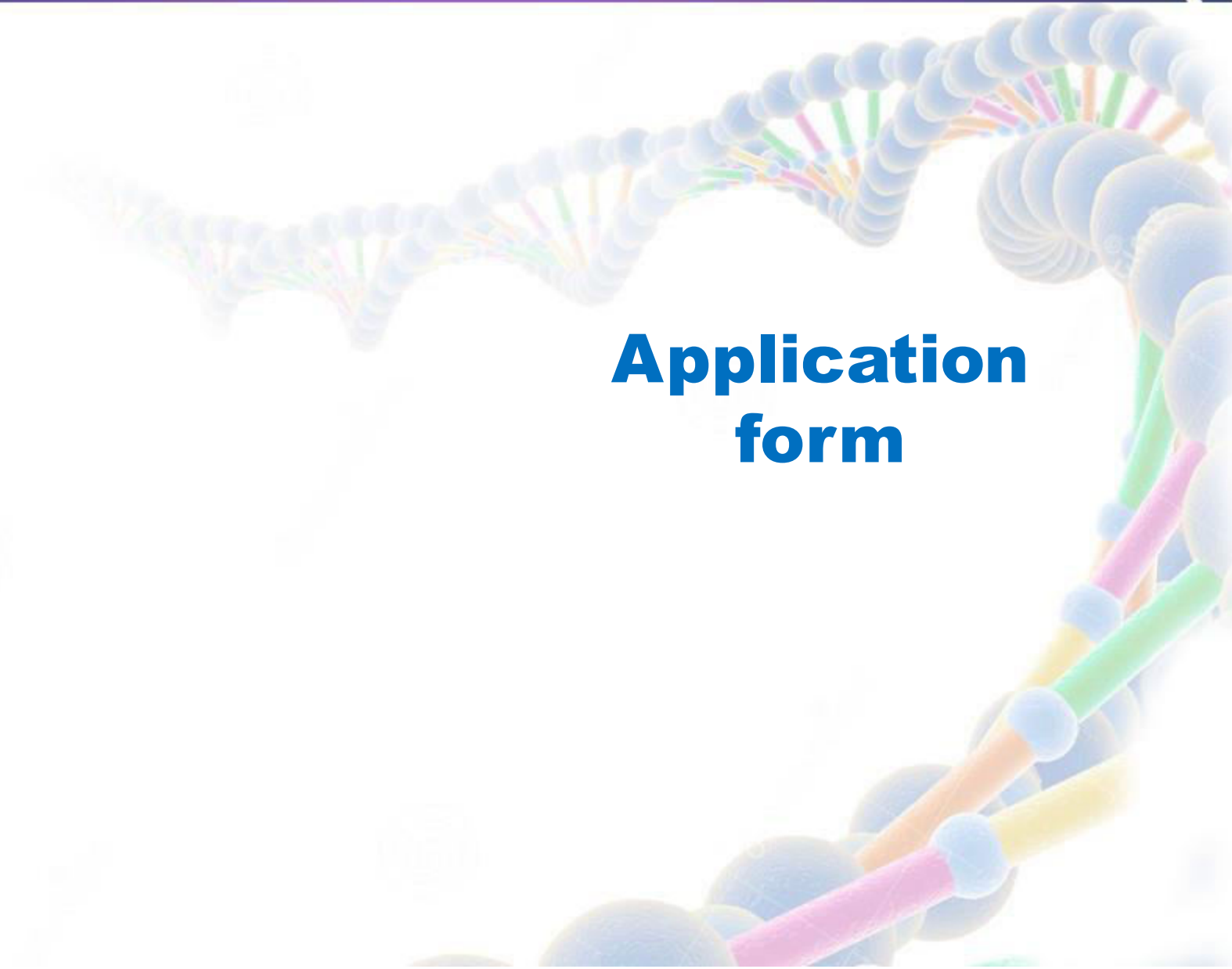
Biosafety



Lesson # 72

Biosafety

**Application
form**



Application form

Form:

- applicants information
- topic covered
- using recombinant DNA
- infectious agents
- toxins
- radioactive material

Application form

Form:

- use of animals
- date of approval
- description of experiments
- sources of DNA
- nature of DNA sequences/attempt gene expression

Application form

Form:

- biosafety levels
- enlist biohazardous materials
- biosafety equipments
- emergency procedures
- biohazardous material storage

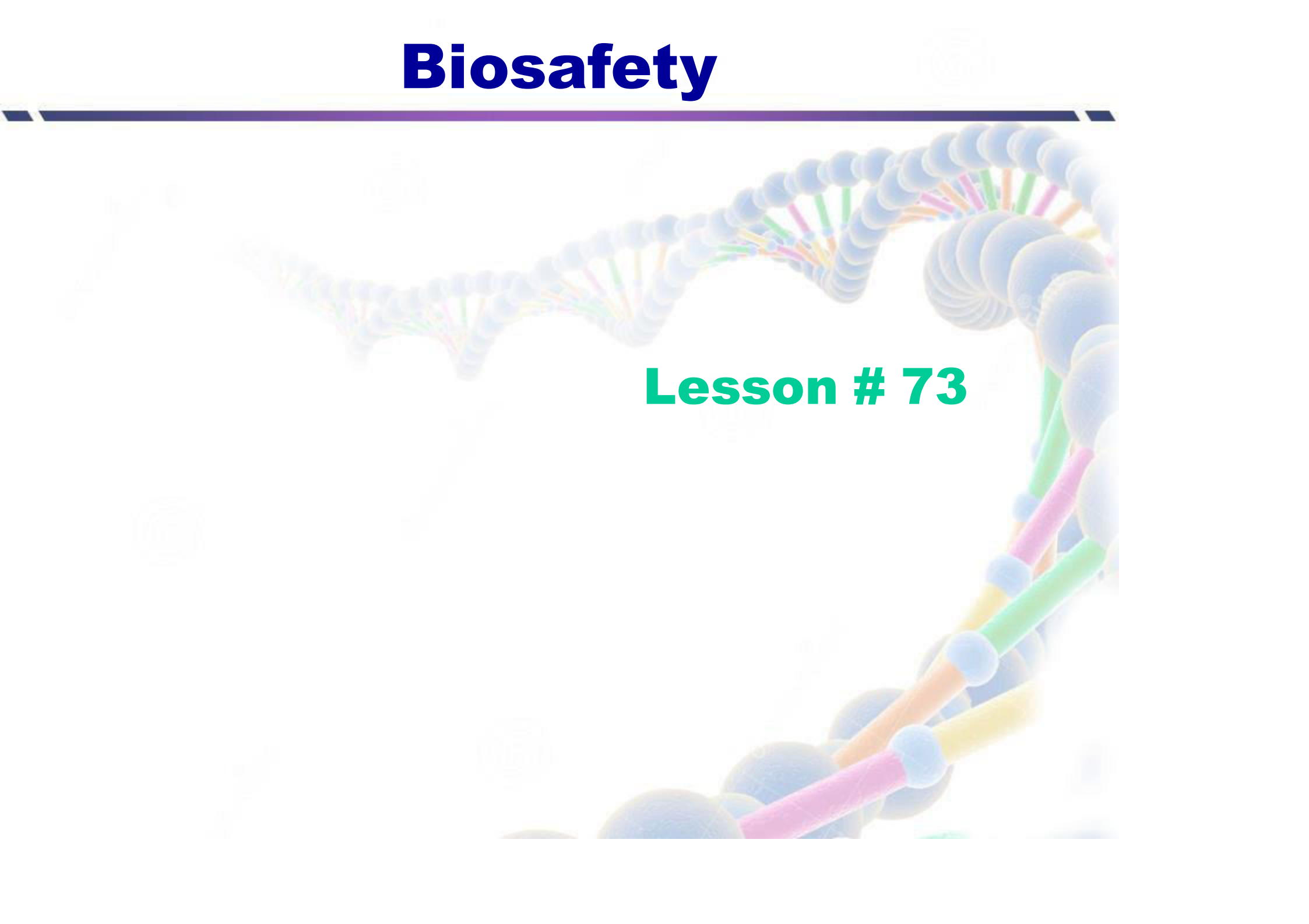
Application form

Form:

- waste disposal
- committee decision



Biosafety



Lesson # 73

Biosafety



**Lab safety
protocols**

Lab safety protocols

Lab safety:

- safety glasses
- closed-toed shoes
- no food/drink
- long hair must be tied
- lab coat
- open flames-
unattended

Lab safety protocols

Lab safety:

- flammable liquids
- skin contact-rinse off
- proper waste disposal
- liquid waste-labeled containers
- equipment must be cleaned and placed back

Lab safety protocols

Lab safety:

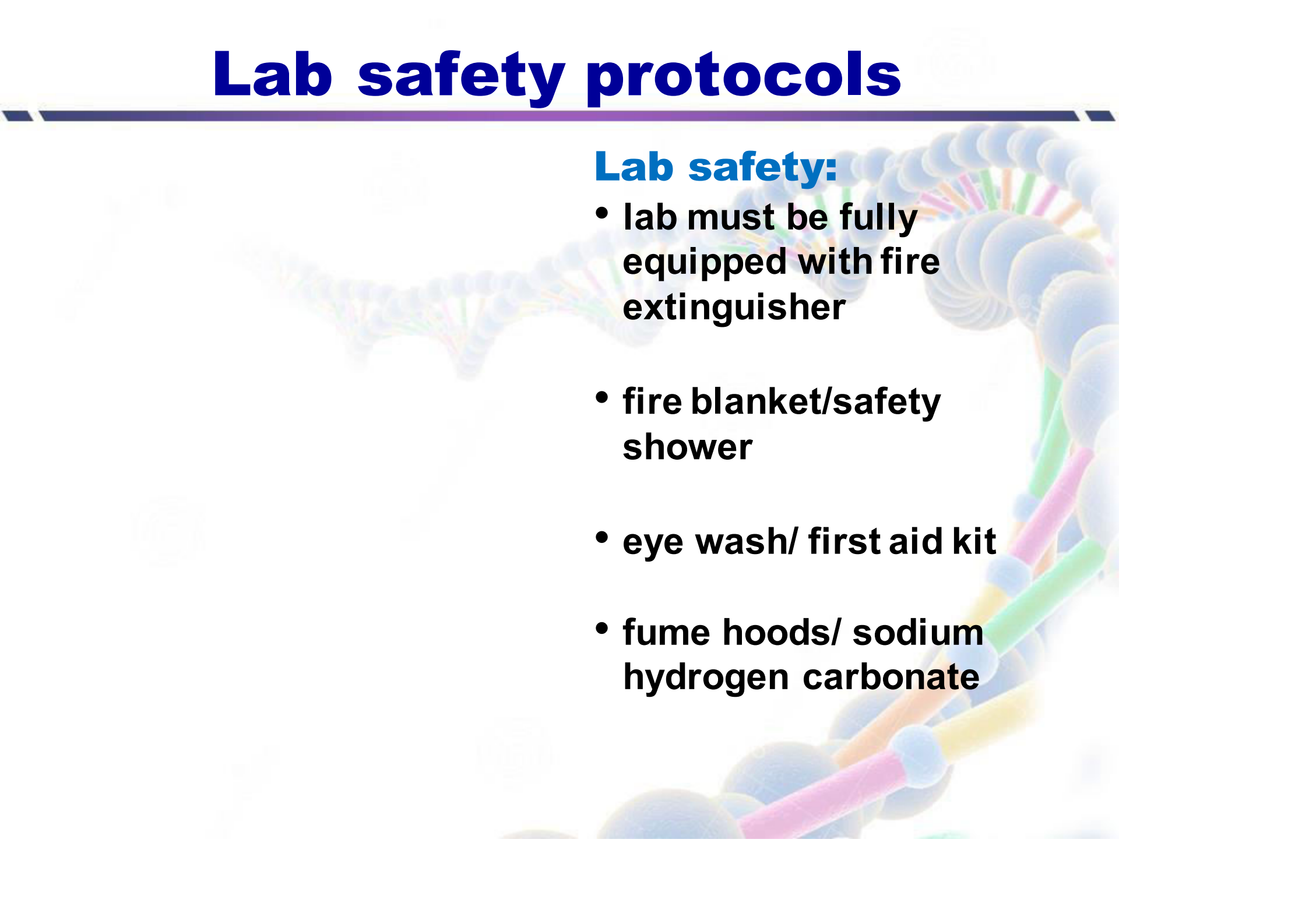
- inform-chemical spill/thermometer breakage
- be careful-handling hot glassware and apparatus
- avoid taking excess amount of chemicals

Lab safety protocols

Lab safety:

- cell phones/ head phones are not allowed
- lab door must be kept close
- chairs not permitted - lab work is in session

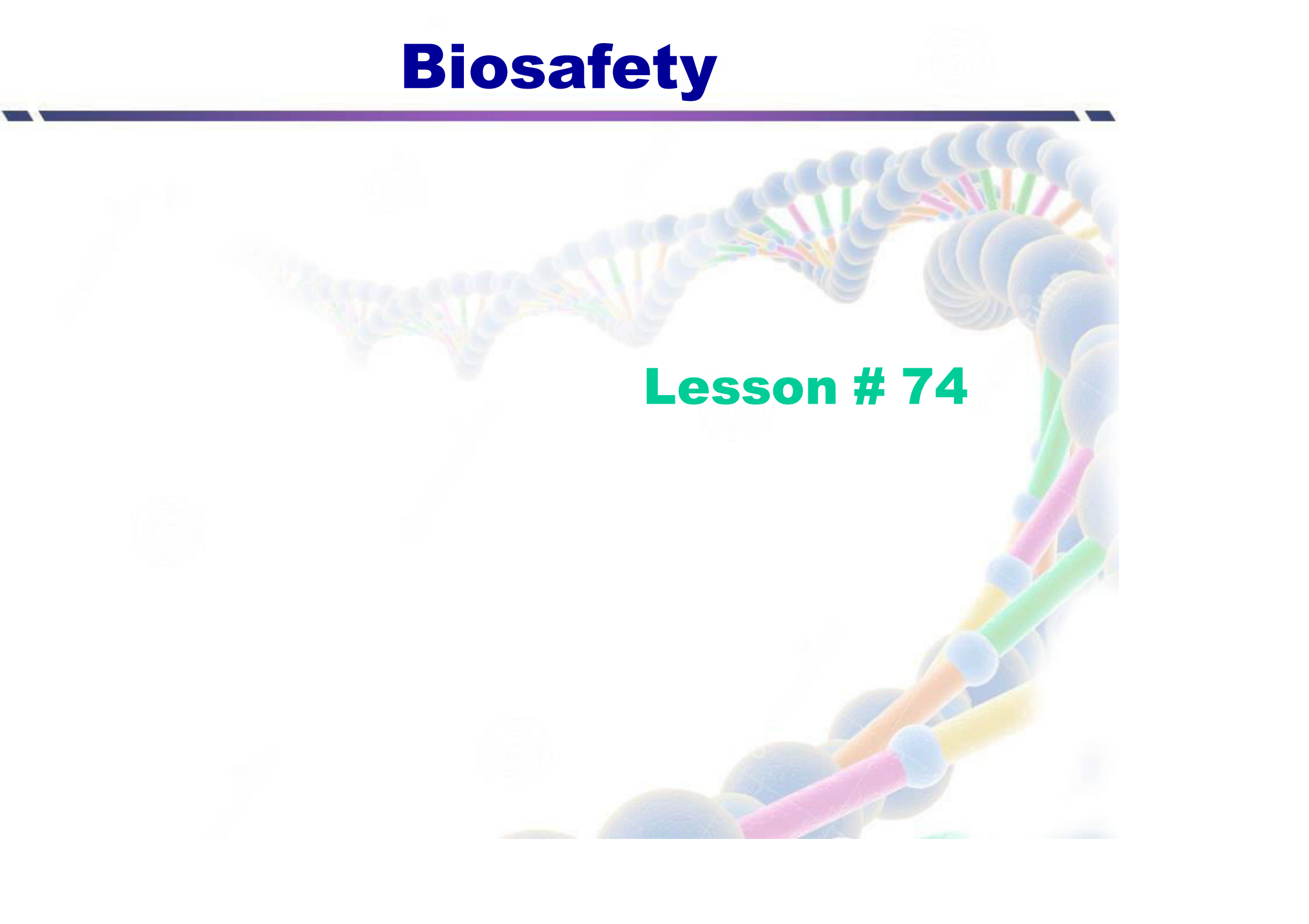
Lab safety protocols



Lab safety:

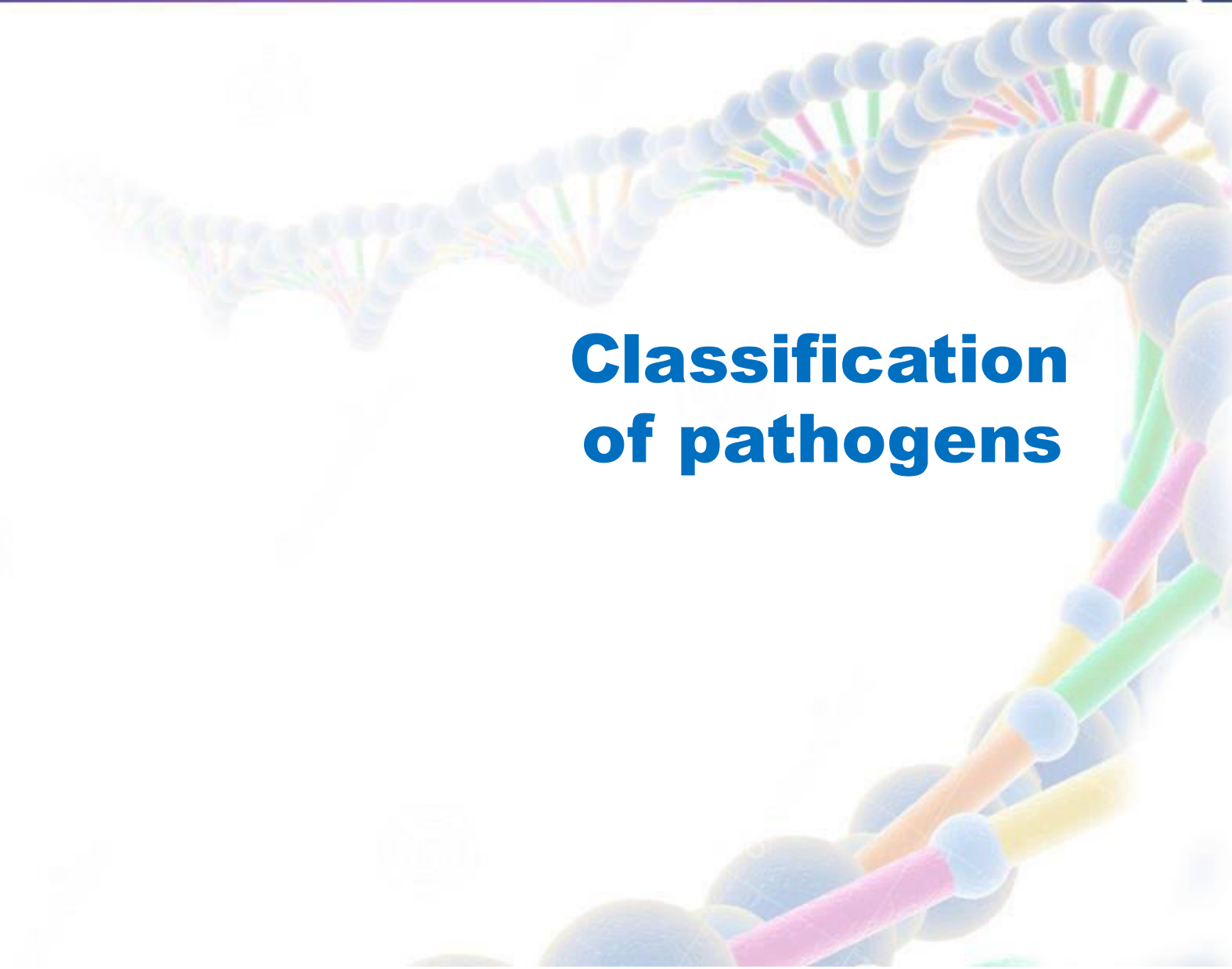
- lab must be fully equipped with fire extinguisher
- fire blanket/safety shower
- eye wash/ first aid kit
- fume hoods/ sodium hydrogen carbonate

Biosafety



Lesson # 74

Biosafety

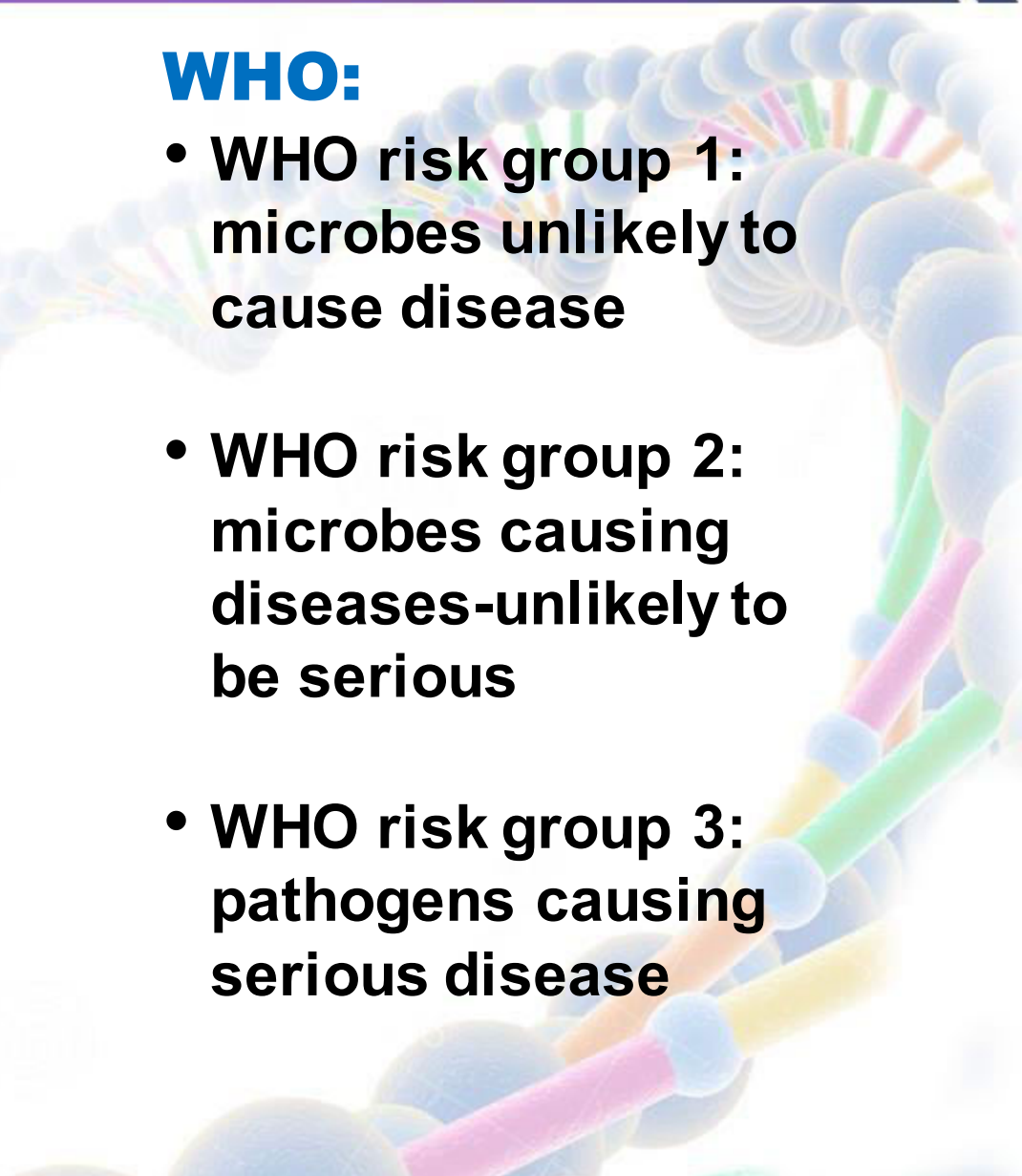


Classification of pathogens

Classification of pathogens

WHO:

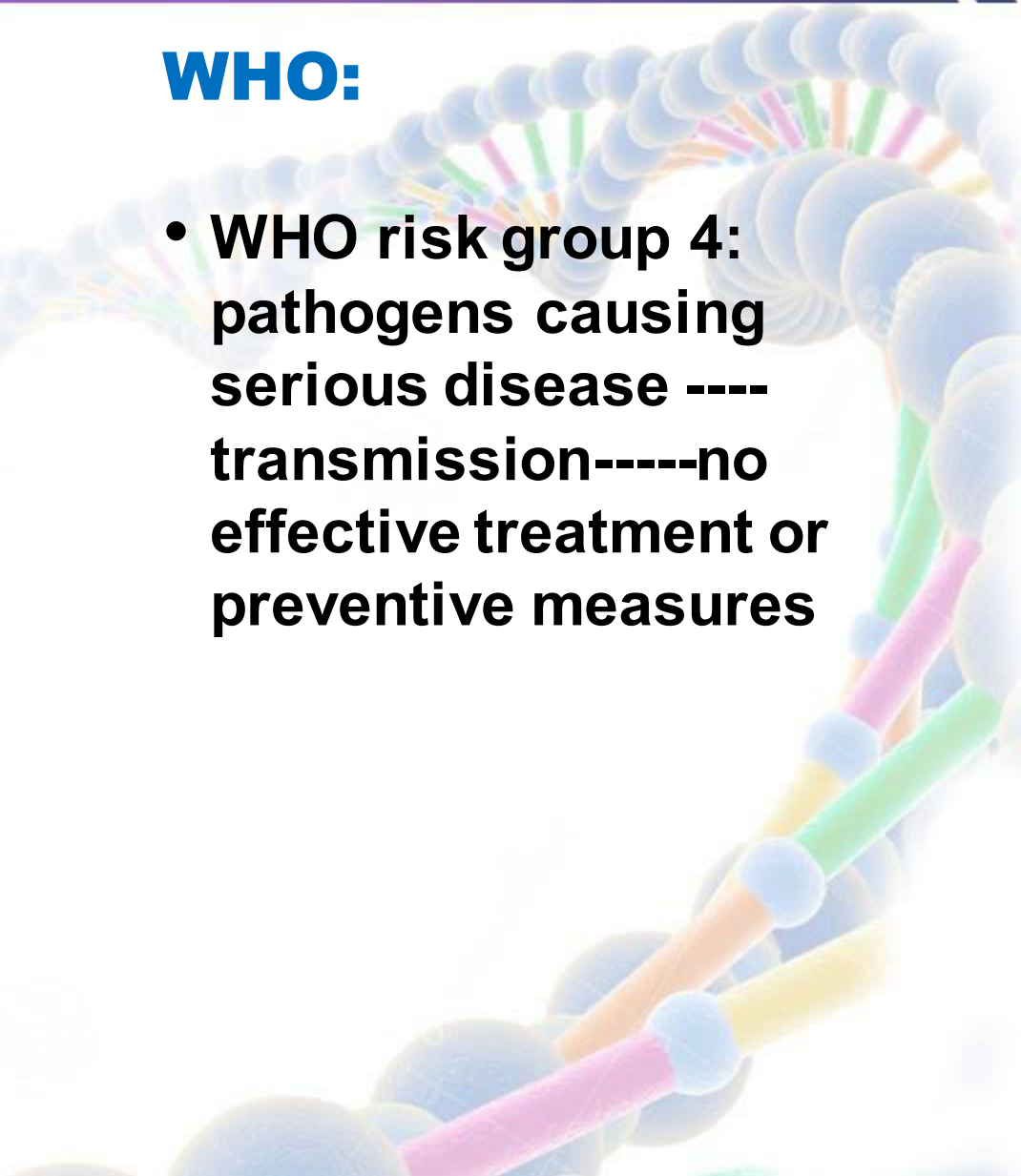
- **WHO risk group 1:**
microbes unlikely to cause disease
- **WHO risk group 2:**
microbes causing diseases-unlikely to be serious
- **WHO risk group 3:**
pathogens causing serious disease



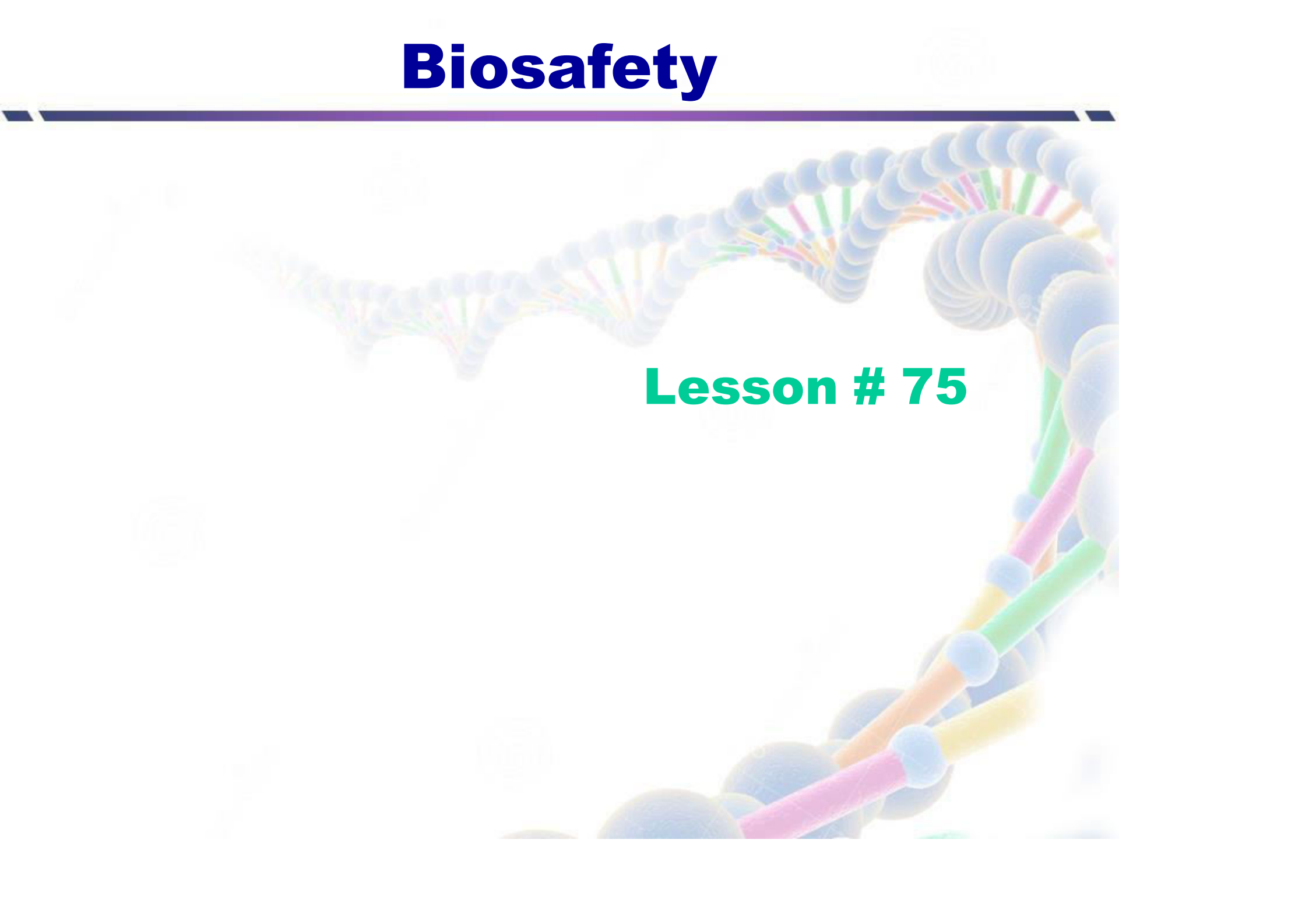
Classification of pathogens

WHO:

- **WHO risk group 4:**
pathogens causing
serious disease ----
transmission-----no
effective treatment or
preventive measures

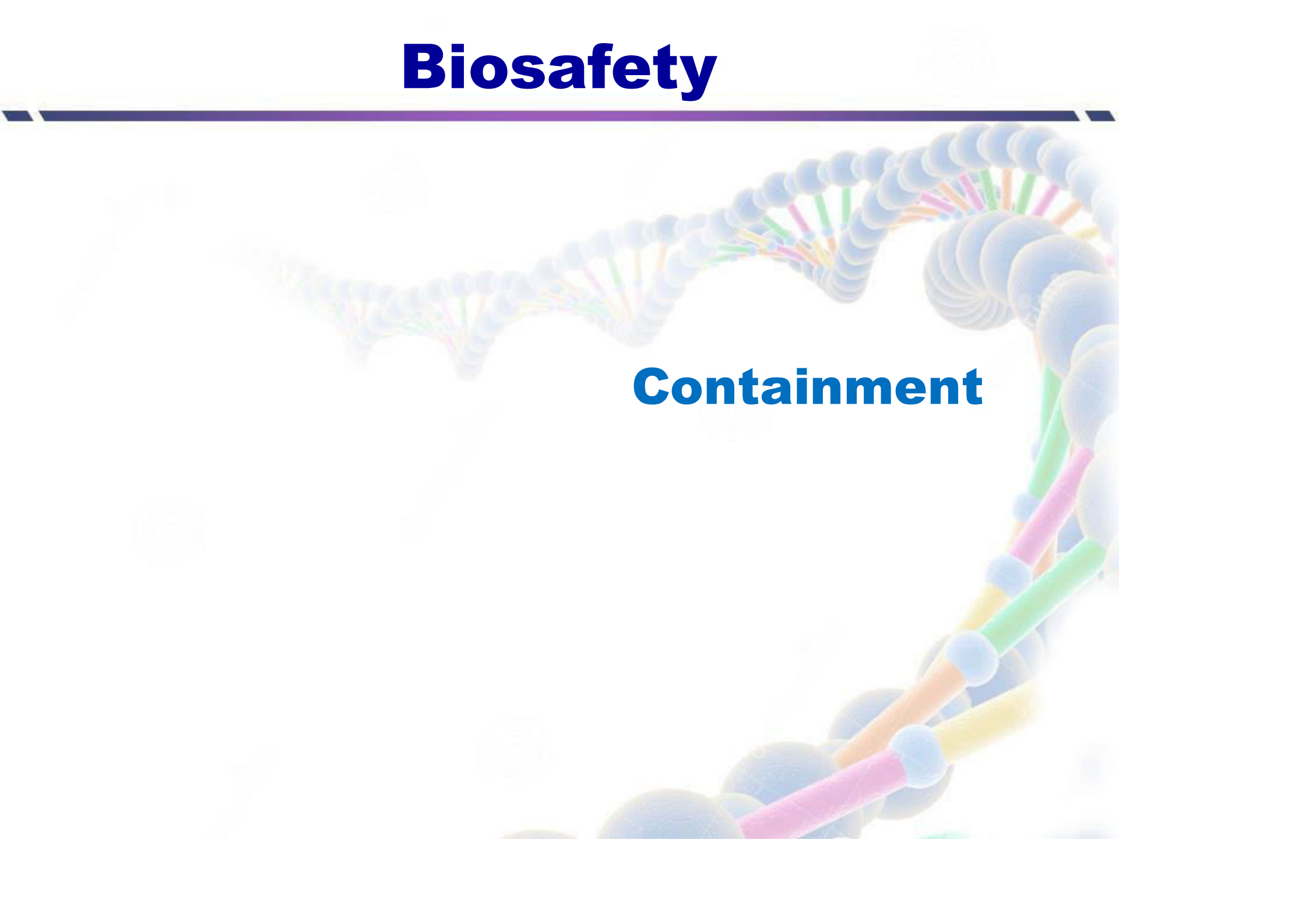


Biosafety



Lesson # 75

Biosafety



Containment

Containment

Definition:

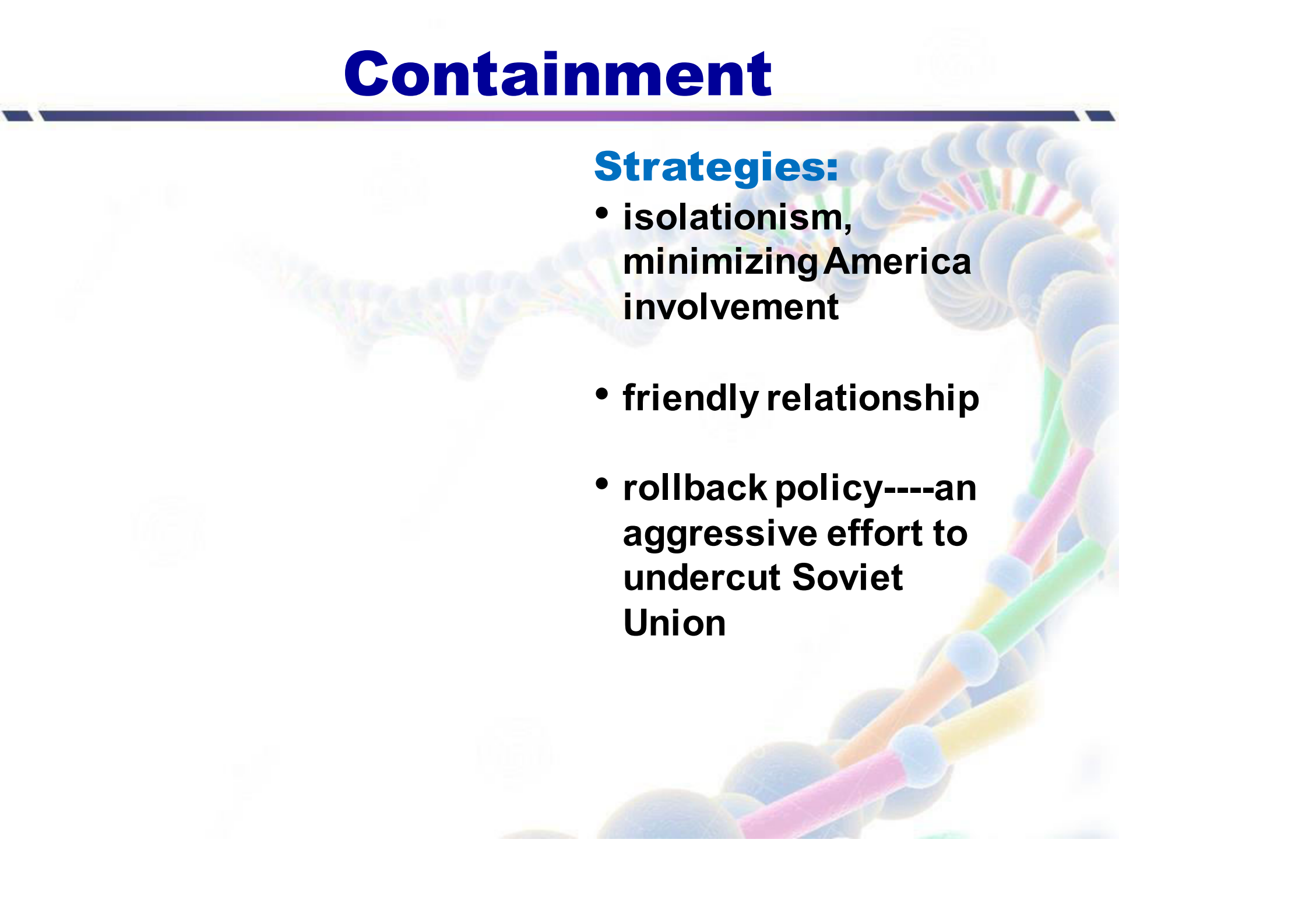
- **military strategy**
- **stop the expansion of an enemy**
- **USA-Cold War policy**
- **prevent the spread of communism abroad**

Containment

History:

- 1850s-anti-slavery forces developed containment strategy
- stop the expansion of slavery and forcing its collapse
- 1941 during World War II-policy was rollback to destroy Japan and Germany

Containment



Strategies:

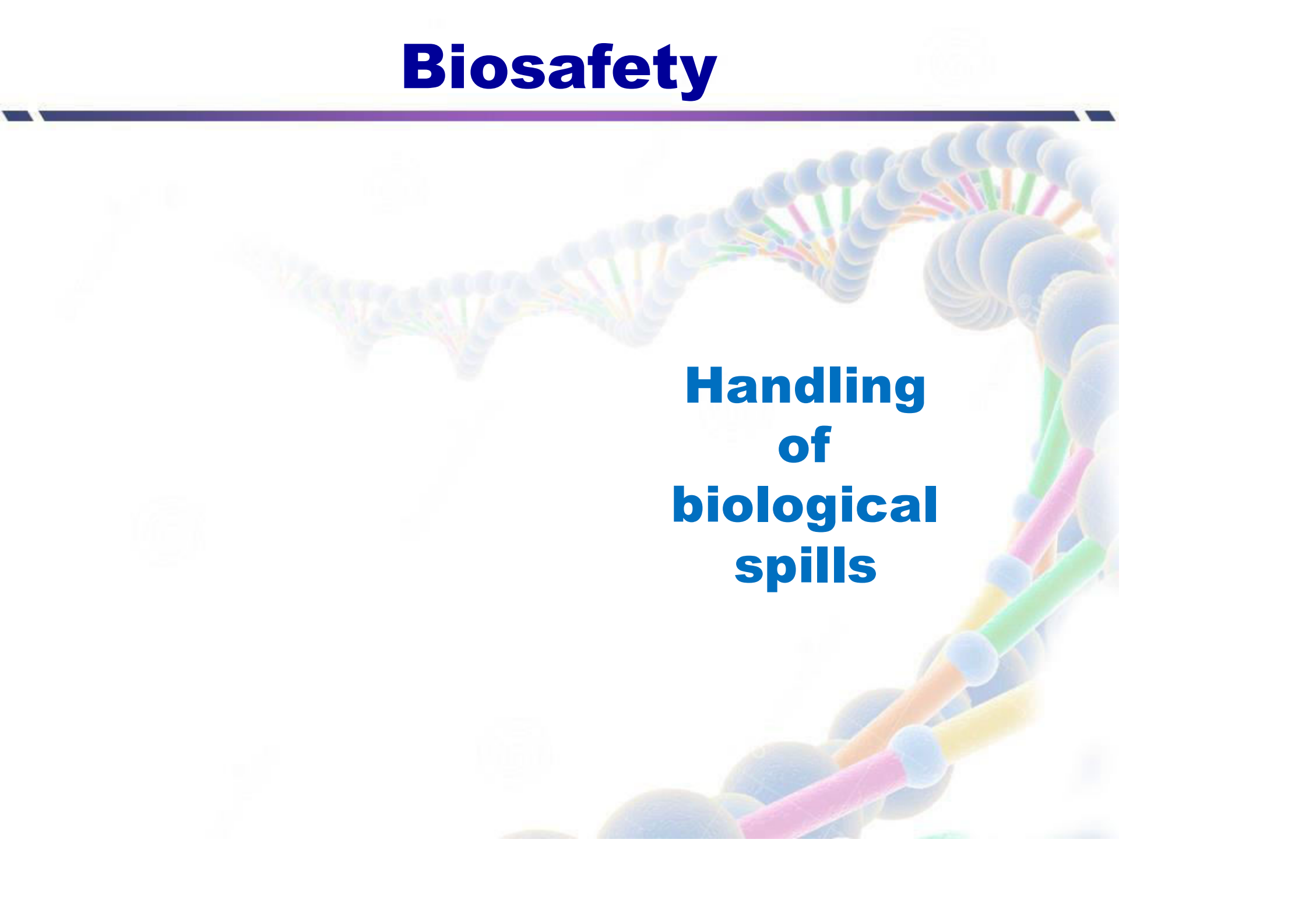
- **isolationism, minimizing America involvement**
- **friendly relationship**
- **rollback policy----an aggressive effort to undercut Soviet Union**

Biosafety

Lesson # 76



Biosafety



Handling of biological spills

Handling of biological spills

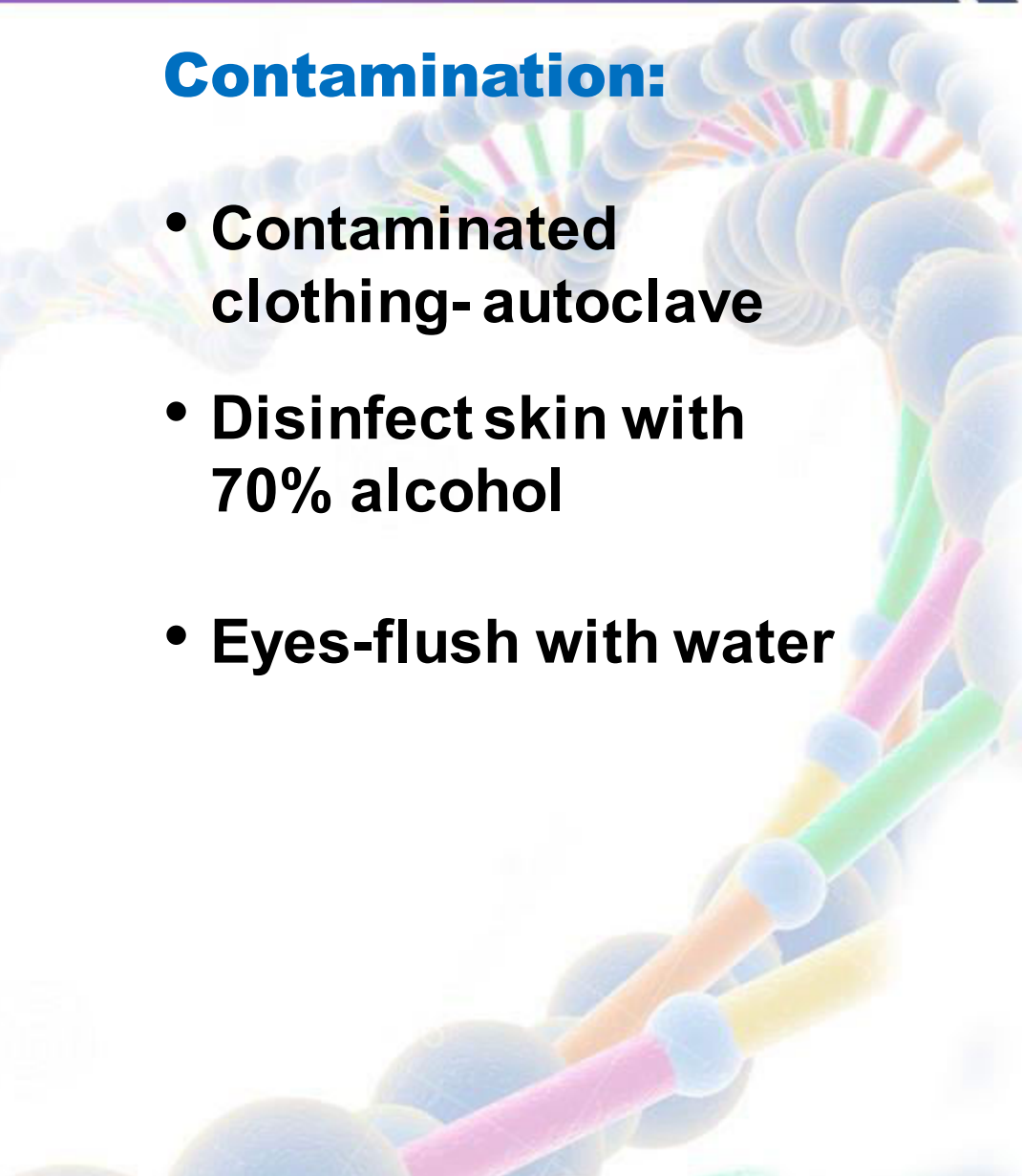
Basics:

- first worker injury-second-spill clean up
- alert others in the vicinity
- clean up and decontaminate
- dispose of clean up waste
- report incident-safety officer

Handling of biological spills

Contamination:

- Contaminated clothing- autoclave
- Disinfect skin with 70% alcohol
- Eyes-flush with water



Handling of biological spills

Biological spill kit:

- household bleach
- 70% alcohol
- spray bottle
- sterilization bags
- absorbent paper towels
- yellow trash bags
- disposable gloves

Handling of biological spills

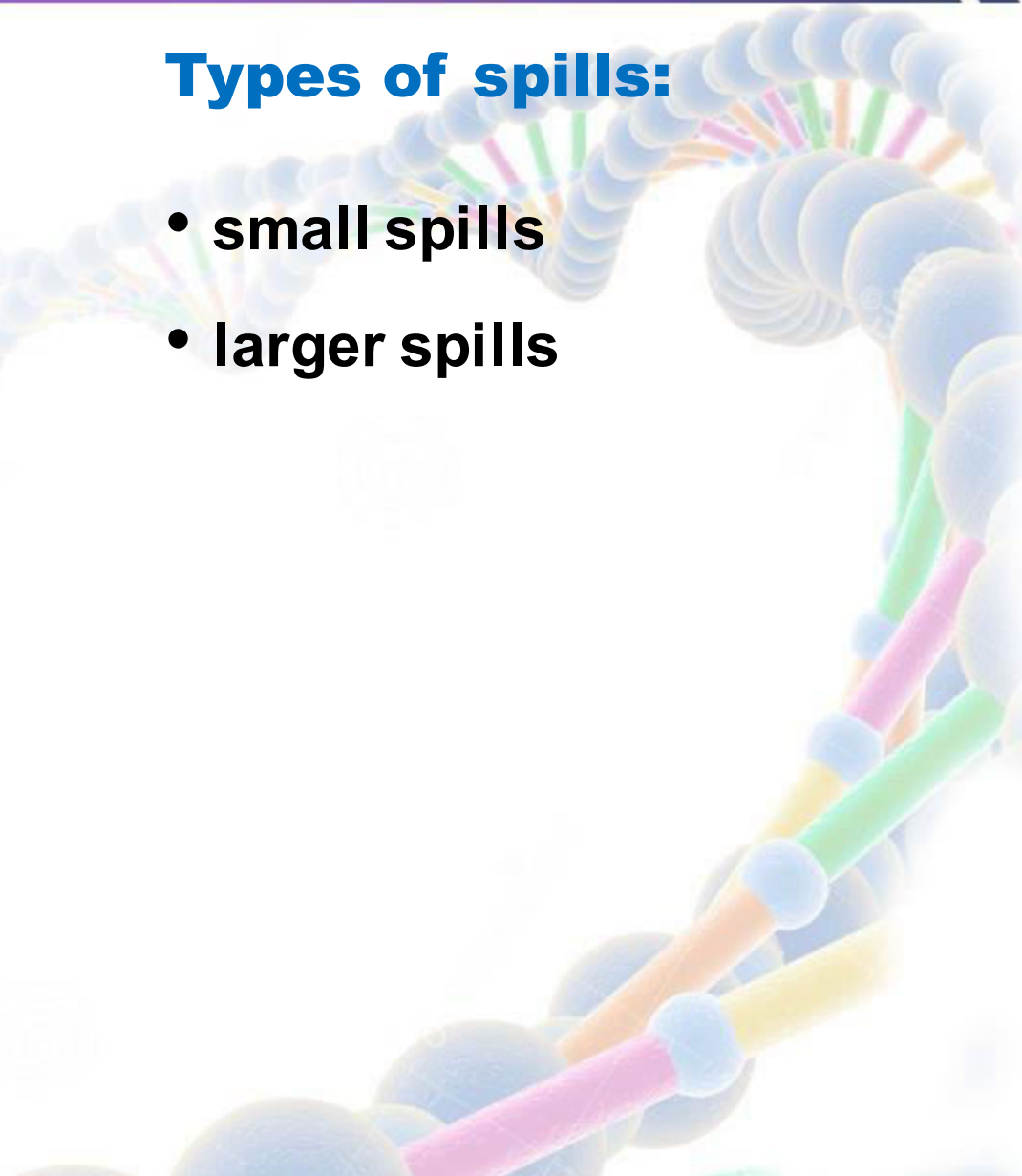
Biological spill kit:

- **metallic tongs**
- **surgical masks**
- **safety goggles, shoe covers and face mask**
- **spill control and cleanup procedures**

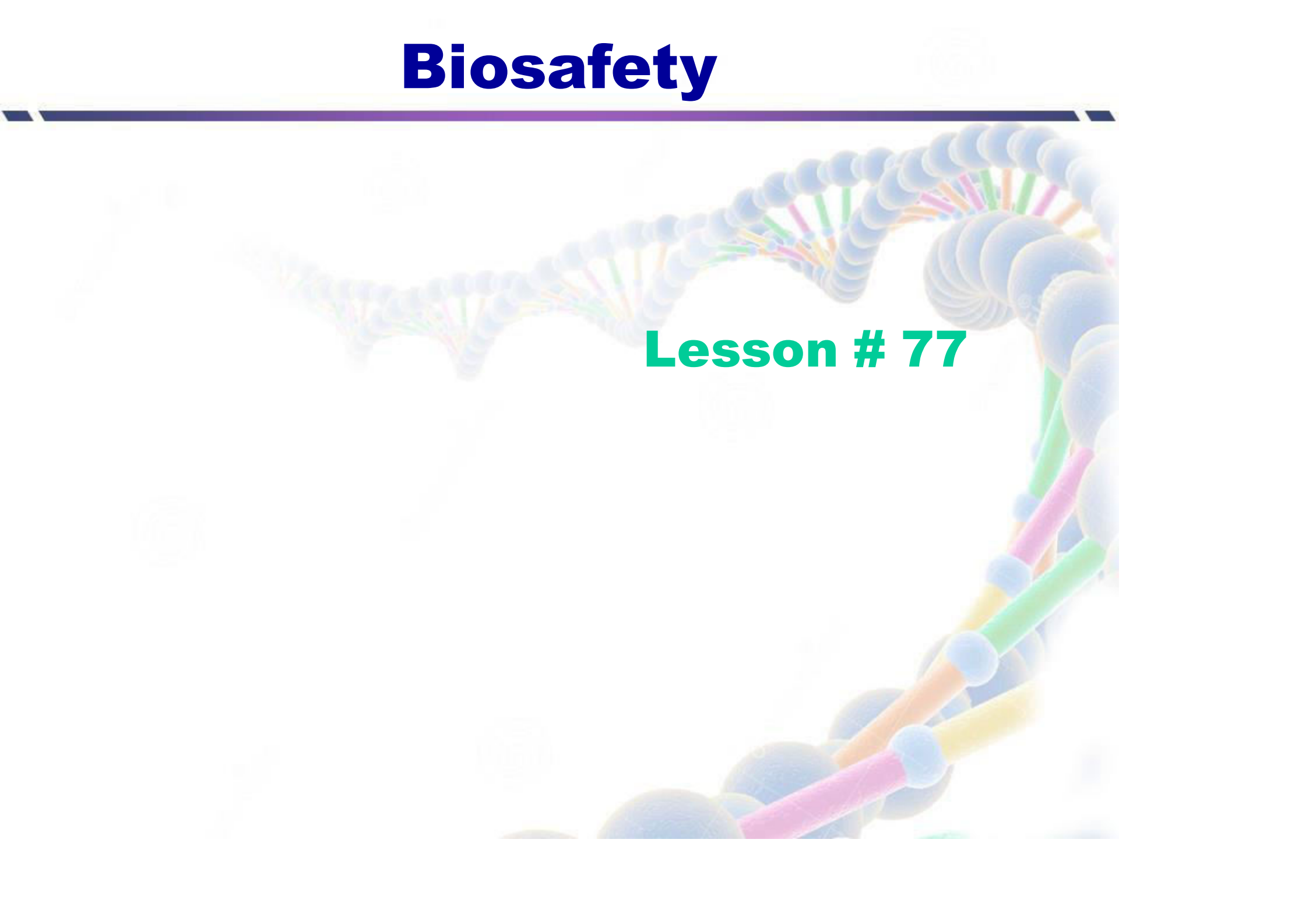
Handling of biological spills

Types of spills:

- small spills
- larger spills



Biosafety



Lesson # 77

Biosafety



Sterilization and disinfection

Sterilization and disinfection

Definition:

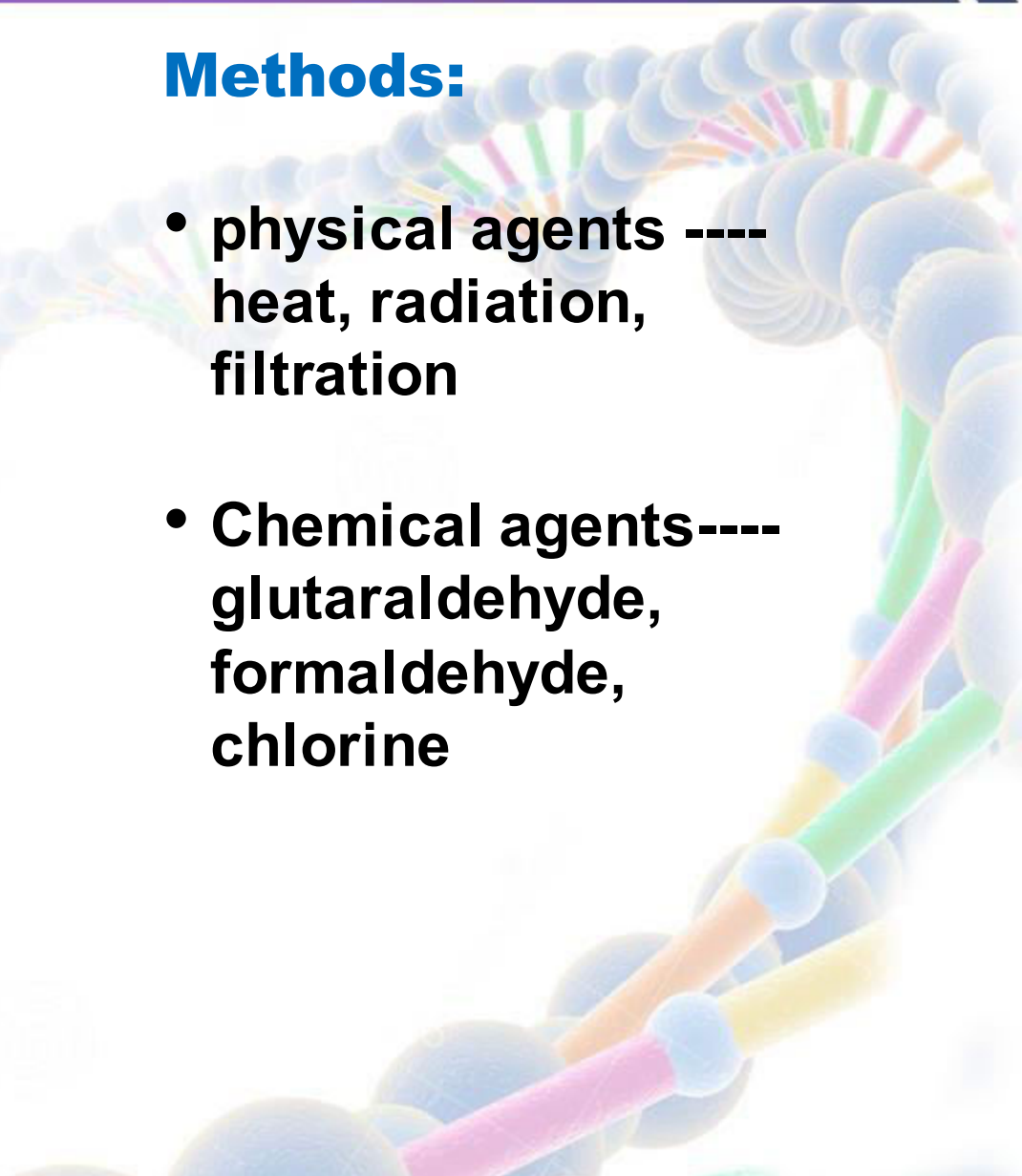
- **sterilization--removal of microbes including bacterial spores**
- **disinfection-killing many microbes**



Sterilization and disinfection

Methods:

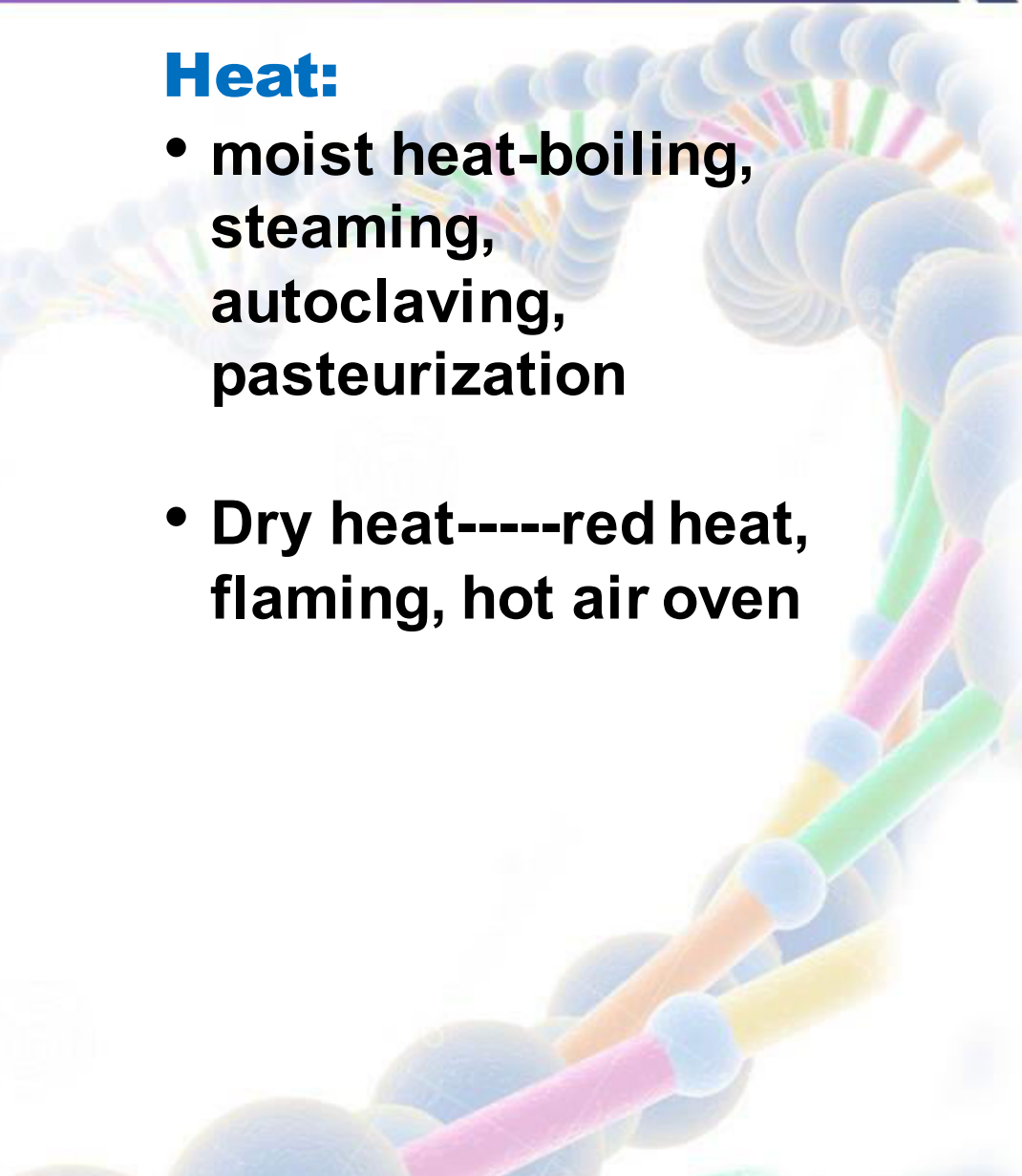
- **physical agents ----**
heat, radiation,
filtration
- **Chemical agents----**
glutaraldehyde,
formaldehyde,
chlorine



Sterilization and disinfection

Heat:

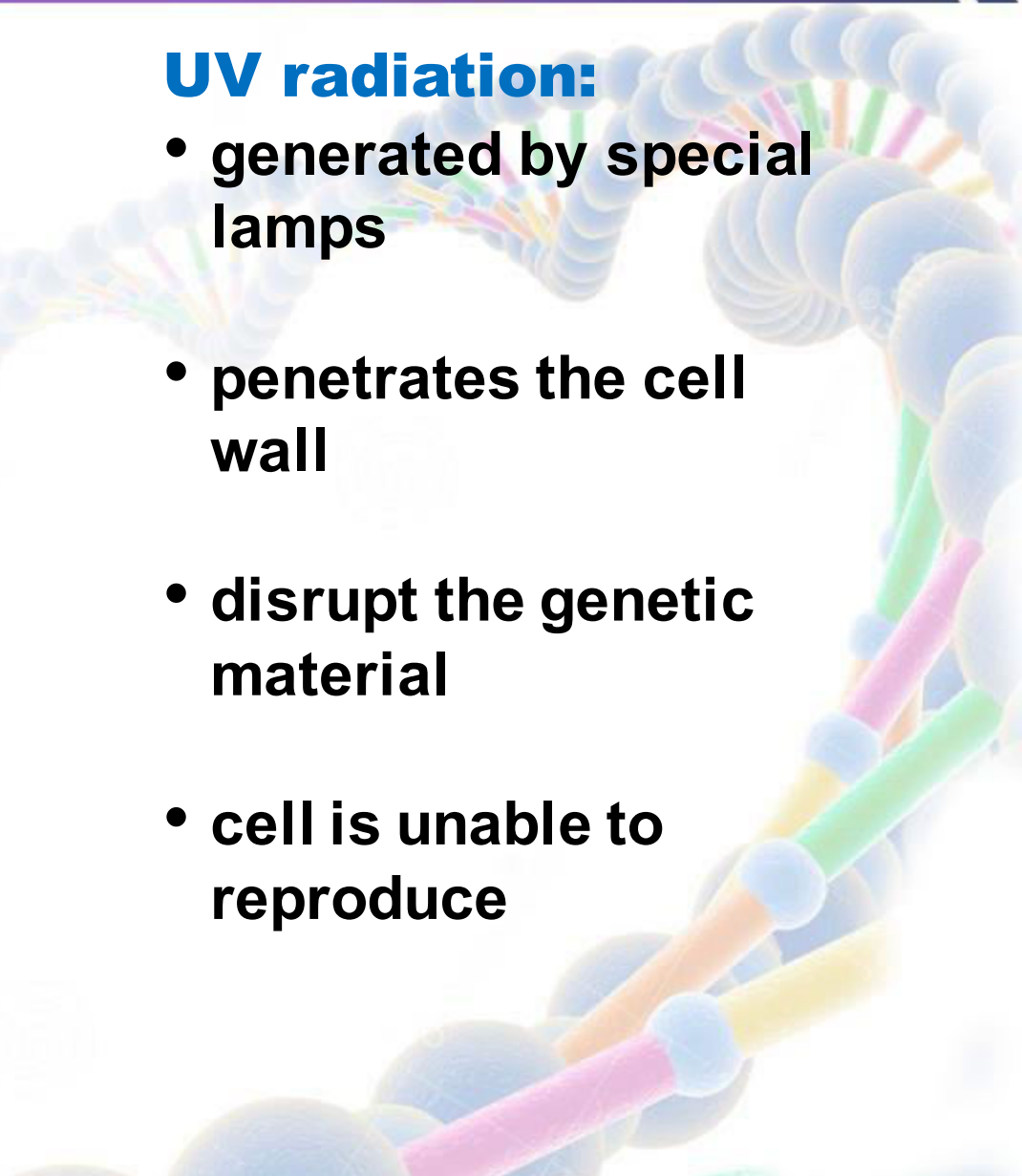
- moist heat-boiling, steaming, autoclaving, pasteurization
- Dry heat-----red heat, flaming, hot air oven



Sterilization and disinfection

UV radiation:

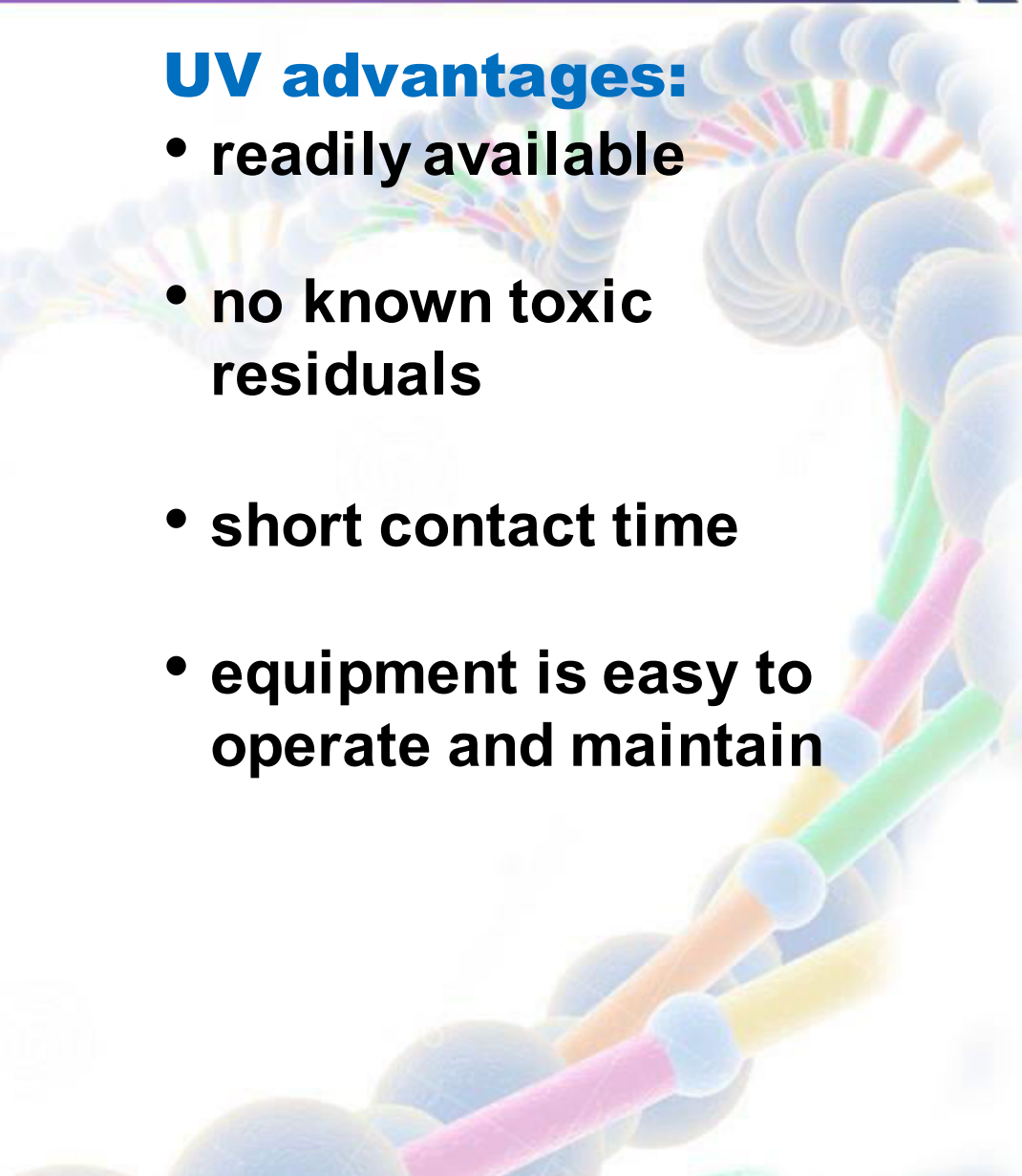
- generated by special lamps
- penetrates the cell wall
- disrupt the genetic material
- cell is unable to reproduce



Sterilization and disinfection

UV advantages:

- readily available
- no known toxic residuals
- short contact time
- equipment is easy to operate and maintain



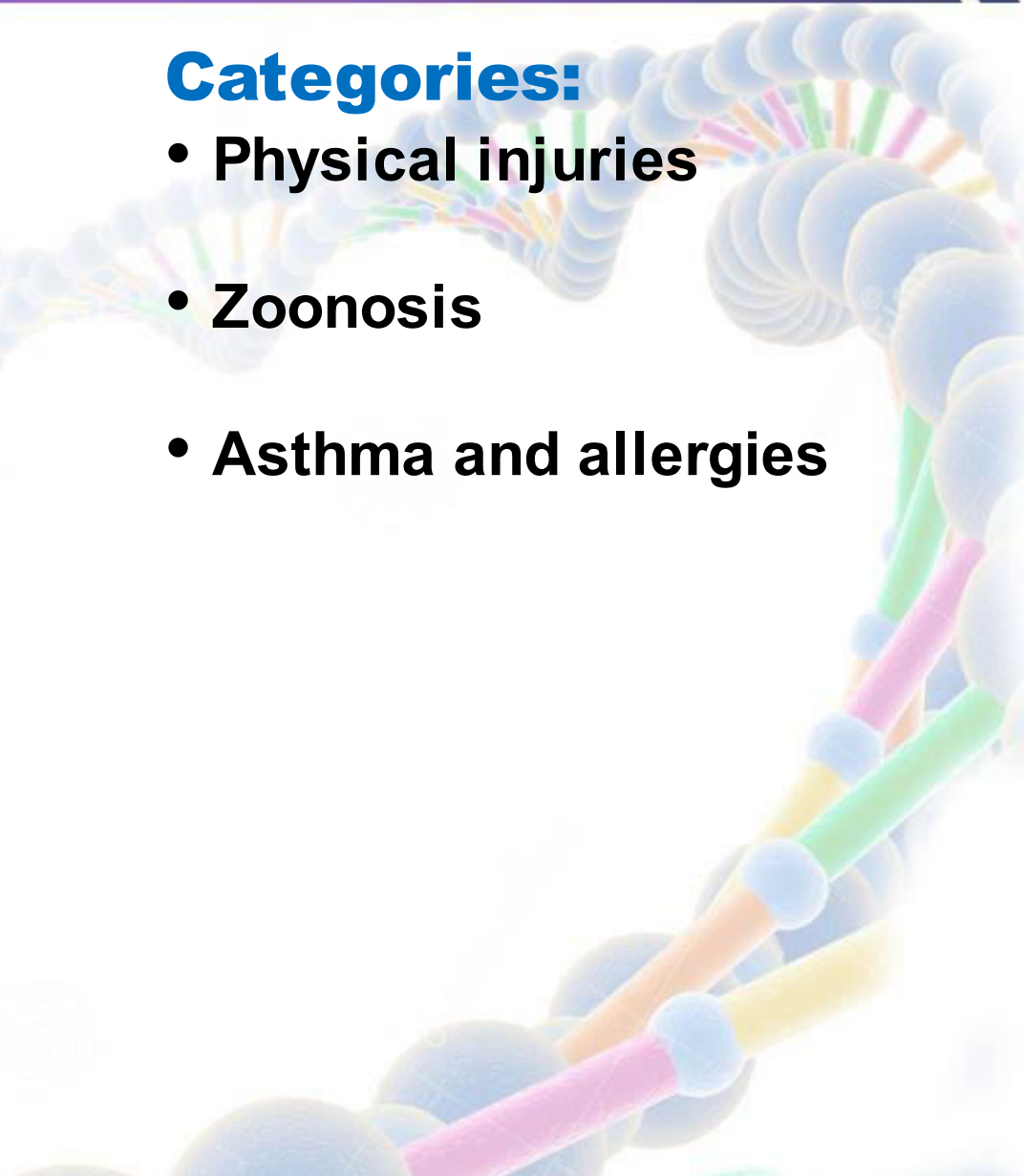
Biosafety

Lesson # 78
Biohazards-
animal
handling

Biohazards-animal handling

Categories:

- Physical injuries
- Zoonosis
- Asthma and allergies



Biosafety

Lesson # 79



Biosafety



**Handling-Lab
equipments**

Handling-Lab equipments

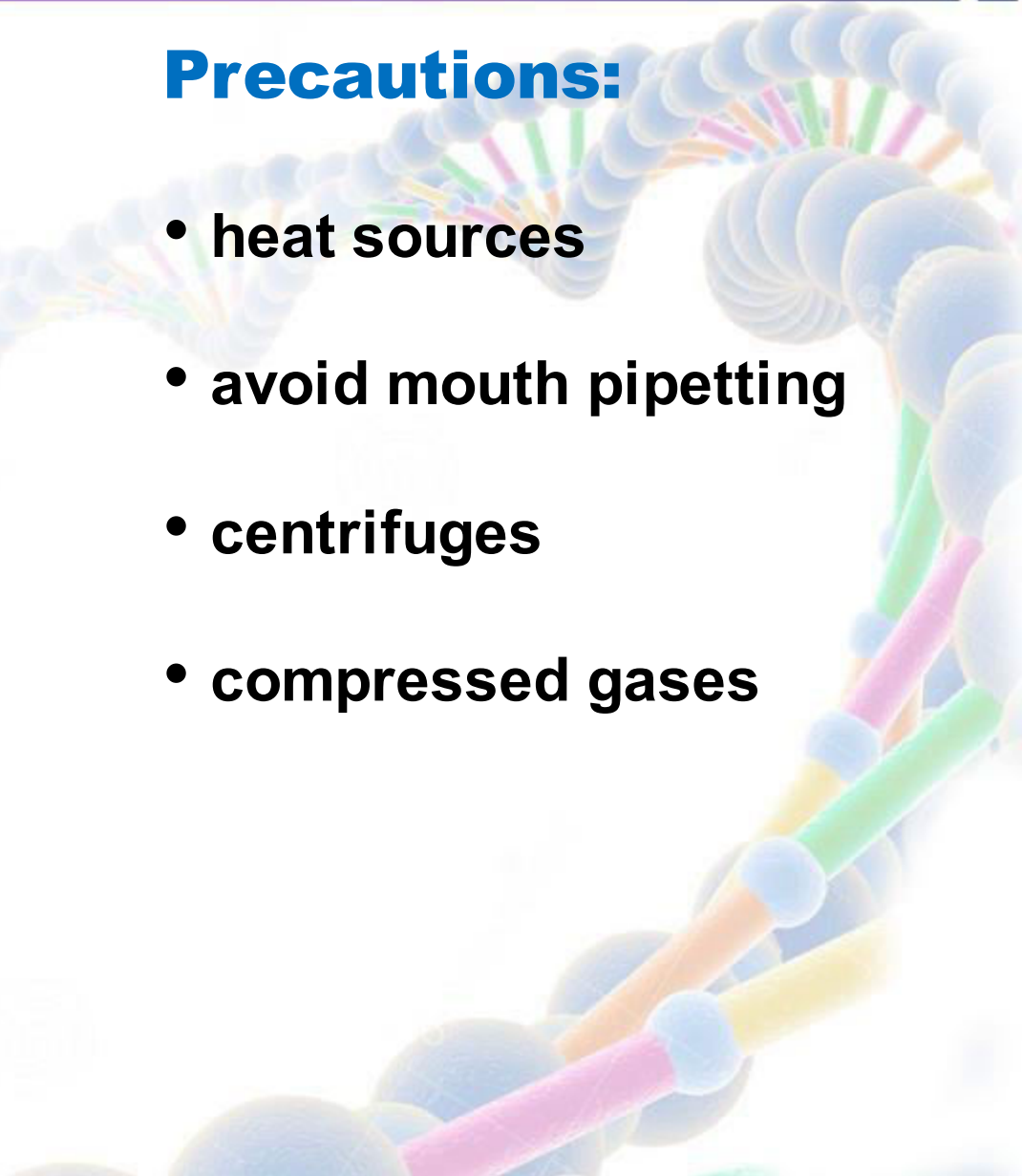
Glassware:

- borosilicate glassware
- corks for sealing organic solvents
- thermometer-stirring device
- thermometer bulb-heat

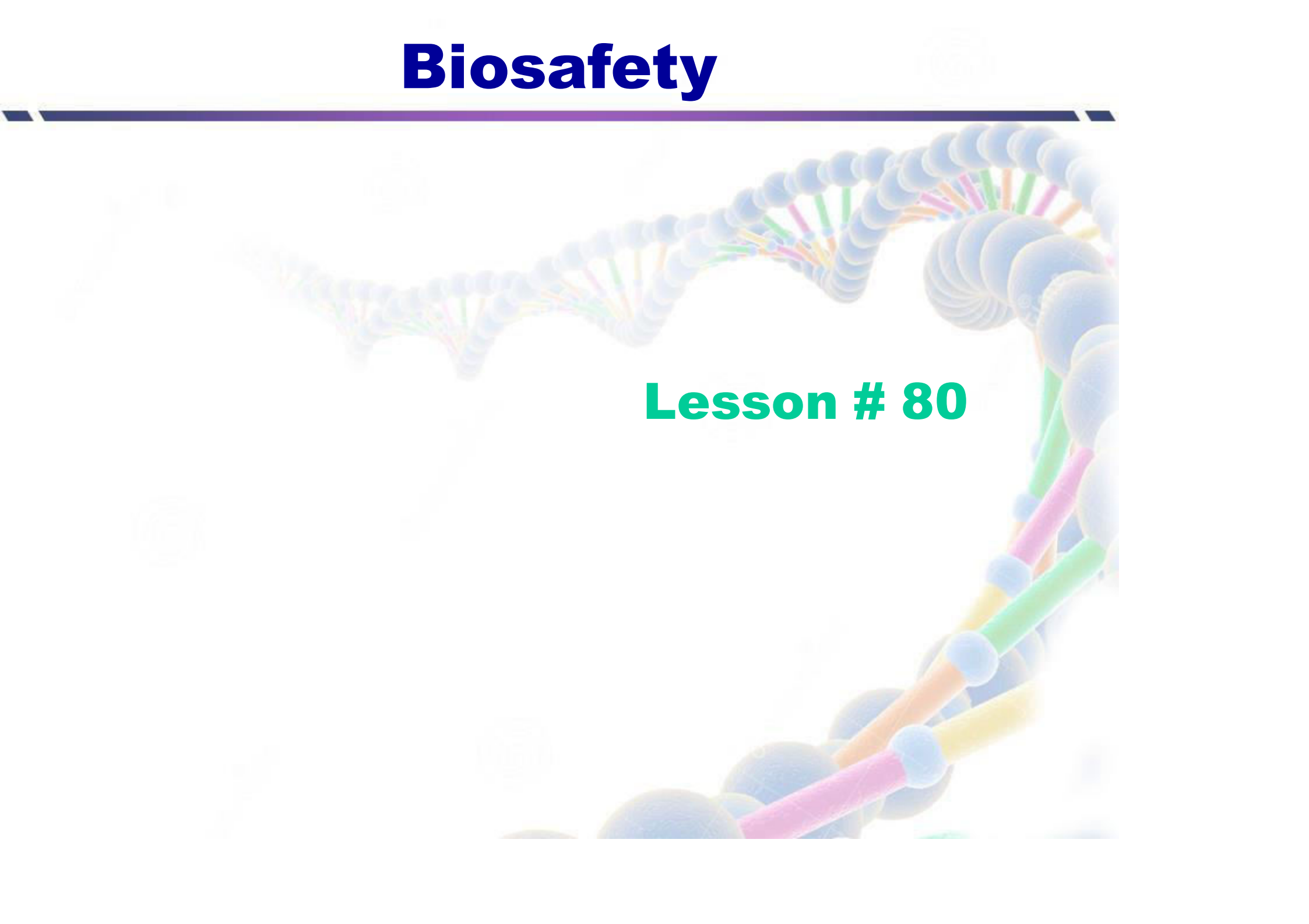
Handling-Lab equipments

Precautions:

- heat sources
- avoid mouth pipetting
- centrifuges
- compressed gases



Biosafety



Lesson # 80

Biosafety



Report of accidents

Report of accidents

Report:

- Accident report must be prepared-five days of accident
- What is accident?
- Person involved
- Witness to the accident

Report of accidents

Case study:

- investigator or supervisor
- multiple causes
- not intended to assign blame
- improve safety protocols

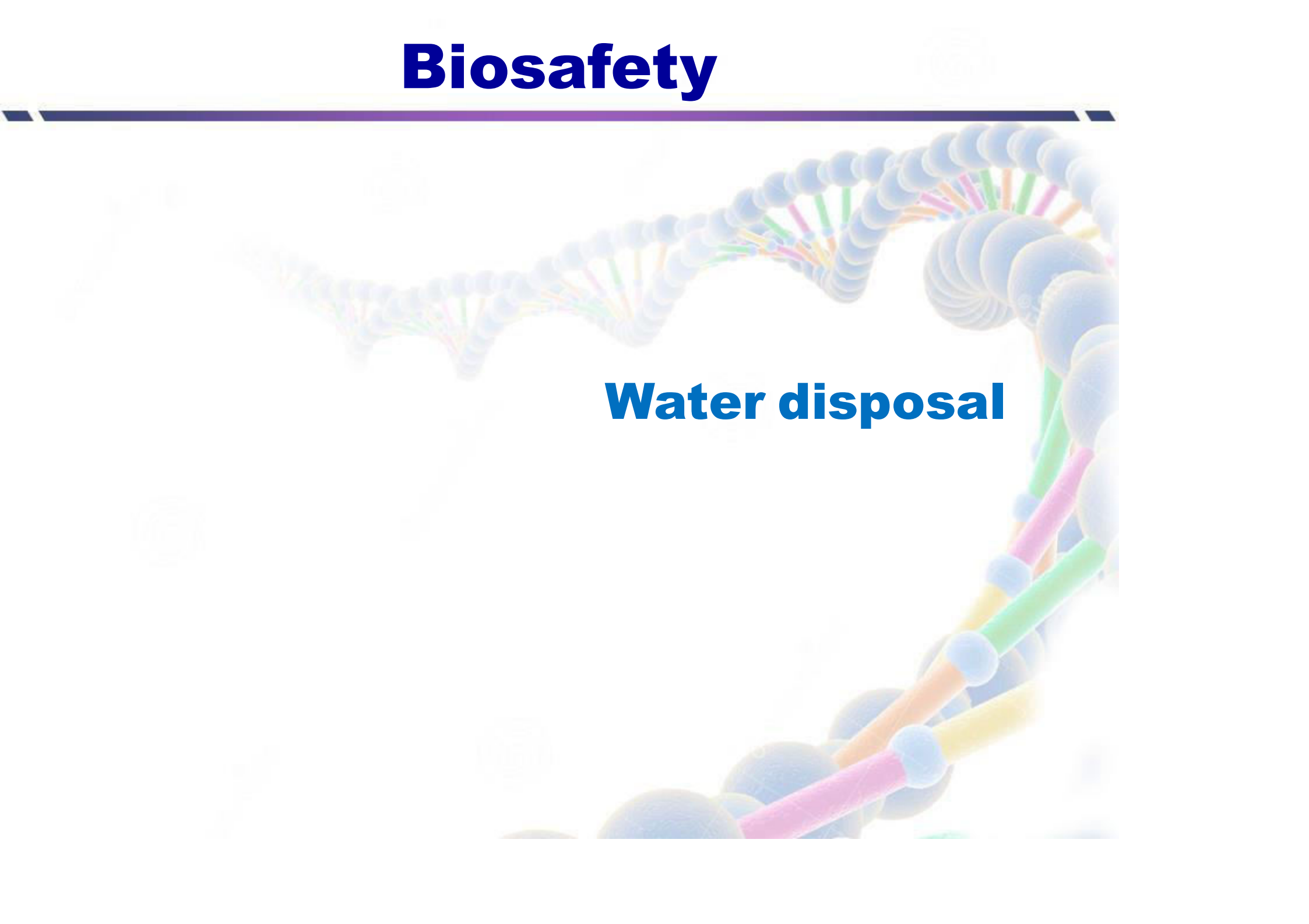
Biosafety

Lesson # 81



Biosafety

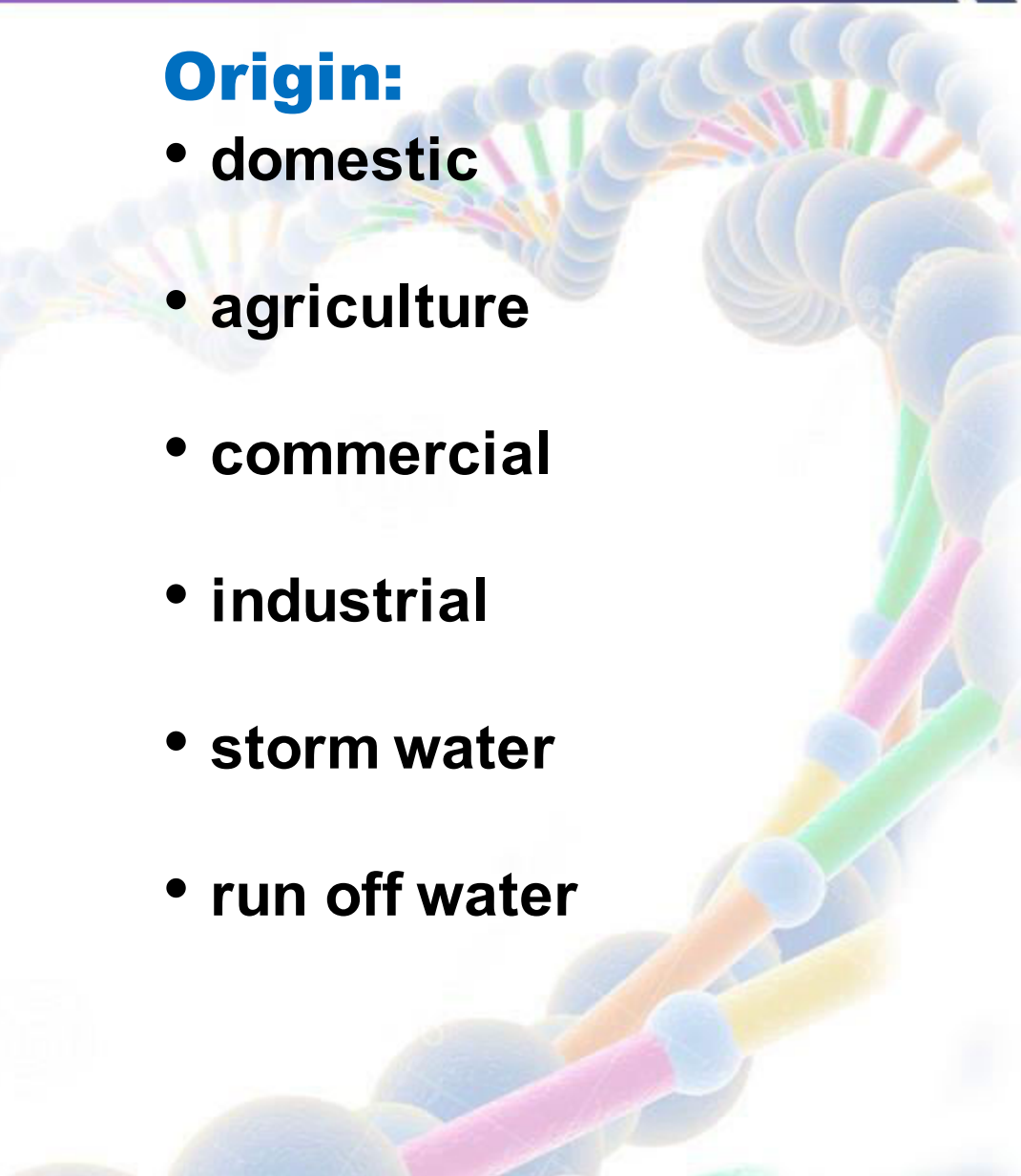
Water disposal



Water disposal

Origin:

- domestic
- agriculture
- commercial
- industrial
- storm water
- run off water



Water disposal

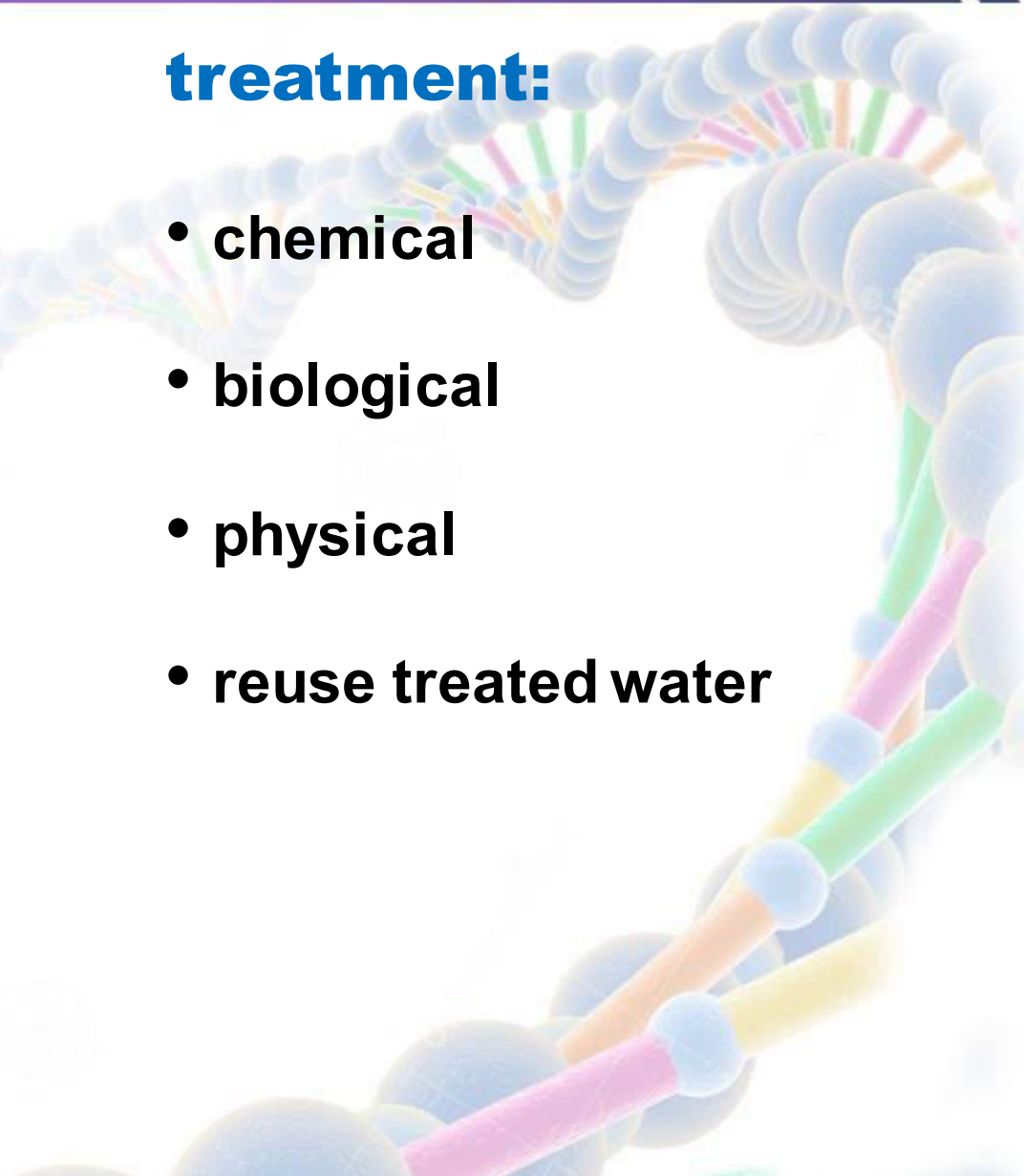
Constituents:

- pathogens
- non-pathogens
- organic/soluble
organic/inorganic
particles
- animals
- gases/emulsion/toxin

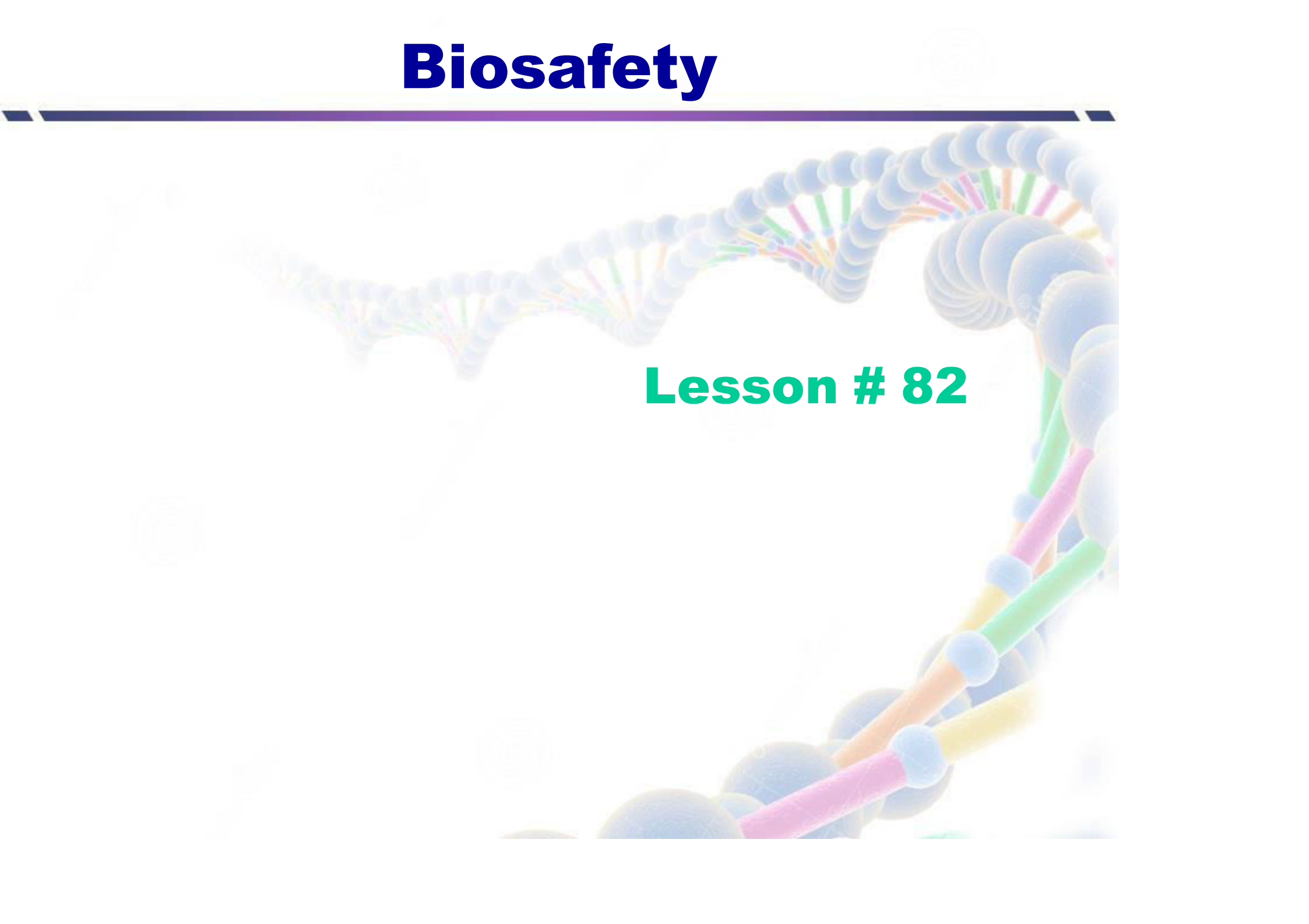
Water disposal

treatment:

- chemical
- biological
- physical
- reuse treated water



Biosafety



Lesson # 82

Biosafety



**Lab biosafety
level criteria**

Lab biosafety level criteria

Definition:

- level of containment precautions
- isolate dangerous biological agents
- enclosed laboratory facility
- containment level (BSL-1 to BSL-4)

Lab biosafety level criteria

History:

- **USA- CDC/European union-directives**
- **biosafety cabinet, 1943—Hubert**
- **biological warfare labs ----- 1955**
- **American Biological Safety Association -- 1984**

Lab biosafety level criteria

Containment zone:

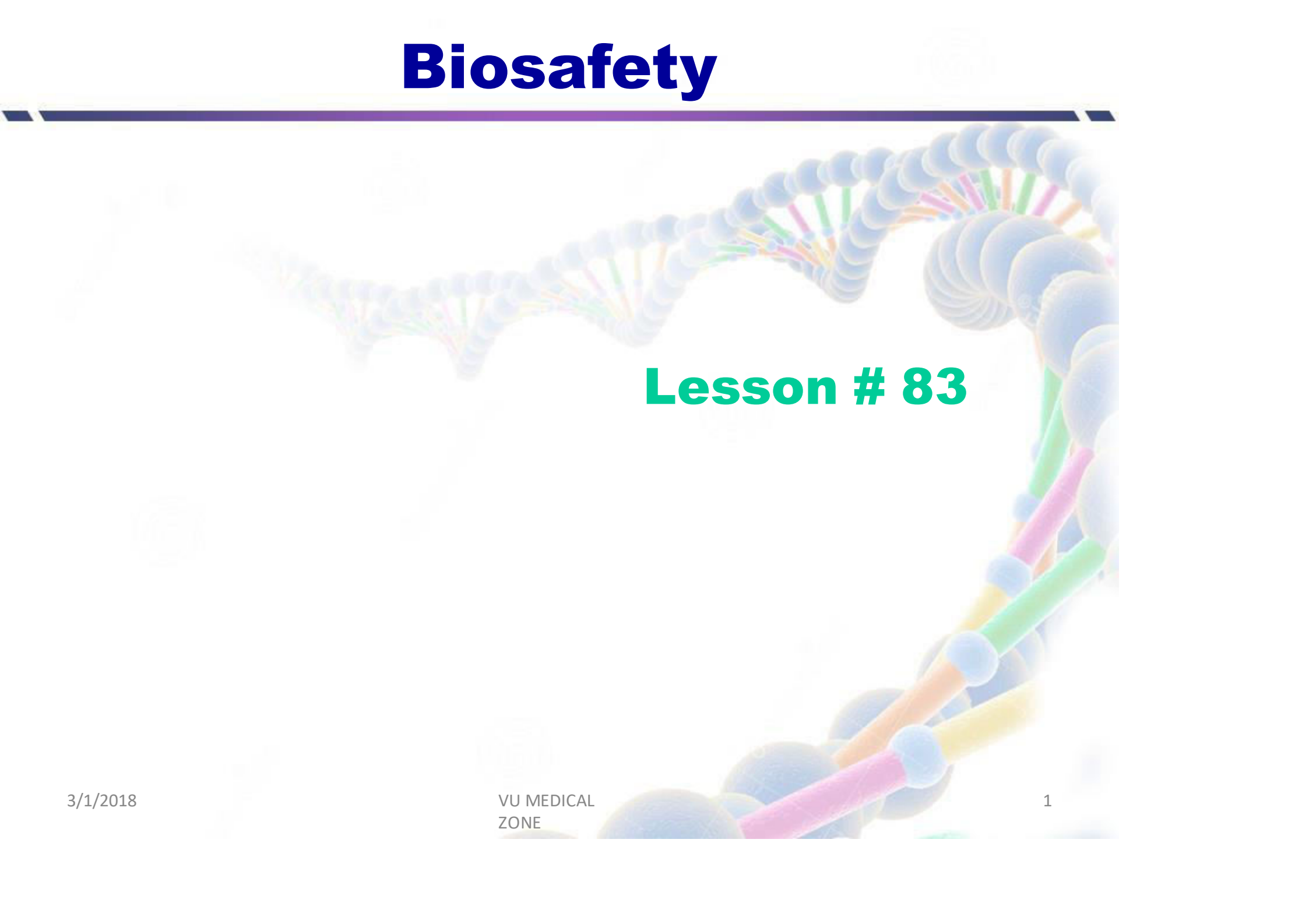
- only be a chemical fume hood
- isolation of microorganisms
- building systems, sealed rooms, sealed containers and personnel suits

Lab biosafety level criteria

Procedures:

- entering the room
- decontamination procedures for leaving the room
- high security
- “hot zone”

Biosafety



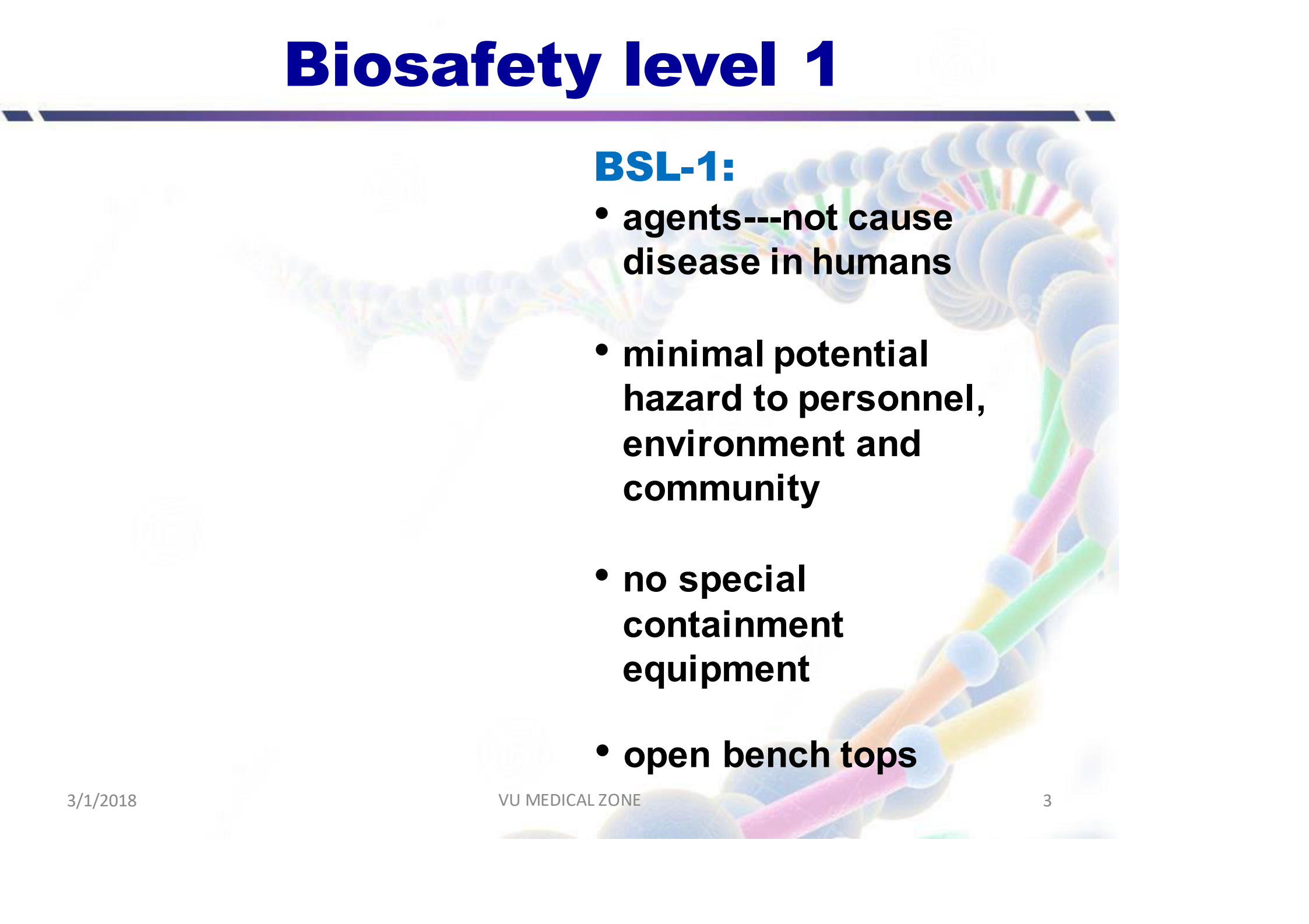
Lesson # 83

Biosafety



Biosafety level 1

Biosafety level 1



BSL-1:

- **agents---not cause disease in humans**
- **minimal potential hazard to personnel, environment and community**
- **no special containment equipment**
- **open bench tops**

Biosafety level 1

Microbiological practices:

- workers must be trained
- supervisor enforce institutional policies
- workers must wash their hands
- eating, drinking, smoking

Biosafety level 1

Microbiological practices:

- mouth pipetting is prohibited
- policies for the safe handling of sharps
- procedures to minimize aerosols and splashes
- decontaminate work places

Biosafety level 1

Microbiological practices:

- decontaminate cultures
- biohazard symbol
- pest management programs
- special practices not required

Biosafety level 1

Safety equipments:

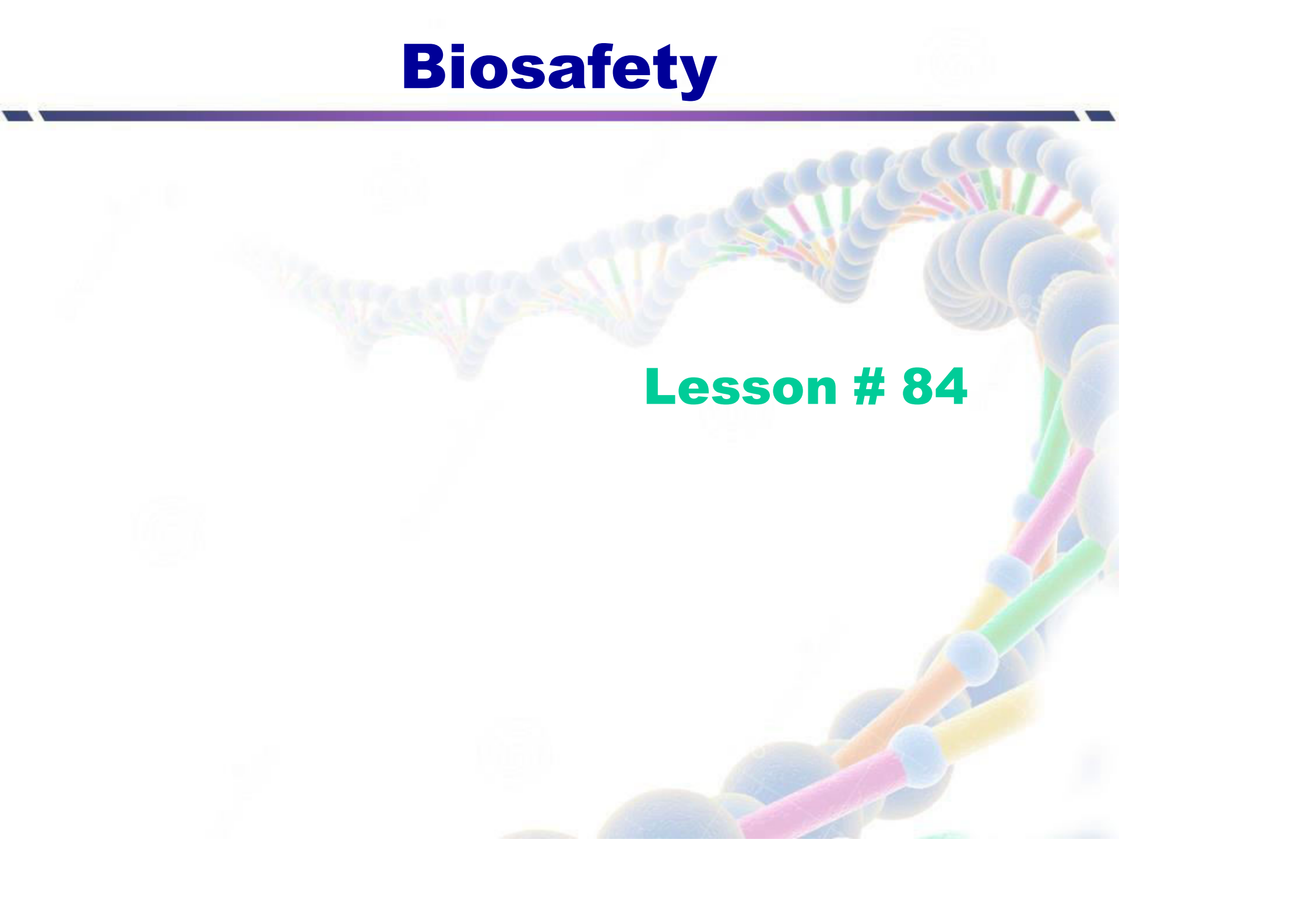
- **Gloves**
- **lab coats**
- **protective eyewear**

Biosafety level 1

Laboratory facilities:

- doors for access controls
- sink for hand washing
- bench tops-resistant
- chairs-easy to disinfect
- lab windows fitted with screens

Biosafety



Lesson # 84

Biosafety

**Biosafety level
2**



Biosafety level 2

BSL-2:

- moderate hazards to personnel and environment
- microbiological practices-same to BSL-1
- special equipments and practices required



Biosafety level 2

Special practices:

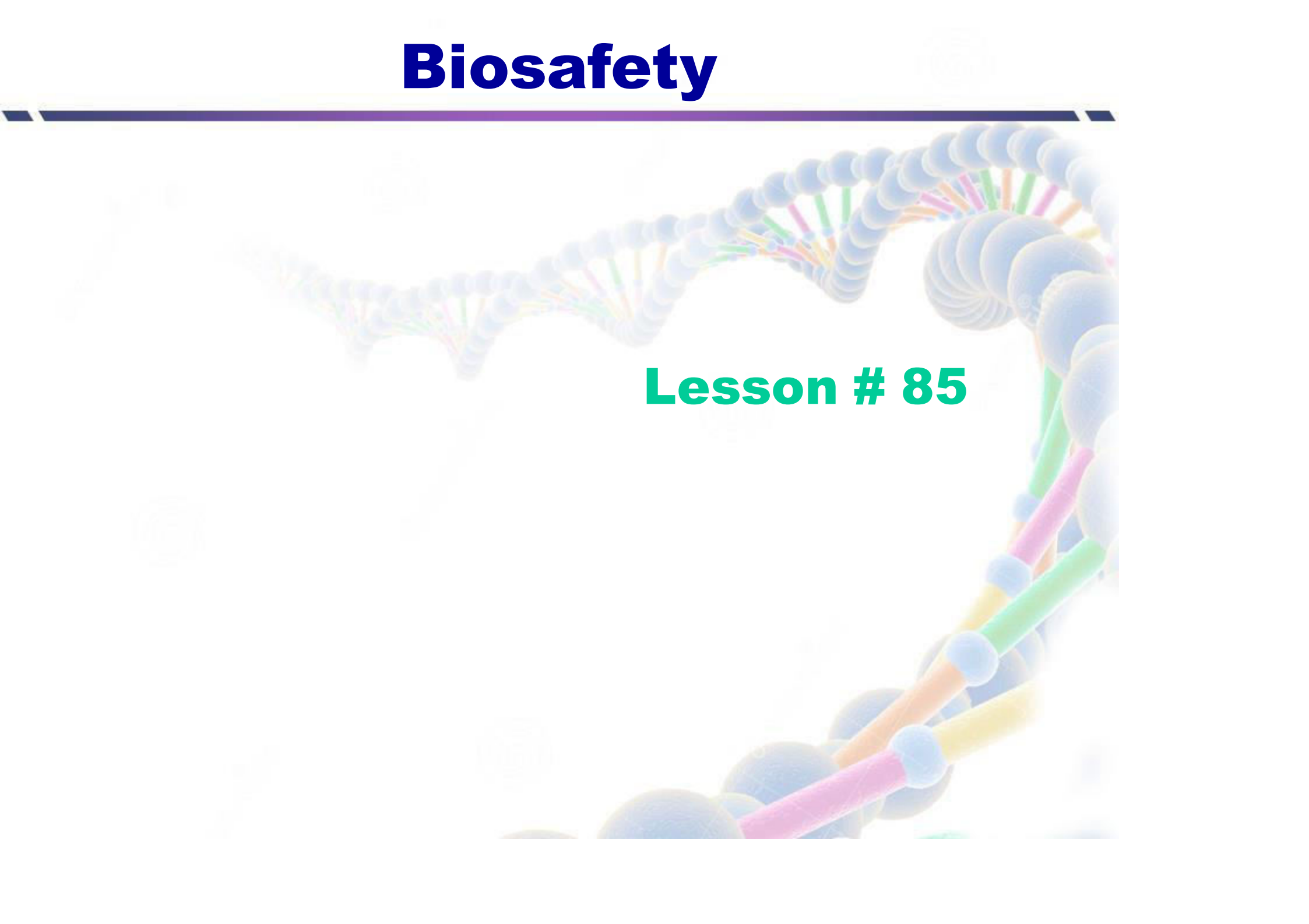
- **meet specific entry and exit requirements**
- **workers must be immunized**
- **biosafety manuals must be available**
- **proper collection, handling, processing, storage or transport**

Biosafety level 2

Special practices:

- lab equipments must be decontaminated
- incidents must be informed/eye, face, hand protection
- animals and plants should not be permitted in the lab
- aerosols-physical containment equipment

Biosafety



Lesson # 85

Biosafety

**Biosafety level
3**



Biosafety level 3

BSL-3:

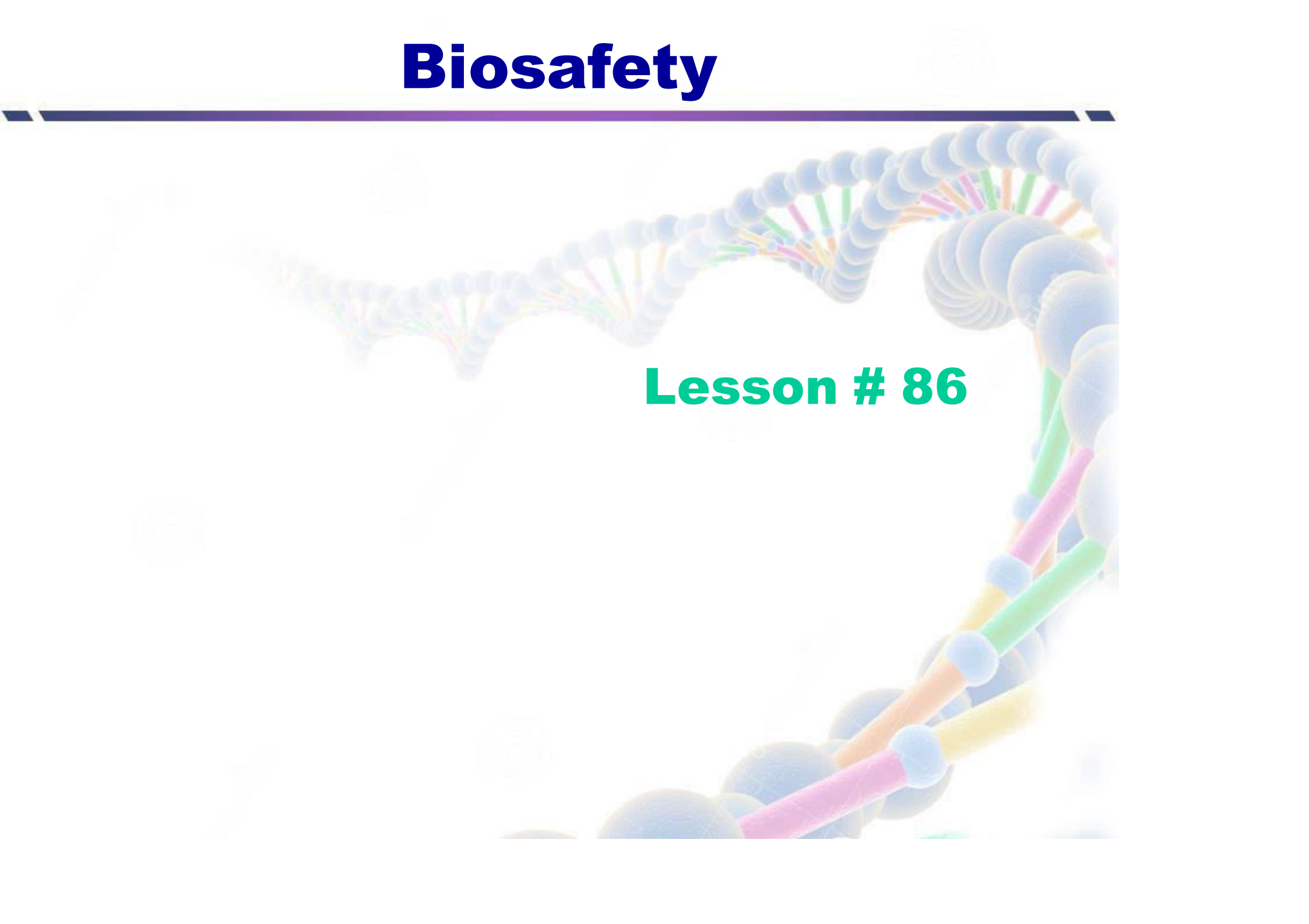
- applicable to diagnostic / clinical/ research/ production/ teaching facilities
- potentially lethal disease through the inhalation route
- all procedures must be performed in a biosafety cabinet

Biosafety level 3

Equipments:

- vacuum lines must be protected with HEPA filters
- ducted air ventilation system
- HEPA filter exhaust air
- BSL-3 facility design, operational/parameter/procedures must be documented

Biosafety



Lesson # 86

Biosafety

**Biosafety level
4**



Biosafety level 4

BSL-4:

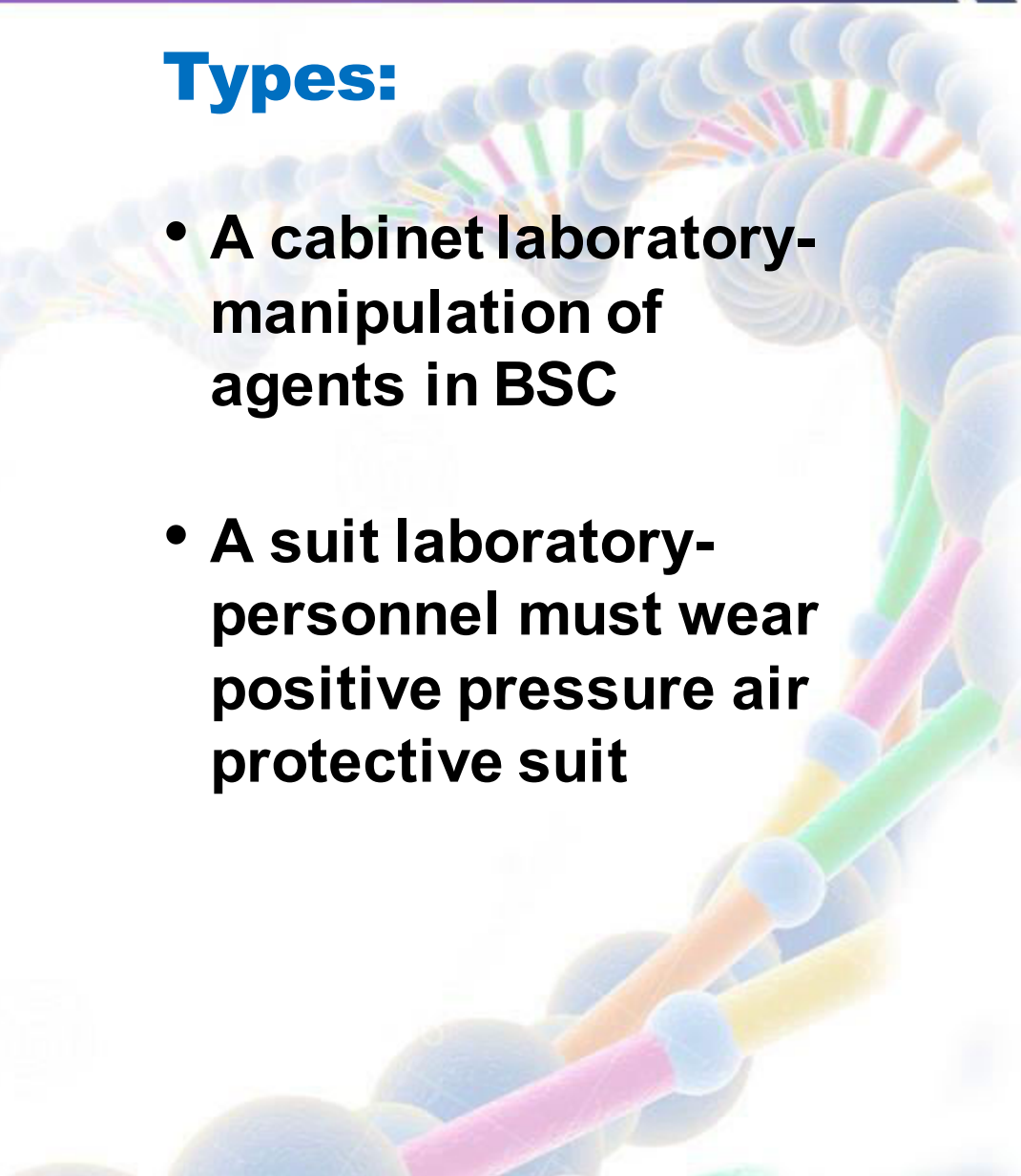
- **dangerous exotic agents**
- **aerosol transmitted lab infections**
- **no vaccine/treatment**
- **unknown risk of transmission**



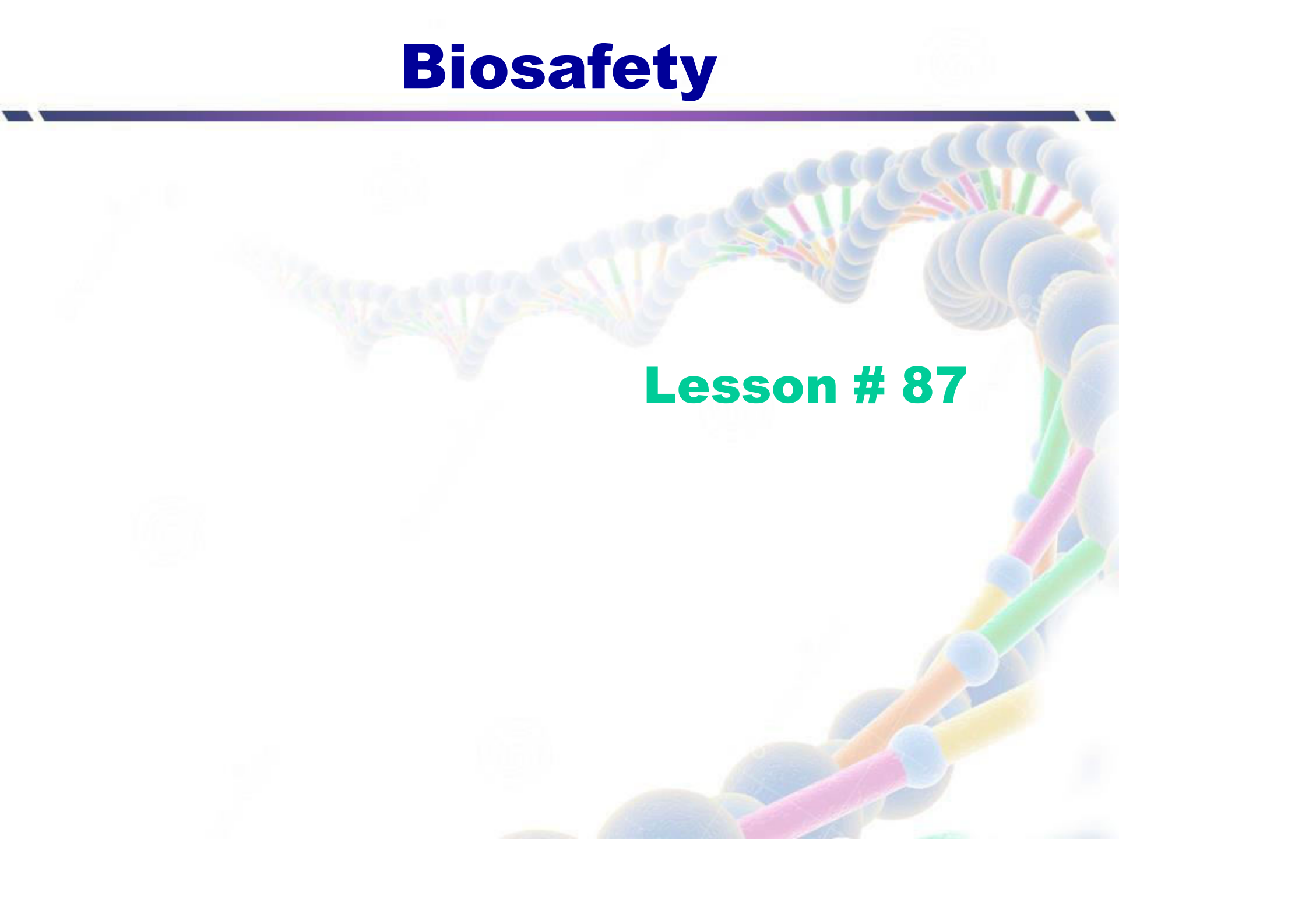
Biosafety level 4

Types:

- A cabinet laboratory-manipulation of agents in BSC
- A suit laboratory-personnel must wear positive pressure air protective suit



Biosafety



Lesson # 87

Biosafety



Biosafety measures for TB lab

Biosafety measures for TB lab

Biosafety measures:

- codes of practice
- equipment
- lab design and facilities
- health surveillance
- training
- waste handling

Biosafety measures for TB lab

Concepts:

- lab access
- responsibilities of lab manager
- personnel protective equipment
- work areas
- equipment

Biosafety measures for TB lab

Concepts:

- waste handling
- incineration
- autoclaving
- disinfection



Effort By

Amaan Khan